Deadjectival verb formation in Indo-European and beyond

Edited by

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Preface

Most of the contributions collected in this volume are based on papers presented at the workshop “Deadjectival verb formation in Indo-European: Historical and theoretical perspectives” held at the Institute of Iranian Studies of the Austrian Academy of Sciences in Vienna from March 10–11, 2023 (https://deadjectivals.wordpress.com/). We wish to express our gratitude to the colleagues who reviewed the abstracts for the workshop, namely Chiara Bozzone, Hannes Fellner, Peter Hallman, Andrew Koontz-Garboden, Marek Majer, Melanie Malzahn, Birgit Anette Olsen, Alexander Pfaff, Velizar Sadovski, David Sasseville, Michiel de Vaan, Michael Weiss, and Martina Werner. At the Institute of Iranian Studies, we wish to thank the institute’s director Florian Schwarz for providing the space and infrastructure to host this workshop and Bettina Hofleitner and Nurdan Şentürk for their support in organizing it.

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Chapter 1

Introduction: Deadjectival verb formation in Indo-European and beyond

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This volume presents a collection of papers discussing the morphological, syntactic and semantic properties of deadjectival verbs, defined here as verbs formed from property concept roots or from adjectival stems. Although the focus is on older Indo-European (IE) languages, the volume also includes insights from a theoretical perspective using data from modern IE as well as non-IE languages. This chapter serves as an introduction to the volume and provides a precise definition of deadjectival verb formation and an overview over the current state of the art both from the perspective of general and of historical linguistics. We begin with a discussion of different strategies for lexicalizing property concepts, which cross-linguistically constitute the core of the morphosyntactic class "adjective" and various diachronic developments this class can undergo, moving on to describing two types of deadjectival verb formation (root- vs. stem-derivation) and their differentiation from non-deadjectival change-of-state verbs. Subsequently, we introduce the "Caland system" and its implications for the reconstruction of property concept adjectives and the verbal system of the Indo-European proto-language, followed by an outline of recent advances and open questions in the reconstruction of verbal derivational morphology. We conclude with a short summary of each of the respective chapters collected in this volume, a general conclusion, and an outline of open issues.

1 Introduction

1.1 Background: Adjective classes

The papers collected in this volume discuss the morphology, syntax, and semantics of deadjectival verbs and related formations, with a focus on the older Indo-European (IE) languages. The goal of this chapter is to provide a definition and

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overview of the current research on deadjectival verbs and the expression of property concepts in general, with a focus on theoretical morphosemantic studies on the one hand and recent advances in the reconstruction of property concept-associated lexical classes in Indo-European on the other. Ultimately, our goal is to show that while these two strands of literature are often blissfully ignorant of one another, each could profit from a serious engagement with the evidence discussed in the other and the conclusions drawn from this evidence – especially since these conclusions converge more often than not. We also point out some promising avenues for future research in the domain of Indo-European derivational morphology and morphological reconstruction more generally.

Before discussing deadjectival verbs, we need to give a definition of what we mean, which in turn hinges on the definition of "adjective". The literature on this enigmatic lexical category is vast and we cannot do it justice here (see Fábregas & Marín 2017; Panagiotidis & Mitrović 2022; Beck 2023 for recent overviews). For our purposes, we focus on the adjectival classes in ex. (1), which pick out a syntactically and morphologically clearly defined class in the older Indo-European languages: syntactically, members of this class are prototypical noun modifiers (Croft 1991; Alfieri 2014) and morphologically, they are formed with specifically class-defining (adjectival) stem-forming morphology and agree with their head noun.[1] We illustrate these with examples from English and selected older IE languages; note that most of the adjectival suffixes in these examples can be used in more than one class.

(1) Three types of adjectives in Indo-European
  a. Property concept adjectives[2] (e.g., Engl. *tall*, *smart*, *red*, *soft*):
    i. Vedic Sanskrit: *náva-* 'new', *gurú-* 'heavy', *bṛhánt-* 'high, great', *pṛthú-* 'broad', etc.
    ii. Ancient Greek: *kratús* 'strong', *néos* 'young, new', *eruthrós* 'red', etc.
    iii. Latin: *gravis* 'heavy', *clārus* 'clear, bright', *ruber* 'red', etc.
  b. Relational (possessive, genitival, etc.) adjectives (e.g., Engl. *wood-en*, *water-y*, *father-ly*):
    i. Vedic Sanskrit: *aśv-ín-* 'having horses', *āyas-á-* '(made) of metal', *paśu-mát-* 'consisting of cattle', etc.

[1] We cannot discuss other adjectival categories such as participles here for reasons of space; on participles in Indo-European see, e.g., Lowe 2015; Fellner & Grestenberger 2018; Grestenberger 2020.

[2] Also called qualitative (or qualifying) adjectives in the literature.

   ii. Ancient Greek: *khrū́s-eos* ‘golden’, *odunē-rós* ‘painful’, *rhodó-eis* ‘rosy; of roses’, etc.
   iii. Latin: *barbā-tus* ‘bearded’, *patr-ius* ‘of the father, fatherly’, *argent-eus* ‘silvery, of silver’, *mar-īnus* ‘of the sea, marine’, etc.

c. (De)verbal adjectives (e.g., Engl. *wash-able*, *break-able*):
   i. Vedic Sanskrit: *kām-ín-* ‘desiring’, *cá-kr-i-* ‘making’, *krīḍ-ú-* ‘playing’, etc.
   ii. Ancient Greek: *tom-ós* ‘cutting; sharp’, *do-tós* ‘granted, given’, *main-ás* ‘raving’, etc.
   iii. Latin: *trem-ulus* ‘trembling’, *fīd-us* ‘trusting’, *aud-āx* ‘daring’, *amā-bilis* ‘lovable’, etc.

The classes in (1b–1c) do not usually serve as the input to deadjectival verb formation (though see Sections 2.2 and 3.4 for some circumscribed exceptions to this generalization), so we focus on (1) in the following.

1.2 The semantics & lexicalization of property concepts

The core problem of the definition of “adjective” (and hence “deadjectival”) centers on the nature of property concepts (PCs) as defined by Dixon (1982) and illustrated in Table 1.

Table 1: Dixon’s property concepts (Dixon 1982: 16; Rau 2009a: 79)

DIMENSION	big, small, long, tall, short, etc.
PHYSICAL PROPERTY	hard, soft, heavy, wet, rough, strong, etc.
COLORS	black, white, red, etc.
SPEED	fast, quick, slow, etc.
AGE	new, young, old, etc.
VALUE	good, bad, lovely, atrocious, perfect, proper, etc.
HUMAN PROPENSITY	jealous, happy, kind, clever, generous, cruel, etc.

As is well known from the typological literature (Dixon 1982, 2004; Thompson 1989, Alfieri 2014, a.m.o.), the lexicalization of PCs differs cross-linguistically in terms of category: Some languages express these concepts as “primary” (root-derived or monomorphemic) adjectives, as illustrated in (2a), while others use a “verbal” strategy, that is, they lexicalize these roots in the same way as eventive ones, namely with verbal (including subject agreement) morphology, cf. (2b). A third strategy that has received less attention but which is highly relevant for the problem of PCs in Indo-European is the lexicalization of PCs as nouns, illustrated in (2c). That "quiet" in (2c) is a noun is suggested by the fact that it takes its own nominal class marker (class 5) rather than agreeing with the nominal class of the subject (class 19).

(2) Cross-linguistic variation in the lexicalization of PCs (Hanink & Koontz-Garboden 2025: 1)

a. I'm **happy**.

b. ma:-č m-**ahan**-kəm
   2-SBJ 2-**be.good**-INCMPL
   'You're good.' — Yavapai (Yuman), Kendall 1975: 90

c. hí-nuní híí hí yé li-**mugɛ̂**
   19-bird 19.that 19.SUB be 5-**quiet**
   'That bird is quiet.' — Basaá (Bantu), Jenks et al. 2018: 650

Cross-cutting the distinction in terms of lexical category is the difference between what Francez & Koontz-Garboden (2015, 2017) term the "predicative" and the "possessive" strategy: While the predicative strategy makes use of canonical predication devices available in a given language (such as the copula), (3a), the possessive strategy uses possessive morphology to express property concepts, (3b).

(3) Predicative vs. possessive lexicalization of property concepts (Francez & Koontz-Garboden 2017: 22)

a. Pierre is hungry. (English)

b. Pierre a faim. (French)
   Pierre has hunger

Based on this observation, Francez and Koontz-Garboden formulate the generalization that property concept lexemes that use possessive predication denote qualities while property concept lexemes that use non-possessive predication characterize individuals, as in (3a) – The Lexical Semantic Variation Hypothesis (Francez & Koontz-Garboden 2017: 77; but see below for an important diachronic qualification of this generalization). They derive this generalization from the differing denotation of property concepts depending on whether they are lexicalized as nouns or as adjectives: While nouns can denote either qualities or characterize individuals, adjectives can only characterize individuals, but cannot denote qualities. Concretely, this means that (4a) can never have the same meaning as (4b), namely that of expressing the modification of the denotation of the noun.[3] In the same vein, the hypothetical deadjectival verb in (4c) can be individual-characterizing like (4b) meaning "Indra is/becomes strong", but not quality-denoting (Francez & Koontz-Garboden 2017: 83–84). We discuss some equivalents of (4c) in Section 3.3.

(4) a. Indra is strength
  b. Indra is strong
  c. Indra strong-s

While (3b) as such is not used in the older IE languages (though see the discussion of (5) below), (3a = 4b) is amply attested, and with morphology that is actually called "possessive" because it is also used to form denominal relational adjectives from substantives in the older IE languages (though not all languages preserve both functions), e.g., (proterokinetic) *-*u*-, *-*(e/o)nt*-, *-*u̯ont*-, *-*ro*-, *-*mo*-, *-*no*-, etc. (cf. Rau 2009a: 77–78). Examples for the denominal-possessive vs. the property concept root-derived use of these are given in Table 2.

Table 2: Possessive and PC adjectives in Indo-European

Suffix	a. Possessive adj.		b. PC adjective	
-(e/o)nt*-	Hitt.	*perun-**ant**-* 'rocky'	Ved.	*bṛh-**ánt**-* 'high'
base:		*peruna-* 'rock'		√*bṛh* 'high'
*-*ro*-	Ved.	*aṃhu-**rá**-* 'constricted'	AG	*eruth-**ró**-s* 'red'
base:		*aṃhu-* 'constriction'		√*ereuth-*/*eruth-* 'red'
*-*u*-	Ved.	*āy-**ú**-* 'lively'	OAv.	*pərəθ-**u**-* 'broad'
base:		*ā́y-u-* n. 'life (force)'		√*pərəθ* 'broad'
*-*o*-	Ved.	*pīvas-**á**-* '(having) fat'	Ved.	*śvet-**á**-* 'bright'
base:		*pī́vas-* 'fat' n.		√*śvit* 'bright'

In terms of diachrony, this functional overlap could be taken to suggest that the PC adjectives in column b. were originally *denominal* adjectives derived from quality-denoting nouns. Hence one way of creating individual-characterizing adjectives like (4b) is by adding possessive morphology to the quality nominal, (5a),

[3] It *can* mean "Indra represents strength" or something similar, of course.

or morphology with broadly "instrumental" meaning (e.g., prepositions meaning "with" or, in the case of the older IE languages, instrumental case endings, cf. Section 3.2).

(5) a. Indra is strength-POSS
  b. Indra is with strength/strength-INSTR

Under this view, the b. examples in Table 2 actually instantiate strategy (5a), though whether this is only true diachronically or also synchronically is to some extent a matter of analysis that also touches on the problem of root denotation: are PC roots in Indo-European adjectival in the sense of adnominal/stative (as argued for Proto-Indo-European by, e.g., Nussbaum 1976, 2021a, 2022), quality-denoting (as argued for PC roots in Ulwa by Koontz-Garboden & Francez 2010, Francez & Koontz-Garboden 2017) and hence implicitly nominal (in the sense of "substantival"), or verbal[4] in the sense that they behave in the same way as eventive and other types of stative roots (see RAU, this volume)?

The case of Ulwa, an endangered Misumalpan language spoken in Nicaragua, is especially relevant in this context. In Ulwa, like in the examples in Table 2, the same possessive morphology, namely a suffix *-ka*, is found on possessed nouns, where it marks a 3sg. possessor, (6a), and on PC roots both in predicative, (6b), and in attributive/adnominal position, (6c).

(6) Possessive predication of PCs in Ulwa (Koontz-Garboden & Francez 2010: 200; Francez & Koontz-Garboden 2017: 31)

  a. Alberto pan-**ka**
   Alberto stick-3SG.POSS
   'Alberto's stick'

  b. yang as-ki-na minisih-**ka**.
   1SG shirt-<1SG.POSS> dirty-3SG.POSS
   'My shirt is dirty.'

[4] As implicitly assumed in LIV^2, where most of the reconstructable Indo-European PC roots are listed as forming primary, i.e., root-derived verbs, e.g., **k̑u̯ei̯t-* 'light up brightly', LIV^2 340; **leu̯k-* 'become bright, light', LIV^2 418–419; **peh₂ĝ-* 'become stiff, fixed', LIV^2 461; **$h_1$reu̯d^h-* 'redden', LIV^2 508–509; **su̯eh$_2$d-* 'become tasty, sweet', LIV^2 606–607; **tep-* 'be warm, hot', LIV^2 629–630; etc. That PCs in Proto-Indo-European (PIE) could be verbal is also explicitly stated by Balles (2009: 8) ("it was obviously possible in PIE to encode "adjectival" concepts by primary verbs") and argued for by Rau (2009a, 2013, 2016, this volume) and Bozzone (2016), cf. Section 2.2.

In addition to forming primary verbs, PCs also share a number of other derivational properties with eventive roots in (P)IE, such as the formation of root noun and "*tómos*-type" abstracts (Alan Nussbaum, p.c.; see also Nussbaum 2022 on the properties of "adjectival roots" in IE).

c. Al adah-**ka** as tal-ikda.
man short-3SG.POSS INDEF see-1SG.PAST
'I saw a short man.'

The parallel is not exact (the Ulwa marker is an agreement marker, whereas the suffixes in Table 2 are primary categorizing and/or derivational suffixes which cannot be used in "possessive genitive" contexts such as (6a)), but it does not seem trivial that the adjectival equivalents of (6b–6c) in Table 2 use morphology that also forms relational (and very often specifically possessive) adjectives. In fact, as Hanink & Koontz-Garboden (2025) argue based on Menon & Pancheva (2014), there is cross-linguistic evidence that "categorizers of property concept roots are tied up with possession in the creation of nominal, verbal, and adjectival property concept words" (Hanink & Koontz-Garboden 2025: 5). In these cases, the categorizer lexically categorizes the root and at the same time introduces a possessive relation: Possessive categorizers take a property *P* (a PC root) and return the set of individuals *x* that are in possession of this property (Hanink & Koontz-Garboden 2025: 6).

An interesting diachronic problem arises from this proposal that remains to be explored, namely how this type of root-derived possessive categorization is related to possessive morphology that selects already categorized nominal bases, as in column a. of Table 2. At least in the older IE languages, there is ample evidence that the latter can give rise to root-derived possessive categorizers through a reanalysis of the categorized base as an uncategorized root (or a root with a zero categorizer), cf. Grestenberger (2020, 2021), Grestenberger et al. (2023). We discuss this further in Section 3.1. First, we want to emphasize that the existence of forms like those in Table 2, that is, possessive and PC-related adjectival morphology, which can be securely reconstructed for PIE, provides a strong argument against the view that PIE did not have adjectives as a distinct morphosyntactic class, as proposed by Balles (2006) and more recently in a different form by Alfieri (2016). Balles argues that adjectival/PC root-derived nominals, especially zero-derived "root nouns" and *i*-stems, were categorially underspecified and could be used both as adjectives (e.g., in attributive modification contexts) and as nouns (e.g., as arguments). But the existence of a productive process of zero derivation of substantives from adjectives (not just PC adjectives) does not mean that the categorial distinction is moot, since the substantivizations are morphosemantically clearly secondary, i.e., based on the adjectival form and meaning (e.g., Lat. *datum* 'given' (neuter adj.) → 'what is given, gift', neuter substantive). The existence of denominal possessive adjectival suffixes in turn suggests that the categorial distinction was both derivationally and syntactically relevant, as Balles herself admits (2006: 286). That specifically PC related adjectival morphology can be reconstructed for the oldest layer of the proto-language (as well as the comparative and superlative suffixes **-i̯os-* and **-is-tó-* for at least the inner Indo-European[5] stage) is another strong argument against "word class indifference".[6]

Alfieri, on the other hand, has argued in a series of articles (Alfieri 2016, 2019, 2021; Alfieri & Gasbarra 2021) that PIE was an adjective-less language by classifying PC roots as "verbal roots of stative-unaccusative meanings" (Alfieri 2016: 159) or "verbal root[s] meaning a quality" (2021: 314). Adjectives based on such roots are hence treated as verbal adjectives and actually grouped with verbal stem-derived participial forms, though with a different range of functions than the latter. Alfieri's treatment of these forms is primarily based on his interpretation of the Vedic evidence and acknowledges that the Vedic system differs from the Part-of-Speech (PoS) systems of, for example, Ancient Greek and Latin. But even if there were compelling reasons for privileging the Vedic PoS system over that of the other daughter languages for the purposes of reconstruction, this approach ignores the cross-linguistic evidence for a systematic morphosemantic difference between PC roots and other types of CoS-associated roots for which correlates are also found in the older IE languages (see Sections 2.2 and 3.3 for a more detailed discussion) as well as the specifically adjectival morphology associated with the "Caland system" (= PC roots and their derivatives in (Proto-)Indo-European, cf. Section 3.1) and with (denominal) relational adjective formation, all of which can be securely reconstructed for PIE.[7]

Finally, strategy (5b) has also played an important role in the scholarship on PCs in Indo-European in the context of the interpretation of the "*cvi* construction", an Old Indic predicative construction consisting of an indeclinable form of a noun (sometimes described as adverbial in the literature) and a finite form of a copula or light verb, usually *as-* 'be', *bhū-* 'become', and *kr̥-* 'make', with which it is sometimes univerbated (Schindler 1980; Balles 2000, 2006, 2009). The name of the construction, *cvi*, is the label used by the ancient Indian grammarians (e.g., Pāṇini, *Aṣṭādhyāyī* 5.4.50). Some examples of this construction are given in Table 3.

While Old Indic and the Indo-Aryan languages in general are no strangers to periphrastic constructions consisting of a verbal noun + light verb, the problem

[5]I.e., the branches that descended from a common ancestor (≈ Late PIE) after the ancestral languages of the Anatolian and Tocharian branches left the family.

[6]See Meier-Brügger 2014, Rau 2014, Pinault 2018 on the prehistory of the comparative suffix and its role in the Caland system.

[7]See further Acedo-Matellán 2022b for arguments against the existence of "adjective-less languages".

Table 3: Old Indic *cvi* forms (ex. from Balles 2006: 45–46)

krūrá-	'sore'	*krūrī́ kr̥-*	'make sore'
náva-	'young, new'	*navī kr̥-*	'renew, rejuvenate'
phála-	'fruit'	*phalī́ kr̥-*	'winnow, clean'
yúvan-	'young'	*yuvībhū-*	'become young'

of the *cvi* construction lies in the interpretation of the *-ī* of the nonfinite part of the construction. The interpretation as an instrumental singular **-i-*h_1 of an adjectival abstract noun, effectively a construction of type (5b), is appealing in light of the cross-linguistic evidence of languages using a possessive strategy for lexicalizing property concepts. Given that PIE did not have a HAVE lexeme, it was most likely an S-POSSESSOR and/or S-POSSESSEE language (evidence for both strategies exist in the older IE languages), in which possessive predication was expressed through an intransitive construction in which either the possessor (S-POSSESSOR) or the possessee (S-POSSESSEE) was coded as the subject (Heine 1997, Stassen 2009). With its reconstructed deinstrumental origin, the *cvi* construction would fit the S-POSSESSOR pattern, specifically the COMITATIVE-POSSESSIVE type (Creissels 2023) also seen in, e.g., Hausa, (7), cf. the "*with*-possessive" in Stassen (2009: 54–57).[8]

(7) Hausa (Chadic), Newman 2000: 222 (cit. after Creissels 2023: 38)
Yarồ yanằ dà fensìr̃
boy 3SG.M.ICPL with pencil
'The boy has a pencil.', lit. 'The boy is with pencil.'

Although there are solid structural parallels to the *cvi* construction in other IE languages that point towards an inherited strategy of possessive predication with instrumental nouns (see Sections 3.2 and 3.3), as well as robust typological parallels, its diachronic connection to the expression of PCs in Indo-European in general and Old Indic in particular awaits further elucidation.[9]

[8]Cf. also Heine (1997: 53–57). The S-POSSESSEE strategy with genitive- and dative-marked possessors also existed in the older IE languages.

[9]The antiquity of the construction has been called into question because it is practically absent in the oldest Vedic text (the Rigveda) and is not specifically associated with inherited PC roots or lexemes (Jamison 2023). Rather, it expresses predicative possession in general, and the derivational basis can be adjectival or nominal (see Balles 2006: 45–75; 151–157 for a detailed discussion), as also seen in the examples in Table 3. In that sense, the *cvi* construction falls

With this introductory background in mind, we can now turn to the question of deadjectival verbs. That is, our focus in the following as well as in the papers in this volume is on the verbal strategy for expressing PCs, i.e., the one illustrated in example (2b) and the hypothetical (4c). However, most of the contributions in this volume are not restricted to the discussion of deadjectival verbs in this strict sense, but also address how such verbs arise diachronically and how they interact with and differ in distribution from the other strategies discussed above. In the next section, we provide a more detailed definition of deadjectival verbs and a discussion of the different types thereof.

2 Deadjectival verbs cross-linguistically

2.1 The name of the game

In the previous section, we have focused on deadjectival verbs formed to roots that express "adjectival meaning", i.e., property concepts (PCs). This is the definition commonly found in the theoretical literature (e.g., Hale & Keyser 2002; Koontz-Garboden 2005, 2014; Bobaljik 2012; Beavers & Koontz-Garboden 2020) and also adopted in some recent Indo-Europeanist literature on the "Caland system" (cf. Section 3.1), e.g., Rau, this volume. Some examples are given in (8a).

There is a second, less salient use of the term that can be paraphrased as in (8b) and essentially defines deadjectival verbs primarily via their morphology, that is, as based on an adjectival *stem* whose adjectival categorizing morphology is contained in the derivative.

(8) Two types of deadjectival verbs
   a. Root-derived verbs from PC roots
      i. Latin
         - *mac-ēre* 'be thin' (Pl.), *mac-ēscere* 'become thin' (Pl.+) ← **mak̂*- or **meh₂k̂*- 'long, thin' vs. adj. *macer*, *mac**r**-* (Nussbaum 1976: 51; NIL 478–479)
         - *rub-ēre* 'be red', *rub-ēscere* 'become red' ← **h₁reu̯dʰ*- 'red' (LIV[2] 508–509) vs. adj. *ruber*, *rub**r**-*

into the general late Old Indic/post-Vedic Sanskrit tendency to increasingly use analytic "light verb" constructions to express complex verb meanings, such as the periphrastic causative, perfect, future, habitual, etc. (Zehnder 2011; Ittzés 2015, 2022; Lowe 2017; Grieco 2023). Note that the structurally related *gúhā bhū-/kr̥-* construction ('become/make hidden/with hiding') is of Rigvedic age, cf. further Section 3.2.

- *tep-ēre* 'be warm' ← **tep-* 'be warm, hot' (LIV^2 629–630) vs. adj. *tep**idus***

ii. Slavic (PSl.)

- **rŭd-ěti sę*, *rŭdi-*[10] 'grow red, blush' ← *$h_1re\underset{\frown}{u}d^h$- 'red' vs. adj. **rŭd**rŭ***
- **top-iti*, *topi-* 'to heat' ← **tep-* 'be warm, hot' (compare Ved. *tāpáyati* and Yav. *tāpaiieiti*, LIV^2 629–630) vs. adj. **tep**lŭ***

iii. Hebrew (Arad 2003: 749; 743)

- *xizeq* 'strengthen' ← √*xzq* vs. adj. *xazaq*
- *hišmin* 'grow fat, fatten' ← √*šmn* vs. adj. *šamen*

b. Verbs derived from overtly marked/categorized adjectives

i. Latin

- *mac**r**-ēscere* 'become thin' (Hor.+) ← *macer*, *mac**r**-* 'thin'
- *pig**r**-ēre* 'to be reluctant' (Acc.) ← *piger*, *pig**r**-* 'torpid, inactive' (EDL 464)
- *crūd**ēl**-ēscere* 'become cruel' (Aug.) ← *crūd**ēl**is* 'cruel' (← *crūdus* 'raw, bloody, rough, cruel')

ii. Slavic (OCS)

- *ost**r**iti*, *ostri-* 'sharpen' ← *ost**rŭ*** 'sharp' (ESJS 601)
- *ob-lĭg**ŭč**iti*, *ob-lĭgŭči-* 'lighten, make light' ← *lĭg**ŭk**ŭ* 'light' (ESJS 447–448)
- *sŭ-tep**l**iti*, *sŭ-tepli-* 'to warm, to heat' ← *tep**l**ŭ* 'warm, hot' (ESJS 955)

iii. Hebrew (Arad 2003: 752, fn. 11)

- *hitqamc**en*** 'act miserly' ← *qamc**an*** 'miser' (√*qmc*)
- *hištaxc**en*** 'act arrogantly' ← *šaxc**an*** 'arrogant' (√*šxc*)

While the verbs in both (8a) and (8b) are semantically deadjectival and display meanings expected of deadjectivals ('be X', 'become X', 'make X'), only the latter ones are *morphologically* adjective-derived and contain the stem-forming morphology of the adjectival base. The difference is especially clear when comparing "doublets" like the first of the Latin examples, where the verbal morphology

[10] The citation forms used for Slavic verbs are usually the infinitive and the first person singular, e.g., **rŭždǫ sę*. For ease of comparison (in particular with Lithuanian, where the citation forms are the infinitive and third person), we use the present stem here instead of the first person form. The past forms are omitted.

of *mac-ēscere* in (8a-i) attaches directly to the root, while *mac-**r**-ēscere* in (8b-i) clearly shows the -*r*- of the adjective stem that is kept in the derivative, and the same holds for *pig**r**ēre*. The productivity of the latter strategy of stem-derived formations is confirmed by *crūd**ēl**ēscere* from *crūd**ēl**is*, which is itself a derivative of the adjective *crūdus*.

A similar situation is encountered in the Slavic examples. In (8b-ii) we see adjectives that are derived from typical Indo-European "Caland" roots (*h_2eḱ̂-*,[11] *$h_1$leng^{uh}-* and **tep-*) by using the associated adjectival suffixes **-ro-* (which is also the source of the Latin -*r*- in the examples above), **-u-* and **-lo-*. Those suffixes surface in the respective derived verbs, i.e., they are clearly derived from the adjective rather than from the root. In this way, we get *sŭtep-**l**-iti* as opposed to the root-derived causative **top-iti* in (8a-ii), or Latin *tep-ēre* in (8a-i). Particularly illustrative of this pattern is *oblĭgŭčiti* in (8b-ii): Old **u*-adjectives were regularly extended by the suffix **-ko-* in Slavic. This extension can famously be dropped in derivation (e.g., ChSl. *ǫz-iti* 'to narrow' ← *ǫzŭkŭ* 'narrow' < **$h_2$emĝ^h-u-*) and in doing so behaves like other Caland suffixes (cf. Majer 2017: §4.2 for a detailed discussion with references). However, the -*č*- of *oblĭgŭčiti* is the reflex of the palatalized -*k*- from *lĭgŭkŭ*, which means that the verb must be derived from the whole synchronic *stem* rather than from the root. Further evidence from the individual IE languages as well as comparative reconstruction suggest that the latter strategy in (8b) is an innovation in the PC context, becoming more and more productive and replacing the root-derived principle in (8a); see, e.g., Watkins (1971) and Nussbaum (1976) for more examples from various Indo-European branches, and Majer (2017) and Reiter (2023) for (Balto-)Slavic.

The contrast between root- and adjective-derived verbs is not restricted to Indo-European languages. In the Hebrew examples in (8a-iii), the root-derived verbs are created by inserting certain vowel combinations into the root consisting of three consonants (and, in the second example, a prefix), the same strategy as is used for the creation of deradical adjectives. On the other hand, (8b-iii) shows verbs derived from the adjectives rather than from the root, as the adjective-building suffix -*an* is retained (albeit with a change of the vowel quality). The "act like" semantics are a manifestation of the broader semantics of stem-derived verbs as opposed to root-derived ones (see Kastner 2020: 194–195 for further examples).

In Indo-European studies, the morphology-oriented use of the term "deadjectival" has been the dominant one so far, i.e., it is used almost exclusively for

[11]But see Nussbaum (2021a: 10) for potential problems of an analysis of PIE **h_2eḱ̂-ro-* as historical PC adjective due to the unusual accent of Lith. *aštrùs* and AG *ákros*, which points to **h_2éḱ̂-ro-* rather than expected **h_2eḱ̂-ró-*.

verbs of type (8b), while (8a) is usually discussed under the label "Caland system" (cf. Section 3.1). Since verbs like those in (8b) are stem-derived, they are traditionally labeled "secondary" as opposed to "primary" formations that are derived directly from the root (cf. above).[12] As a consequence, primary verbs are usually not labeled "deadjectival" (or more broadly "denominative") because they are not deadjectival from a morphological point of view, even if they are semantically. For example, PSl. **rŭděti sę*, *rŭdi-* 'grow red, blush' is usually taken as a primary verb in the literature as it is morphologically deradical and builds an *i*-present as opposed to morphologically unambiguous deadjectivals that form a different type of present stem (e.g., OCS *o-slaběti*, *o-slaběje-* 'become weak' ← *slabŭ* 'weak'; cf. Villanueva Svensson, this volume). Consequently, it is not discussed in the literature on deadjectivals (but see Villanueva Svensson 2021: 275 for an exception). Of course, when analyzing a verbal form, a clear-cut decision on the primary vs. secondary status is sometimes difficult to make (cf., e.g., Boldt, Höfler & Stifter and Sasseville, this volume, or again Villanueva Svensson 2021) and the analysis might differ depending on whether one takes a synchronic or diachronic viewpoint: As is shown by various contributions in this volume, historical secondary formations can at some point be reanalyzed as primary, and a synchronic (re)association of an originally independent derivate with a nominal formation is possible, too.[13] This can lead to a situation where a verb is labeled differently across publications (within one branch as well as across different daughter languages of Indo-European[14]), which makes a comparison somewhat difficult – especially when the criteria for classifying a given example as (synchronically or historically) primary or secondary are not made transparent.

As these examples and the elaborations in Section 1.2 show, an equation of "deadjectival" with "secondary" in the morphological sense of (8b) would exclude an important amount of evidence, especially in the case of the "Caland" phenomenon, as will be further discussed in Section 3.1, and is at odds with the use of the term in theoretical and typological studies. We therefore use the term "deadjectival" in both meanings defined above to encompass both secondary verbs

[12] Because it is common practice in the discipline to set up the root and its meaning as verbal wherever possible (cf. footnote 4), the term "deverbal" or "deverbative" is sometimes inaccurately equated with "deradical" or "primary" in morphological terms. This should be avoided.

[13] Compare Jasanoff (2004: 147–148), who calls the distinction between denominative and deradical in the context of Latin *ē*-verbs "illusory, or at least epiphenomenal" because historically, the ultimate source of the verbal formation itself is denominative, cf. Section 3.2.

[14] E.g., PSl. **rŭděti sę* is equated with Latin *rubēre*, Old Irish *ruidid* 'blush' and OHG *rotēn* 'be reddish, become red', cf. Höfler & Stifter, this volume.

built on overtly marked adjectival stems as well as morphologically primary but semantically deadjectival verbs derived from Property Concept roots (cf. Section 4 for further advantages of this use).

Finally, note that in languages with little or no adjectival categorizing morphology like English, the matter of the distinction between (8a–8b) becomes a serious theoretical problem: Are there verbs which are morphologically deadjectival, but happen to be derived from adjectives in which the adjectival categorizer is zero? Would we expect such verbs to be semantically distinguishable from genuinely root-derived verbs as in (8a)? This also depends on whether we expect verbs of type (8a) to systematically differ semantically from verbs of type (8b), as is suggested *inter alia* by the Hebrew examples and the iteratives discussed in Section 3.4. Further, the problem of root- vs. stem-based derivation links to the question of why some deadjectival CoS verbs are transparently built on the comparative while most others are not, e.g., Engl. *to worsen*, *to lessen*, *to better* vs. *to broaden*, *to clean*, etc., a phenomenon that is found in the older IE languages as well.[15] The same question arises for the verbs of type (8b), which also show overt adjectival (though positive rather than comparative) morphology. We must leave this to future research; in this volume, the contributions by Sasseville, Merritt, Iacobini & De Rosa and Tanza-Kamil treat the complex relationship between verbs that contain overt adjectival morphology (at least historically) and primary/root-derived verbs.

2.2 Two types of change-of-state verbs

In discussing deadjectival CoS verbs, we must moreover be careful to distinguish verbs formed from PC roots from other types of CoS verbs whose base is not adjectival in one of the two senses defined in Section 2.1. Beavers & Koontz-Garboden (2020: 58) observe that "there is a split between two types of change-of-state verbs in terms of whether the entailment of change is introduced solely by the template or an entailment of change is (also) present in the root itself", a difference that can be paraphrased as that between adjectival and eventive (in the broad sense) roots. The first ("adjectival") class that has no entailments of change gives rise to deadjectival CoS verbs, (9). These correspond to the PC states of Dixon (1982) and "Caland" adjectives in the older IE languages (Section 3.1).

[15]Semantic arguments against taking deadjectival degree achievement verbs such as *flatten*, *widen*, *straighten*, etc., from the positive adjective are discussed by Beavers & Koontz-Garboden (2020: 42–45 & fn. 24), etc.; see also Bobaljik 2012: 169–207; Beavers & Koontz-Garboden 2020: 46, fn. 26; Fábregas 2023: 80–90. On the lack of zero-derived deadjectival verbs see Acedo-Matellán 2022a.

The second class of CoS verbs is derived from "states that are the result of some action" (Dixon 1982: 50), (10); cf. also Rau 2013 for a more fine-grained version of this distinction.

(9) Deadjectival CoS verbs (Levin 1993: 245, after Beavers & Koontz-Garboden 2020: 58)
awaken, brighten, broaden, cheapen, coarsen, dampen, darken, deepen, fatten, flatten, freshen, gladden, harden, hasten, heighten, lengthen, lessen, lighten, loosen, moisten, neaten, quicken, ripen, roughen, sharpen, shorten, sicken, slacken, smarten, soften, stiffen, straighten, strengthen, sweeten, tauten, thicken, tighten, toughen, weaken, widen, …

(10) Non-deadjectival CoS verbs (Beavers & Koontz-Garboden 2020: 58)
   a. Levin's (1993: 241) *break* verbs: break, chip, crack, crash, crush, fracture, rip, shatter, smash, snap, splinter, split, tear
   b. Levin's cooking verbs (Levin 1993: 243): bake, barbecue, blanch, boil, braise, broil, charbroil, charcoal-broil, coddle, cook, crisp, deepfry, fry, grill, hardboil, poach, …
   c. Verbs of killing (Levin 1993: 230–233): crucify, electrocute, drown, hang, guillotine, …
   d. *inter alia*

Crucially, these two classes systematically differ with respect to their morphology and semantics across a wide sample of genetically unrelated languages, as Beavers et al. (2021) show: While the verbs in ex. (9) form both basic and derived statives (= primary and secondary/deverbal adjectives, in Indo-Europeanist terminology), the verbs in ex. (10), or "result roots", can only form derived statives which always entail a change of state.[16] This difference is illustrated in Table 4 for English (cf. also Rau 2009a: 162–163 on the same contrast in Vedic Sanskrit and Latin).

The same contrast is found in languages that use the verbal strategy (or a mix of strategies) for statives. In Hebrew, for example, both adjectival and result

[16]The "result stative" category in Table 4 and Table 5 conflates different subtypes of (participial) states, e.g., the distinction between target and resultant states of Kratzer (2001), Anagnostopoulou (2003), Anagnostopoulou & Samioti (2014), Alexiadou et al. (2015), etc. What matters for our purposes is that this (composite) category differs systematically from that of the "basic statives". For a differing view under which "statives" (including underived adjectives) and adjectival passives (≈ Kratzer's target states) are grouped together to the exclusion of "resultatives" (≈ Kratzer's resultant states) see Embick (2004).

Table 4: Basic and derived statives in English

Adjectival root	Basic stative	Verb	Result stative
√FLAT	*flat*	*flatten*	*flatten-ed*
√RED	*red*	*redden*	*redden-ed*
Result root	Basic stative	Verb	Result stative
√CRACK	–	*crack*	*crack-ed*
√COOK	–	*cook*	*cook-ed*

statives are categorially *verbal*. While simple statives in Hebrew do not entail a change of state, result statives always do, both for adjectival and for result roots, and they also differ morphologically, cf. the examples in Table 5 (a similar pattern of primary and secondary verbal stative formation is discussed by Tanza-Kamil, this volume, for Akkadian).

Table 5: Basic and derived statives in Hebrew (ex. from Beavers et al. 2021: 457)

Root		Simple stv	Inchoative	Causative	Result stv
Adjectival	√*gdl* 'large'	*gadol*	*gadal*	*higdil*	*mu-gdal*
Result	√*švr* 'break'	–	*ni-šbar*	*šavar*	*šavur*

Given how widespread the morphosemantic differentiation between adjectival and non-adjectival/result CoS verbs is cross-linguistically, this suggests that the (often informal) distinction between "verbal" and "adjectival" roots found in some of the Indo-Europeanist literature (e.g., Nussbaum 2022) is indeed theoretically and typologically valid. In other words, adjectival/PC roots have different combinatorial morphological properties than eventive/result roots, and this reflects their underlying semantic difference. In order to terminologically distinguish between verbs derived from PC vs. eventive/result roots, we therefore use the terms *fientive* (Lat. *fieri* 'become') and *factitive* for, respectively, intransitive and transitive CoS verbs from PC roots and *inchoative* and *causative* for intransitive vs. transitive CoS verbs from eventive/result roots from here on (cf. Villanueva Svensson, this volume). Note, however, that these two groups can also interact and sometimes even share the same verbal morphology (cf. Rau 2013 & this volume; Sasseville, this volume; as well as fn. 4 and Table 8). Moreover, reanalysis of secondary adjectives as primary adjectives, lexicalization, etc., can blur the lines between these two classes for individual lexical items, so that "variation in the interpretation of particular adjectival forms among both simple and derived adjectives is actually expected owing to lexical drift or idiosyncratic semantic factors." (Beavers et al. 2021: 449; cf. also Beavers & Koontz-Garboden 2020: 71–73). For example, the adjective *broken* in (11), also from Beavers et al. (2021: 449) does not entail a change of state, which therefore can be denied. This is unexpected given that *broken* is formed from a result root.

(11) The window is broken but it never broke. It was manufactured that way.

Beavers & Koontz-Garboden (2020: 72) suggest that *broken* could have "undergone lexical drift for some speakers to have a specialized meaning like "not functioning" or "not canonical in a way that suggests defects," which could be an inherent property rather than one derived through a change". In other words, new property concept lexemes can arise through reanalysis or lexical drift, a development that has also been observed in the context of the Caland system in the older IE languages (e.g., Nussbaum 1976, 2021a, 2022; Rau 2009a: 118–126; Alfieri 2016). Reiter (2023: 127–138) discusses some examples in Slavic, in which formations that are traditionally explained as historical participles or resultative deverbal adjectives function as synchronic adjectives with a PC reading that can in turn serve as base for further deadjectival verbs, cf. Table 6. This is also the

Table 6: Slavic deadjectivals from former participles (Reiter 2023: 128; 138). Participial morphemes are bolded.

Base verb	Ptcp. → adj.		Deadjectival verb
PSl. **zelti*[a]	OCS *zel**en**ŭ* 'green'	ChSl.	(*pro-*)*zel**en**ěti* (*sę*), -*ěje-* 'become green'
		Russ.	*zel**en**ét'* 'id.'
		Russ.	*zel**en**ít'* 'make green'
OCS *čгŭviti* 'dye red' (← *čгŭvĭ* 'worm')	OCS *čгŭv(lj)**en**ŭ* 'red'	Cz.	*červ**en**ěti* 'become red'
		Cz.	*červ**en**iti* 'paint red'

[a] cf. Lith. *žélti, žẽlia/želi* 'green, sprout'. Note, however, that for the first example there might also be an alternative explanation since the PIE root *$\hat{g}^h elh_3$- means 'green, yellow, pale', i.e., it has PC semantics. Therefore, the apparent base verb and its "participle" could also be independent derivations.

case for many prototypical PC adjectives in English, e.g., *old*, which goes back to a past participle of the PGmc. verb **alan-* 'grow, bring up' (EDPG 20; Beavers et al. 2021: 449).

All in all, these facts suggest that deadjectival verbs need to be distinguished from non-deadjectival CoS or result verbs, from which they differ semantically (PC vs. result/eventive roots). Morphologically, they are often treated as primary (root-derived) in much of the theoretical and Indo-Europeanist literature, but there are also instances in which they appear to contain overt adjectival stem-forming morphology, and in these cases one must carefully distinguish between instances in which the adjectival morphology has been reanalyzed and essentially acts as verbalizing morphology (cf. Section 3.3) and instances which are derivationally transparent and which raise the question why adjectival morphology is contained in the derived verb and what semantic function (if any) it contributes compositionally.

Since the discussion here and in the following focuses largely on CoS contexts, it is important to point out that deadjectival verbs can also be states and activities. Fábregas (2023) distinguishes between the three broad classes of deadjectival verbs given in ex. (12) in his study of derived verbs in Spanish.

(12) Three types of A > V derivation (Fábregas 2023: 63)

a. Change of state verbalisation, where the adjectival base defines the set of properties used to evaluate the change undergone by an argument (*claro* 'clear' > *aclarar* 'to make clear')

b. Activity property verbalisation, where the adjective defines a set of properties exhibited by an argument when performing an event (*holgazán* 'lazy' > *holgazanear* 'to act lazily')

c. Stative property verbalisation, where the adjective defines the properties exhibited by an argument, and not subject to change or transformation (*transparente* 'transparent' > *transparentar* 'to be transparent')

For reasons of space, we focus on the first class in the following, though see Section 3.4 for a brief discussion of derived activity verbs as in (12b).[17]

Having laid out the theoretical foundations of the topic as well as the definition of deadjectival verbs used in this volume, in the next section we move on to discussing the encoding of PCs in the Indo-European languages in more detail.

[17] See also Oltra-Massuet & Castroviejo 2014 and Acedo-Matellán & Gibert-Sotelo 2022 on this type.

3 Property Concepts and deadjectival verbs in Indo-European

3.1 The "Caland system"

In the older Indo-European languages, property concepts are usually discussed under the label "Caland system", named after Willem Caland (1859–1932), who first observed the peculiar morphological behavior of primary adjectives and their derivatives in Indo-Iranian (Caland 1892, 1893). Caland noticed that primary adjectives, especially those formed with the Indo-Iranian suffixes *-*ra*-, *-*ma*-, and *-*ant*-, seemingly "lost" their adjectival suffix in comparatives, superlatives, and in compounds, often replacing it with a suffix *-*i*- otherwise associated with *nominal* stem formation. This replacement pattern is illustrated in Table 7 with examples from Vedic, Avestan (both Indo-Iranian) and Ancient Greek (AG). Essentially, while comparatives, superlatives, and abstracts from primary ("Caland") adjectives are semantically deadjectival or "secondary", they seem to be morphologically root-derived or "primary" formations.

Table 7: The Caland system in Indo-Iranian and Ancient Greek

	Positive adj.		Comparative	Superlative	Abstract
Ved.	*tig-má-*	'sharp'	*téj-īyaṃs*	*téj-iṣṭha-*	*téj-as-*
	pr̥th-ú-	'broad'	*práth-īyaṃs-*	*práth-iṣṭha-*	*práth-as-*
OAv.	*ug-ra-*	'strong'	*aoj-iiah-*	*aoj-išta-*	*aoj-ah-*
YAv.	*bərəz-ant-*	'high'	*barəz-iiah-*	*barəz-išta-*	*barəz-ah-*
AG	*kūd-rós*, *-nós*	'glorious'	*kūd-íōn*	*kū́d-istos*	*kūd-os*
	krat-ús	'strong'	*kréssōn* (< **krét-jōn*)	*krát-istos*	*krát-os*

Subsequent research has shown that this descriptive replacement of the adjectival morphology with *-*i*- in composition, termed "Caland's Law", is neither confined to Indo-Iranian nor to the suffixes discussed by Caland (see Dell'Oro 2015 for an overview of the research history). Building on earlier works by Wackernagel (1897) and Risch (1974), Nussbaum (1976) showed in his seminal study that the alternating adjective- and abstract-forming suffixes must be conceptualized as forming a complex derivational *system* to roots with "inherently adjectival semantics" (Nussbaum 1976: 6). While some of them are more frequent and therefore "central" to the system, as the aforementioned *-*ro*-, *-*u*-, *-*(e/o)nt*- and

*-*i*- (and, added later, simple thematic *-*o*-, cf. Nussbaum 2021a, 2022),[18] others are less frequently used for forming PC adjectives and therefore “marginal” or “peripheral”, like *-*mo*-, *-*no*-, *-*lo*- and *-*to*- (see Rau 2009a: 71–74). A combination of suffixes, like *-*i-no*- or *-*u-mo*- is possible as well. The reason for the synchronic substitution principle is to be found in the derivational base of the respective formations: Since for a number of roots displaying a Caland system a root noun (that is, a noun built deradically with a zero suffix/categorizer) with abstract semantics is either attested or reconstructable, Nussbaum (1976: 100–105) argues that the seemingly primary formations with alternating suffixes are ultimately parallel derivations from such a root noun and consequently (historically) *secondary* formations, as outlined in ex. (13).

(13) Derivation of Caland adjectives from root nouns (Rau 2009a: 129–130)
  a. root noun **krúh$_2$-Ø-* ‘gore, raw or bloody flesh’
     → adj. **kruh$_2$-ró-* ‘bloody, gory’
  b. root noun **h$_1$rud^h-Ø-* ‘redness’
     → adj. **h$_1$rud^h-ró-* → ‘red’,
     → adj. **h$_1$re/o̯ud^h-ó-* ‘possessing redness’ → ‘red’

The question of whether Caland formations should be classified as primary or secondary (called “*the* basic problem” by Nussbaum himself in 1976: 101) has been a crucial one ever since, as it is rarely possible to decide on the derivational base using morphological criteria. However, while the path described in (13) is most likely the ultimate source of the morphological behavior, Caland systems can also develop without attested root nouns. This suggests a reanalysis of the derivational base as being not a (non-overtly marked) nominal formation, but the root itself, giving way to the deradical derivational principle of the system.[19]

Several attempts have been made to trace the diachronic development of PC lexicalization even further back in time. Jasanoff (1978, 2004) discusses the integration and subsequent extension of Indo-European “*ē*-verbs” in the Caland system (see Table 9 and Sections 3.2 and 3.3 below) and the formation of stative-intransitive verbs more generally, arguing for a nominal, (de-)instrumental origin of the *ē*-suffix (cf. Grestenberger 2023 with refs.).

[18]Note that except for *-*i*- these are the suffixes illustrated in Table 2.

[19]The origin of the Caland substitution rule, its connection to the independently observed *-*o*- → *-*i*- replacement rule, and the original function and distribution of the latter are discussed in, e.g., Schindler 1980; Nussbaum 1999, 2022; Rasmussen 2002; Balles 2006: 269–287, 2009; Pinault 2018; Grestenberger 2021. For reasons of space, we do not discuss this rule here.

As mentioned above, Balles (2006, 2009) has argued that PIE did not have a categorial noun/adjective distinction and that PCs were encoded either as verbal (cf. 4) or through underspecified PC nominals as discussed above in Section 1. The Caland system evolved only after the need to differentiate the adjectival from the abstract reading of these PC nominals arose, which according to her also explains why Caland adjectives "are semantically primary, but formally secondary, namely derived" (Balles 2009: 10–11). In other words, "Caland's Law" is an epiphenomenon of the emergence of adjectives as a distinct word class, which was only in its beginnings when the Anatolian branch left the family. The *i*-stem of the *cvi*-construction in Table 3 would then be a relic from the time before this development started.

While in Balles' analysis PIE could encode adjectival semantics both via a verbal and nominal strategy (2006: 271) and the CS is the result of the emergence of an adjectival word class from the nominal strategy, Bozzone (2016) argues that the CS is a relic of an older *verbal* strategy of lexicalizing PCs. She argues that Caland adjectives were originally inflected and behaved syntactically like verbs, based on a comparison with Japanese PCs. According to Bozzone, PC-associated verbal formations were specifically root aorists with fientive semantics, remnants of which can be found in some Vedic aorist participles. These verbs were thus effectively constructions of type (4c) above, which were productive in pre-PIE but became obsolete in the history of PIE due to a shift from a verb-like adjective class to a noun-like one and correspondingly to differences in adjectival predication. Before this shift, pre-PIE could encode adjectival meanings either through Caland root verbs or via predicative instrumentals of nouns that were *not* part of the CS. After the shift, Caland roots were able to form both root verbs and root nouns, whose predicatively used instrumentals became verbalized and led to the aforementioned stative *ē*-verbs. The root verbs, on the other hand, were specifically associated with perfective aspect (aorists). This would explain why the competing predicatively used instrumentals became associated with *present* stem formation and would fit the typologically common pattern of encoding fientive/CoS semantics as verbal and stative semantics as nominal. Finally, the proposed resulting "switch adjective system" is, per Bozzone, typical of the last stage of a shift from verb-like to noun-like adjectives.

Though innovative, both accounts are problematic for several reasons. Balles' account wrongly dismisses the Anatolian evidence for an old (inherited) class of primary adjectives and at the same time overestimates the age of adjectival *i*-stems both in Anatolian and Indo-Iranian, which are likely to be secondary, independent innovations (Grestenberger 2009: 8–10, 2014: 99; Pinault 2018: 316).

Bozzone, on the other hand, places too much emphasis on Vedic participial formations in her reconstruction of pre-PIE PC aorists, many of which are usually analyzed as Caland *adjectives*, e.g., Ved. *bṛh-ánt-* 'high', which may have been originally denominal to a root noun (cf. Table 2 and the discussion there; see also Lowe 2012, 2015: 283–294). Bozzone takes them as fossilized participles from root aorists without providing independent evidence for *finite* forms of such aorists.

Despite these issues, these accounts provide the most detailed extant proposals for how originally nominal (instrumental) forms came to be integrated into the verbal system in the attested IE languages and thus constitute an important starting point for further studies, even though the internal reconstruction of the prehistory of the Caland system must necessarily remain speculative to some extent. One of the main problems that emerges from this work is the nature of primary verb formation of PC roots vs. non-adjectival CoS roots (cf. Section 2.2), especially with respect to what types of primary verbs these two classes could form (e.g., presents vs. aorists) and to what extent the morphological strategies for forming such verbs overlapped (see Rau 2009a, Rau 2013, this volume).

Before we fully turn to the verbal side of the system, it is moreover important to stress that while numerous Caland roots correspond to Dixon's (1982) property concepts in Table 1 (anticipated by Watkins 1971 and Nussbaum 1976 and subsequently shown by Balles 2006, 2009 and Rau 2009a), "it is not true that a CS guarantees an adjectival root and *vice versa*" (Nussbaum 2022: 207). That is, not every root with a PC meaning necessarily forms a Caland system (the most prominent example being PIE **néu̯o-* 'new') and not every root displaying a Caland system necessarily has PC meaning from the beginning, e.g., **pleh$_1$-* 'fill (up)', whose meaning is not prototypically adjectival but whose derivatives across the family show that a securely reconstructable system has developed over time (see Nussbaum 2022: 208, fn. 7 for a possible path of this development). As already mentioned in Section 2.2, the nominal as well as the verbal morphology involved is not *exclusively* found in PC/Caland roots, and consequently, some derivatives from roots that entail a CoS are discussed under the label "Caland" in the literature as well (e.g., in Rau 2009a, Rau 2009a, Rau 2013, this volume; Bozzone 2016; Majer 2017).

A frequently quoted example is **b^{h}eu̯d^h-* 'awake, become alert' where a (possibly) primary stative verb next to a **ro*-adjective is attested. Table 8 shows a comparison of the respective derivatives with those of a (canonical) PC root in Baltic and Slavic. Pairs like these deserve future work, especially concerning criteria for a potential differentiation of the two types both for specific languages and cross-linguistically.

Table 8: Shared morphology in CoS vs. PC root derivatives in Baltic and Slavic (ex. from Majer 2017)

Suffix		a. CoS root *$b^h e\underset{\frown}{u}d^h$-		b. PC root *$h_1 re\underset{\frown}{u}d^h$-
*-*ro*-	Sl.	*bŭdrŭ* ‘alert, vigilant’	Sl.	*rŭdrŭ* ‘red’
	Lith.	*budrùs* ‘id.’	Lith.	–
*-*ē*-	Sl.	*bŭdĕti*, *bŭdi*- ‘be alert’	Sl.	*rŭdĕti*, *rŭdi*- ‘become red’
	Lith.	*budė́ti*, *bùdi* ‘id.’	Lith.	?[a]

[a]Watkins 1971: 64 gives Lith. *rudė́ti* ‘become reddish’ as cognate (compare also Fraenkel 1962–1965: 745 s.v. *rùdas*), but according to ALEW s.v. *rūdìs* and LKŽ s.v. *rūdė́ti*, this verb does not show an *i*-present but a present in -*ė́ja*, which is expected of productive denominatives. The shape of the root vowel, which is long (i.e., *recte* ***rū****dė́ti*, as quoted by Majer 2017: 34), and the present point to a synchronic denominative from *rūdìs* (so ALEW). However, LIV[2] 508 gives a first singular *rūdžiù*, and LKŽ also indicates a present variant in -*i*. Furthermore, Latvian *rudêt* shows the historically expected root vowel quantity. Therefore, it is possible that the present in -*ė́ja* is a later formation. See also Pakerys 2006: 67, with fn. 3.

In other words, although PC and eventive roots seem to be able to diachronically switch class, they need to be carefully separated synchronically in order to make the argument that PCs were originally encoded as verbs of a particular morphological class (or classes).

3.2 Verbal periphrasis in the Caland system

While the morphosemantics of the Caland system in the nominal domain are increasingly well understood thanks to Nussbaum (1976) and much subsequent work on various branches of Indo-European (e.g., Meiser 1993; Nussbaum 1999; Balles 2006, 2009; Rau 2009a; Dell’Oro 2015; Bozzone 2016; Majer 2017, 2021), the verbal domain has received less attention so far. Watkins (1971) was one of the first to point out the close association between Caland adjectives and stative verbs in (*)-*ē*-, arguing that such verbs were derived from adjectives via the same substitution principle as in the nominal domain. That is, the verbal suffix -*ē*- does not select the full adjectival stem but the root, with apparent deletion of the adjectival suffix. This analysis was adopted by Nussbaum (1976) and subsequently investigated further, amongst others, by Jasanoff (1978) and (2004) (see (8a-i) for some examples). The ultimate source of the suffix is again to be sought in root nouns,[20] but unlike in (13), the derivational base was not the noun stem, but a

[20]The alternative account which reconstructs *-*(e)*h_1- as primary verbal stem-forming suffix (Harðarson 1998; LIV[2] 25) is problematic for several reasons; cf. Jasanoff 2004, García Ramón

case form, namely an instrumental in *-eh_1 (> -ē). Such instrumentals seem to have been able to become the basis for "decasuative" verbal derivation[21] using the denominal verbalizing suffix *-i̯e/o-, whence the sequence *-eh_1-i̯e/o- found in present stem formations across the IE languages (Watkins 1971, Rasmussen 1993, Harðarson 1998, Jasanoff 2004, Yakubovich 2014; cf. fn. 20), for example, in the Latin 2nd conjugation presents of the type *clār-**ē**-re* 'be(come) clear' (*clārus* 'clear') or *sen-**ē**-re* 'be old' (*senex* 'old') etc.; cf. Table 11 below. As these examples show, these verbs are synchronically deadjectival rather than built on a root noun and use the same Caland replacement pattern discussed in Section 3.1.

The second relevant strategy is illustrated in Table 9 and involved periphrastic constructions using finite light verbs or auxiliaries, similar to the *cvi* constructions in Table 3.

Table 9: Light verb constructions with the root noun instrumental *h_1rud^h-$éh_1$ 'with redness'

Base	Light verb/AUX	Meaning	Semantic type
*h_1rud^h-$éh_1$	h_1es- 'be'	'be red'	stative
*h_1rud^h-$éh_1$	b^huH- 'become'	'become red'	fientive
*h_1rud^h-$éh_1$	d^heh_1- 'put', 'make'	'make red'	factitive

Univerbations of these constructions are found in the individual languages and have given rise to productive fientive and/or factitive verb formation types, e.g., the Latin types *rub**e**-fīō* 'become red' (< … *b^huH- 'become') and *rub**e**-faciō* 'make red, redden' (< … *d^heh_1- 'put, place'), but there are also remnants of the older analytic construction, e.g., Vedic *gúh**ā** bhū-/kr̥-* 'become/make hidden/with hiding' (*gúh-* 'hiding place', -*ā* INSTR.SG) and Latin analytic *facit ar**ē*** 'make hot/with heat' besides the more common *ār**ĕ̄**-faciō* (Hahn 1947; Weiss 2020: 138, fn. 18). This construction has played a crucial role in the debate on the diachronic integration of PC roots into the IE verbal system (e.g., Schindler 1980; Jasanoff 2004; Balles 2006, 2009; Bozzone 2016; see also MERRITT, this volume). However, since these (univerbated) light verb constructions were arguably denominative in origin and thereby morphologically secondary (note here the discrepancy to the synchronic primary status), it remained unclear whether it was possible to derive verbs with

2014: 156, Grestenberger 2023: 20–21 for discussion and criticism.

[21] See PINAULT and GARNIER, this volume, on decasuative verbs, and Fortson 2020 for an evaluation of decasuative derivation in IE and in general.

primary verbal suffixes from Caland roots in the proto-language. Important evidence for this question was added to the Caland canon only relatively recently by Rau (2009a, 2013), who argues for a systematic association of primary verbs like full-grade thematic presents, nasal-infix formations and iterative-causatives in **-éi̯e/o-* with PC roots (see also the discussion of Bozzone 2016's account in Section 3.1).

Still, some scholars argue that these verbs must be later, parallel innovations of the individual languages and the only strategy of the proto-language to derive verbs from PCs was via a nominal instrumental. Given that the root-based derivational pattern could have been triggered by the reanalysis of the derivational base from a nominal with a zero suffix to an uncategorized root as outlined above (Section 3.1), this potential diachronic development could also have given rise to the derivation of primary verbs from Caland roots. Thus, more research into the verbal correlates of Caland adjectives/PC roots is needed: Although productive means of deriving deadjectival verbs exist in all older Indo-European languages, these have for the most part not been the subject of systematic study and analysis from the perspective of comparative reconstruction or theoretical linguistics, with the exceptions mentioned above. There are several reasons for this fact: On the one hand, verbal secondary derivation is in general less well studied than nominal derivation or primary verb formation (cf. Villanueva Svensson, this volume) and on the other hand, the synchronic productivity of various strategies makes it difficult to distinguish forms that are *einzelsprachlich* from inherited strategies. Within the Caland system, the morphological "replacement" rule of the adjectival stem-forming suffix usually makes it impossible to distinguish primary from secondary formations morphologically, requiring the development of additional diagnostics for establishing derivational directionality that hold cross-linguistically (cf. Marchand 1964; Arad 2003; Balteiro 2007; Lohmann 2017; Grestenberger & Kastner 2022) in order to distinguish between synchronic and diachronic patterns of derivation, especially in cases in which erstwhile adjectival morphology has become reanalyzed as verbal stem-forming morphology.

In the next sections, we briefly discuss some patterns with potential claim to antiquity and also address the issue of voice morphology in deadjectival verbs, which has not been studied systematically so far.

3.3 Deadjectival verbs, voice, and the factitive-fientive alternation

As we saw in Section 2, there is a close derivational link between CoS verbs and primary/property concept adjectives (Hale & Keyser 1998, 2002; Harley 2005, 2011; Koontz-Garboden 2005; Francez & Koontz-Garboden 2017, etc.), and this has also been shown to be the case in the older Indo-European languages (Rau 2013; Yakubovich 2014; Sasseville 2021). This raises the question of the diachronic relationship between deadjectival CoS verbs and other types of statives and CoS verbs (e.g., of the causative alternation, and compare also Table 8), as well as their interaction with voice marking. Thus, in Ancient Greek fientives and factitives (recall from Section 2.2 that these are our terms for intransitive vs. transitive CoS verbs from adjectival roots) are usually formed from the same verbal stem, but alternate between active endings in the transitive-factitive and nonactive or "middle" endings in the intransitive-fientive forms, independent of the verbalizer that is used, cf. Table 10.[22]

Table 10: Alternating deadjectival CoS verbs in Ancient Greek

Adj.	Act. factitive	Nonact. fientive
leukós	*leukó-ō*	*leukóo-mai*
'bright, white'	'make white, whiten'	'turn white'
orthós	*orthó-ō*	*orthóo-mai*
'straight'	'set up straight, straighten'	'sit up, become straight'
kratús	*kratū́n-ō*	*kratū́no-mai*
'strong'	'make strong, strengthen'	'become strong'
barús	*barū́n-ō*	*barū́no-mai*
'heavy'	'make heavy, weigh down'	'be(come) heavy'
thermós	*thermaín-ō*	*thermaíno-mai*
'warm, hot'	'make hot'	'warm up, become hot'
leukós	*leukaín-ō*	*leukaíno-mai*
'bright, white'	'make white, whiten'	'turn white'

However, there is a second pattern in which both the fientive and the factitive alternant are morphologically active and which has not been systematically studied in the older IE languages. There are two variants of this pattern: One in which the fientive and the factitive alternants share the same stem (also called the "labile" pattern) and one in which they form different verbal stems. The "labile" pattern is sometimes discussed under the label "unmarked anticausatives"

[22]The diachrony of these verb classes and in particular the extension of the deadjectival verbalizers *-óe/o-*, *-ū́ne/o-*, *-aíne/o-* to new adjectival bases through reanalysis of the adjectival stem-forming morpheme as part of the verbalizer is discussed in Tucker 1981, 1990; Rau 2009a: 115–120, 152–155; Villanueva Svensson 2024; on the theoretical aspects of this reanalysis cf. Grestenberger 2022, 2023.

in the literature (e.g., Schäfer 2008; Heidinger 2010; Alexiadou et al. 2015) because it is part of a broader pattern by which anticausatives can be either marked (e.g., by a reflexive or detransitivizing affix or particle) or unmarked with respect to the transitive-causative alternant. In Modern Greek, the unmarked strategy is in particular associated with deadjectival CoS verbs, as noted by Alexiadou & Anagnostopoulou (2004: 125), who cite *asprizo* 'whiten', *kokinizo* 'redden', *mavrizo* 'blacken', *katharizo* 'clean', *stroggilevo* 'round', *klino* 'close', *anigo* 'open', *plateno* 'widen', *stegnono* 'dry', *stenevo* 'tighten', *skureno* 'darken', *kathistero* 'delay', *alazo* 'change' and *ksepagono* 'defreeze' as examples of verbs in which both the causative and the anticausative variants obligatorily take the active endings, i.e., verbs which do not alternate.[23] This pattern arose during the Hellenistic period and seems to have begun among non-deadjectival anticausative/CoS verbs, judging by the examples discussed by Lavidas (2009: 112–116), who links the rise of unmarked anticausatives to the more general restructuring of the distribution of active vs. nonactive morphology that took place in Greek around that time (though the details, especially with respect to the extension to deadjectival verbs, require further elaboration). A similar rise of morphologically active intransitivizations among alternating CoS verbs is discussed by Gianollo (2014) for late Latin. However, it remains unclear to what extent such a "labile" pattern existed in the older IE languages and whether it can be reconstructed for PIE.

In light of the diachrony of the thematic active endings, which are formally related to the reconstructed *middle* set of endings and explicitly treated as reanalyzed former middle endings in Jasanoff's "*h_2*e*-conjugation" framework (Jasanoff 1978, 2003, 2004, 2017a, 2019, 2023; cf. also Watkins 1969; Hollifield 1977; Villanueva Svensson 2003, 2021 on the origin of the thematic conjugation), apparently "labile" forms in which the active-marked alternant appears in intransitive contexts could also be seen as remnants of a stage in which the intransitive-fientive alternant took the nonactive endings as well as a different verbal stem formant than the transitive-factitive alternant (see Meiser 1993, Luraghi 2012 for different versions of this view) – that is, verbs that failed to undergo the "formal update" to the synchronically productive middle endings. This is, for example, the standard explanation since (at least) Watkins 1969 for why some *active* thematic aorists such as AG *édrakon* 'saw' and *étraphon* 'grew

[23]There is also a class in which the anticausative alternant can take either the active or the nonactive endings, that is, it can form both unmarked and marked anticausatives. The difference seems to be partially conditioned by the animacy of the subject in the anticausative variant, though the degree of spontaneity or completion of the change of state have also been adduced as possible factors. See Alexiadou & Anagnostopoulou 2004, 2013; Alexiadou et al. 2015: 88–90 for further discussion.

(strong)’ have formally *middle* presents (*dérkomai* and *tréphomai*, respectively; see also Rau 2013, 2016 on this type of development). A “labile” stage at the PIE level is therefore unlikely.

The second relevant pattern, in which both the transitive and the intransitive alternant are formally active, but built from different verbal stems, is better studied, especially in Latin and Hittite. In Latin, the suffix *-*ē*- that is also found in the Ancient Greek passive aorist and a number of other verbal categories across the IE languages (see Section 3.2) gave rise to a subclass of the Latin 2nd conjugation presents in the form of the stative(/fientive) verbs in -*ē-re* which obligatorily take the active endings. In Old Latin, their transitive-factitive alternants are active 1st conjugation presents, as illustrated in Table 11 (later replaced by the *rubefaciō*-type, cf. Section 3.2). The -*ē*-stem was moreover also used as the base for (likewise originally active-marked) fientives in -*ē-sce-re*.

Table 11: Deadjectival statives, fientives, and factitives in (Old) Latin (Watkins 1971: 47)

Factitive	Stative	Fientive	Adjective
*clār-**ā**-re*	*clār-**ē**-re*	*clār-**ē**-**sce**-re*	*clār-us, -a, -um*
‘make clear’	‘be clear’	‘become clear’	‘clear’
*-alb-**ā**-re*	*alb-**ē**-re*	*alb-**ē**-**sce**-re*	*alb-us, -a, -um*
‘make white’	‘be white’	‘become white’	‘bright, white’
*-nigr-**ā**-re*	*nigr-**ē**-re*	*nigr-**ē**-**sce**-re*	*niger, -ra, -rum*
‘make black’	‘be black’	‘become black’	‘dark, black’
*liqu-**ā**-re*	*liqu-**ē**-re*	*liqu-**ē**-**sce**-re*	*liqu-idus, -a, -um*; *lĭ̄qu-ēns*
‘make fluid’	‘be fluid’	‘become fluid’	‘fluid, liquid’

The alternation in Table 11 could be taken to suggest that -*ē*- and its factitive “alternant” -*ā*- are different “flavors” of the verbalizer *v* in the sense of Folli & Harley (2005, 2007); Harley (2013, 2017) associated with stative (BE) and fientive (BECOME) inner aspect (*Aktionsart*) on the one hand, and factitive/agentive (DO) on the other. In other words, the Latin alternation would be the synthetic equivalent of the *analytic* deadjectival factitive-fientive alternation discussed in Sections 1.2 and 3.2, with which it is also connected etymologically via the “stative” stem formant *-*ē*-.

In Ancient Greek, the cognate suffix -*ē*- (and its allomorph -*thē*-) is restricted to the perfective stem, unlike in Latin. However, the original use in Greek is for the most part that of a stative/fientive verbal stem-forming suffix (similar to the Latin examples in Table 11) rather than a Voice marker. Furthermore, it obligatorily takes the active set of endings, again like in Latin. Thus *-*ē*- is associated with the formation of "unmarked anticausatives" both in Latin and in Ancient Greek. In Ancient Greek, however, it is not specifically associated with PC roots/roots that form primary adjectives (though a few such forms are also found), but with CoS verbs more broadly. Examples are given in Table 12.

Table 12: Homeric CoS *ē*-aorists

*e-pág-**ē**-n* PST-become.fixed-**V.PFV**-1SG.PST.ACT	'became fixed, coagulated'
*e-mán-**ē**-n* PST-rage-**V.PFV**-1SG.PST.ACT	'went mad, became enraged'
*e-ág-**ē**-n* PST-break-**V.PFV**-1SG.PST.ACT	'broke' (intr.)
*e-tárp-**ē**-n* PST-enjoy-**V.PFV**-1SG.PST.ACT	'enjoyed, delighted in'
*e-phán-**ē**-n* PST-appear-**V.PFV**-1SG.PST.ACT	'appeared'

Crucially, the transitive factitive/causative alternants corresponding to these fientive/inchoative aorists use different stem formants (and the active endings) both in the imperfective (present) and the perfective (aorist) stem, cf. Table 13.

Table 13: Ancient Greek intransitive CoS *ē*-aorists and their transitive presents & aorists; stem formant = **bold**

Intr. aor.	Meaning	Tr. pres.	Tr. aor	Meaning
*e-ág-**ē**-n*	'broke'	*ág-**nū**-mi*	*é-ak-**s**-a*	'break/broke'
*e-tárp-**ē**-n*	'enjoyed '	*térp-ō*	*é-terp-**s**-a*	'cause(d) joy'
*e-tráph-**ē**-n*	'grew up '	*tréph-ō*	*é-trep-**s**-a*	'cause(d) to grow'
*e-pág-**ē**-n*	'became fixed'	*pḗg-**nū**-mi*	*é-pēk-**s**-a*	'fix/ed'
*e-rrág-**ē**-n*	'break'	*rhḗg-**nū**-mi*	*e-rrḗk-**s**-a*	'break/broke'

The same pattern – the intransitive-fientive and the transitive-factitive are formed from different stems, but both take the active endings – is found in Hittite, suggesting that this pattern is indeed of some antiquity (see Sasseville, this volume). Table 14 gives some examples from Hittite, in which the fientive is formed with the verbal stem formant -*ēšš*-, which has been etymologically linked to Latin statives in -*ē-re* and their associated fientives in -*ē-sce-re* at least since Watkins 1971. The corresponding factitive uses a nasal suffix -*nu*- that has been historically linked to nasal infix-presents from adjectival *u*-stems (Koch 1973, 1980; Sasseville 2021: 482–487, this volume). Synchronically, however, these are formed to adjectives with adjectival stem formants other than -*u*- as well, as Table 14 shows.

Table 14: Hittite deadjectival fientives & factitives (Rau 2009a: 112–113)

Adj.		Fientive	Factitive	
palḫi-	'broad'	*palḫ-**ēšš**-zi*	*palḫ-(a)**nu**-zi*	'broaden'
daluki-	'long'	*daluk-**ēšš**-zi*	*daluk-**nu**-zi*	'lengthen'
daššu-	'strong'	*daššau̯-**ēšš**-zi*	*daš(ša)-**nu**-zi*	'strengthen'
parkui-	'pure'	*parku-**ēšš**-zi*	*parku-**nu**-zi*	'become pure/purify'

Crucially, both the fientive and the factitive stem take the active *mi*-endings.[24] Together with the Latin and Ancient Greek evidence discussed above, this suggests that at least certain classes of stative and fientive deadjectival CoS verbs originally took the active **mi*-conjugation endings and were hence "unmarked anticausatives" in the sense of Schäfer (2008), as argued by Grestenberger (2023) based primarily on the Ancient Greek facts. Since there is a wealth of evidence that suggests that intransitive CoS verbs from result/eventive roots were originally associated with the *h_2*e*-conjugation or "proto-middle" endings (e.g., the "stative-intransitive" aorists of Jasanoff 2003, the intransitive nasal stems discussed by Gorbachov 2007 and Rau, this volume, the simple thematic presents of Rau 2013, 2016, among others), this could point to an original situation in which deadjectival stative and CoS verbs canonically took the active (**mi*-conjugation) endings, whereas CoS verbs from result/eventive roots took the (proto-)middle (*h_2*e*-conjugation) endings, a distribution that was subsequently given up or restructured in the attested daughter languages. This restructuring resulted in the

[24]The deadjectival factitive *nu*-stems take the *ḫi*-conjugation endings in the Luwic branch of Anatolian, which Sasseville (2021: 486) takes to be the original conjugation based on the parallel with the factitive -*aḫḫ*-type, for which *ḫi*-conjugation endings are reconstructed for Proto-Anatolian. He argues that this is due to both types being derived by conversion, but this is difficult to accept at least for the *nu*-stems, which are not instances of conversion in the usual sense since there is an overt verbal stem formant, -*n(u)*-. Moreover, it is not clear why conversion should automatically lead to *ḫi*-conjugation endings. See also Sasseville, this volume, for further discussion of *nu*-causatives and factitives in Anatolian.

mix of strategies that we see in the attested languages as well as in the final, "hybrid" type which uses both different stems and an alternation in voice marking, as we see especially often in Indo-Iranian pairs of full grade thematic middle fientives/inchoatives besides active nasal infix or *-*e̯ie/o*-factitives/causatives (Rau 2009a, 2013, 2016), cf. Table 15.

Table 15: Middle-marked intransitive, active-marked transitive CoS verbs with concomitant stem change in Vedic (Rau 2016: 358–359)

Intr. middle (full grade thematic)		Tr. active (nasal infix)	
*jáva-**te***	'speed'	*junā́-**ti***	'make speed'
*śóbha-**te***	'appear beautiful'	*śumbhá-**ti***	'make beautiful'
*páva-**te***	'become pure, purify oneself'	*punā́-**ti***	'purify'

Rau (2016, this volume) argues that the pattern in Table 15 can be reconstructed for alternating CoS verbs from Caland and non-Caland (eventive) roots at least for the inner Indo-European stage and that its further development, especially in terms of the voice marking of the alternants, is particularly relevant for understanding the development of factitive/causative alternation verbs in Indo-Iranian and Ancient Greek. Taken together, these case studies suggest that the interaction of voice marking and stem formation in deadjectival and result CoS verbs in Indo-European deserves further study.

Another deadjectival factitive class that has recently garnered attention is the "*nēwaḫḫi*-type", named after a class of Hittite factitives to adjectival bases formed with a stem formant -*aḫḫ*- < *-*eh₂*- and the *ḫi*-conjugation endings (Jasanoff 2003: 139–141; Hoffner & Melchert 2008: 175–176; Sasseville 2015, 2021), cf. Table 16. Formal comparanda of this type have been identified in Ancient Greek (Rau 2009b) and Baltic (Jasanoff 2003: 140–141, 2017b: 200–201; Villanueva Svensson 2020), which, however, diverge semantically: While the Anatolian -*aḫḫ*-verbs are factitives, the Baltic comparanda are iteratives (the Greek evidence is too scarce to be conclusive).

The verbal stem formant *-*eh₂*- has been identified with the abstract- and collective-forming *nominal* suffix *-*eh₂*- (Sasseville 2015, 2021, followed by, e.g., Villanueva Svensson 2020), suggesting a derivational chain as in (14), in which a deadjectival verb 'renew' is derived from a deadjectival abstract noun 'newness' via zero derivation or "conversion", as argued by Sasseville (primarily based on Lycian, where this pattern is productive).

Table 16: Hittite deadjectival *nēwaḫḫi*-factitives (Hoffner & Melchert 2008: 176)

Adjective		Factitive	
ḫappinant-	'rich'	*ḫappin-aḫḫ-*	'make rich'
ikuna-	'cold'	*ikun-aḫḫ-*	'make cold'
nakki-	'important'	*nakkiy-aḫḫ-*	'regard/treat as important'
nēwa-	'new'	*nēw-aḫḫ-*	'make new'

(14) **néu̯-o-* 'new' →
**néu̯-eh$_2$-* 'newness, news' →
**neu̯-eh$_2$-Ø-* 'make new/with newness', 3sg. **neu̯-eh$_2$-Ø-e(i)* 'makes new'

A formal parallel for deriving deadjectival factitives from a nominal rather than adjectival surface form comes from English, cf. *long* – *length-en*, *strong* – *strength-en*, though with overt verbalizing morphology. This type of derivation would then have given rise to the Hittite and Baltic forms of this type, in which *-aḫḫ-*/**-ā-* is synchronically a verbal stem formant and the formal and semantic basis of this class are PC adjectives of various stem types (Hittite) or verbal adjectives (≈ "result statives" from non-deadjectival roots, as in Table 4) in Baltic.[25] However, this depends on whether one accepts that conversion was a productive verbal derivational process in PIE[26] (so also Villanueva Svensson 2021) and requires further explanation of the semantic discrepancy (factitive vs. iterative) and the reason for the association of this type with **h$_2$e*-conjugation endings (for a novel proposal concerning the latter point see Pinault, this volume).[27]

[25] Though synchronically the type is "deverbative", cf. Villanueva Svensson (2020: 392–395, this volume).

[26] Adjectives in English do not participate in conversion, e.g., **a*/**to white/flat/smart* vs. *a*/*to hammer/table/walk*, cf. Borer 2013: 371–378. Acedo-Matellán (2022a) argues that this is because they are structurally complex, consisting of a preposition and a noun. If this turns out to be a systematic gap cross-linguistically, this would speak against the conversion account of Villanueva Svensson (2021). However, it is not yet clear to what extent Borer's observation holds cross-linguistically or (if it does) why such a complex structure should be compatible with overt but not with zero verbalizers (assuming that that is what conversion is; see Grestenberger & Kastner 2022: 47–52 for further arguments).

[27] Jasanoff (2003: 146) tentatively proposes that the factitive value of the type in Hittite is due to secondary transitivization – the formation of an oppositional active factitive to an older intransitive fientive with **h$_2$e*-conjugation endings. However, middle-marked fientive *aḫḫ*-verbs are rare in Hittite, and the proposal has not been taken up further.

3.4 Deadjectival activity verbs and iterativity

Although deadjectival factitive-fientive alternation verbs have received most of the attention in the Indo-Europeanist literature on deadjectival verb formation so far, there are also other verb classes that seem to be semantically and/or morphologically associated with adjectival stems. One class was already briefly mentioned in Section 2.2, namely deadjectival activity verbs like Sp. *holgazanear* 'to act lazily' (*holgazán* 'lazy') or Hebr. *hištaxcen* 'to act arrogantly' (*šaxcan* 'arrogant'), in which the base adjective seems to act as a modifier describing a property of the surface subject. These exist in the older IE languages as well, but have not received much attention so far. Marescotti (2024) discusses this class in Ancient Greek under the label "quality verbs", with examples such as *stōmúllō* 'be talkative, chatter' (*stōmúl-os* 'talkative') and *aióllō* 'shift rapidly back and forth' (*aiól-os* 'quick, nimble'). In Latin, examples include *grātor* 'rejoice' (*grāt-us* 'pleasant, agreeable') and *misereor* 'feel pity for' (*miser* 'wretched, pitiful'), though with a slightly different semantic relationship between the base and the derived verb. This class tends to be derived from morphologically complex adjectives and shares semantic and morphological similarities with denominal verbs from animate agent-like bases ("pseudo-agent verbs") such as AG *basileú-ō* 'be king, rule over' (*basileús*), *tektaín-omai* 'be a carpenter, work as a carpenter' (*téktōn* 'carpenter') and Latin *ancillor* 'serve as a maid' (*ancilla* 'maid, female servant'), *interpretor* 'explain, interpret' (*interpres*, *-pret-*) 'intermediary'), *philosophor* 'act like a philosopher, be philosophical' (*philosophus* 'philosopher'), etc.,[28] including the ambiguity between a state and an activity reading.

A related class we want to briefly discuss here consists of verbs that are usually classified as iteratives or pluractionals and describe a plurality of events. Specifically, these verbs are treated as event-internal pluractionals, in which a plurality of subevents is grouped into a single event (e.g., Tovena & Kihm 2008; Tovena 2010; Greenberg 2010). Because they are synchronically derived from (non-PC) roots or verbal stems, they are usually called "deverbal" or "deverbative", though morphologically they are often root-derived (cf. fn. 4 and 12 above). However, from the point of view of their derivational morphemes, such iteratives (used here as a cover term) often seem to be associated with *adjectival* morphology at least diachronically, specifically with "verbal adjectives" from eventive roots rather than PC roots.[29] This pattern is attested in Latin, for example, which has a

[28]See for example Flobert (1975: 666–674), who treats both the deadjectival and the denominal verbs of this types as "predicative" verbs. On "pseudo-agent verbs" see also Xu et al. 2007: 137–139; Bleotu 2019: 163–168; Marescotti & Grestenberger 2026.

[29]Though Rau (2009a, 2013) also discusses iteratives that are associated with the Caland system.

synchronically deverbal class of verbs that forms “repetitive verbs” using a suffix -*tā*-. The -*t*- is generally agreed to be identical to the perfective passive participial suffix at least synchronically (Weiss 2020: 424),[30] -*ā*- marks conjugation class I, cf. ex. a. in Table 17. This class forms event-internal pluractionals, whereas the “frequentatives” in -*it*-*ā*- in ex. b., which are also based on what seems to be the participial -*t*-stem (cf. -*t*-*us*), form multiple-event pluractionals.

Table 17: Latin repetitives and frequentatives

a. Latin repetitives			
cap-e-re	‘seize’	*cap-**t**-ā-re*	‘capture’
hab-ē-re	‘have’	*habi-**t**-ā-re*	‘inhabit’
b. Latin frequentatives			
ag-e-re	‘do’	*ac-**t**-**it**-ā-re*	‘do repeatedly’
can-e-re	‘sing’	*can-**t**-**it**-ā-re*	‘sing repeatedly’

A second pertinent example is found in Germanic, for example in Old High German (OHG), which has a class of synchronically deverbal iteratives in -*ar*-. These inflect as weak class II verbs independently of the conjugational class of the base, cf. Table 18.

Table 18: OHG -*ar*-iteratives

Base		Iterative	
wach-ēn	‘be awake’	*wach-**ar**-ōn*	‘to stand guard’
wīg-an	‘fight’	*weig-**ar**-ōn*	‘refuse, be obstinate’
gang-an	‘go’	*gang-**ar**-ōn*	‘walk around’

[30]Diachronically, the situation is less clear – this class may have originated in denominal verbs based on abstract nouns to *s*-desideratives, cf. Nussbaum 2021b: 18 & fn. 49 (for an opposing view cf. de Vaan 2012). The repetitives of the type *ag-it-ā-re* (< **ag-et-*) ‘shake, agitate’ (*ag-ere* ‘drive’), which are not based on the synchronic participial stem, on the other hand, seem to be derived from agent(ive) nouns of the type *eques*, *equit-* ‘horserider’, whence *equit-āre* ‘to be a rider, to ride’ (see Nussbaum 2017: 260–263 on the functions of nouns in *-*(e)t*-). In that case, this class of repetitives would be derivationally similar to the “activity verbalizations” discussed in the main text above. Cf. also Weiss 2020: 424–425 for a recent discussion of the origin of the Latin repetitives and frequentatives.

The handbooks locate the source of the *-ar*-suffix in adjectival *r*-stems such as *wach-ar* 'awake', *weig-ar* 'obstinate', etc. (e.g., Wilmanns 1896: 92–95), which are usually formed to result roots, though comparatives in *-(e)r-* may also have played a role.

The Baltic continuants of the "*nēwaḫḫi*-type" have already been mentioned in this context in the previous section (Section 3.3). In Baltic, this class forms deverbal iteratives using a suffix *-*ā*-. The diachronic derivational basis of these stems seems to consist of verbal adjectives in *-*to*- (cognate with the Latin -*t*-participles) and *-*o*-. This is implied by the root *o*-grade and the remnants of adjectival morphology in the iteratives while the origin of the *-*ā*-verbalizer is debated (cf. Section 3.3). Table 19 shows some examples from Lithuanian.

Table 19: Lithuanian iteratives (citation form = 1SG)

Iterative		Base verb		Verbal adjective	
***glóst**-au*	'stroke'	*glódž-iu*	'smooth, polish'	***glós-t**-as*	'smooth'
***laik**-aũ*	'hold'	*liek-mì*	'remain'	*-**laik**-as*	'remaining'

What these iterative verb classes have in common is that 1) they inflect according to the synchronically productive inflectional class for forming denominal verbs (e.g., the first conjugation in Latin), independent of the inflectional class of the base, 2) they are atelic iterative/pluractional verbs, and 3) their derivational morphology can plausibly be linked to that of root-derived verbal adjectives at least diachronically. Unlike deadjectival verbs from property concept or "adjectival" roots, which usually form degree achievement verbs of the factitive-fientive alternation (Section 3.3), these verbs are formed to adjectives from eventive roots which are not typically gradable or scalar. The derived iterative verbs themselves are atelic activity verbs rather than achievements (cf. also Tovena & Kihm 2008; Tovena 2010; Grestenberger & Kallulli 2019 on pluractionality and the semantics of "verbal diminutives").

As this brief survey shows, different types of adjectives give rise to different types of deadjectival verbs, and this is reflected both in the morphology and the semantics of the resulting verb classes. However, more work is needed to explain why verbal adjectives become the input to verbal derivation in the first place, how iterativity arises compositionally from such derived verbs, and how they are connected to other types of deadjectival activity verbs.

3.5 Summary

In this section, we have briefly summarized the current state-of-the-art on the Caland system, the lexicalization of PCs in the older Indo-European languages, and the debate around the nature and origin of the CS. We have also provided a survey of different types of periphrastic and synthetic verb forms associated with deadjectival verb formation in these languages, focusing on the factitive-fientive alternation and the formation of different types of statives. This discussion is far from exhaustive, and we refer to the articles in this volume for a more in-depth discussion – for individual branches in particular the contributions in part I.

In the next section, we provide a brief outline the structure of the volume and summarize the contributions.

4 Contributions

Because of the intentionally broad definition of "deadjectival" we have proposed in Section 2.1 and the "categorial instability" that is cross-linguistically observable with respect to the lexicalization of PCs, the contributions collected in this volume cover not only the factitive-fientive alternation, but also deadjectival verbs in which the semantic derivational basis appears to diverge from the morphological basis, as in the famous case of the "*nēwaḫḫi*-type" (cf. Section 3.3 and the contribution by Pinault). This volume is therefore also intended to connect to the existing literature on denominal verb formation in Indo-European more generally, which usually treats verbal derivation from both adjectival and substantival nominal bases (e.g., Tucker 1988, 1990; Insler 1997), as for instance in the contributions by Merritt and Pinault, as well as to more recent work on factitive-fientive alternations related to the Caland system and the discussion concerning the potential deadjectival origin of the PIE *u*-presents and simple thematic presents (Villanueva Svensson 2021; Jasanoff 2023).

The contributions are divided into three parts: Part I contains contributions that aim to fill empirical gaps in languages or subbranches of Indo-European in which deadjectival verbs are understudied, namely Boldt for Old Norse, Höfler & Stifter for Celtic, Sasseville for Anatolian, and Villanueva Svensson for Balto-Slavic.

In "To be or to become, that is the question: The morphosemantics of the weak verbs formed to color adjectives in Old West Norse", Ramón Boldt discusses the morphology and semantics of the verbal derivatives of eight Basic Color Terms (BCTs) in Old West Norse, thus filling a gap in the literature on Old Norse verbal derivation. His examination of the derived weak verbs shows that they display the semantics expected of deadjectival formations, namely factitive and fientive/anticausative meanings. However, in some cases the translations offered by the dictionaries lead to unclear or even contradictory readings both concerning their sense and the mapping between morphology and semantics. Therefore, the verbs in question require a new philological analysis of the source texts. Boldt argues for determining the semantics as precisely as possible for these problematic cases in order to identify the original semantics of the different verb classes and their diachronic development. He also discusses their associated stem formation types.

Stefan Höfler and David Stifter's article "Caland verbs in Celtic" discusses deadjectival verbs in an Indo-European branch that has not yet been the subject of a comprehensive treatment within Caland-related studies, especially regarding verbal derivation. The authors show that at least 20 verbal stems associated with the Caland system can be found in Celtic. While half of them have identifiable cognates in other branches and are therefore likely to be inherited, the other half lack extra-Celtic cognates. However, within those probable inner-Celtic formations, based on the evidence of Old Irish and Middle Welsh the authors identify verb types that are clearly related to the Caland system, such as nasal-infix presents and iterative-causatives as well as a stative and a simple thematic present. They conclude that this evidence speaks in favor of a mild productivity of the root-deriving principle of the Caland system in Proto-Celtic.

David Sasseville discusses the development of the denominal/deadjectival morpheme -*nu*- in the Anatolian branch in his contribution "The relationship between *u*-adjectives and *nu*-causatives in Anatolian". The Anatolian evidence points to a similar derivational pattern as in Old Indic and Ancient Greek, namely the reanalysis of *n*-infixed verbs derived from *u*-adjectives as containing a *nu-suffix*. Though it can be shown that the infixation strategy precedes suffixation, synchronically the forms are often ambiguous regarding their derivational base, i.e., they could be analyzed as being deadjectival or as derived from a root verb (a verb with a zero verbalizer). As Sasseville argues, this situation can lead to yet another process of reanalysis, namely one that gives rise to a "*u*-extension" of roots. While previous research has focused mainly on Hittite, the addition of data from other Anatolian languages provides new insights into the the diachronic interaction of root verbs, *u*-adjectives and *nu*-formations, and demonstrates that the loss of the adjective is not a requirement for a reanalysis of the *u*-extension type. Thus, some verbs in Luwian, Lycian and Palaic can be explained as derived from precisely such *u*-extended roots.

In "Balto-Slavic denominatives from an Indo-European perspective: An update", Miguel Villanueva Svensson provides a timely survey of denominative verb formation in Balto-Slavic, an urgent desideratum both from the perspective of Indo-European and of Balto-Slavic studies. As the author himself states, treatments of this topic are either missing completely or are present only in a short and often outdated way in the literature. This paper serves as an overview that will pave the way towards further research on secondary verb formation in Balto-Slavic. Specifically, Villanueva Svensson argues that despite being notoriously innovative in some regards, the verbal system of Balto-Slavic provides valuable insights into the reconstruction of denominative verb formation in Indo-European. By surveying the different morphological means (i.e., suffixes) attested in Baltic and Slavic, he first identifies the inherited as well as innovated strategies for deriving verbs from nominal bases and then connects them to formations in different branches of the family such as Vedic, Ancient Greek, and, in the case of the *nu*-factitives, Anatolian (cf. the immediately preceding chapter by Sasseville).

The contributions in part II focus on the reconstruction of derivational patterns of deadjectival verbs and property concept roots in Proto-Indo-European and their development into the attested languages.

Romain Garnier discusses "Delocatival adjectives surfacing as participles" in a contribution that addresses the difficult question of categorial affiliation of several participial and adjectival PIE suffixes containing the morpheme *-*en*-, which he argues emerged as delocatival formations based on root nouns. This paper thus contributes to understanding how nominal and adjectival suffixes can become integrated into the verbal system and eventually be reanalyzed as verb forms.

Andrew Merritt's paper "De-adjectival and De-"adjectival": Greek κρύπτω 'cover' and Proto-Germanic **χreuðanq̨* 'deck, equip, adorn"' deals with questions of internal derivation and complex derivational chains in the domain of deadjectival verb formation. Merritt shows that within such chains, originally secondary formations such as *-*i̯e/o*-presents from adjectives could be reanalyzed as primary verb formations (and even lead to the creation of new roots), which in turn gave rise to new (verbal) adjectives functioning as derivational bases for new secondary formations. Decasuative light verb periphrases with *$d^h eh_1$- and *$b^h uH$- to adjectival formations can also serve as bases for the creation of new roots and adjectives. Understanding the diachrony of the Caland system thus provides a tool for linking different synchronic reflexes of various branches (Greek, Tocharian, Germanic, Balto-Slavic) to a derivational "bigger picture".

In "Reassessing the emergence of the PIE *-eh_2-factitive present type", Georges-Jean Pinault addresses a verbal class that has only recently become the focus of attention in the context of the question whether conversion (the transformation of a nominal stem into a verbal stem, and vice versa, without overt affixation) played a role in verbal derivation in Proto-Indo-European or in specific daughter branches thereof, such as Anatolian. By questioning the exact nature of this process within Proto-Indo-European, the paper discusses three examples for which conversion has been proposed and suggests alternative explanations, partially based on inner-IE derivational processes that are already known, but also by introducing a novel analysis of the *nēwaḫḫi*-type, namely as a reanalyzed dative singular case form of abstract nouns used in predicative function. Contrary to previous explanations of this enigmatic verbal stem class, this approach explains the persistent *ḫi*-inflection of this type in Anatolian.

Jeremy Rau's article "A late Nuclear Proto-Indo-European verbal type: Intransitive thematic nasal-infix present actives" identifies a new verbal derivational class in "late Nuclear PIE" (the ancestor of the remaining IE languages after Anatolian, Tocharian, and Italo-Celtic branched off), namely thematic nasal-infix presents to state-oriented roots. Rau argues that at least some of these presents are "deadjectival" in the sense of being derived from Caland-associated property concept roots, and often associated with the causative alternation (or factitive-fientive alternation) synchronically. He further identifies this intransitive state-oriented type with the "Northern IE" intransitive/anticausative *h_2e-conjugation nasal presents of Gorbachov (2007). This is a significant finding that pushes back the date of reconstruction for this class and at the same time clarifies further aspects of primary verbal derivatives to Indo-European PC/Caland roots and state-oriented roots more generally.

Part III concludes with two contributions that address deadjectival verb formation from a quantitative diachronic perspective (Iacobini & De Rosa) and a theoretical perspective (Tanza-Kamil), thus situating the Caland phenomenon in the broader empirical landscape.

In their article "Diachronic trends in the formation of Italian deadjectival verbs", Claudio Iacobini and Maria Pina De Rosa report the result of a diachronic corpus study on the morphological expression of deadjectival verbs in Italian, based on a dataset of ca. 900 verbs from the 13th to the 20th century. The authors identify three main verb formation processes (suffixation, conversion, parasynthesis), whose productivity varies diachronically, with suffixation becoming the dominant mode of forming deadjectival verbs in the history of Italian. They provide a detailed analysis of the different strategies and their distribution according to adjective type (e.g., simplex vs. morphologically complex) as well as aspectual-semantic class, confirming that the majority of deadjectival verbs are change-of-state verbs. This study therefore provides an important quantitative argument in favor of treating deadjectival/"Caland-associated" verbs as the regular expression (and diachronic source) of the factitive-fientive alternation, as argued in several other articles in this volume for the older IE languages.

In "Deriving the semantics of the Akkadian stative: Perspectives from the adjectival stem", Iris Tanza-Kamil discusses the semantics of the Akkadian stative, a deadjectival construction based on verbal adjectives that express a resultant state from (different types of) CoS roots, but also from unergative, unaccusative, and transitive eventive roots. These are called primary statives, besides which the author identifies "secondary statives" derived from nominal and adjectival stems. She argues that all statives are verbal in terms of their category and resultative in terms of their semantics. This article provides an important typological and theoretical addition to the other contributions in the volume and adduces crucial evidence from a category that is otherwise underrepresented in the discussion of deadjectival verbs, namely verbal adjectives as the input to deadjectival verb formation.

5 Conclusion

In this introductory chapter, we have outlined current research avenues in deadjectival verb formation from the perspective of theoretical linguistics, linguistic typology, and historical Indo-European linguistics. We have discussed the cross-linguistic lexicalization patterns of PCs, the connection between PCs and possessive predication, the nature of lexical categorizers associated with PC roots, and different types of change-of-state verbs. We have also provided a working definition of deadjectival verbs that will hopefully be useful for future investigations of these verb classes.

Our main focus rested on the introduction to and state-of-the-art of research on the "Caland system", that is, the lexicalization and derivational patterns associated with PC roots in (Proto-)Indo-European, followed by a survey of the various verbal derivational patterns in the older IE languages. We have pointed out desiderata as well, such as a stronger engagement with the theoretical and typological literature on PCs cross-linguistically, the need to better understand the difference between root- and stem-based deadjectival verb formation, the role of different types of adjectival morphology in verbal derivation, and the interaction of deadjectival verb formation with various voice and argument structure alternations. Together with the articles collected in this volume, we thus hope that this introduction will serve as a starting point for further work on deadjectival verb formation, the lexicalization of property concepts, and the various ways in which adjectival and verbal morphology interact.

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Abbreviations

Act.	Active
Adj.	Adjective
AG	Ancient Greek
ChSl.	Church Slavic
CoS	change of state
CS	Caland system
Cz.	Czech
Fact.	Factitive
Hitt.	Hittite
IE	Indo-European
Intr.	Intransitive
Lat.	Latin
Lith.	Lithuanian
MG	Modern Greek
Nonact.	Nonactive
OAv.	Old Avestan
OCS	Old Church Slavonic
OHG	Old High German
PC	Property concept
PGmc.	Proto-Germanic
PIE	Proto-Indo-European
PoS	Part of speech
PSl.	Proto-Slavic
Russ.	Russian
Sg.	Singular
Sl.	Slavic
Tr.	Transitive
Ved.	Vedic
YAv.	Young Avestan

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Part I

Deadjectival verbs in the older Indo-European languages

Chapter 2

To be or to become, that is the question: Word formation of the weak verbs formed from color adjectives in Old West Norse

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This article examines the semantic properties of a group of twelve Old West Norse deadjectival weak verbs, namely those derived from the eight Basic Color Terms that are attested in this language. The data is drawn from five major Old West Norse dictionaries. This way, it can be shown that the three class I *jan*-verbs which are of Proto-Germanic origin share factitive semantics. The six genuinely Nordic class II *nan*-verbs share anticausative-inchoative semantics. The three class II *ōn*-verbs, however, are diverse in semantics and age. It can further be shown that the lack of context typical for dictionaries nurtures semantic imprecisions and inconsistencies, which can be met by investigating the original sources.

1 Introduction and aims

In what follows, I will present some initial considerations as part of an ongoing project on Old West Norse (OWN) verbal word formation. As the Vienna workshop focused on deadjectival verb formation, the topic of this paper is the formation of weak verbs to color adjectives. For this purpose, I have mainly evaluated some of the larger Old West Norse dictionaries (Section 3). Underlining the shortcomings of this method in comparison to working with corpora with regard to issues of *Aktionsart* (hence "to be or to become" in the title) has also been a goal of the talk and, subsequently, of this article.

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2 Basic Color Terms

With respect to color adjectives, it is not possible to cover all words for colors that are attested in Old West Norse texts for reasons of space. Therefore, the selection will be limited to the so-called Basic Color Terms, a term well established since Berlin & Kay 1969 (*Basic Color Terms: Their Universality and Evolution*). In the most recent investigation, Jackson Crawford's dissertation *The Historical Development of Basic Color Terms in Old Norse-Icelandic* (Crawford 2014), the author provides a selection of color terms that can be considered basic in Old West Norse.[1] Crawford himself refers to the latest comprehensive study of the topic, Biggam (2012). Here is not the place to discuss their concept of Basic Color Term (BCT) in length. Instead, I will present the most important criteria according to Crawford (2014: 125–153) and Biggam (2012: 21–42) in order to motivate the focus on a certain class of color adjectives in this article. In principle, I follow both authors, but not always rigidly, and I deviate from their criteria when it serves my purpose of elucidating matters of verbal word formation.

a) The color term in question must be monolexemic, i.e., a simplex. It must therefore not be deducible from the sum of its parts. In other words, it must be non-predictable. Most of the complex color terms in Old West Norse seem to be *ad hoc* formations, attested only once or twice, e.g., *móbrúnn* 'dark brown' or *iðjagrønn* 'green again (?)'. However, it can be argued that those compounds could only have been formed on the assumption that readers must have been able to predict their meaning.[2]

b) The color term in question must not be a hyponym. Crawford (2014: 59) is "hesitant" in ascribing basic status to OWN *brúnn* 'brown'. He rather considers it a contextually restricted hyponym for *svartr* 'black', as it is used only to describe a certain dark hue of hair, especially horse hair. In other words, a horse with *brúna* hair can also be described as having *svarta* hair. Nonetheless, I will make some brief remarks on OWN *brúnn* here as well, because it is important for issues related to word formation.

[1]For a similar approach, cf. Wolf (2013), Brückmann (2012), Wolf (2006a). For the Non-Basic Color Terms in Old West Norse, cf. Wolf (2015).

[2]Crawford (2014: 126) discusses the compound *skolbrúnn*, which he translates as "dirty water brown" or perhaps "dirty-water (eye)browed". In light of similar compounds like *léttbrúnn* 'light-browed (i.e., happy looking)' and *hvítbrúnn* 'white-browed', it seems to me that this can not be taken as a good example here.

c) Related to b) is the idea that the use of a color term in question must not be restricted to a class of objects. A good example can be seen in German and English *blond* which can only be used to describe the human hair color.[3]

d) BCTs are morphologically more productive than other color terms. For example, in contemporary English, we see that only *red*, *black*, and *white* can serve as bases for anticausative-inchoative derivations: *redden*, *blacken*, and *whiten*. This could, as we will see, be true for Old West Norse as well in that at least one derivative is attested for every BCT.

e) Finally, a lexeme in question must contain information about hue, that is "the spectrum of visible light, parts of which, according to their wavelength or frequency, are perceived by humans to differ from others" (Biggam 2012: 3). Lexemes that only contain information on saturation or tone, that is "the admixture of white or black with a hue" (ibid.), are excluded. An example from OWN would be *døkkr* 'dark', which is neutral in respect to hue. However, OWN *bleikr*, often translated as 'pale' in the dictionaries or conceptualized as "macrocolor covering, at least partly, the category of pale of light colors" (Wolf 2013: 143), is part of Crawford's investigation as he favors it over *gulr* as the BCT for 'yellow'. I follow him in that regard.

In light of what we have established so far, the following color terms will be considered basic in Old West Norse: *blár* 'blue, black'; *bleikr* 'yellow'; *brúnn* 'brown'; *grár* 'grey'; *grǿnn* 'green'; *hvítr* 'white'; *rauðr* resp. *rjóðr* 'red'; *svartr* 'black'. Accordingly, the verbal word formation of those color adjectives will be the focus of this article.

3 Sources

As mentioned in Section 1, I focus on dictionaries here in order to highlight the shortcomings of this method in comparison to working with corpora, namely the sometimes imprecise determination of the semantics of the respective verbs. The following dictionaries were used:

- Cleasby's and Vigfússon's *Old Norse to English Dictionary* (from now on CV),

[3] In contrast to OWN *brúnn* : *svartr*, English and German *blond* is usually not considered a hyponym of, say, *gelb* or *yellow*, respectively.

- Baetke's *Wörterbuch zur altnordischen Prosaliteratur* (WAP),
- Fritzner's *Ordbog over det gamle norske Sprog* (OgnS),
- and the not yet completed *Dictionary of Old Norse Prose* (ONP).

The last one is certainly more than a mere dictionary in that it contextualizes the words in question and provides scans of the relevant editions of those texts in which the words are attested. Weak verbs from the poetic language recorded in the *Lexicon Poeticum* (LP) do not provide any substantial additional information regarding the questions discussed here. The LP will therefore be quoted only occasionally.

4 History of research

In comparison to the other Old Germanic languages, we know relatively little about verbal word formation in Old (West) Norse, although researchers have long taken a keen interest in the language.[4] In the early Neogrammarian period, research focused mainly on formal aspects of verbal inflection. Noreen (1923) – considered the standard reference grammar for Old West Norse until today – does not mention word formation at all. The same is true for other grammars and surveys from that era (Wimmer 1871; Brenner 1882; Heusler 1932; Krause 1948; Andersen 1954). In contrast, Holthausen (1895) seems to be the first to discuss the verbal word formation of Old West Norse. He does so on one page in a chapter entitled *Bildungslehre* (131). Here, Holthausen is mostly concerned with the *jan*-suffix that, according to him, primarily forms causatives from strong ablauting verbs, for example, *føra* 'to lead, transport, move' from *fara* 'to go, travel, fare'.[5] Additionally, the class II *ōn*-suffix forms denominatives, e.g., *tala* 'to speak' from *tal* 'talk, conversation; speech, language'. Also, class II contains *nan*-verbs of the former class IV, e.g., *vakna* 'wake up' in contrast to *vaka* 'being awake, waking'. Holthausen describes those verbs as passive and inchoative, a view still upheld by some researchers.[6]

[4] For Gothic see Marti Heinzle (2023), García García (2005); for Old English see Bammesberger (1965); for Old High German see Marti Heinzle (2019), Riecke (1996); for Middle High German see Leipold (2006); for Early New High German see Habermann (1994); for Old Frisian see Marti Heinzle (2014); for a comparative study on Gothic and Continental West Germanic see Schwerdt (2008).

[5] Still visible today in Nynorsk *å føra* : *å fara* or German *führen* : *fahren*.

[6] In light of more recent research, it is argued that the class-forming feature of the *nan*-verbs is not inchoativity (i.e., the verb denotes the (gradual) beginning of a situation) and/or pas-

Class IV exists as such only in Gothic. In West Germanic, these verbs joined class II or class III (in Old High German). A few *nan*-verbs may be attested in Proto-Norse, but their readings and interpretations must be considered uncertain.[7] Moreover, it is difficult to determine whether they formed a separate class.

The path of research that began with Holthausen (1895) seems to have reached its climax with Wessén's (1965) *Svensk språkhistoria: Ordbildningslära*. Wessén was comprehensive in his analysis, writing for almost 20 pages on Norse verbal formation (90–107). However, we are still rather limited in what we know compared to other older Germanic languages (see fn. 4). Except for the field's current focus on the former class IV verbs, the state of research seems to be condensed in Nedoma's (2010: 118–120) *Kleine Grammatik des Altisländischen*. According to Nedoma, the three weak classes of Old West Norse entail the following semantic properties:

- class I *(i)jan*-verbs: deverbal causatives, denominal factitives, and some intensive-iteratives
- class II *ōn*-verbs: denominals of various kinds, as well as some intensive-iteratives
- class III *ēn*-verbs: mainly intransitive duratives.

5 Weak verbs attested in Old West Norse

5.1 General remarks

In what follows, I will evaluate the semantics of the Old West Norse weak verbs formed to BCTs as given in the dictionaries listed above and examine whether their translations correspond to what is discussed in the reference grammars and handbooks. In OHG, we can witness a decreasing correlation between semantic

sivity, but anticausativity (i.e., the verb denotes a situation where the subject is non-agentive, often relating to internal and external nature), cf. Scheungraber (2014); Ottósson (2013); also Villanueva Svensson (2011). Haspelmath (1987: 16–17) seems to be the first to explicitly refer to the Gothic *nan*-verbs as anticausative. Gorbachov's (2007) study on the nasal presents in Germanic, Slavic, and Baltic does not make use of the term *anticausative* but calls this class *inchoative*. See the introduction of this volume on the different uses of these terms in the context of deadjectival verb formation in general.

[7] Cf. PN *hleunan** 'to become warm' (cf. Boldt 2024), *sufnan** 'to start moving', *dwenan** 'to dwindle'.

classes and certain morphemes, a process coming to an end in MHG, where the suffixes merely function as purely morphological markers of inflection.[8]

I will present the BCTs (as mentioned in Section 2) one after another in alphabetical order. I will also present their meaning, cognates in the other Old Germanic languages, basic etymological information, and finally the weak verbs derived from them. I will then give the meaning of those verbs as posited in the dictionaries as well as representative examples of their actual occurrences. In most cases, the meanings provided by the dictionaries agree with one another. However, there are some instances where the provided meanings are inconsistent. In examples of the latter, I will try to provide a solution based on the actual texts.

5.2 *blár* 'blue, blue-black; black; dark'

The first BCT to be presented here is *blár* 'blue; blue-black; black; dark'.[9] It is well attested in the other Old Germanic languages, cf. OE in *blǽ-hǽwen* 'of a blue hue, bluish' and *blǽwen* 'light blue', OHG *blāo*, OFris. *blāu*, OS *blāo* < PG **blēwa-* 'blue', and connected to Lat. *flāvus* and OIr. *blā*, although the exact circumstances are not entirely clear (cf. EWA II 160; EDPG 68; EWGP 135; **Luhr2000**).

Regarding verbal derivations of *blár*, only one weak verb is attested in Germanic in general, and in OWN in particular. It is the verb *blána* that belongs to the former class IV, but synchronically to class II. It is also attested in OSwed. *blana*. According to ONP, the verb is attested at least 13 times in OWN prose. With regard to the meaning of this verb, the dictionaries agree: They all imply anticausative-inchoative semantics.[10]

- ONP: 'to turn black-and-blue, take on a livid color'
- CV: 'to become black, livid'
- WAP: 'dunkel werden, eine dunkle Farbe annehmen, blauschwarz werden'
- OgnS: 'blive blaa' ('to become blue')

An example for this use of *blána* is given in (1):

8 MhdGr III: 494–496; Boldt (2023: 109–111).

9 For a detailed study on the use of *blár*, also in comparison to *svartr*, in Old Icelandic, cf. Wolf (2006b).

10 As stated in fn. 6, in more recent research the anticausative character of (former) *nan*-verbs is favored over the earlier assumption of primarily inchoative semantics. As this issue is not decisive here, I will stick with the term anticausative-inchoative, cf. Villanueva Svensson (2011).

(1) Ágrip af Noregs konunga sǫgum 5,6[11]
Var þá hvata-t bál-i ok hon
was then prepare.hastily-ptcp.prf.nom.sg pyre-dat.sg and she
bren-d ***blána-þi*** *þó áþr all-r*
burn-ptcp.prf.nom.sg **turn.dark**-3sg.pret but before complete-nom.sg
líkam-inn ok ull-o ór orm-ar ok
body-nom.sg.det and swarm-3sg.pret out worm-nom.pl and
eþl-or …
serpent-nom.pl
'Then a pyre was hastily prepared, and she was burnt, but before that, the body **turned dark** completely and there swarmed out worms and serpents, …'

5.3 *bleikr* 'pale; yellow'

The second BCT in our list is *bleikr* 'pale; (light) yellow'. In light of OE *blác*, *blǽce*, OHG *bleih*, and OS *blēk*, a preform PG **blaika-* must be reconstructed (EWA II 176; EDPG 66; EWGP 127).

For *bleikr*, three verbal derivations are attested, the first one of which is the class I verb *bleikja*. It is also attested in OE *blǽcan*, OHG *bleichen*, MLG *blēken*, and must therefore be reconstructed as PG **blaikijan-*. The dictionaries once again agree on the meaning:

- ONP: 'to bleach, make lighter'
- CV: 'to bleach'
- WAP: 'bleichen'
- OgnS: 'blege' ('to bleach')

Compare example (2):

(2) Snorra Edda 130,38
Brynhild-r ok Gvðrvn gengv til vat-z at
Brunhild-nom.sg and Guðrun.nom.sg went to water-gen.sg to
bleik-ia *hadd-a sina*
bleach-inf hair-acc.pl their
'Brunhild and Guðrun went to the water in order to **bleach** their hair.'

[11]All sources are cited according to the editions that are used and given in the ONP.

The second weak verb derived from *bleikr* is OWN *blikna*. Just like *blána*, it is a class II verb with anticausative-inchoative semantics, earlier associated with the fourth class. As with *blána*, it is attested only in OWN (at least 30 times according to ONP, twice according to LP).[12] In addition, *blikna* is one of the zero-grade derivations from a strong verb well attested in OWN,[13] namely from OWN *blíkja* 'to glisten, glint' (< **bleikan-* with secondary palatalization, see AnGr §362). This makes it likely that this is a purely Norse innovation.

In this case, the dictionaries again agree on the semantics of the verb in question:

- ONP: 'to turn pale, turn sallow'
- CV: 'to become pale'
- WAP: 'blass werden, erbleichen'
- OgnS: 'blegne, blive hvid eller bleg' ('to become pale, become white or sallow')

(3) Óláfs saga helga 173,13

þa	***blicna-ði***	*hann*	*oc*	*varð*	*faul-r*	*sem*	*nár*
then	**turn.sallow**-3SG.PRET	he	and	become.3SG.PRET	pale-NOM.SG	like	corpse

'He then **turned sallow** and became pale like a corpse.'

Apart from *blikna*, we have another class II verb belonging to *bleikr* in OWN, namely *blika*. In contrast to *blikna*, it is an old *ōn*-verb, going back to PG **blikōn-*, as attested in OSwed. *blika*, OE *blician* (AEW 44), another zero-grade formation of **bleikan-*. In contrast to the aforementioned verbal formations, the dictionaries disagree with each other in this case:

- ONP: 'to glisten, glint, shine'
- CV: 'to gleam, twinkle'
- WAP: 'blinken, funkeln' but also 'erglänzen, aufblitzen'

[12] Cf. Middle English *bliknen* 'to become pale' is a loan from OWN showing anticausative-inchoative semantics as well.

[13] On that type, cf. Ottósson (2013: 344) with literature.

- OgnS: = *blíkja* ‘shine, gleam’
- LP: ‘gleam, flash’

Already Krahe & Meid (1969: 240) stated that the old deverbal *ōn*-verbs are predominantly of intensive-iterative character, which would be confirmed by the derivation’s meaning ‘to glisten, glint’ with regard to a verbal base **bleikan-* ‘to gleam, shimmer’. The entry in WAP, however, would suggest an additional inchoative or rather semelfactive meaning ‘to flash’. To address this problem, I have examined the actual attestations. According to ONP, *blika* is attested at least 10 times in the corpus of Old West Norse prose. Strikingly, in 9 out of those 10 attestations *blika* refers to *skildir* ‘shields’, and the combination of both might be regarded as a fixed expression.[14] In light of this information, a classification of *bleikr* ‘pale; (light) yellow’ as BCT might seem questionable, because it contradicts the fact that a BCT must not be restricted to a class of objects. As I cannot discuss this matter in detail here, I will keep the focus on issues of verbal word formation and therefore quote two passages:

(4) Njáls saga 210,27
... *Skild-ir* ***blik-a****...,* *er* *sól-in* *skin-n* *á* ...
shield-NOM.PL **blika**-3PL.PRS which sun.NOM.SG-DET shine-3SG.PRS on
‘The shields ***blika***, on which the sun shines ...’

(5) Heiðarvíga saga 99,16
... *skip* *skrið-r,* *skilld-ir* ***blik-a***
ship.NOM.SG glide-3SG.PRS shield-NOM.PL **blika**-3PL.PRS
sol-skin *sne* *leg-ir*
sun-shine.NOM.SG snow.ACC.SG melt-3SG.PRS
‘[Whoever breaks an oath is outlawed as long as Christians go to church, fires blaze, fields green], ship glides, shields ***blika***, sunshine melts snow.’

Neither of the two text passages can help us in determining the semantic value of *blika* as intensive or semelfactive, since both are decidedly ambiguous.

The second text passage in (5) is cited here additionally in order to draw attention to the shortcomings of working with very limited contexts. The entry given in the ONP consists of nothing more than *skilldir blika.* Though this is more than

[14] A fixed order of the two components could also account for the verb fronting in example sentences (6) and (7). On the other hand, the three attestations of *blika* given in LP from the poetic texts are not combined with *skildir*, but with *merki* ‘banners’, *vôpn* ‘weapons’, and *reiði* ‘harness’.

what we get from the other dictionaries, it still can hardly be called context at all. Luckily, this problem is resolved by the ONP, which provides detailed information on where to find the passage in question in the edition as well as a scan of that very passage.

The other passages are not helpful in further determining the semantic value of *blika*. To illustrate this claim I will quote another two of them:

(6) Egils saga 159,17
Menn sa af þinginu at flock-r manna
one saw from thing.DAT.SG.DET at crowd-NOM.SG man.GEN.SG
reið neðan … ok ***bliku-ðu*** *þar skilld-er við*
ride.3SG.PRET down and **blika**-3PL.PRET there shield-NOM.PL with
'One saw from the Thing a crowd of men riding down … and there their shields ***blikuðu*** …'

(7) Bolla þáttr 299,18
Eða hvat sé ek þar upp kom-a, ***blik-a*** *þar eigi*
but what see.1SG I there up come-INF **blika**-3SG.PRS there not
skild-ir við?
shield-NOM.PL with
'But what do I see coming up there, aren't there shields ***blika***?'

In the second example, *upp koma* 'to come up, appear' in the first of the two asyndetically connected main clauses conveys an inchoative sense of the situation. In my view, this might even speak against a semelfactive reading of the second main clause and therefore of *blika*. As I can see coming up a thunderstorm, but not the lightning that comes with it, I can see coming up a troop of soldiers, but not the flashing of their shields.

Whatever one might think about this question of detail, it seems clear that an unambiguous choice between reading *blika* as intensive-iterative 'to glisten, glint' or as inchoative resp. semelfactive 'to flash' is hardly possible on grounds of the texts that have come down to us. We find ourselves amidst, on the one hand, interpretations that are read into the texts retrospectively and, on the other hand, semantic overlaps that are an integral part of the actual language studied and every language in general.

5.4 *brúnn* 'brown, dark brown; brownish-violet'

As there is no class III weak verb of *bleika* attested, I shall proceed to the next color adjective on the list, i.e., *brúnn* 'brown, dark brown; brownish-violet'.[15] With OE, OS, OHG, and OFris. *brūn*, it can be traced back to PG **brūna-* (EWA II 374; EWGP 143).

A class I verb is not attested in OWN, but in OHG there is *brūnen* 'to make brown-violet, purple; to give colorfulness' (only in Notker), to be read as /brynen/ with not yet graphically realized *i*-Umlaut, cf. MHG *briunen*, German *bräunen*.

As for class II in OWN, the expected OWN *brúna** 'to become brown' is not actually attested. Interestingly, the adjective *brúnaðr* 'brown colored' is attested at least 17 times in the OWN prose texts (according to ONP). The existence of this adjective presupposes that at some earlier point in time *brúna** must have been a verb that either vanished or just happens not to occur in the texts we know.[16]

5.5 *grár* 'grey; hostile'

Derivations of OWN *grár* 'grey', also 'hostile',[17] are comparably sparse. Because of OHG *grāo*, OS *-grē* (only in *appulgrē* 'grey spotted, piebald [about a horse]'), Langob. **grāus*, Goth. **grēwa*, we know that there must have been PG **grēwa-* (with **grēw-ija-* > OE *grǣg*), another *wa*-adjective just like the words for 'blue' and 'yellow' (EWA IV 591; EWGP 259; **Luhr2000**; EDPG 189).

No class I verb is attested to this adjective. OHG *grāwēn* is a class III verb that surfaces only in the glosses. Typical for OHG verbs of this class, it vacillates between inchoative and durative semantics and therefore would have to be translated as 'to become grey' and/or 'to be grey; shimmer'.

In OWN, again, there is a class II verb attested only here, namely *grána*. According to the dictionaries, it shows predominantly metaphoric semantics:

- ONP: 'to develop badly (between someone), take a serious turn'
- CV: 'to become grey, metaph. to be coarse and spiteful'
- WAP: '(eigentlich grau werden) unfreundlich werden'

[15] For a detailed study on the use of *brúnn* in Old Icelandic, cf. Wolf (2017).

[16] Crawford (2014: 132, Fn. 30) states: "A verb *bruna* does exist in ModNorw, but it does not have inchoative meaning". This is indeed the case, but the way Crawford puts it is rather misleading. There is a verb *brune/-a*, but according to ordbønene.no, it has factitive semantics 'gjøre brun'.

[17] For a detailed study on the use of *grár* in Old Icelandic, cf. Wolf (2009).

- OgnS: ‘blive graa, antage en graa Farve’ (‘to become grey, take on a grey color’)
- LP: ‘fade away’

The semantics in the ONP are given based on the following example:

(8) Sturlunga saga 20,7
Ok af þess-um ákǫst-um tek-r heldr at
and from this-DAT.PL provocation-DAT.PL begin-3SG.PRS rather to
gran-a *gaman-it, ok kom-a*
take.a.serious.turn-INF pleasure.NOM.SG-DET and come-INF
kviðling-ar við svá
mocking.verse-NOM.PL with so
‘… and because of these provocations the pleasure began to rather **take a serious turn** and mocking verses were added.’

5.6 *grǿnn* ‘green’

Somewhat surprisingly, only one verbal derivation to OWN *grǿnn* ‘green’ is reflected in the texts. The adjective itself has parallels in OE, OFris. *grēne*, OS *grōni*, OHG *gruoni* and can therefore be said to continue an older PG **grōni-* (EWA IV: 667; EWGP 260–261; EDPG 191–192).

It is the second time that we encounter a class I verb: *grøna*. According to ONP s.v. *grǿna*, it is attested only once, as medio-passive *grønask*. It surfaces in the Flórents saga as a 3sg. preterit form <græniz> with the ending *-iz* from earlier (but in this particular case unattested) *-isk* (AnGr §544,2). Perhaps because it is attested only once in the prose texts (and not at all in the poetic texts), ONP and CV agree in their translation, while OgnS does not list the verb:

- ONP and CV: ‘blive grøn’ (‘to become green’)

Mindful of the fact that we are dealing here with a medio-passive class I verb, we could expect a reflexive factitive meaning ‘to make oneself green’. But who or what is capable of doing such a thing? The solution becomes apparent if we have a look at the actual paragraph:

(9) Flórents saga 197,53
… þa leinga-z dag-ar, ok batna-r
then lengthen-3SG.PRS.REFL day-NOM.PL and improve-3SG.PRS

vedratta **græni**-*z* *jord* ...
weather.NOM.SG **make.green**-3SG.PRS.REFL earth.NOM.SG
'... the days get longer, and the weather improves, the earth **makes itself green** (= becomes green) ...'

If one believes that the earth is referred to as an animate entity here and is therefore not only the subject of the clause, but also the agent, then the use of *græniz* would be strictly reflexive. If, on the other hand, this is not the case, it would point to anticausative semantics, since *jord* 'earth' is still the subject of the clause, but not the agent. As the patient, becoming green is something that happens to it.[18]

Class III verbs are attested only for OE *grēnian* 'to become green (about foliage); flourish', and OHG *gruonēn* 'to become / to be green, blossom, flourish' again with the vacillation between inchoative/ingressive and durative meaning so characteristic for OHG class III verbs.

5.7 *hvítr* 'white'

Continuations of PG *h^weita-* 'white' are widely attested. Next to OWN *hvítr* we have West Germanic cognates OE *hwīt*, OHG (*h*)*wīz*, OFris. (*h*)*wīt*, *wit*, and even Goth. *hveits* (EWGP 316–317; EDPG 267).

A Proto-Germanic factitive *h^weitijan-* can be assumed based on OE *hwītan* 'to make white, to whiten; to polish', OHG (*h*)*wizen* '*idem*; also: to wash white', and Goth. *gahweitjan* 'to make white, to whiten'. It is, however, not attested in OWN.[19]

According to the findings so far, we have yet another class II *nan*-verb *hvítna* that is restricted to OWN. Again, the dictionaries agree on the meaning:

- ONP: 'blive hvid' ('to become white')
- CV: 'to become white'
- WAP: 'weiß werden'
- OgnS: 'blive hvid' ('to become white')

[18] This matter cannot be decided on the basis of the text. There are references in Old Norse sources (especially the translation of the Elucidarius, and the Edda) to the equation common in the Middle Ages of a man as a microcosm with the macrocosm, whereby the four major elements earth, water, air and fire correspond to human flesh, blood, breath and body heat, cf. Simek (1990: 94–95) and *passim*.

[19] An online search in the ONP for the definitions 'to make white' or 'whiten' does not yield a single result.

(10) Snorra Edda 49,22

hann	*herþ-i*	*hendrn-ar*	*at hamar-skapt-i-nv,*	*sv*
he	clasp.firmly-3SG.PRET	hand-NOM.PL	at hammer-shaft-DET-DAT.SG	so

at	***hvitnv-þv***	*knv-ar-nir*
that	**become.white**-3PL.PRET	knuckle-NOM.PL-DET

'He clasped the hammer shaft so firmly with his hands that his knuckles **became white**.'

Interestingly, the dictionaries contradict one another with respect to verbs attested in OE *hwītian* (class II) and OHG *wīzēn* (class III). According to AEW 273, neither of them is inchoative. According to BT s.v. *hwítian*, however, the English verb vacillates between 'to be / to become white'. The same is true for the OHG verb. While **Schutzeichel2012** translates it as inchoative 'weiß werden', Heidermanns (EWGP 317) suggests 'weiß werden, sein'. This, again, illustrates that, when it comes to semantics, working with dictionaries can be a venture that should be approached with caution.

5.8 *rauðr* and *rjóðr* 'red'

We shall now deal with derivations of the words for 'red'. In addition to OWN *rauðr*, a second lexeme *rjóðr* is attested in the texts as well. While *rauðr* is the common word (at least 248 instances according to ONP, and 112 according to LP), *rjóðr* is "extremely uncommon [and] used only for the color of human faces" (Crawford 2014: 120).

OWN *rauðr* can formally be traced back to PG **rauda-* 'red', as shown by OE *rēad*, OFris. *rād*, OS *rōd*, OHG *rōt*, and Goth. *rauþs*. OWN *rjóðr* on the other hand belongs to the ablauting PG **reuda-* 'red', in turn shown by OE *rēod*, and Goth. *gariuþs* 'honorable' (< 'shameful' < 'with a red face'), EWA VII 653–654; EWGP 438–439, 448; **Luhr2000**.[20]

If we now take a closer look at the weak verbs, we find *roðna* and *roða* (both class II). The first one is translated by the dictionaries as having anticausative-inchoative semantics:

[20]It has been suggested (EWGP 448), that PG **reuda-* 'red' is a Germanic innovation to the present stem of a strong verb PG **reuda-* 'to make red', but Ved. *róhita-* 'red (adj.); red horse (subs.)', Av. *raoδita-* 'red', and Lat. *rūbidus* < **$h_1$re/oud^h-i-* (EWA VII: 655) show that there was an old *e*/*o*-ablauting adjective of which both forms survived in Germanic and were therefore directly ancestral to **reuda-* as well as **rauda-*. Moreover, there must have been a zero-grade *ró*-derivative **$h_1$rud^h-ró-* (> Latin *ruber*, Greek ἐρυθρός 'red') that gave rise to PG **rudra-*. As this only survives as an *ōn*-stem feminine noun OWN *roðra* 'blood', it can be disregarded here.

- ONP: –
- CV: ‘to redden, become red (of the face), to blush’
- WAP: ‘rot werden, erröten’
- OgnS: ‘rødme, blive rød; blive rød i Kinderne’ (‘to blush, become red; to turn red in the cheeks’)

An example for the use of *roðna* (as 3SG.PRET *rodnaþi* ‘(she) blushed’) can be seen below in (12).

Less clear, on the other hand, is the derivational history of this word. No cognates in the other Germanic languages suggest that it might be an OWN innovation. According to Wessén (1965: 100), all it takes for a word *roðna* to be built is the need for finite verb forms with the meaning ‘to become red’ and a sample group (“Mönstergrupp”) that can serve as a model. As we have seen, the other OWN weak *nan*-verbs formed to color adjectives have no correspondences in the other Germanic languages. It can thus be assumed that the pattern was productive in OWN.

Seebold (1970: 379), however, states that *roðna* ‘rot werden, sich röten’ is derived from the strong verb PG **reuda-* ‘röten’. Heidermanns’ etymological dictionary of the Germanic primary adjectives does not contain *nō*-derivatives neither s.v. *rauda-* ‘rot’ (EWGP 438–439) nor s.v. *reuda-* ‘gerötet’ (EWGP 448). He therefore seems to assume that OWN *roðna* is of deverbal origin as well.

If we were to assume that OWN *roðna* continues a *voreinzelsprachliche* formation, we could not decide if it was deverbal from **reuda-* ‘redden’ or deadjectival from either **rauda-* ‘red’ or **reuda-* ‘reddened’ on formal or semantic grounds. On the formal side, all three possibilities would require a zero-grade root and the subsequent development **rud-nō-* > **rud-na-* > *roðna*.

On the semantic side, if we recall the anticausative-inchoative function of the *nō*-formations, a deverbal formation to **reuda-* would yield ‘to become reddened’,[21] and a deadjectival formation would yield ‘to become red’ or ‘to become reddened’.

Given that there are no cognates in the other Germanic languages and that former *nō*-formations were productive in OWN, the assumption that *roðna* is indeed an innovation seems to be the most plausible.

In addition to *roðna* there is, as stated above, another class II weak verb: *roða*. No translation is given in the ONP yet, but the other dictionaries do not contradict one another regarding the meaning of the verb:

[21] Cf. Goth. *us-brukan* ‘to become broken off’ : *brikan* ‘to break’, OWN. *sofna* ‘to fall asleep’ : *sofa* ‘to sleep’, see Krahe & Meid (1969: 253) for further examples.

- CV: 'to gleam red'
- WAP: 'roten Schein geben, rot leuchten'
- OgnS: 'kaste et rødt Skin, en rød Farve fra sig eller over noget' ('to cast a red light, a red color from oneself or at something')

In the Germanic etymological dictionaries, *roða* is usually reconstructed as a class III verb **rudēn-* (> OHG *rotēn* 'to be or become red', AEW 450; EDPG 416; EWA VII 658). While this is formally possible, the attested Old West Norse forms, however, synchronically show class II inflection. According to ONP, the verb is attested at least twice in the Old West Norse prose literature. There are, however, no attestations in the poetic texts. Two of the attestations are given in examples (11)-(12) below.

(11) Sverris saga 92,29

Sva var at sia i fiall-it upp sem i log-a
so was to look.INF at rock.NOM.SG-DET up like in fire-ACC.SG
sę̨i, er ***roþa-ði*** *sciolld-unum*
look.3SG.PST.SBJV that **gleam.red**-3SG.PRET shield-DAT.PL

'Looking up at the rock that **gleamed red** from the shields was like looking into a fire.'

(12) Karlamagnúss saga 199,16

… Kong-s dottir rodna-þi miǫg vid
king-GEN.SG daughter.NOM.SG blush-3SG.PRET very.much at
þess-a sǫgu ok vard þvi likuz sem þa
this-ACC.SG story.ACC.SG and become.3SG.PRET therefore like as then
er ***roda-r*** *fyrir vpprennandi solu i*
when **gleam.red**-3SG.PRS while rise.PTCP.PRS.DAT.SG sun.DAT.SG in
hinv fegursta heidi
the.ART.DAT.SG beautiful.SUPERL clear.sky.DAT.SG

'The king's daughter blushed very much at this story and (she) therefore then became as when it **gleams red** while the rising of the sun in its most beautiful clear sky.'

The two forms in question are *roþaði* '(the rock) gleamed red' (3SG.PRET) in (11) and the impersonal *rodar* 'it gleamed red' (3SG.PRS) in (12). In Old West Norse, only the class II weak verbs show a vowel -*a*- in the middle syllable of the present and preterit forms, whereas the class III verbs have no such vowel in the preterit nor -*i*- in the present, cf. *vakti* (3SG.PRET), *vakir* (3SG.PRS) from *vaka* 'to be awake'.

Formally, *roða* could continue both a class II and class III verb, but the durative/essive semantics rather point to a class III verb that secondarily has adopted class II inflection.

5.9 *svartr* 'black, dark'

Let us finally have a look at the verbal derivations of the word for 'black, dark'. OWN *svartr*, Goth. *swarts*, OE *sweart*, OHG *swarz* indicate that it continues PG **swarta-* (EWGP 574).

While an already PG factitive **swartijan-* 'to make black/dark, blacken/darken' can be reconstructed on the basis of Goth. *swartjan**, OHG *swerzen* as well as ODan. *swertæ* and OSwed. *sværta*, the West Norse word is attested no earlier than in ModNorw *sverte* (Bjorvand & Lindeman 2019: 1222).

In contrast, class II verbs are attested in OWN only. Here we have another *nan*-verb with zero-grade root, i.e., *sortna*. Again, the ONP does not offer a translation, but the other dictionaries agree on anticausative-inchoative semantics:

- CV: 'to grow black'
- WAP: 'schwarz, dunkel werden'
- OngS: 'sortne, antage en sort Farve' ('to become black / blacken, take on a black color')

(13) Cecilíu saga 284,4
… *dagr-inn* *fly-di,* *ok* *sol* ***sortna-di,*** *ok*
day.NOM.SG-DET leave-3SG.PRET and sun.NOM.SG **blacken**-3SG.PRET and
ger-di *myrk-t* *i* *heim-inum* *aull-um*
make-3SG.PRET dark-N in home-DAT.PL.DET all-DAT.PL
'… the day left and the sun **blackened** and made all homes dark.'

More interesting, however, is *sorta*. According to ONP and LP this class II verb is sparsely attested in Old Norse prose, but at least four times there and once in the poetic texts. While WAP and ONP, again, offer no translation, a look in the other dictionaries does not reveal contradictions:

- CV: 'to dye black'
- OsgN: 'formørke' ('to turn sth. black/dark; to eclipse') (tr.)
- LP: 'to tar'

I will still discuss this word here because class II verbs are not usually associated with what seems like causative/factitive semantics.

From the four instances that are listed in the ONP, I will discuss two instructive examples here. The first one in example (14) describes people living in Ethiopia:

(14) Stjórn 96,18

… *hueria*	*er*	*sol-in*	*brenn-ir*	*ok*	***sorta-r***
which.ACC.PL	who	sun.NOM.SG-DET	burn-3SG.PRS	and	**sorta**-3SG.PRS

'…which the sun burns and ***sortar***.'

The first component of the binomial *brennir* 'burns' (3SG.PRS) (< PG **brannjan-* 'to make burn' tr.), which in OWN falls together with its derivational basis PG **brinnan-* 'burn' (intr.; AEW 56),[22] conveys causative semantics. This seems to be true also for the second one, *sortar* 'makes black, blackens' (3SG.PRS) – at least at first glance. However, this would be rather unexpected since it is the class I verbs that are known to be causatives/factitives, an observation that is confirmed by the material discussed here as well.

We might assume based on the aforementioned forms ODan. *swertæ*, OSwed. *sværta*, and ModNorw. *sverte* that there was in fact once an OWN factitive **sverta* 'to make black/dark, blacken' that is coincidentally not attested in the texts. If *sorta* also had factitive semantics, as suggested by CV, OsgN and LP, we would have to assume that these two forms were in competition with one another and that *sorta* did not emerge victorious. This is far from impossible. However, this issue cannot be solved with the material the dictionaries provide, but only by means of a thorough investigation of the actual texts in their local and temporal context.

At second glance, however, another option seems more promising. We can see that Heidermanns connects *sorta* to the zero-grad nouns OWN *sorti* 'blackness, darkness' and OWN *sorta* 'black color' (EWGP 574). Since Germanic denominal class II verbs usually exhibit the same ablaut grade in the root as their derivational bases (Krahe & Meid 1969: 238–240), and since no strong verb that could have functioned as derivational base exists, assuming a denominal formation for *sorta* is formally appealing. Reconsidering the formal aspects of denominal *ōn*-formations enables us to better understand the role *sortar* plays in Stjórn 96,18 in example (14). A case in point is OHG *salbōn*, OS *salvon*, OE *sealfian* < West Germanic **salbōn-*, a derivative of **salbō-* 'ointment'. The verb is best paraphrased

[22]Something similar has happened or is happening in most Germanic languages. For example, until Early New High German the strong verb *brinnen* was distinct from its derivative *brennen*. The former is in use today only in some Upper German dialects and is being replaced by the latter (EWA II 342).

as ‘to provide with ointment’. The same is true for Goth. *gapaidōn*, a derivative of *paida* ‘garment’, best paraphrased as ‘provide with garment’ (Krahe & Meid 1969: 239) or OWN *dúka*, a derivative of *dúkr* ‘cloth, veil, scarf’ and therefore ‘to provide with a *dúkr*’ (Iversen 1923: 178). These class II verbs have been termed ‘ornative’ (recently Filatkina 2018: 250; Schaefer 1984: 376–378), and this is key to understanding *sorta* as well. For example, in contrast to OWN *døma* ‘to judge, pronounce a judgement’, a class I derivative of *dómr* ‘judgement, verdict’, ornative verbs do not denote the realization or materialization of what is denoted in the derivational basis, but that something or someone is provided with the feature or property that the derivational basis has. This means that *sorta* conveys a meaning ‘to provide with blackness’, rather than a meaning ‘to make black, to blacken’. As the difference is not as clear cut as, for example, in **salbōn-* (‘to provide with ointment’ vs. hypothetical *‘to make/turn into ointment’), constellations like these might have contributed to the collapse of the morphosyntactic system of the Germanic weak verbs.

Let us now see if this interpretation can hold up in the light of another example:

(15) Konungs skuggsjá 60,29
… En þo ero pannzar-ar hofuð vapn til
and but be.3PL.PRET shield-NOM.PL main weapon.NOM.SG for
lifð-ar a skip-um gorv-ir af blaut-um
protection-GEN.SG on ship-DAT.PL make-PTCP.PRF.NOM.PL of soft-DAT.PL
lerept-um oc vel ***sortu-dum****, goð-ar*
canvas-DAT.PL and well **sorta**-PTCP.PRF.DAT.PL good-NOM.PL
hialm-ar oc hang-anða stalhufur.
helmet-NOM.PL and hang-PTCP.PRS.NOM.PL steel.hood-NOM.PL
‘But shields are the main protective weapons on the ships, made of soft and well-**blackened** canvas, good helmets and low-reaching steel hoods.’

It is a reasonable assumption that the canvas is the one that is ‘provided with blackness’ and that the participle consequently means ‘blackened’.

6 Conclusion

Let us now wrap up the most important findings. Class I verbal derivatives are attested for three out of the eight BCT that I have discussed here: OWN *bleikja*, *grønask*, and – in lack of an OWN cognate – ODan. *swertæ* resp. OSwed. *sværta*.

All of them are at least of Proto-Germanic origin and, according to the dictionaries, express factitive semantics.

The situation is the opposite with regard to the class II *nan*-verbs. For OWN *blána*, *grána*, *hvítna*, *roðna*, *sortna*, and *brúna** (which can be assumed to have existed based on the adjective *brúnaðr* 'brown colored'), no Germanic cognates are attested. This group was highly productive in OWN, and it is doubtful that any of these words are old.[23] All of them express anticausative-inchoative semantics.[24]

The remaining three verbs *blika*, *roða*, and *sorta*, despite being class II verbs, differ in terms of meaning: The meaning of *blika* and *sorta* is disputed according to the dictionaries consulted in this article. Even when looking at actual attestations, an unambiguous determination in terms of semantics is not possible for *blika*, at least not based on the contexts I investigated. They vacillate between intensive and inchoative (or semelfactive) semantics.

For *sorta*, the two dictionaries consulted in this study that offer a translation (CV and OgnS) are not precise enough in determining its semantic value. With this in mind, the translations suggested in CV and OgnS 'to dye black, blacken' should be refined to 'to provide with blackness'. While this might be considered an artificial differentiation and while it is probably handy to stick to 'to dye/make black' in a translation within a wider context, with respect to the derivational history of this word it is preferable to be as precise as possible in order to be able to understand how the semantic change may have come about.

Finally, synchronically *roða* is a class II verb as well, but its essive/durative semantics suggest that historically it is an old *ēn*-verb that was integrated into class II in OWN.

None of the weak verbs formed to adjectives for BCT in Old West Norse is a member of the third class, which has become highly productive in OHG in forming inchoative verbs (e.g., *fūlēn* 'to rot or to decay', *altēn* 'to become or to grow old').

[23] According to Gorbachov (2007: 114–115) it is unclear how and if OWN *hvítna* is related to Lith. *šviñta* (Inf. *švìsti*) 'become light, start shining'. The long vowel in the OWN form could be explained as the result of leveling with its base *hvítr*, but the consonants do not match. No mention is made of this connection in ALEW s.v. *švìsti*. Gorbachov (2007: 116) also shows that OWN *roðna* is a perfect formal match to OIr. *roindid* [sic!] 'reddens, dyes (with blood)' and Ved. *ruṇáddhi* 'restrains, keeps back', but the latter may very well represent a different PIE verbal category altogether since they have a different (transitive-terminative) function. As Stefan Höfler (p.c. 17.01.25) has furthermore informed me, *roindid* is rather Middle Irish with analogous palatalization of the cluster *-nd-*. For the OIr. form see Höfler & Stifter, this volume, Section 2.2.

[24] Ottósson (2013: 346): "At least the bulk of the deadjectival [verbs] seems to have denoted types of events that were typically agentless […]."

We know from corpus studies on the other Old Germanic languages that the word forming verbal suffixes exhibit a functional variety already in the earliest texts (see fn. 4). Along these lines, Schwerdt (2008) has shown that the Old High German verbal classes are often ambiguous with respect to their semantic roles. This is the case for about 60% of *jan*-verbs, 50% of *ōn*-verbs, and 50% of *ēn*-verbs. This ambiguity, however, is not found within Old West Norse, at least to this degree.[25] In other words: The system of the Old West Norse weak verbs was intact (although it is clear that it could never have been completely consistent) to a degree that change of state verbs could not function as stative verbs and *vice versa* as they can in Old High German, i.e., they either denote 'to be x' or 'to become x'. However, if and to which degree this might be an artifact arising from working with dictionaries – which do not describe the semantic differences adequately – rather than corpora, is something that deserves to be fleshed out in future studies.

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Abbreviations

Av.	Avestan	OFris.	Old Frisian
Goth.	Gothic	OHG	Old High German
Langob.	Langobardic	OIr.	Old Irish
Lat.	Latin	OS	Old Saxon
Lith.	Lithuanian	OSwed.	Old Swedish
MHG	Middle High German	OWN	Old West Norse
MLG	Middle Low German	PG	Proto-Germanic
ModNorw.	Modern Norwegian	PIE	Proto-Indo-European
ODan.	Old Danish	PN	Proto-Norse
OE	Old English	Ved.	Vedic

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Chapter 3

Caland verbs in Celtic

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This study offers a comprehensive examination of Caland verbs in the Celtic languages, an area that has previously received little attention within the broader investigation of the Caland system across Indo-European languages. By identifying and analyzing twenty Celtic verbs associated with the Caland system, we determine that eleven likely represent inherited forms, while the remaining verbs appear to be inner-Celtic innovations. Through detailed philological, phonological, and morphological analysis, we assess the productivity of root-derived deadjectival verb formations in Proto-Celtic. Despite the relatively limited number of verbs and their distribution across various verbal stem types, the findings suggest that the Caland system retained a marginal but discernible degree of productivity in the verbal domain of the Celtic languages. This conclusion is further supported by the existence of at least two neo-roots in the nominal domain, demonstrating parallel processes of morphological innovation.

1 Introduction

When it comes to the systematic study of the Caland system and Caland morphology in the Indo-European languages, most of the relevant research so far has been conducted on the basis of Indo-Iranian, Greek, Hittite, and Latin (e.g., Nussbaum 1976; Rau 2009; Rau 2013; Lowe 2016, etc.), Balto-Slavic (e.g., Majer 2017), or has been focused on the reconstruction of the system in the proto-language (e.g., Bozzone 2016; on the history of scholarship see Dell'Oro 2015).

The Celtic languages have played a subordinate role in establishing the dynamics of the system. When Celtic material has been mentioned at all, its function has been ancillary rather than constitutional, serving as a means to provide evidence for an already established framework rather than being exploited to gain

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new insights into the Caland complex as a whole. And while a couple of book chapters and conference talks on Celtic such as De Bernardo Stempel (1999: 529–537, 2003), Höfler (2016), and Kahl (2018) (on Gaulish) have in recent years been devoted to the investigation of the nominal domain, the Caland-associated verbs still lack a comprehensive treatment.[1]

Very broadly speaking, Caland verbs are verbs that belong to Caland roots, or roots with a "Caland profile" (see Nussbaum 2022). These roots possess an adjectival meaning that typically falls into one of the categories known as 'property concepts' (i.e., dimensions, physical properties, colours, speed, age, value, human propensities, etc.), such as, for instance, 'big', 'small', 'red', 'fast', 'slow', 'old', 'new', 'good', 'bad', etc. (see Rau 2009: 78–79; Balles 2003: 18–20; Balles 2006: 269–287; see also the introduction to this volume, Sections 1.2 and 3.1). Despite their adjectival meaning ('X'), these roots do not (or at least not normally) surface as root adjectives. If a root noun is attested, it typically has the meaning of an adjectival abstract ('X-ness') or a concrete noun ('thing/person that is X'). Compare the root *_bʰerĝʰ-_ 'high' with the adjective *_bʰr̥ĝʰ-(é)nt-_ 'high' (Gaulish, Old British ethnonym sg. _Brigans_, pl. _Brigantes_, Βρίγαντες, fem. _Brigantiae_ (dat.), OIr. _Brigit_ < *_bʰr̥ĝʰ-n̥t-ih$_2$-_ (see also Stifter forthcoming); cf. Vedic _br̥hánt-_, _br̥hatī́-_ 'big, high, noble', Young Avestan _bərəzant-_ 'high') and the root noun *_bʰr̥ĝʰ-_ 'height, high spot' (Old Irish _brí_, gen.sg. _breg_ f. 'hill'; cf. Young Avestan _bərəz-_, nom.sg. _barš_ 'height, mountain; high', Proto-Germanic *_burg-_ 'castle' > Gothic _baurgs_ f. 'city, tower', Old High German _burg_ etc. 'castle'), or the root *_(h$_1$)reu̯dʰ-_ 'red' with the adjective *_(h$_1$)re/ou̯dʰ-o-_ 'red' (Old Irish _ród_, _rúad_, Welsh _rhudd_; cf. Latin _rūfus_, _rōbus_, Proto-Germanic *_rauda-_, Lithuanian _raũdas_, Old Church Slavonic _rudъ_) and the root noun *_(h$_1$)rudʰ-_ 'redness' (Old Irish _rú_, dat.sg. _roid_ f. 'redness, red dye plant'). As with *_bʰr̥ĝʰ-(é)nt-_ 'high' and *_(h$_1$)rou̯dʰ-o-_ 'red', the corresponding base adjective _X_ usually exhibits derivational suffixes such as *_-ro-_, *_-u-_, *_-o-_, *_-nt-_, *_-mo-_, *_-no-_, a fact that in and of itself would not be overly surprising. However, all (seemingly) deadjectival derivatives are consistently root-derived, not stem-derived. This is true for typical deadjectival formations such as the comparative and superlative (e.g., *_tép-n̥t-_ 'hot' > Old Irish _tee_, _té_, nom.pl. _téit_, but comparative *_tep-i̯ōs_ > Old Irish _téo_, _teo_, _teou_), deadjectival abstracts (e.g., *_treks-no-_ 'strong' > Old Irish _trén_, but *_treks-i̯ā-_ 'strength' > Old Irish _treise i̯ā_, f. 'strength, vigour'; cf. Jasanoff 1991), as well as for deadjectival verbs. This synchronic rule of suffix deletion is what characterises the Caland system.

There are numerous verbal formations with deadjectival semantics that are associated with the Caland system, mostly characterised present stems involving

[1] A comprehensive study of Caland formations in all lexical domains of Celtic will be published in Höfler & Stifter (in preparation).

the suffixes **-eh$_1$i̯e/o-*, **-(eh$_1$)sk̑e/o-*, **-i̯e/o-*, **-éi̯e/o-*, nasal infix presents, and regular thematic presents (cf. Rau 2009: 136–160). Unlike Bozzone (2016), we see no convincing evidence for Caland-associated primary root aorists (see also Section 3.1 of the introduction to this volume), but, with Rau (2013), there seems to have been an association with the perfect. The present stems are attested in all three meanings expected for a deadjectival verb: stative, fientive, and factitive. While some of the involved suffixes are predominantly associated with one of the three meanings (e.g., **-eh$_1$i̯e/o-* and stative semantics), this specialization might reflect a later development. In Celtic, some types are attested in more than one meaning.

Celtic preserves only a small, yet exquisite number of Caland verbs. Four of the present stem types listed in the previous paragraph are attested, as well as one Caland-associated perfect. There are two roots for which more than one present stem is found. While there is evidence for a certain amount of productivity (see below), none of these types becomes (or remains) the go-to template for deadjectival verb formation in Celtic.

When collecting the verbal stems, we applied the label "Caland root" relatively generously. As is clear from the introduction to this volume (see especially Section 3.1), it is often not easy to define such a root based on morphological or semantic criteria alone. On the one hand, the typical Caland suffixes in the adjectival and nominal domains also appear with verbal roots (e.g., **ĝneh$_3$-* 'recognise, know' and **ĝn̥h$_3$-ró-* 'knowing, known' > Latin *gnārus* 'knowing, skilled', Greek γνωρίζω 'make known'), and on the other hand, some roots allow both a verbal and an adjectival interpretation, such as **seĝ^h-* meaning either 'overpower, overcome, subdue' (e.g., in the thematic present **séĝ^h-e-t(o)i* > Vedic *sáhate* 'overcomes', Greek ἔχει 'has', or the agent noun **séĝ^h-tor-* > Vedic *sā́ḍhar-* m. 'victor', Greek Ἕκτωρ) or '(be/make/become) strong, victorious' (e.g., in the perfect **se-sóĝ^h-e* with an intransitive meaning in Vedic *sāsā́ha* 'is victorious', or the comparative **seĝ^h-i̯os-* > Ved. *sáhyas-* 'stronger'). In such a case, it is conceivable that adjectival '(be/make/become) strong, victorious' was the original meaning and that a verb **séĝ^h-e-toi̯* 'is strong, victorious' (cf. Greek ἐρεύθεται 'is red') was construed with an object accusative, from which the meaning 'is strong, victorious (sc. in relation to somebody or something)' > 'overpowers (somebody or something)' could then have developed. The same pathway can essentially be observed for **d^{h}ers-* '(be/make/become) bold, courageous' in the case of Greek θαρσέω, intransitive 'be confident' (*Il.*+) and transitive 'dare' (*Od.*+) < 'be bold/confident in regard to something' (cf. θάρσει τόνδε γ' ἄεθλον 'take on this contest with courage' *Od.* 8.197) and similarly for Vedic pres. *dhr̥ṣṇóti*, perf. *dadhárṣa* intransitive 'be courageous' and transitive 'dare'.

This is why we include some roots in our collection that are usually seen as verbal roots (e.g., *h_2*e̯id*^{h}- or *h_1*ai̯d*^{h}- 'burn, ignite') if typical Caland morphology is attested in Celtic or elsewhere (e.g., *h_1*ái̯d*^{h}*-u-* > Old Irish *aed* 'fire', *h_1*(a)i̯d*^{h}*-ró-* in Greek αἴθρη f. 'the clear sky', ἰθαρός 'serene, clear', αἰθί-οψ 'burnt-face') and if an originally adjectival meaning may therefore be postulated (e.g., '(be/make/become) burning, bright').[2] We are aware of the potential risk in applying such a broad definition of what a Caland root (and, thus, a Caland verb) is, but we believe that for the sake of completeness and a more comprehensive understanding of the system, it is justified to be more inclusive. In any event, we will address the potential verbal meaning of these ambiguous roots individually in the commentary section of the respective roots and make sure they do not unduly influence our overall conclusions.

In total, Celtic exhibits twenty Caland-associated verbal stems. Eleven of these have cognates in other languages, which means that they are probably inherited. The remaining nine do not have cognates elsewhere and are therefore perhaps inner-Celtic creations. In these cases, it is an interesting question to explore whether there is evidence for a corresponding adjective that could serve as the basis for the verb. Since the verb is nonetheless root-derived, not stem-derived (i.e., lacks the adjectival suffix), this would point to at least some productivity of the Caland system in the verbal domain in Celtic (see Section 7).

2 The 'stative' type

2.1 Background

The suffix of the so-called 'stative' type (as per Jasanoff 1978, Jasanoff 2004; see also Peters 2007, **Kummel2012**, Malzahn 2014, Peters 2016, Jasanoff 2021) is reconstructed as *-*eh$_1$i̯e/o-* (> *-*ei̯h$_1$e/o-* ?) and usually takes a zero-grade root. Structurally, the statives are interpretable as decasuative present stems in *-*i̯e/o-* derived from adnominally used instrumental singular forms in *-*eh$_1$* of root nouns, for example *h_1*rúd*^{h}- 'redness' (OIr. *rú* f. 'redness, red dye plant') → *h_1*rud*^{h}*-eh$_1$* 'with redness; red' → *h_1*rud*^{h}*-eh$_1$-i̯e/o-* (Latin *rubeō* 'I am, become red', Old High German *rotēn* 'flush').[3] If this analysis is correct, the 'stative' type would be expected to have typical deadjectival semantics, which means that the stative mean-

[2] For *h_2*ei̯d*^{h}- (*h_1*ai̯d*^{h}-) as a Caland root, see Nussbaum (1976: 35): "*h_2*ei̯d*^{h}- 'burn, shine' […] makes a well-attested Caland system".

[3] LIV^2 508–509 sets up an "Essiv" **rud*^{h}*-h$_1$i̯é-* here. For the "Essiv" type see further LIV^2 25. Schumacher (2004: 41–42), who otherwise follows LIV^2 closely, remarks that both Celtic and Latin require *-*eh$_1$-i̯e/o-*, which may be secondarily introduced.

ing ('be red') is only one of three predictable meanings of such a present stem, viz. besides fientive ('become red') and factitive ('make red'; cf. Malzahn 2014: 94, n. 43).

In Celtic, the suffix(oid) **-eh$_1$-i̯e/o-* develops to Proto-Celtic **-ī-* and is one of the sources of Old Irish W2 verbs with a palatal root final consonant and a 3sg. conjunct ending *-i* (e.g., *ruidid*, *·ruidi* 'flush, blush'). In Middle Welsh, the stative type results in verbs with raising in the root vowel of the 3sg. and an ending absolute *-it* (archaic), conjunct -Ø. This type is attested for four Caland roots in Celtic (cf. Schumacher 2004: 41–42), a transitive-factitive meaning is well-represented beside the stative or fientive, though it is not clear whether this is an archaism or an innovation. Three of those four verbs have cognates in other branches of Indo-European.

2.2 **(h$_1$)reu̯d^h-* 'red'

**(h$_1$)rud^h-eh$_1$-i̯e/o-* 'be/make/become red' > PCelt. **rud-ī-* > OIr. *ruidid*, *·ruidi* 'flush, blush; make red' (cf. Schumacher 2004: 552–553).

(1) Old Irish (*Críth Gablach* 548)
ataat trí toichnedai frisná ***ruide*** *tocrád*
are three fastings at.which.not **blush**.3SG.PRS.IND vexation
ríg
king.GEN.SG
'There are three fastings at which the offended majesty of a king does not blush.'

(2) Old Irish (*Anecd.* I 12.26)
a rí ***ruides*** *gréin*
oh king **redden**.3SG.PRS.IND.REL sun
'oh king who reddens the sun' (var. lect. *reithes* 'moves')

Cognates Latin *rubēre* 'be/become red', Old High German *rotēn* 'be/turn red', Latvian *rudêt* 'rust', Russian Church Slavonic *rъděti sja* 'become red' (LIV^2 508–509).

Adjective Old Irish *rúad*, archaic and poetic *ród* 'red' < **(h$_1$)re/ou̯d^h-o-* (cf. Gaul. *roudo-*, OW, OCorn., OBret. *rud*, ModW *rhudd*, MCorn. *ruth*, *ruyth*, NBret. *ruz* 'red'; also Latin *rūfus*, *rōbus*, Proto-Germanic **rauda-*, Lithuanian *raũdas*, Old Church Slavonic *rudъ*).

Commentary The preponderant, and probably primary, meaning of the verb in Early Irish is fientive ‘turn red’ (namely in the face, i.e., ‘to flush, blush’, usually as a consequence of being shamed), or stative ‘be red’. Possible examples of factitive ‘make red’, as a verb or as a verbal noun, are very rare. They are only found in poetry, unfortunately often in philologically difficult contexts. The form *ruides* in the example cited from the tale *Scéla Cano meic Gartnáin* is attested in the single manuscript copy of the tale. Its varia lectio *reithes* ‘who moves, makes run’ occurs only in the philologically less reliable *Nebenüberlieferung* of the *Mittelirische Verslehren*; the causative meaning ‘move, make run’ of *reithid* is in itself unusual.

2.3 **k⁽ᵘ̯⁾su̯ei̯bʰ-* ‘swift (?)’

**k⁽ᵘ̯⁾su̯ibʰ-eh$_1$-i̯e/o-* ‘be/make/become swift’ > PCelt. **ksu̯ib-ī-* >> MIr. *scibid*, *·scibi* (with sporadic metathesis of the initial cluster?) ‘flow, swim (of a vessel); move; move sth. jerkily’, MW *chwyfu* (3sg. *chwyf*) ‘move to and fro, budge’, MBret. *fifual* ‘move’ (cf. Schrijver 2003; Schumacher 2004: 422–424).

(3) Middle Welsh (*WM 455$_{36–8}$ = CO 80*; 14th c.)
Ni ***chwy[u]ei*** *ulaen blewyn arnaw*
not **move**.3SG.IMPF tip hair on.him
‘Not a tip of a hair would stir on him.’

Cognates Old High German *swebēn* (amongst other meanings) ‘be agitated (of the sea), be in high and low tide’.

Adjective W *chwith* ‘left, awkward, sinister, wrong’< **k⁽ᵘ̯⁾su̯ibʰ-tó-* (compare English *swift*).

Commentary The basic meaning of the root is not quite clear. It seems to refer to some sort of motion, not a single jerky movement, but rather a state or property or an iterative kind of motion. Based on Caland morphology, especially in Iranian (cf. Young Avestan *xšuuiβra-* ‘quick, vibrant’, *xšuuiβi.išu-* ‘with vibrant arrows’, *xšuuaēβa-* ‘moving quickly’, on which see Schrijver 2003), we assume an adjectival rather than a verbal basic meaning.

A potential Caland adjective **k⁽ᵘ̯⁾su̯ibʰ-tó-* (compare English *swift*) is attested in W *chwith* ‘left, awkward, sinister, wrong’, though the meaning is quite removed from the purported meaning of the root. Schrijver (2003) has argued that

ModIr. *ciotach* (the unattested OIr. form would be **cittach*) 'left-handed, awkward, clumsy', which could be compared with *chwith* under the assumption of a sporadic, irregular loss of initial *s*-, is unrelated. The Irish verb is found both with forms that point to lenited and unlenited *b*; forms with the latter preponderate in the later period. Besides, in ModIr. the verb *sciobaidh* 'snatch, sweep away' is found (eDIL s.v. *scipaid*). The reason for the variation in the status of lenition and palatalisation in these forms is obscure. Contamination with Old Norse *skip* 'ship' may have been a factor or sound-symbolism may be at play (unlenited **scipaid* presupposes a geminate *-*bb*-, for the sound-symbolic character of which see Stifter 2024, especially Section 11).

2.4 ****temk*- 'hard'**

**tm̥k-eh*$_1$*-i̯e/o-* 'be/make/become hard' > PCelt. **tank-ī-* > OIr. *·téici* 'congeal, become solid; make solid'.

(4) Old Irish (ZCP 7, 1910. 484.33 = *Enchiridion Sancti Augustini*)
con*-*<ro>*·*théced (gl. *adiunctus atque concretus*)
<AUGM>·**thicken**.3SG.PRET.PASS
'was solidified'

Cognates –

Adjective OIr. *técht* '(of liquids) thick, frozen; calm, peaceful' < **tm̥k-to-* (cf. Pashto *tat* 'close, thick').

Commentary McCone (1998: 469–470 and 474–475) and LIV^2 625–626 see OIr. *·téici* as the remodeling of a nasal infix present **tm̥-né/n-k-*. Schumacher (2004: 615–616), on the other hand, argues that the continuant of a nasal infix present ought to be OIr. **·tic* and reconstructs a thematic present **tm̥k-e/o-* instead. The weak inflection would then be secondary. However, nothing speaks against a regular stative **tm̥k-eh*$_1$*-i̯e/o-* > PCelt. **tank-ī-* > OIr. *con·téici* (now also Schumacher, p.c.).

The meaning 'calm, peaceful' of the adjective OIr. *técht* could be metaphorical and thus secondary within Irish. On the other hand, the connotation 'established *status quo* < lack of fluidity' is supported by OW *tagc*, MW *tanc* 'peace', Gaul. PN *Tanco-rix* (if 'peace-ruler'[4]) < **tm̥k-o-*.

[4]This provisional translation is agnostic as to what precisely is meant by 'peace', a concept

2.5 ***tu̯em-** ‘thick’

**tum-eh₁-i̯e/o-* ‘be/make/become thick’ > PCelt. **tum-ī-* > MW *tyfu* (3sg. *tyf*) ‘grow; make grow; heal’, MBret. *teñvañ*, *tiñvañ* ‘grow together, develop sap, heal’ (OBret. 3sg. *ni-tum*), MCorn. *tevi* ‘grow’ (cf. Schumacher 2004: 646–648).

(5) Middle Welsh (*J 1* 1082; c. 1400)
Tyuit *maban*, *ny* *thyf* *y* *gadachan*
grow.3SG.PRS.IND infant not **grow**.3SG.PRS.IND his swaddling-clothes
‘An infant grows, his swaddling-clothes grow not.’

Cognates Latin *tumēre* ‘be/become swollen’, Lithuanian *tumė́ti* ‘become thick, coagulate’ (LIV[2] 654 s.v. **tu̯em-*).

Adjective There is no corresponding simplex adjective, but we find a compound MW *hydwf* ‘full-grown, thriving’, OBret. PN *Hutum* < **su-tum-V-*. It is formally possible to equate these with Ir. *sothaim* ‘amusing, agreeable’, *dothaim* ‘morose, grim, surely’ < **su-/du-tum-i-*, perhaps in the sense ‘has grown/become good/bad’.

Commentary Based on the cognates in Latin and Lithuanian, PCelt. **tum-ī-* can be considered an inherited Caland verb. A *-*ró*-adjective is attested in Vedic (*túmra-* ‘strong, stout’).

3 Nasal infix presents

3.1 Background

Nasal infix presents are a well-attested class in Celtic (cf. McCone 1991; Schumacher 2004: 42–45). Five such present stems are attested from Caland roots, three of which have factitive (and only factitive) meaning. This seems to support the widely held notion that the original function of this present stem class was to create factitives (see, e.g., Rau 2009: 143–146; this volume). Yet, the evidence of the individual languages is ambiguous and might indicate that this need not necessarily have been true in the proto-language. Fientive meaning is attested, for example, in Old Russian *svьnuti* ‘become bright, day’, Lithuanian *švintù* ‘become bright’ (**k̑u̯ei̯t-* ‘bright’; cf. the ‘stative’ Lithuanian *švitė́ti* ‘shine brightly’),

that is historically and culturally very ambivalent, and whose interpretation in the context of Gaulish society may differ significantly from 21st century ideas of peace.

in Lithuanian *senkù* ‘drop (of water level)’, Old Church Slavonic *i-sęčetъ* ‘will dry out’ (**sek-* ‘dry’), and stative meaning in Latin *langueō* ‘be slack, languid’, Ancient Greek λαγγών ‘coward’ (**sleĝ/g-* ‘slack’), or in Ancient Greek ἁνδάνω (+DAT) ‘please, delight’ < ‘be tasty’ (**su̯eh$_2$d-* ‘tasty’; cf. the factitive meaning of Vedic *s$_{(u)}$vádati* ‘make tasty’ from **suh$_2$-n̥-d-* as per LIV^2 606, n. 2b).[5] The remaining two nasal infix presents from Celtic might corroborate this.

3.2 **(h$_1$)reu̯dʰ-* ‘red’

**(h$_1$)ru-n(e)-dʰ-* ‘make red’ >> PCelt. **rund-e/o-* > Old Irish *rondaid*, *·roind* ‘dye red’, *fo·roind* ‘stain’ (cf. Schumacher 2004: 553–554, LIV^2 508–509).

(6) Old Irish (*Ält. Ir. Dicht.* i 39 § 3 = Rawl. 115a25)
gáeth rúad ***rondad*** *for fáebur fuilchiaid*
wind red **redden**.3SG.IMPF on edge blood.mist
‘A red wind which coloured sword-edges with a mist of blood.’

Cognates –

Adjective OIr. *rúad* ‘red’ < **h$_1$rou̯dʰ-o-* (see Section 2.2 above).

Commentary For the vocalism *o* < **u*, see Stüber (1998: 87). *Rondaid* and its compound *fo·roind* occur mostly in contexts of fighting and battles with respect to things that get coloured by blood.

3.3 **h$_2$ei̯dʰ-* or **h$_1$ai̯dʰ-* ‘burning, light’

**h$_{2/1}$i-n(e)-dʰ-* ‘make burning’ >> PCelt. **ind-e/o-* > MW *enn-ynnu* (3sg. *enn-ynn*) ‘kindle, light a fire’ (cf. Schumacher 2004: 374–375; LIV^2 Addenda s.v. **h$_2$ei̯dʰ-*).

(7) Middle Welsh (WML 26; 14th c.)
Ef a ***enyn*** *y ganhwyll gyntaf rac bron y brenhin*
he REL **light**.3SG.PRS.IND the candle first before breast the king
‘He lights the first candle before the king.’

Cognates Vedic *indhé* ‘lights a fire’ (LIV^2 259 s.v. **h$_2$ei̯dʰ-*)

[5] On this phenomenon, see Gorbachov (2007) (for Balto-Slavic and Germanic); Rau (2009: 146, n. 68); this volume.

Adjective –

Commentary This root may be a verbal root (see Section 1 for a discussion). Other Celtic material from this root includes OIr. *áed u*, n. 'fire', Bret. *oaz* m. 'rivalry, jealousy' (sceptical LEIA A-19) and the Gaul. ethnonym *Aeduī* (in Latin transmission). An etymological connection with the root **and-* represented in OIr. *andud* 'act of kindling, lighting' and *ad·andai* 'kindle, light' (suggested, for example, in LEIA A–75) is formally excluded (Schumacher 2004: 375).

3.4 ****pleh***$_1$**- 'full'**

**pl̥-n(e)-h*$_1$- 'be/become full' > ? PCelt. **φli-ni-* > OIr. *do·lin* 'flow, abound' (cf. McCone 1991: 18, 21, 34–35; Schumacher 2004: 374–375).

(8) Old Irish (Sg. 158a1)
do·linim (gl. *māno*)
flow.1SG.PRS.IND
'I flow.'

(9) Old Irish (Ml. 56a14; 83c7)
du·linat (gl. *mānantium* and *pollentes*)
flow.3PL.PRS.IND
'which flow'

Cognates Vedic *pr̥ṇā́ti* 'fills', Old Avestan *pərənā* 'fill!', Latin *polleō* 'be strong, able', Armenian *lnwom* 'I fill', Albanian *m-blon* 'fills' (LIV^2 482–483 s.v. **pleh*$_1$-).

Adjective OIr. *lán* 'full' < **pl̥h*$_1$*-no-* (cf. W *llawn*, B *leun* 'full').

Commentary The root features prominently in Caland-associated nominal formations: **pl̥h*$_1$*-no-* > OIr. *lán*, W *llawn* B *leun* 'full', **pelh*$_1$*-u-* > Gaul. *elu-*, OIr. *il-* 'many, much', W *el-*, comp. OIr. *lia* 'more', equative *lir* 'as much as', or W *llanw* 'tide, flow, flood, influx', OBret. *lanu*, NBret. *lanv* 'flood' < **φlanu̯o-* < **pl̥h*$_1$*-n-u̯o-*. On its status as a verbal or adjectival root, see Nussbaum (2022: 208, n. 7) and Section 3.1 of the introduction to this volume.

Determining the precise inflectional class of *do·lin* 'flow, abound' (thus the headword in eDIL dil.ie/18026; later continued as the simple verb *tuilid*) is difficult, as is the demarcation against the verb that is set up in eDIL as *do·lína* 'increase, add to' (dil.ie/18028). The 3pl. *do·linat* displays a non-palatalised present-stem-final *n*, which is *a priori* not compatible with the reconstructed stem **φli-ni-*.

It rather points to *φ*li-nu-*, perhaps with a transfer to the class of *nu*-verbs; unless the alternation in the palatalisation of the stem-final consonant is analogical to the S1 verbal class, the old *e*/*o*-thematic verbs. The augmented 3sg. preterite *do·rulin* gl. *manasse* (Ml. 64c18) shows that the nasal was synchronically felt to be part of the root, as if it were a weak verb. To complicate matters, the root of OIr. *do·lin* may alternatively be from or be influenced by **lei̯h$_x$-* 'pour' (MW 3sg *dillyð* 'flows'; see below 4.5), or a merger of both roots may have occurred (Stifter 2023: 176–177). If this is the case, and if any credence can be given to the synchronic behaviour of the verb in Old Irish, the laryngeal of that root may be specified as **h$_3$*, i.e., PCelt. **linū́-* < **linō-* < **li-n-e-h$_3$-*.

3.5 **seh$_1$(i̯)-* 'long'

**seh$_1$-n(e)-i-* 'make long' > PCelt. **sī-ni-* >> **sīnī-* > OIr. *sínid*, *·sín*(*i*) 'stretch (oneself), extend'.

(10) Old Irish (LU 4927)

sínithi	*iarom*	*co*	*memdatar*	*in*	*da*	*liic*
stretch.3SG.PRS.IND.himself	then	so.that	break.3PL.PRET	the	two	stones
ro	*batár*	*immi*				
AUGM	were	around.him				

'He stretches himself so that the two stones around him broke.'

Cognates –

Adjective OIr. *sír* 'long' < **seh$_1$-ro-* (cf. W, Corn., Bret. *hir* 'long, tall, lengthy').

Commentary The verb is only attested in sources from the Middle Irish period onwards. All attestations are compatible with the analysis as a weak W2a verb. This implies that the inherited strong nasal-infix verb was re-interpreted as a weak *ī*-verb at some point in the prehistory of Irish. The root features prominently in Caland-associated nominal formations: comparative OIr. *sia*, *sía*, MW *hwy* 'longer', superlative OIr. *síam*, MW *hwyaf*, OBret. *huiam* 'longer', all of which demonstrate the absence of the adjectival suffix **-ro-* vis-à-vis the underlying adjective **seh$_1$-ro-* > OIr. *sír* 'long', W, Corn., Bret. *hir* 'long, tall, lengthy'.

3.6 *$\underset{\frown}{u}$elh$_x$- 'powerful, in power'

*$\underset{\frown}{u}$$\underset{\circ}{l}$-n(e)-h$_x$- 'be powerful' >> PCelt. *$\underset{\frown}{u}$al-na- >> *$\underset{\frown}{u}$all-nā- > OIr. *follnaithir* 'rule' (cf. Schumacher 2004: 655–656).

(11) Old Irish (Tec. Corm. §2)
recht ***fallnathar*** *for* *talman* *tuind*
right **rule**.3SG.PRS.IND.REL over earth.GEN.SG surface.DAT.SG
'The right that rules on the surface of the earth.'

Cognates –

Adjective *$\underset{\frown}{u}$$\underset{\circ}{l}h_x$-o- 'powerful, in power' as second compound member *-$\underset{\frown}{u}$alo- in Gaul. *Katouualos*, OW *Catgual*, OIr. *Cathal* (*katu-$\underset{\frown}{u}$alo- 'battle king' *vel sim.*) or PrimIr. *Cunouali*, Ogam *CUNAVA[LI]*, archaic OIr. *Conual*, OIr. *Conall* < *kuno-$\underset{\frown}{u}$alo- 'hound king'.

Commentary Other Caland material belonging to this root: Latin *ualeō* 'am strong, have power', *ualidus* 'strong, healthy', Toch. B *walo*, A *wäl* 'king' (< *$\underset{\frown}{u}$$\underset{\circ}{l}h_x$-ont-). The majority of verbal stems attested in the Indo-European languages point to a verbal meaning 'to rule', however (see LIV^2 676–677 s.v. 1. *$\underset{\frown}{u}$elH- 'stark sein, Gewalt haben'). According to McCone (1991: 16), the OIr. inflection speaks in favour of setting up the root-final laryngeal as *h_2 (see also LIV^2 676 s.v. 1. *$\underset{\frown}{u}$elH-, n. 1).

It is doubtful if the rare glossatorial and poetic word OIr. ^{2}*fál* 'king' (eDIL dil.ie/21232) continues the uncompounded substantivised adjective *$\underset{\frown}{u}$$\underset{\circ}{l}h_x$-o- 'powerful, in power'. Its long vowel, which is secured by rhyme, is incompatible with the proposed etymology. Perhaps ^{2}*fál* 'king' is only a figurative use of ^{1}*fál* 'wall'.

A number of sound developments in the OIr. verb are unexpected. The preponderant *o* of the initial syllable of *follnaithir* in Old Irish, beside occasional *fallnaithir*, is probably due to sporadic rounding of **a* caused by the preceding labial consonants. The vocalisation PCelt. *$\underset{\frown}{u}$aln- instead of *$\underset{\frown}{u}$lin- < *$\underset{\frown}{u}$$\underset{\circ}{l}$n- is probably based on the influence of *$\underset{\frown}{u}$alo- < *$\underset{\frown}{u}$$\underset{\circ}{l}h_x$-o-. The *-lln-* of *follnaithir* could either be the result of contamination with *follán* 'sound, full', or due to the regular change of *-ln- > *-ll- (Stifter 2024: Section 2.1; cf. 3pl. pass. *follatar* Ml. 77b4), with subsequent reintroduction of the present-tense morpheme *-n-*.

If the interpretation as a Caland nasal infix present is correct, the stative meaning 'be powerful' (instead of factitive 'make powerful') is noteworthy.

4 Plain thematic presents

4.1 Background

Plain thematic presents typically have a full grade in the root and are attested in stative and fientive (active or middle) and, less frequently, in factitive meaning (active) (see Rau 2009: 146–157; Rau 2013: 258–265; Villanueva Svensson 2021). Compare Vedic *rócate* 'shines', Young Avestan act. ptcpl. *raociṇt-* 'shining', Tocharian B *lyuśtär* 'illuminates' (: **leu̯k-* 'bright'), Vedic *tápati*/*tápate* 'be hot, burn', *tápati* 'make hot, burn (sth.)', Khotanese mid. *ttavāre* 'are hot' (: **tep-* 'hot'), or Greek ἄγχω 'squeeze, throttle', Latin *angō*, *-ere* 'bind, throttle' (: **$h_2$emĝ^h-* 'narrow').

In Celtic (cf. Schumacher 2004: 36–37), there are two verbs with a full-grade root and two verbs with zero-grade root (one of which is uncertain). Rather than being continuants of *tudáti* presents, the latter owe their root vocalism to analogical influence from nominal formations that (in both cases) have a zero grade throughout. Two of these verbs are factitive, one is stative/fientive, and one is stative; only one has cognates in other Indo-European branches.

4.2 **g$^{u̯h}$er-* 'hot'

g$^{u̯h}$ér-e/o-* 'make hot' > PCelt. **g$^{u̯}$er-e/o-* > OIr. *geirid, *fo·geir* 'heat, inflame; irritate' (cf. Schumacher 2004: 372–373).

(12) Old Irish (Metr. Dinds. iv 334)
aní ***fo·geir*** *mo menma dom·beir fri degra*
the.DEICTIC **heat**.3SG.PRS.IND my spirit brings.me to *degra*
duibe.
darkness.GEN.SG
'The thing that frets my spirit brings me to the *degra* (fire?) of darkness.'

Cognates Ancient Greek θέρομαι 'become warm/hot', θέρω 'heat',[6] Alb. *zien* 'cook (tr.)' (LIV^2 219–220).

Adjective OIr. *gor* 'pious' < **g$^{u̯h}$or-o-* (cf. Bret. *gor* 'warm', W *gwâr* 'civilised, gentle, kind').

[6] In this type of Greek verbs, the middle inflection with intransitive meaning is often attested earlier than the active inflection with transitive meaning (cf. θέρομαι 'become warm/hot' Hom.+ : θέρω 'heat' A.R.+); see Rau (2009: 155).

Commentary The simple verb *geirid* is only attested in a single Middle Irish text, in the form *gerthiut* 'it warms thee' (BDD² 317). This is most probably an analogically created form.

4.3 **th*$_2$***eu̯s-*** 'silent'

**th*$_2$*eu̯s-e/o-* > 'be/become silent' > PCelt. **taus-e/o-* > OIr. *tóaid* 'be silent', remodeled as Late Proto-British **tau-i-* > MW *tewi* (1sg. *tawaf*, 3sg. *teu*) 'be/become quiet, silent', MBret. *teuell* 'be/become quiet, silent', MCorn. *tewel* 'be quiet, silent' (cf. Schumacher 2004: 621–623 without *tóaid*).

(13) Old Irish (*Ériu* iv 30.22)
no ***thoad*** *in sluag*
PTCL **be.silent**.3SG.IMPF the host
'The host was/became silent.'

(14) Middle Welsh (GMB 76; 12$^{\text{th}}$ c.)
Py ***dawant*** *anant,* *na* *phrydant* *wawd?*
why **be.silent**.3PL.PRS.IND musicians that.not compose.3PL.PRS.IND praise
'Why are the musicians silent that they do not sing praise?'

Cognates –

Adjective OIr. *tó* < **th*$_2$*e/ou̯s-o-* and *tóe, túae* < **th*$_2$*e/ou̯s-i̯o-*, both 'silent'.

Commentary OIr. *tóaid* may continue the primary verb or be a secondary, denominal formation from OIr. *tó* 'silent'. The interjection OIr. *tá* 'quiet! hush!' has been tentatively explained as the outcome of the imperative **th*$_2$*eu̯s-e!* (cf. LEIA T1–2), but this is phonologically implausible.

4.4 ***meh**$_2$***k-*** 'big'

**mh*$_2$*k-e/o-* 'make big' > PCelt. **mak-e/o-* > OIr. (*do·for*)*maig* 'increase, add', MW *magu* (3sg. *mac*) 'rear, breed, nourish, foster', MBret. *maguaff* 'feed, nourish, rear, suckle', MCorn. *maga* 'rear, nourish' (cf. Schumacher 2004: 466–470).

(15) Old Irish (Fél. Ep. 195)
do·formaig *a ánae*
increase.3SG.PRS.IND his wealth
'It increases his wealth.'

(16) Middle Welsh (MA² 241_{a47-8}; 12^{th}–13^{th} c.)
or a ***fag magu*** *daear hi*
if REL **feed**.3SG.PRS.IND earth her
'if the earth nourishes her'

Cognates –

Adjective Old Irish *machtae* 'great', MW *meith* 'long, large' < **mh₂kt(i)i̯o-*; MW *maeth* m. 'sustenance, nourishment, fosterage' < **mh₂k-to-* (a substantivised adjective?).

Commentary OIr. *machtae* 'great', MW *meith* 'long, large' may originally have been the participle (**mh₂kt(i)i̯o-*) of the verb PCelt. **mak-e/o-* 'raise, increase', but both phonologically as well as semantically these forms and MW *maeth* could also belong with the root **meĝ-* 'big, great' qua **mₔĝt(i)i̯o-* and **mₔĝ-to-*, respectively.

4.5 ****lei̯hₓ-*** 'liquid (?)'

**lihₓ-e/o-* 'be liquid (?)' > PCelt. *°*lii̯-e/o-* > MW 3sg. *dillyð* 'flow, abound' (cf. Schumacher 2004: 372–373).

(17) Middle Welsh (T 21_{24}; 14th c.)
Gogwn pan ***dillyd***, *gogwn pan wescryd.*
I.know when **flow**.3SG.PRS.IND I.know when ebb.3SG.PRS.IND
'I know when it flows, I know when it ebbs away.'

Cognates –

Adjective Not attested directly, but the following nouns may be substantivisations of Caland adjectives: W *llin* 'flow of blood, pus, discharge' < **lihₓ-nó-*, OIr. *lië* 'flood, spate', W *lli*, pl. *lliant* 'flood(s)' < **lihₓ-n̥t-*, OIr. *ler*, W *llŷr* 'sea, ocean' < **lihₓ-ró-*.

Commentary Despite the evidence for a set of typical Caland-derivatives (see above), the root **lei̯h_x-* may be a non-Caland verbal root with a meaning 'to pour' (cf. LIV^2 405–406). See also Section 3.4 on the question of the merger of this root with **pleh$_1$-* 'fill'.

5 Causatives/iteratives

5.1 Background

The causatives/iteratives are characterised by an *o*-grade in the root and the suffix **-éi̯e/o-*, and also feature among the typical Caland present stems (cf. Rau 2009: 141–143). It is disputed whether they constitute a veritable root-derived verbal stem type in the proto-language or if they are rather denominal verbs derived from **R(ó)-o-* verbal nouns or adjectival abstracts (e.g., **g$^{u̯h}$óro-* 'heat' → **g$^{u̯h}$or-e-i̯e/o-* 'apply heat'[7]).[8] These two viewpoints are not necessarily mutually exclusive. A set of frequent denominal verbs could be seen as the diachronic source of a subsequently abstracted verb-formation type, which then encouraged the creation of new root-derived verbal stems by formal analogy.

As root-derived, true causatives, we would expect factitive semantics to be predominant. If they were originally denominal verbs based on an adjectival abstract 'X-ness', however, all three possibilities (stative, fientive, factitive) may be expected, although a secondary specialization of one of them would not be surprising. Within the Indo-European languages, the factitive type prevails, with some notable exceptions. Compare the polysemy of Latin *lūcet* (if **lou̯kei̯eti*) 'is light, is clear, shines' and 'lights (sth.)', which, however, might be due to a conflation with the zero-grade stative.

In Celtic, this type is attested for six roots, four of which have extra-Celtic cognates. Originally, the meaning of all of them was likely factitive, though some of the attested verbs have undergone substantial semantic change. An *o*-grade thematic stem is attested for three of them (one is uncertain), and could be the derivational basis in a not-so-distant past for at least two of them.

5.2 **$b^h er\hat{g}^h$-* 'high'

**$b^h or\hat{g}^h$-ei̯e/o-* 'make high' > PCelt. **borg-ī-* > OIr. *·díbairg* (suppletive prototonic stem of *do·bidci*) 'throw, shoot, hurl', MW *bwrw* 'throw, cast, strike' (cf. Schrijver 1995: 55–56; Schulze-Thulin 2001: 120–121).

[7] It is also thinkable that the basis was the instr. sg. of the underlying noun: **g$^{u̯h}$óreh$_1$* 'with heat' → **g$^{u̯h}$or-eh$_1$-i̯e/o-* (> **g$^{u̯h}$or-ei̯h$_1$e/o-*?) 'be/become/make with heat'.

[8] On this question, see Peters (2016); noncommittal Bozzone (2020).

(18) Old Irish (RC iii 178 (LL))
*ro·**díbairg*** *inn gai* *dó*
AUGM·**throw**.3SG.PRET the spear at.him
'He hurled the spear at him.'

(19) Middle Welsh (LIDW 63_{10}; 13th c.)
*o **buru*** *y* *mor betheu yr* *tyr* *neu yr* *traeth hunnv*
if **throw**.3SG.PRS.IND the sea things to.the land or to.the beach that
byeuyt y *brenhyn*
owns the king
'But if the sea throws anything on the land or on that beach, they belong to the king.'

Cognates Vedic (*sám*) *barháya-* 'strengthen', YAv. (*us*) *barəzaiia-* 'rear' (LIV^2 78–79).

Adjective **bʰr̥ĝʰ-(é)nt-* 'high' (see Section 1), but semantically distant from the synchronic meaning of the verbs.

Corresponding *o*-grade thematic stem –

Commentary The semantic development underlying the Celtic verbs can be imagined as 'make high' > 'raise' > 'throw' and must predate the diversification of the Insular Celtic languages.

5.3 **g^u̯h er-* 'warm, hot'

**g^u̯h or-ei̯e/o-* 'make warm, heat' > PCelt. **g^u̯ or-ī-* > OIr. *guirid* 'warm, hatch', MW *gori* 'brood, hatch, fester' (cf. Schulze-Thulin 2001: 146–147; LIV^2 219–220).

(20) Old Irish (Ml. 39c24)
guirit *són* (gl. *fouent*)
warm.3PL.PRS.IND this
'they warm'

(21) Middle Welsh (AL i. 734; 13th c.)
teithi iar *yw dodwi* *a* ***gori***
traits hen is lay.eggs.VN and **brood**.VN
'The nature of a hen is it to lay eggs and to brood.'

Cognates Alb. *n-xeh* ‘warms’ (?) (LIV^2 219–220).

Adjective See Section 4.2 above.

Corresponding *o*-grade thematic stem OIr. *gor o* m. ‘inflammation, pus; the brooding of hens’, W *gôr* ‘pus; hatching’, MBret. *gor* ‘pus, furuncle’ (**g$^{u̯h}$or-o-*).

Commentary The evidence is somewhat ambiguous: Since the continuants of the thematic noun do not mean ‘warmth, heat’ any longer, the meaning ‘to warm’ of OIr. *guirid* points to an inherited formation (cf. Alb. *n-xeh* ‘warms’). On the other hand, the ‘brood, hatch’ semantics that are attested for the verbs and the thematic verbal nouns in both Irish and Welsh may suggest an origin as a denominal verb.

5.4 **$h_2$ek̂-* ‘sharp, pointed’

**$h_2$ok̂-ei̯e/o-* ‘make sharp, pointed’ > PCelt. **ok-ī-* > MW *hogi* (3sg. *hyc*) ‘sharpen, whet’, NBret. *koñvog*, *kougañ* ‘sharpen, roughen’ (cf. Schumacher 2000: 158 and Schulze-Thulin 2001: 171–173).

(22) Middle Welsh (B XXII 121; 15th c.)

ef	*a*	*ddaw*	*baedd*	*ac*	*a*	***hyc***	*y ddannedd*
PTCL	REL	come.3SG.FUT	boar	and	REL	**sharpen**.3SG.PRS.IND	its teeth

‘There comes a wild boar and sharpens its teeth.’

Cognates Old High German *eggen* ‘harrow’ (cf. LIV^2 261).

Adjective OIr. *ér* ‘noble, great’ < **$h_2$ek̂-ro-* (cf. Ancient Greek ἄκρος ‘utter, highest’, Old Lithuanian *ašras* ‘sharp’), but semantically distant.

Corresponding *o*-grade thematic stem W *og* ‘harrow’ < **$h_2$ok̂o-* or **$h_2$ok̂eh_2-* (but semantically distant).

Commentary The meaning of the verbs in Welsh and Breton is clearly factitive. NBret. *koñvog*, *kougañ* ‘sharpen, roughen’ corresponds to W *cyfogaf*, *cyfogi* ‘make keen or pointed, roughen’. The word-initial *h-* in Welsh is unetymological (Schumacher 2000: 158).

5.5 ******temk*- ‘hard’

**tomk-ei̯e/o-* ‘make hard’ > PCelt. **tonk-ī-* > OIr. *tocaid* ‘destine’ (via Proto-Goidelic **toggī-*), MW *tynghu* ‘destine, affirm’, MBret. *tonquaff* ‘destine’ (cf. Schumacher 1995; Schulze-Thulin 2001: 158).

(23) Old Irish (Thes. ii 293.6–7)
*ma rom·**thoicther**-sa in so …*
if AUGM.me·**destine**.3SG.PRS.SUBJ.PASS-me the this
*manim·**rotchaither***
if.not.me·AUGM.**destine**.3SG.PRS.SUBJ.PASS
‘if this be destined for me … if it not be destined for me’

(24) Middle Welsh (R 1036.43, CL1H II 21.a; c. 1400)
*truan a dynghet a **dynghwyt** y lywarch*
miserable of destiny REL **destine**.IMPERS.PRET to Llywarch
‘It was a wretched destiny that was destined to Llywarch.’ (Schumacher 1995: 53)

Cognates Old Saxon *ā-thengian*, *an-thengian* ‘carry out, complete’ (cf. LIV^2 625–626).

Adjective OIr. *técht* ‘(of liquids) thick, frozen; calm, peaceful’ < **tm̥k-to-* (but semantically distant). See Section 2.4 above.

Corresponding *o*-grade thematic stem –

Commentary The expected OIr. outcome of Proto-Goidelic **toggī-* is **tucaid* (thus also Schumacher 1995). The actually attested forms *tocaid* etc. owe their vocalism probably to influence from the verbal abstract *tocad* ‘fortune’ < **tomk-eto-*. The Brythonic verbs have often been treated as belonging to PCelt. **tung-e/o-* ‘swear’ (OIr. *tongaid*, W *tyngaf*, *tyngu*, MBret. *toeaff*). However, Schumacher (1995) has shown that they have to be kept separate.

5.6 ******u̯elk*- ‘wet’

**u̯olk-ei̯e/o-* ‘make wet’ > PCelt. **u̯olk-ī-* > OIr. *folcaid* ‘wash the head’, MW *golchi* (3sg. *gwylch*) ‘wash’, MCorn. *gol(g)hy*, MBret. *guelchiff*, NBret. *gwalc’hañ* (see Schulze-Thulin 2001: 141; 159–160; LIV^2 679).

(25) Old Irish (Sg. 146b3)
folcaimm (gl. *lavo*)
wash.1SG.PRS.IND
‘I wash’

(26) Middle Welsh (T 41, 6–7; 14th c.)
golchettawr *ei lestri, bid groyw ei vrecci*
wash.IMPERS.IMPV his vessels be.3SG.IMPV clear his wort
‘Let his vessels be washed, his wort be clear.’

Cognates –

Adjective PIE *u̯l̥k-u- > PCelt. *u̯liku-. This may be continued directly in OIr. *fliuch* ‘wet’. OW *gulip*, ModW *gwlyb*, fem. *gwleb* ‘wet’, Corn. *gleb*, OBret. *gulip*, NBret. *gleb* looks like a thematisation *u̯liku̯o- > *u̯lik$^{u̯}$o- from this (cf. **Kummel2022**).

Corresponding *o*-grade thematic stem The corresponding noun is *u̯ólk-o- > OIr. *folc o*, m. ‘heavy rain, wet weather’, perhaps W *golch* ‘a washing, cleansing’ (cf. Latvian *val̂ks* m. ‘creek, brook, small stream of water’). Since W *golch* is attested rather late, it is probably a backformation and not a descendant of *u̯ólk-o-.

Commentary The root *u̯elk- (cf. ?*u̯elk- in LIV² 679 ‘feucht sein/werden’) is a doublet of *u̯elg- ‘wet’ (cf. 2.*u̯elg- in LIV² 676 ‘feucht werden’). For the latter, a causative *u̯olg-ei̯e/o- ‘make wet’ is attested in Serbo-Croatian *vlȁžiti* ‘moisten’, Russian *volóžit’* ‘id.’, Lithuanian *válgyti* ‘eat’. It is formally conceivable that *u̯elk- is a regular Celtic development of *u̯elg-sk̂-. In this case the Caland behaviour of this neo-root *u̯elk- would have to be an inner-Celtic development. However, a root-final voiceless *k is also necessary to explain Lith. *valkà* f. ‘puddle’, Latv. *val̂ka* ‘creek, brook, wet meadow’.

5.7 **meĝ-* ‘big’

*moĝ-ei̯e/o- ‘make big’ > PCelt. *mog-ī- > OIr. **mogaid/moigid*, *·mogai/·moigi** (?) ‘make great, increase, magnify’ (>> *móaigid*), MW *moi* ‘to foal’ (cf. Joseph 1987: 136; Schumacher 2004: 469).

(27) Old Irish (*Ériu* xvi 65.47, *Celtica* v 136)

rod-bet	*o*	*gallaib*	*bann[ach]aib*(?)	*barca* …
AUGM.you-be.3PL.PRS.SUBJ	from	foreigners	zealous	ships

maínib	***mogatar***	(ms. *mogetair*)
treasures	**magnify**.3PL.PRS.IND.PASS.REL	

‘From industrious (?) foreigners may you have (receive) … boats that are magnified (enriched) with treasures.’

(28) Middle Welsh (*WM* 31_{1-4}; 14^{th} c.)

a	*phob*	*nos*	*Calanmei*	*y*	***moei***,	*ac*	*ny*	*wybydei*	*neb*
and	every	night	1^{st}.of.May	PTCL	**foal**.3SG.IMPF	and	not	knew	anyone

un	*geir*	*e wrth*	*y*	*hebawl.*
one	word	about	her	colt

‘And every May-eve [the mare] foaled, but no one knew a word about her colt.’

Cognates –

Adjective OIr. *maige* ‘great’, Gaul. *Magio-rix*, *Magius* < **m̥ₔĝ-i̯o-*.

Corresponding *o*-grade thematic stem –

Commentary Most attestations of this verb in OIr. are philologically difficult. Variants in *mog-*, *moig-* and *moaig-* or *móaig-* occur seemingly in free variation in the manuscript tradition of texts. *Moaig-*/*móaig-* could be a reanalysis or even a recent secondary formation based on the comparative *mó* (of *már, mór*) ‘bigger, greater’ + the productive verbal suffix *-aig-*. Furthermore, a grave shadow of doubt is cast on the inherited character of this verb by the fact that the regular outcome of PCelt. **mog-ī-* ought to be ***mugaid* in Old Irish (cf. PCelt. **logī-* ‘lay’ > *do·lugai* ‘forgive < *‘lay down’). This is perhaps an indication that the vowel of the first syllable is long in the first place.

The meaning of MW *moi* ‘to foal’, on the other hand, is difficult to reconcile with ‘to make big’. In short, the cognacy between the two verbs is just as uncertain as their derivation from **moĝ-ei̯e/o-* ‘make big’.

6 Perfects: **temh$_x$-* ‘faint, dark’

**te-tomh$_x$-* ‘has become faint’ > PCelt. **tetom-* >> **tetam-* > OIr. *tathaim*, *tathamair* ‘died’ (cf. Schumacher 2004: 68–79; 634–635).

(29) Middle Irish (Metr. Dinds. iv 34.52)
conid *'sin* *tsléib* ***tathamair***
so.that.it.is on.that mountain **die**.3SG.PRET
'So that it is on that mountain that he died.'

Cognates Vedic *tatāma* 'fainted' (cf. LIV[2] 624; LIV[2] Addenda s.v. **temH*-).

Adjective OIr. *tem*, *teim* 'dark' < **temh*$_x$*-o-*.

Commentary On perfects and the Caland system, see Rau (2013). Besides 'dark' (cf. OIr. *tem*, *teim* 'dark'), the root also had a meaning '(physically) faint' (cf. IEW 1063–1064 "geistig benommen, betäubt" and "dunkel"; LIV[2] 624 "ermatten, ohnmächtig werden"); compare English *to black out* 'to faint'.

According to Schumacher (2004: 634–635), the irregular vocalism in the reduplication syllable can be explained as based on an unattested present **taimid*, **·taim* 'dies' (cf. pres. *cainid* 'sings' and pret. *cachain*); alternatively, it can have been generalised from the weak stem **tetm̥h*$_x$-, which regularly yielded PCelt. **tetam*-. The earliest, albeit Middle Irish, attestations of the verb have deponent endings, perhaps influenced by other suffixless preterites of roots in *-m* such as *lámair* (*ro·laimethar* 'dare') and *dámair* (*daimid* 'endure, suffer').

7 Discussion

To find out whether the derivation of root-based deadjectival verbal stems remained at least to some extent productive in the prehistory of the Celtic languages, it will first be necessary to divide the total number of twenty verbal stems into those with extra-Celtic cognates (A) and those without extra-Celtic cognates (B). Group (A) consists of eleven verbs.

(A)
1. **(h*$_1$*)rud*^{h}*-eh*$_1$*-i̯e/o-* 'be/make/become red' > PCelt. **rud-ī-* > OIr. *ruidid*, *·ruidi* 'flush, blush; make red' (cf. Latin *rubēre* 'be/become red', Old High German *rotēn* 'be/turn red', Latvian *rudêt* 'rust', Russian Church Slavonic *rъděti sja* 'become red')
2. **k*$^{(u̯)}$*su̯ib*^{h}*-eh*$_1$*-i̯e/o-* 'be/make/become swift' > PCelt. **ksu̯ib-ī-* > MIr. *scibid*, *·scibi* 'flow, swim (of a vessel); move; move sth. jerkily', MW *chwyfu* (3sg. *chwyf*) 'move to and fro, budge', MBret. *fifual* 'move' (cf. Old High German *swebēn* 'be agitated (of the sea), be in high and low tide')

3. *tum-eh_1-$i̯e/o$- 'be/make/become thick' > PCelt. **tum-ī-* > MW *tyfu* (3sg. *tyf*) 'grow; make grow; heal', NBret. *teñviñ* 'grow, increase' (OBret. 3sg. *ni-tum*), MCorn. *tevi* 'grow' (cf. Latin *tumēre* 'be(come) swollen', Lithuanian *tumė́ti* 'become thick, coagulate')
4. *$h_{2/1}$*i-n(e)-dʰ-* 'make burn' >> PCelt. **ind-e/o-* > MW *enn-ynnu* (3sg. *enn-ynn*) 'kindle, light a fire' (cf. Vedic *indhé* 'lights a fire')
5. **pl̥-n(e)-*h_1- 'make full' > ? PCelt. **φli-ni-* > OIr. *do·lin* 'flow, abound' (or stative/fientive 'be/become full'?) (cf. Vedic *pr̥ṇā́ti* 'fills', Old Avestan *pərənā* 'fill!', Latin *polleō* 'be strong, able', Armenian *lnwom* 'I fill', Albanian *m-blon* 'fills')
6. **g̑ᵘʰér-e/o-* 'make hot' > PCelt. **g̑ᵘer-e/o-* > OIr. *geirid**, *fo·geir* 'heat, inflame; irritate' (cf. Ancient Greek θέρομαι 'become warm/hot', θέρω 'heat', Alb. *zien* 'cook (tr.)')
7. **bʰorĝʰ-ei̯e/o-* 'make high' > PCelt. **borg-ī-* > OIr. *·díbairg* (suppletive prototonic stem of *do·bidci*) 'throw, shoot, hurl', MW *bwrw* 'throw, cast, strike' (cf. Vedic (*sám*) *barháya-* 'strengthen', YAv. (*us*) *barəzaiia-* 'rear')
8. **g̑ᵘʰor-ei̯e/o-* 'make warm, heat' > PCelt. **g̑ᵘʰor-ī-* > OIr. *guirid* 'warm, hatch', MW *gori* 'brood, hatch, fester' (cf. Alb. *n-xeh* 'warms' ?)
9. *h_2*ok̑-ei̯e/o-* 'make sharp, pointed' > PCelt. **ok-ī-* > MW *hogi* (3g. *hyc*) 'sharpen, whet', NBret. *koñvog*, *kougañ* 'sharpen, roughen' (cf. Old High German *eggen* 'harrow')
10. **tomk-ei̯e/o-* 'make hard' > PCelt. **tonk-ī-* > OIr. *tocaid* 'destine' (via Proto-Goidelic **toggī-*), MW *tynghu* 'destine, affirm', MBret. *tonquaff* 'destine' (cf. Old Saxon *ā-thengian*, *an-thengian* 'carry out, complete')
11. **te-tomh*$_x$- 'has become faint' > PCelt. **tetom-* >> **tetam-* > OIr. *tathaim*, *tathamair* 'died' (cf. Vedic *tatāma* 'fainted')

The existence of corresponding verbal stems in the other branches suggests an inherited status of the eleven verbs of group (A). Note that items 4. and 5. might belong to verbal roots 'to ignite' and 'to fill', respectively.

Group (B) contains nine verbs, none of which has exact correspondences in the other Indo-European branches. This may suggest that these verbs were created within Celtic. Interestingly, all of them have adjectives (or potential substantivisations of such adjectives) beside them.

(B)

1. **tm̥k-eh₁-i̯e/o-* 'be/make/become hard' > PCelt. **tank-ī-* > OIr. *con·téici* 'congeal, become solid; make solid' (cf. OIr. *técht* '(of liquids) thick, frozen; calm, peaceful' < **tm̥k-to-*)
2. **(h₁)ru-n(e)-dʰ-* 'make red' >> PCelt. **rund-e/o-* > OIr. *rondaid*, *·roind* 'dye red', *fo·roind* 'stain' (cf. OIr. *rúad*, archaic and poetic *ród* 'red' < **(h₁)re/ou̯dʰ-o-*)
3. **seh₁-n(e)-i-* 'make long' > PCelt. **sī-ni-* >> **sīnī-* > OIr. *sínid*, *·sín*(*i*) 'extend, stretch (oneself)' (cf. OIr. *sír* 'long' < **seh₁-ro-*)
4. **u̯l̥-n(e)-hₓ-* 'be powerful' > PCelt. **u̯al-na-* >> **u̯all-nā-* > OIr. *follnaithir* 'rule' (cf. **u̯l̥hₓ-o-* 'powerful, in power' as the second compound member *-*u̯alo-* in Gaul. *Katouualos*, OW *Catgual*, OIr. *Cathal*, etc.)
5. **th₂eu̯s-e/o-* > 'be(come) silent' > PCelt. **taus-e/o-* > OIr. *tóaid* 'be silent', >> Late Proto-Brit. **tau-i-* > MW *tewi* (1sg. *tawaf*, 3sg. *teu*) 'be(come) quiet, silent', MBret. *teuell* 'be(come) quiet, silent', MCorn. *tewel* 'be quiet, silent' (cf. OIr. *tó* 'silent' < **th₂e/ou̯s-o-*; OIr. *túae* 'silent' < **th₂e/ou̯s-i̯o-*)
6. **mh₂k-e/o-* 'make, become big' > PCelt. **mak-e/o-* > OIr. (*do·for*)*maig* 'increase, add', MW *magu* (3sg. *mac*) 'rear, breed, nourish, foster', MBret. *maguaff* 'feed, nourish, rear, suckle', MCorn. *maga* 'rear, nourish' (cf. OIr. *machtae* 'great', MW *meith* 'long, large' < **mh₂kt(i)i̯o-*; MW *maeth* m. 'sustenance, nourishment, fosterage' < **mh₂k-to-*, a substantivised adjective?)
7. **lihₓ-e/o-* 'be liquid (?)' > PCelt. **°li̯i-e/o-* > MW 3sg. *dillyð* 'flow, abound' (cf. the potential substantivisations of Caland adjectives: W *llin* 'flow of blood, pus, discharge' < **lihₓ-nó-*, OIr. *lië* 'flood, spate', W *lli*, pl. *lliant* 'flood(s)' < **lihₓ-n̥t-*, OIr. *ler*, W *llŷr* 'sea, ocean' < **lihₓ-ró-*)
8. **u̯olk-ei̯e/o-* 'make wet' > PCelt. **u̯olk-ī-* > Old Irish *folcaid*, *·folcai* 'wash the head', MW *golchi* (3sg. *gwylch*) 'wash', MCorn. *gol(g)hy*, MBret. *guelchiff* (cf. OIr. *fliuch* 'wet' < PCelt. **u̯liku-* < **u̯l̥k-u-*; OW *gulip*, ModW *gwlyb*, fem. *gwleb* 'wet', Corn. *gleb*, OBret. *gulip*, NBret. *gleb* < **u̯likᵘo-* < **u̯liku̯o-*)
9. **moĝ-ei̯e/o-* 'make big' > PCelt. **mog-ī-* > OIr. **mogaid*/*moigid*, *·mogai*/*·moigi** 'make great, increase, magnify' (>> *móaigid*), MW *moi* 'to foal' (cf. OIr. *maige* 'great', Gaul. *Magio-rix*, *Magius* < **mₔĝ-i̯o-*)

Caution is advised regarding items 4., 7., 9. as they are beset with philological, phonological, and morphological difficulties. In addition, 4. and 7. may have a verbal root at their core. Item 8., the causative/iterative **u̯olk-ei̯e/o-* 'make wet' > 'wash', not only has an adjective (**u̯l̥k-u-* 'wet') but also an *o*-grade thematic stem **u̯ólk-o-* (> OIr. *folc o*, m. 'heavy rain, wet weather') beside it, which may have served as the derivational base for a denominal (rather than a deadjectival) verb at some point in the common prehistory of the Insular Celtic languages.

While it cannot be excluded that each of the verbs in group (B) stems from a distant past and once had extra-Celtic cognates that happen to have been lost in the non-Celtic branches of Indo-European, it is also possible to assume that these verbs were created within Celtic based on the corresponding adjective. The five secure examples, items 1., 2., 3., 5., 6., consist of one stative, two nasal-infix presents, and two plain thematic presents. The causative/iterative 8. is ambiguous (see the paragraph above).

Under such an interpretation, it is necessary to conclude that at this point in time, the language could still derive deadjectival verbs from roots and not from adjectival stems, as the respective adjectival suffixes (e.g., **-to-*, **-o-*, **-ro-*) are seemingly deleted and the generated verbs follow the primary formation patterns of deradical verbs. In other words, these examples give the impression that Caland morphology continued to be productive to a limited extent in Celtic. Note that this analysis is possible even for the problematic items 7. and 9.; for the third one, 4. OIr. *follnaithir* 'rule', it is necessary anyway to assume that the root vocalism was secondarily adjusted to its potential derivational base **u̯l̥h*$_x$*-o-* 'powerful, in power' > **-u̯alo-* (attested as a second compound member), as discussed in Section 3.6.

Support for the assumption of such a limited vitality of Caland morphology in Celtic in general, and a suffix deletion rule in deadjectival derivation in particular comes from the nominal domain. Here, there is evidence for at least two inner-Celtic neo-Caland roots. The first example is the adjective OIr. *trén* 'strong' < **treksno-* (cf. also the personal names Gaul. *Trenus*, Ogam *TRENA-GUSU*), which probably continues a denominal **-nó*-adjective of the type **leu̯k-s-no-* 'having light' (YAv. *raoxšna-* 'bright', OHG *liehsen* 'bright') based on an **-s*-stem **trég-os* 'strength' (cf. Old Norse *þrek* n. 'id.'). From this **treksno-* a secondary Caland root **treks-* 'strong' was derived by suffix deletion, which is at the core of formations such as **treks-i̯ā-* 'strength' > Old Irish *treise i̯ā*, f. 'strength, vigour', **treks-n̥t-* 'id.' > MIr. *treisset* 'strength, might', the comparative OIr. *tressa*, MW *trech*, MBret. *trech*, NBret. *trec'h* 'stronger', and the superlative **treks-(is)m̥mo-* > OIr. *tressam*, W *trechaf* 'strongest' (the British forms have no positive beside them).

An even clearer example is the Celtic neo-root for ‘young’. Like a couple of other branches of Indo-European, Celtic inherited a morphologically complex adjective meaning ‘young’, *_h_2i̯uh$_x$n̥k̂o_- (cf. Ved. _yuvaśá_-, Lat. _iuuencus_, Proto-Germanic *_jungaz_). This formation is not derived from a Caland root, but rather from the noun *_$h_2$ói̯-u_- n. ‘vital force’ (Ved. _ā́yu_-, OAv. _āiiū_-). Its Proto-Celtic descendant *_i̯ou̯anko_- is continued by OIr. _óac_, _óc_, MW _ieuanc_, W _ifanc_, OCorn. _iouenc_, and MBret. _yaouanc_ ‘young’.[9] Based on its status as an essential property concept, this adjective became the basis of its own inner-Celtic Caland system consisting of a comparative and a superlative. In order to generate them, the *-_anko_- part of *_i̯ou̯anko_- (despite not being a Caland suffix) was deleted and a neo-root *_i̯ou̯_- ‘young’ was extracted serving as the basis for a root-derived comparative OIr. _óa_, MW _ieu_, W _iau_, Bret. _iaou_ ‘younger’ and superlative PCelt. *_i̯ou̯-isamo_- > OIr. _óam_, MW _ieuhaf_, W _ieuaf_, _iefaf_, _ifaf_ ‘youngest’.

Taken together, the secure examples of inner-Celtic deradical Caland verbs of group (B) on the one hand, and the nominal neo-roots *_treks_- ‘strong’ and *_i̯ou̯_- on the other seem to corroborate the view that Caland morphology remained mildly productive in Celtic.

8 Conclusion

This study is the first of its kind to provide an exhaustive overview, discussion, and analysis of both inherited and potentially innovative deadjectival verbs in the Celtic languages. The preceding sections reveal that many of these verbs present philological, phonological, and morphological issues, to which one can add the synchronically full or partially weak behaviour of the OIr. verbs _do·lin_ (Section 3.4), _sínid_ (Section 3.5) and _tóaid_ (Section 4.3). In addition, the total number of potential Caland verbs is rather low so that any conclusions drawn from them need to be taken with caution, particularly when attempting to make broad generalisations.

Nevertheless, the investigation has yielded significant findings. Of the twenty Caland-associated verbs, eleven appear to belong to an inherited stock of formations with formal cognates in other Indo-European languages. The remaining nine formations lack exact correspondences in other Indo-European branches, and therefore likely represent innovations within the Celtic languages themselves. Each of these verbs is associated with an adjective (or a substantivisation

[9] Besides PCelt. *_i̯ou̯anko_-, the Gaulish personal names in _Io(u)inc_- such as _Iouinca_, _Iouincillus_, etc. (cf. Delamarre 2003: 191–192) suggest the existence of a variant *_i̯ou̯enko_- (i.e., with full grade of the suffix, perhaps in analogy to a stem *_i̯ou̯en_- < *_h_2i̯uh$_x$en_-, inferable from Latin _iuuenis_ ‘young (man)’?).

of such an adjective), suggesting that they were derived within Celtic based on these adjectives. Since the formation of these verbal stems follows root-based primary patterns, the suffix of the adjective (e.g., *-*to*-, *-*o*-, *-*ro*-) appears to have been systematically deleted in the formation of these deadjectival verbs.

The deletion of adjectival suffixes and the emergence of primary deradical verbs in these cases further support the hypothesis that Caland morphology maintained a degree of vitality in Celtic. This process of suffix deletion in verbal formations is mirrored in the nominal domain, where adjectives like **treksno*- 'strong' and **i̯ou̯anko*- 'young' similarly give rise to neo-Caland roots **treks*- and **i̯ou̯*-, which in turn serve as the basis for further deadjectival derivation.

These patterns, found in both the verbal and nominal domains, suggest the persistence and a certain degree of flexibility of the Caland system within the Celtic languages. Further research could explore whether similar processes of suffix deletion and root creation occurred in other Indo-European branches as well, providing a more comprehensive understanding of the Caland phenomenon across the language family.

Abbreviations

For quotations from Irish and Welsh, we follow the conventions and abbreviations in eDIL (version 2019) and GPC.

Act.	Active	OAv.	Old Avestan
Bret.	Breton	OBret.	Old Breton
Corn.	Cornish	OCorn.	Old Cornish
Dat.	Dative	OIr.	Old Irish
Gaul.	Gaulish	OW	Old Welsh
Gen.	Genitive	PCelt.	Proto-Celtic
Lat.	Latin	Perf.	Perfect
Latv.	Latvian	Pl.	Plural
Lith.	Lithuanian	PrimIr.	Primitive Irish
MBret.	Middle Breton	Ptcpl.	Participle
MCorn.	Middle Cornish	Pres.	Present
MIr.	Middle Irish	Pret.	Preterite
ModW	Modern Welsh	Sg.	Singular
MW	Middle Welsh	Tr.	Transitive
NBret.	Modern Breton	Ved.	Vedic
Nom.	Nominative	W	Welsh

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Chapter 4

The relationship between *u*-adjectives and *nu*-causatives in Anatolian

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The significant number of Hittite causative/factitive formations in -*nu*- attested next to *u*-adjectives has always been striking and led together with comparative evidence from Old Indic and Ancient Greek to a theory in the 70s that the process occurs through *n*-infixation between the root and the adjectival suffix -*u*-, e.g., Hitt. *tēp-u-* 'small' ⟶ Hitt. *tēp-n-u-*^{mi} 'to make small, diminish' (matching OInd. **dabh-u-* (< PIE **d*^{h}*eb*^{h}*-u-*, replaced by OInd. *dabh-rá-* 'small') ⟶ OInd. *dabh-n-ó-ti* 'to harm, deceive'), Hitt. *park-u-* 'high' (< PIE **b*^{h}*erĝ*^{h}*-u-*, beside Hitt. *park-taru* 'must become high') ⟶ Hitt. *parg-n-u-*^{mi} 'to make high', provoking at a certain point a reanalysis of the morpheme -*n*- to -*nu*-. Although this theory never gained much popularity, recent studies on the Luwian language have brought forth more data that have yet to be examined in depth vis-à-vis the morphological process involved, e.g., HLuw. *wa/i-su-* /wasu-/ 'good' (< PIE **h*$_{1}$*u̯es-u-*, beside HLuw. *wa/i-s*° /wass-^{ti}/ 'to be good, pleasing') ⟶ HLuw. *u-sa-nú-* /usnuwa-^{i}/ 'to make good, benefit', and, thus, a re-examination of the Anatolian material is warranted. In addition to retracing the development of the denominal morpheme -*nu*-, its extension to other types of nominal stems, notably to the *i*-stems (e.g., Hitt. *ḫarki-* 'white' (< PIE **h*$_{2}$*erĝ-i-*) ⟶ Hitt. *ḫarg(a)nu-* 'to make white'), as well as the motivation behind it is addressed. Furthermore, the phenomenon of *u*-extended roots that has been considered to be intertwined with the development of the *nu*-suffix is explored and the discussion is expanded with a new set of Anatolian data (cf. HLuw. *wa/i*$_{5}$*-su-* /wasu-/ 'to be good, successful').

1 Introduction

The numerous examples of Hittite denominal factitives in -*nu*- had already been noticed in the 30s, when comparative linguistic work on Hittite was thriving.

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Indo-Europeanists were convinced that the suffix -*nu*- was primarily deverbal and that the denominal function had been secondarily acquired by Anatolian. In his doctoral thesis submitted in 1973, Harold Koch scrutinized the data more than it had been done before and cogently defended the view that Hittite factitives in -*nu*- were originally derived from *u*-adjectives via *n*-infixation. Further, he explained that the co-existence of root verbs, *u*-adjectives and *nu*-factitives all based on the same root would have been the starting point for the replacement of the isolated infixation process with the more common suffixation one.[1] Thus, by reanalysing the derivational base, the deverbal causative suffix would have been created.[2] This theory can be illustrated with the following examples, i.e., first the proto-type and second the reanalysis:

1. CVC-*u*- ⟶ CVC-*n*-*u*- (e.g., Hitt. *ḫuišu*- 'alive, raw' ⟶ *ḫuiš-n-u-*^{mi} 'to keep alive')
2. CVC- ⟶ (reanalysis) CVC-*nu*- (e.g., Hitt. *ḫuiš-*^{mi} 'to live' ⟶ *ḫuiš-nu-*^{mi} 'to keep alive, let live')

It is noticeable that the formation of Hittite *ḫuišnu-*^{mi} is synchronically ambiguous, as one could even switch the example of the proto-type with the one of the reanalysis, arguing for the reverse direction in the development and making the *n*-infixation younger than the *nu*-suffixation.[3] Koch (1973: 34–36), however, insists that the *n*-infixation is older by pointing out that an *n*-infixed verb like Hitt. *ḫarni(n)k-*^{mi} 'to destroy' is later replaced by the more productive suffixed formation Hitt. *ḫarknu-*^{mi} 'to destroy'.[4] The recently discovered Luwian verb

[1]Koch 1973: 25–26, 34–36. Oettinger (1979: 165–166 with fn. 72) points out that the infixation process should be older than the apparent separation between nominal and verbal roots within Proto-Indo-European, adducing PIE **u̯ed*- 'water' ⟶ **u-né-d-/und-´* 'to water' (continued in Ved. *unátti*) as a case in point; cf. NIL 712, fn. 43. However, the nature of Proto-Indo-European roots and the relationship between diverse root formations is a topic that will require more attention in future research.

[2]This approach slightly differs from the theory defended by Strunk (1967: 60–61), who maintains that the suffix -*nu*- originates in the reanalysis of the infix to a root ending in -*u̯* such as PIE **k̂leu̯*- 'to hear' ⟶ **k̂l̥-né-u̯-/k̂l̥-nu-´* 'to cause to hear'. However, since this putative prototype is rare (see also Villanueva Svensson 2025), Strunk had to resort to *u*-extended roots in order to provide further support to this theory (the role of *u*-extensions for this topic is addressed in detail in the last part of this paper). The view presented here also differs from the more recent theory by Steer (2015), who argues that the *nu*-suffix is primary and the *n*-infix secondary. Although his approach is morphologically acceptable, it goes against the evidence of the daughter languages, in which the *n*-infix is older than its various reanalyses as a suffix.

[3]Thus Steer 2015.

[4]Thus also Oettinger 1979: 192, Shatskov 2020: 281–282.

tarnuwa-^{i} ‘to release’ vis-à-vis Hitt. *tarna-/tarn-^{ḫi}* ‘to release’ is another example of such a replacement (cf. Melchert 2022: 109–110), comparable to the relationship between Ved. *str̥̄ṇā́ti* ‘spreads’ and Gk. στόρνυμι ‘id.’ or Ved. *yunákti* ‘yokes’ and Gk. ζεύγνυμι ‘id.’, where the original *n*-infixed verb has been replaced in Ancient Greek by a *nu*-suffixed verb (see Rasmussen 1999: 308–309 and Villanueva Svensson 2025 for a summary of these phenomena). The radical *n*-infixation in the Anatolian branch is indeed an archaism that can still be observed in Hittite, but almost vanished in the other 2$^{\text{nd}}$ millennium BC languages like Luwian and Palaic. In these other two languages, it appears to have been replaced by suffixation after having been reanalyzed with a root-final laryngeal before combining with another morpheme; the result being *-i-na-^{i}* (consisting of verbal *-*i̯e/o*- or nominal *-*i*-, followed by either *-*néh$_2$-/-nh$_2$-´* or *-*néh$_3$-/-nh$_3$-´*).[5] On the other hand, the *nu*-suffix experienced an increasing productivity in Hittite and Luwian, as will be shown in more detail in the next section.

In order to prove that the *n*-infixation among *u*-adjectives is inherited, Koch (1973) cites parallels in Old Indic and Ancient Greek. The Old Indic example for the proto-type that has been cited in the literature for almost a century and was convincingly defended by Oettinger (1979: 164–165) and Koch (1980) is the following one:[6] Hitt. *tēpu-/-aw-* ‘small’ ⟶ *tepnu-^{mi}* ‘to make small, belittle’ matching OInd. **dabh-u-* (replaced by OInd. *dabh-rá-* ‘small’) ⟶ OInd. *dabh-n-ó-ti* ‘to harm, deceive’, both leading to the PIE reconstruction **d^h(é)b^h-u-/-éu̯-* ‘small’ ⟶ **d^h(e)b^h-n-éu̯-/-u-´* ‘to make small’.

Whereas the denominal process has been given up in Old Indic, Koch (1978) and Tucker (1981: 22–28) both argued independently that the process is continued in Ancient Greek factitives in -ύνω, e.g., βαθύς ‘deep’ ⟶ βαθύνω ‘to deepen’,

[5] Sasseville 2021a: 540f., 548.

[6] **GotzePedersen1934**, Kammenhuber 1969: 231, EWAia I 694–696; followed by Oettinger 1979: 164–165, Tucker 1981: 27, Praust 2004: 385–386, Steer 2015: 208–210, Lundquist & Yates 2018: 2162, Oettinger 2022: 157 (adding PIE **méi̯-u-* ‘few’ ⟶ **mi-né-u̯-ti* ‘to make few’). The first objection raised by Shatskov (2020: 277) that Hitt. *tepnu-^{mi}* cannot be old because it is not attested in Old Hittite is invalid. The “Old Hittite” corpus is very small and highly fragmentary, which makes Shatskov’s claim *ex silentio*. Shatskov also argues against the word equation due to the, in his words, “quite different” meanings, although Koch (1980: 225–232) has explained in detail the semantic development of ‘to belittle’ ⟶ ‘to deceive’ and provided sufficient contextual parallels between the Hittite and the Indo-Iranian cognate (e.g., the gods as patient). It is also interesting to notice that the Hittite factitive *tepnu-* (just like OInd. *dabhnóti*) shows the restored full grade of the *u*-adjective. Given that the expected zero-grade of the root would have yielded an unstable consonant cluster, i.e., Hitt. [tbnu-], it may have facilitated a restoration of the full grade in analogy to the *u*-adjective in the prehistory of each language. Only Old Avestan *dəbənaotā* preserves the original zero-grade with the complex consonant cluster (Oettinger 1979: 164).

θρασύς 'bold, courageous' → θαρσύνω 'to embolden', εὐρύς 'wide' → εὐρύνω 'to widen' and *αἰσχύς (replaced by αἰσχ-ρό-ς 'ugly, shameful') → αἰσχύνω 'to make ugly, shame'.[7] However, Ancient Greek must have innovated by replacing the process of infixation by suffixation followed by the suffix *-i̯e/o- in the present stem of the denominative verbs.[8] For a new explanation of the development of this derivational process in Greek and its relationship to the denominative suffix -αίνω, see Villanueva Svensson 2024.

2 Deadjectival Verbs in Anatolian

2.1 Formations in -*nu*-

As is well known, the Hittite language has a significant number of *nu*-formations built from adjectives in -*u*-, e.g., *parku*- 'high' (< *b^herǵ^h-u-*) → *park(a)nu*- 'to make high', cf. the root verb *parktaru* (3sg.imp.) 'must become high'; *āššu*- 'good' (< *$h_1e(h_1)s$-u-*) → *aš(ša)nu*-^{mi} 'to take care of', cf. the stative, mediopassive root verb *āšš*- 'to be dear to'; *ḫatku*- 'narrow' (< *$h_2ed^hg^h$-u-*) → *ḫatg(a)nu*-^{mi} 'to make narrow', cf. the root verb *ḫatk*-^{hi} 'to close'. However, despite what scholars may claim, it is not always clear whether the *nu*-formation is derived from the *u*-adjective or from the root verb.[9] Although some formations may be more transparent than others, it is always possible to argue for the one or for the other direction, which is relevant for the topic of reanalysis. Considering solely the deadjectival type from a synchronic point of view, one may speak of an *n*-infix or of a *nu*-suffix with deletion of the *u*-suffix. The latter process is precisely the one which applies to the adjectival *i*-stems, e.g., Hitt. *šalli-/-ay*- 'big, great' (< *$solh_{2/3}$-i-*) → *šall(a)nu*-^{mi} 'to make big', *ḫarki-/-ay*- 'white' (< *$h_2$erǵ-i-*) → *ḫark(a)nu*-^{mi} 'to make white', *parkui-/-ay*- 'pure' (< *$perk^u$-i-*)[10] → *parkunu*-^{mi} 'to make pure' and *mekki-/-ay*- 'many, much' (< *$meĝh_2$-i-*) → *maknu*-^{mi} (*meknu*-^{mi}) 'to make numerous'.[11]

[7] Followed by Villanueva Svensson 2024. Earlier works considered the -*n*- in the suffix -ύνω to be simply an extension without further explanation (see Palmer 1980: 264).

[8] Chantraine 1947: 276–277, de Lamberterie 1990: 25 ("-ῡνω < *-ῠν-i̯ω, aor. -ῡνα < *-ῠν-σα"). On the denominal nasal suffix in the northern Indo-European languages, see Villanueva Svensson 2016.

[9] Luraghi (1992: 156) sees *aš(ša)nu*-^{mi} as deverbal instead. The etymological rejection of *aš(ša)nu*-^{mi} made by Kloekhorst (2008: 216–218) is rebutted by Sasseville (2021a: 473, fn. 10).

[10] For the root, see eDiAna ID 565.

[11] For a list of examples of the denominal type, see Luraghi 1992: 155–156 and Vanséveren 2014: 70–72.

It is appropriate to recall that there are no clear examples of Hittite *a*-stems, which reflect the merger of **-o-* (unless it was syncopated, in which case the stem became consonantal) and **-eh$_2$-*, that serve as derivational bases of *nu*-formations. There are a few examples that have been adduced in the literature and the first one is *marš(a)nu-^{mi}* 'to desecrate' next to *marša-* 'false(hood)'.[12] For this reason, Koch (1973: 29) finds its interpretation as a derivative from the *a*-stem *marša-* suspect and therefore does not exclude the possibility that it may have been derived from an unattested root verb or even from a prehistoric *u*-stem. Likewise, Oettinger (1979: 239, fn. 1) and Luraghi (1992: 157) distance themselves from the nominal base *marša-* and presuppose instead a deverbal formation from *maršiye/a-^{mi}* 'to be false' with deletion of the verbal suffix *-iye/a-^{mi}* (< **-i̯e/o-*).[13]

A third possibility that has been overlooked is a deadjectival derivation *maršai(ya)-* from an *i*-stem *marši-/-ay-* 'pertaining to desecration, false', if it truly is a Hittite lexeme and not a Luwianism.[14]

Another possible example for an *a*-stem as derivational base is presented in HW Ḫ 87–88, where it is argued that *ḫalluwanu-^{mi}* of uncertain meaning is not derived from *ḫalluwā(i)-^{mi}* 'to fight', i.e., 'to cause to fight', but rather from *ḫalluwa-* 'deep', i.e., 'to deepen' against the previous literature. The argument rests on the understanding of two contexts, which cannot be reviewed within the frame of this article. Moreover, it is worth noting that the adjective *ḫalluwa-* has a *u*-stem allomorph *ḫalluwaw-* visible in the acc. pl. *ḫalluwawuš* and *ḫalluwāmuš* (see HW Ḫ 84–85), which, if this lexeme is indeed the derivational base, may have paved the way for the *nu*-factitive, i.e.,*ḫalluwaw-* ⟶ *ḫalluwa-n-w-*.[15] A similar problematic case concerns Hitt. *ḫatuganu-^{mi}* 'to frighten', which can be derived from the adjective *ḫatuga-* (or *ḫatugi-*) 'frightful' (Luraghi 1992: 155) or from the root verb *ḫatuk-^{mi}* 'to spread fear (intr.)' (Koch 1973: 19f.; HW Ḫ 529), both semantically adequate, i.e., 'to cause to spread fear' or 'to make frightful'.[16] Moreover,

[12]This example is left out in Vanséveren 2014: 70–72.

[13]Note that there is now even less evidence for a synchronic Hittite class of stative verbs in *-e-^{mi}*, since it has finally been recognized that the cuneiform sign E may have the phonetic value of the glide *-i̯-*; see Kloekhorst 2014: 134–161 and the subtle warnings in the older literature such as in Neu 1983: 10 and Melchert 1994: 35, 141–142. Oettinger (1992: 225–226) has already pointed out that only *ar-ša-ne-e-ši* and *mar-še-e-er* remain as probative examples. Nonetheless, the unmistakable formation *-iye/a-^{mi}* in the amply attested *aršaniye/a-^{mi}* as well as the fact that the hapax *mar-še-e-er* could still reflect a stem *maršiye/a-^{mi}* despite not being written **mar-ši-e-er* render the existence of this class synchronically highly questionable.

[14]Cf. eDiAna ID 2869 with further literature.

[15]The hapax *maninkuwanut* (CHDS 3.3, 7') is not a relevant example because its derivational base is not (properly) attested (see CHD L–N 170 **maninkuwa-* 'near'). Therefore, the stem formation of the base is unknown; cf. Oettinger 1979: 246 ("*maninkui-*").

[16]On the few spellings *ḫatuganu-* for expected **ḫatuknu-*, cf. the similar situation of *zalukanu-* attested beside the variant *zaluknu-* (Kloekhorst 2008: 1027–1028).

the adjective *ḫatuga-* shows an *i*-stem allomorph *ḫatugay-* in, e.g., abl. *ḫatugayaz*, nom. pl. *ḫatugayeš* (*ḫa-tu-ga-e-eš*) and acc. pl. *ḫatugauš* (HW Ḫ 525–529), which may have been relevant for the derivational process, if the adjective turns out to be the proper base. Be that as it may, it is safe to posit that there is no unambiguous synchronic evidence that would show that the Hittite factitive *nu*-suffix can derive verbs from *a*-stems.[17]

In the smaller Luwian corpus, we find the several cases of the denominal type in /-nuwa-^{i}/ (cf. Sasseville 2021a: 471–477, 484–485) derived from *u*-adjectives, e.g., (1).

(1) a. CLuw. *wāšu-*, HLuw. *wa/i-su-* ‘good’ (< *h_1u̯os-u-) → HLuw. *u-sa-nú-* /usnuwa-^{i}/ ‘to make good, bless’
b. Luw. */assu-/ ‘good’ (= Hitt. *āššu-* < *h_1e(h_1)s-u-) → HLuw. *a-su-nu-* /ass(u)nuwa-^{i}/ ‘to make good (?)’ (cf. Lyd. *f-ašvo-*^{d} ‘to make good (?)’)[18]
c. HLuw. */habanzu-/ ‘loyal’ (< *$h_2$op-ent-i̯u-[19]) → HLuw. *ha-pa-za-nu-wa-* /habanznuwa-^{i}/ ‘to make loyal’ (cf. CLuw. 𒑱*ḫapanzuwalatar/n-* ‘loyalty’ and HLuw. *ha-pa-zú+ra/i-wa/i-* /habanzulawa(/i)-/ ‘devoted, devotion (?)’)

From a synchronic perspective, these formations can be interpreted either as *n*-infixed or as *nu*-suffixed via deletion of the adjectival suffix *-u-*, although the former is expected to be the inherited process. The rest of the deadjectival derivations with the Luwian suffix /-nuwa-^{i}/ are built on mutated stems that can go back to either a thematic *o*-stem or an *i*-stem, e.g., CLuw. *ura/i-*, HLuw. MAGNUS+*ra/i-* ‘big, great’ (< *$h_1$ur-ó-)[20] → CLuw. *uranuwa-*^{i}, HLuw. MAGNUS+*ra/i-nú-wa/i-* /uranuwa-^{i}/ ‘to make great’; CLuw. *ḫalāl(i)-* ‘pure’ (Semitic borrowing) → *ḫalalanuwa-*^{i} ‘to make pure’; CLuw. *zantal(i)-* ‘low’ (< *ḱm̥t-ó-lo-) → CLuw. 𒑱*zantalanuwa-*^{i} ‘to make low, diminish’.[21] As can be gathered from these examples, there is no process of deletion here. Instead, the suffix *-nuwa-*^{i} is added to the thematic vowel *-a-* (< *-o-), which is the regular derivational vowel in the Luwian *i*-mutated paradigm. In comparison with Hittite, however, one may posit

[17] As pointed out by Sasseville (2021a: 483), since the Common Anatolian thematic suffix *-o- was syncopated in post-tonic position and the formation transferred to another stem class, e.g., *u*-stems, it is always possible that a formation in *-nu- was originally created at the time when the thematic suffix was still present. Unfortunately, there is as of yet no positive evidence for this claim.

[18] See Sasseville 2021a: 479–480.

[19] Cf. eDiAna ID 1580 for various ways of interpreting the morphological chain.

[20] See eDiAna ID 3247.

[21] The lack of a clear example going back to a PIE *i*-stem is simply due to chance.

that the factitive suffix was able to reach the thematic stems in Luwian only because of their merger with the *i*-stems and that the process of deleting the suffix -*i*-, assuming that it existed in Pre-Luwian, was given up, i.e., *-*i*- (with deletion) + *-*nu*- ⟶ *-*o*-/-*i*- + -*nu*- = *-*o*-*nu*-. On comparative grounds, adjectival *u*-stems inherited the *n*-infixation and, as far as we can tell, preserved it.

The next question concerns the path of the extension from *u*-adjectives as the derivational base to *i*-adjectives and the reason behind the reanalysis of the infixation process as suffixation. The two following scenarios require infixation to not only be genuine but also older: The first explanation is that old *u*-adjectives were gradually replaced by younger *i*-adjectives, which motivated the reinterpretation of the factitive formation from infixation to suffixation via deletion of the suffix, i.e., CVC-*u*- : CVC-*n*-*u*- reanalyzed as CVC-*i*- (ousting the *u*-stem) : CVC-*nu*- (Koch 1973: 35, Tucker 1981: 27). On the other hand, it has been claimed that the reanalysis of the derivational process simply happened due to the decline of the infixation process altogether, which led to a new suffix *-*nu*- creating factitives via deletion of the adjectival suffix *-*u*- and then *-*i*- (cf. Oettinger 1979: 245–247, 2022: 157).[22] The reason why the *i*-stems would have been targeted would simply be due to their similarity to the *u*-stems. In the end, the second scenario has the advantage of better explaining why it is the *i*-stems that followed the path of the *u*-stems without having to recur to the loss of an original *u*-adjective.

2.2 New ways of forming verbs from Anatolian *u*-adjectives

The suffix -*nu*- is not the only one that can create deadjectival verbs. At some point, Hittite started using the factitive suffix -*aḫḫ*-^{i}, which originates in stems in *-*eh*$_2$-, to derive *u*-adjectives via their oblique stem, e.g., *tēpu*-/-*aw*- 'small' ⟶ *tepawaḫḫ*-^{i} 'to make small, humble' (vs. *tepnu*-^{mi}), *idālu*-/-*aw*- 'bad, evil' ⟶ *idalāwaḫḫ*-^{i} 'to treat badly (tr.), to act badly (intr.)'.[23] There is also one potential example in Palaic, i.e., *wāšu*- 'good' ⟶ *wašūḫ* (2sg.imp.) 'to act well, be good.[24] Nonetheless, the number of examples is very low, so that the rivalry between -*aḫḫ*- and -*nu*- regarding adjectival *u*-stems is subtle. Shatskov (2020: 282) claims that the two Hittite examples mentioned above (though re-addressing

[22] However, the claim found in, e.g., Oettinger 1979: 246–247 that Hittite has stems in -*ui*- (< *-*u̯*-*ih*$_2$) is considered outdated here; see Kloekhorst 2008: 829.

[23] There is a widespread view that -*aḫḫ*- was mainly derived from stems in -*ant*- via deletion of the suffix (see Oettinger 1979: 240–241, Vanséveren 2014: 147), which would be possible as a secondary phenomenon, but is unconvincing since all examples can instead be derived from the base in -*a*-, which was often extended with the highly productive denominal suffix -*ant*-. See Sasseville 2015: 284–285 for this criticism.

[24] See Sasseville 2021a: 536, fn. 8.

only *tepawaḫḫ-*^{i}) were created by non-native speakers who did not grasp Hittite morphology, but this is not persuasive. Although there is a strong interference from Luwian native speakers in New Hittite, this is certainly not the case here because the Luwian factitive suffix was restricted to conversion and, therefore, there was no such model for deriving *u*-stems in their own language.[25] The Hittite factitive suffix *-aḫḫ-* simply became highly productive and extended its derivational scope to all kinds of nominal stems.

Koch (1973: 29–30) notes that the extension of *-aḫḫ-* to the adjectival *i*-stems is largely restricted to synchronically non-ablauting *i*-stems, e.g., *nakkī-*[26] 'heavy, important' ⟶ *nakkiyaḫḫ-*^{i} '(intr.) to become a concern to', *šannapili-* 'empty' (but cf. acc. pl. [*ša-an-n*]*a-pí-la-a-uš*; CHD Š 159) ⟶ *šannapilaḫḫ-*^{i} 'to empty', *šanezzi(ya)-* 'pleasant' ⟶ *šanezziyaḫḫ-*^{i} 'to indulge oneself', *šarazzi(ya)-* 'upper' ⟶ *šarazziyaḫḫ-*^{i} 'to make prevail'. Regarding the suffix *-zzi(ya)-*, we are dealing with an originally thematic suffix *´-_ti̯o-_ that secondarily underwent contraction or syncope of the thematic vowel, thus inflecting sometimes like an *i*-stem. A similar explanation applies to stems in *-ili*, whose suffix originates in *-_í-lo-_ with syncope of the thematic vowel in post-tonic position followed by a secondary transfer of the stem class.[27] These data indicate that the Hittite factitive suffix *-aḫḫ-* was still closely associated with the *a*-stems (*-_eh_$_2$-, *-_o-_) and, after syncope of *´-_o-_, with the consonantal stems, after which it was even extended to the (secondary) non-ablauting *i*-stems. An exception to this pattern is *šuppi-/-ay-* 'pure' ⟶ *šuppiyaḫḫ-*^{i} 'to purify', which would show that the factitive suffix in the end did reach the genuine ablauting *i*-stems, conflicting once more with the factitive suffix *-nu-*.

Overall, the Hittite factitive suffix *-aḫḫ-* was so productive that it could be derived from any substantival or adjectival nominal stem, whereas the factitive suffix *-nu-*, based on the attested data, was confined to adjectival *u-* and *i*-stems.[28] The reason behind this distribution may be the nature of the process

[25] On Luwian interference in Hittite, see Melchert 2005, Rieken 2006, van den Hout 2006, Yakubovich 2010: 15–73.

[26] For the exceptional origin of this particular *i*-stem, see Widmer 2006.

[27] See Rieken 2008.

[28] It is not entirely clear whether Hittite *-nu-* could derive verbs from substantival bases; cf. Vanséveren 2014: 71, who lists three examples: *ešḫarnu-*^{mi} 'to make bloody', *ēdriyanu-*^{mi} 'to feed' and *šaminu-*^{mi} 'to burn (trans.)', although a deverbal derivation based on the denominative verb *ēdriye/a-*^{mi} 'to feed' is another viable option (thus Hoffner & Melchert 2008: 175, fn. 10; cf. Luraghi 1992: 155) which deserves further investigation. The same ambiguity is found in KAL-*tarnuškanzi* /hadugadarnuskanzi/ (KBo 12.109, 7'), which may be derived from the abstract noun in *-ātar* (i.e., *ḫatugātar*) or from an unattested denominative in *-_iye/a-_ with deletion of the suffix. The derivational base of *šaminu-*^{mi} is problematic and thus debatable (cf. CHD Š 118,

itself: Whereas the former is simply suffixed to the nominal stem in Hittite, the latter uses either infixation or suffixation with deletion of the nominal suffix of the base, which must have been morphologically less stable.

There is a view that verbs in *-uwe/a-*^{mi}, which are mostly deradical to roots ending in *-u-* derived with the suffix *-iye/a-*^{mi}, may also be denominal to *u*-adjectives, e.g. *ḫuišu-/-aw-* 'alive, raw' → *ḫuišuwe/a-*^{mi} 'to live, stay alive'. Dismissing this morphological interpretation, as well as that of a stative in *-e-*^{mi} because of the lack of parallels, Jasanoff (2023: 69) analyzes this example as a thematized *-*u*-present, comparing the reconstructed PIE **g̑ᵘíh₃u̯e/o-* 'live'.[29] As per Jasanoff, the evidence for a denominal verb in *-iye/a-*^{mi} derived from a *u*-adjective, albeit formally possible, is lacking independent support. However, what has not been considered is the formal possibility of a deradical verb in *-iye/a-*^{mi} derived from a *u*-extended root. This topic as well as this example is addressed again in Section 3 of this paper.

Turning to the Luwian language, Sasseville (2021a: 190–199) suggests the existence of a denominal suffix /-(a)u-^{di}/ used to derive *u*-stems in order to explain the strange morpheme /-(a)u-/ found in some Luwian verbs, although a coherent picture of the data could not be successfully provided. Melchert & Yakubovich (2022) take a different approach and convincingly show that the Luwian endings *-unta* and *-aunta* do not stand for the 3pl. preterit, as previously thought, but for the 1pl. preterit ending. Their findings therefore eliminate the evidence for a special denominal suffix /-(a)u-^{di}/ in Luwian. In other words, there is currently no other way to derive adjectival *u*-stems in Luwian except with the suffix /-nuwa-^{i}/. As already mentioned, the factitive suffix *-*eh₂*- is not a potential candidate either because it is restricted to conversion in Luwian. On the other hand, the Luwian suffix /-nuwa-^{i}/ could derive verbs from *a/i*-stems (< *-*o*-, *-*i*-) and perhaps even from *a*-stems (< *-*eh₂*-), thus being more susceptible to encroach on the factitive suffix *-*eh₂*-. In the end, this yields a situation that is the opposite of the Hittite one.[30]

Luraghi 1992: 158). One last possible example of a denominal case is the hapax *aimpanu-*^{mi} 'to burden' (HW A 48), if it is derived from the noun *aimpa-/impa-* c. 'burden' instead of from the denominative verb **(a)impā(i)-*^{mi} 'to burden' (attested solely as a verbal noun *impawar* and as an imperfective *impaiške/a-*^{mi} according to HW I 56). Despite the occasional ambiguity, it does appear as if a desubstantival derivation for *-nu-* was slowly becoming possible, perhaps through reinterpretation of the derivational base.

[29]See also Villanueva Svensson 2021: 273–274 with fn. 33.

[30]Cf. Sasseville 2021a: 471–477. However, due to the size of the Luwian corpus and the lack of data, it is not always clear whether the derivational base was mutated (i.e., *-*o*-, *-*i*-) or not (*-*eh₂*-).

Turning to the Anatolian languages of the 1st millennium BC, Lycian has basically eliminated nominal *u*-stems, transferring them to the paradigm of the *a*-stems (< *-eh_2-), so that it does not offer any evidence whatsoever for the current topic. Lydian, however, seems to have had highly productive "*u*"-stems, so that we might find denominal verbs derived from adjectival *o*-stems (< PIE *-*u*-), e.g., *taso*- 'strong, heavy (*vel sim.*)' (= Hitt. *daššu*- 'heavy', CLuw. *kwanzu*- 'id.') ⟶ *taso*-^{d} 'to impose'.[31] Taking this example seriously, Lydian appears to derive denominal verbs in -*o*-^{d} of the lenited *mi*-conjugation from adjectival stems in -*o*- via conversion, although the mechanism in this case is more likely to be secondary, rather than inherited.[32]

3 Reanalysis and *u*-extensions

According to the analysis of Strunk (1967: 60–108), there is a difference between *n*-infixed verbs to roots ending in -*u* (e.g., **k̑leu̯*- ⟶ **k̑l̥-né-u̯-/-n-u-´*) and *n*-infixed verbs to roots ending in a laryngeal, in which the -*u*-, unlike the laryngeal, is not part of the root; compare Ved. *śr̥ṇā́ti* 'to break apart', *śīrṇá*-, aor. *aśarīt*, *śīrti*- 'breaking' with Ved. *vr̥ṇóti* 'to enclose', *vr̥tá*-, aor. *ā́var*, *vr̥ti*- 'fencing'. Although Strunk argues that the origin of "*nu*-presents" is to be found in roots that end in -*u*, apart from **k̑leu̯*- 'to hear' and **ki̯eu̯*- (Gk. κίνυται 'to be in motion', ἔσσυτο, OInd. *cyuti*- 'rapid motion'), he only provides examples of these so-called *u*-extended roots, e.g., *dabhnóti* (!), *vr̥ṇóti* 'defends' (cf. Gk. ἔρυσθαι), *vr̥ṇóti* 'to enclose' (cf. Gk. εἰλύω 'id.'). This demonstrates at the very least that there is a certain connection between *nu*-verbs and *u*-extensions.

The phenomenon of *u*-extension in the Indo-European languages is an obscure topic, especially vis-à-vis the so-called "*u*-present" suffix.[33] García Ramón (2021)

[31] Sasseville 2021a: 180–190; differently Garnier & Sagot 2020: 174, fn. 20.

[32] See Sasseville 2021a: 203–205.

[33] Cf. Strunk 1967: 60–61, Klingenschmitt 1982: 231–233 w. fn. 4, Steer 2015: 211–217, Jasanoff 2023; cf. also Watkins 1969: 53 on "*u*-endings". The "*u*-present" explanation is used for Hittite *tarḫu*-^{mi} and *lāḫū*-^{ḫi} by Oettinger (1979: 222–223., 424). Contra Jasanoff (2023: 67 with fn. 25), Hitt. *tarḫu*-^{mi} can only be interpreted as a root verb of the *mi*-conjugation and the lack of assimilation to **tarruzzi* is due to the generalization of the weak stem, where the accent was not on the full-grade root, across the paradigm (cf. Hitt. *walḫ*-^{mi} 'to strike' << *$u̯élh_2$-/$u̯l̥h_2$-´). The two attestations *tar-ḫu-id-du* (KUB 36.75 iv 10') and *tar-ḫu-e-zi* (KBo 38.126, r.col. 10'), which he uses as evidence for an inherited thematic root verb, are not decisive: Both attestations recall the occasional marking of an extra -*i*-/-*e*- vowel following a -*u*-, in this case combined with a marker of the fortis quality of the ending in the first example breaking the consonant cluster, i.e., [tárhh^{w}ₔttu] and [tárhh^{w}ₔtsi] respectively; cf. *tar-aḫ-ḫu-i-iš-ke/a*- [tarhh^{w}ₔské/á-] (KBo 61.221 obv.$^{?}$ r.col. 2'), *pa-aḫ-ḫu-*⸢*i*⸣*-na-li-az* (VBoT 58 iv 36 to $^{\text{DUG}}$*paḫḫunal(l)i*- n. 'fire

has recently attempted to differentiate the two by using a semantic approach. He posits that the morpheme -*u*- should be seen as an extension, whenever there is no semantic distinction between the root verb with -*u*- and the one without. However, if the two happen to differ semantically, he claims that the morpheme acts as an aspectual marker (imperfective/perfective), insofar as it appears only in the present stem. If it is not restricted to the present stem, the morpheme is seen as a marker of *Aktionsart*.[34] This hypothesis definitely needs more evidence from a larger set of data, which we cannot pursue in its entirety here. The present chapter will primarily be concerned with the morphological aspect of the problem, taking Anatolian "*u*-verbs" as one category. Nevertheless, attention will also be paid to the semantics of the formations in question.

In the Anatolian languages, there are adjectival *u*-stems attested next to root verbs, to formations in *-*nu*- as well as to *u*-extended verbs, all built on the same root, and it is tempting to see a connection. A similar picture has already been described for Hittite and Indo-Iranian. Koch (1980: 235) argues based on Old Indic *dabhnóti* 'to harm, deceive' next to *ádbhuta*- 'wonderful' that the deadjectival *n*-infixed formation could at some point be mistaken by native speakers as an *n*-infixed verb. In this way, the morpheme -*u*- was reanalyzed as part of the root, i.e., **debʰu*- or **dbʰeu̯*-, and, in a similar way, the Old Indic 8th class with, e.g., *kru*- 'to do' may have originated as, e.g., *kr̥*- 'to do' ⟶ *kr̥ṇóti/kr̥ṇvánti* ⟶ reanalysis of the root as *kru*- 'to do' ⟶ *karoti/kurvanti* (cf. Thumb & Hauschild 1959: 265–268 on the problems of this class and its examples).[35] Independent evidence for

container'), *wa-at-ku-it-ta* (KUB 30.67, 6') and *wa-at-ku-at-ta* (KBo 13.137, 11' to *watku*-*ᵐⁱ* 'to jump'; pace Oettinger 1979: 337, fn. 161) and similarly with *i*-glide insertion between *u* and another vowel [*te*]-*pa-u-i-e-eš-ta* (KBo 58.8+ rev. 23'), *ša-an-ḫu-u-i-en* 'we swept' (KBo 38.257, 19'), [*ar-k*]*u-i-an-zi* (KBo 37.70, 6'; HW A 311) and [*a*]*r-ku-ia-a*[*n*? -*zi*] (KBo 40.328, 10' to *arku*-*ᵐⁱ* 'to chant' attested beside the denominative *arkuwā(i)*-*ᵐⁱ* 'to pray, chant'). On the other hand, if one wishes to take these spellings with -*i*-/-*e*- seriously, a deradical verb in *-*i̯e/o*- derived from the *u*-extended root with the loss of the *i̯*-glide remains a viable alternative. In any case, assuming the existence of an additional verbal stem formation which is not securely attested in the first place based on these two forms is highly dubious.

[34]Unfortunately, the Hitt. verb †*tarḫ*-*ᵐⁱ* 'to overcome' used by García Ramón (2021: 157–158) is a ghostword (outdated Oettinger 1979: 222–223), since only a *u*-extended root verb is truly attested in Hittite, i.e., *tarḫu*-*ᵐⁱ*; see already Kloekhorst (2008: 837) who convincingly shows that all spellings of *tar-Vḫ*- should be read *tar-uḫ*- not ***tar-aḫ*-.

[35]Contra Hoffmann (1976), who proposes a special phonological development to derive the *u*-verb from the *nu*-verb, showing on philological grounds that the former has fully replaced the latter over time. One should also note the ambiguity in another example of the same class, i.e., *tan*- 'to stretch' ⟶ *tanóti/tunvánti* < **tn̥-néu̯-ti/tn̥-nu̯-énti* or < **ten-éu̯-ti/t(e)n-u̯-énti*. Assuming the first formation to be original, the morpheme boundary between root and suffix may have been redefined and the derivational process reinterpreted, i.e., as *tan-ó-ti/tun-v-ánti*; cf. Thumb & Hauschild 1959: 265 with further literature. Differently, Jasanoff (2023: 63) takes the middle "*u*-present" to be original (Ved. *tanute*) and the "*nu*-present" to be a backformation thereof.

the validity of the reanalyzed root *db^heu̯- comes from Old Avestan *dəbaoman-* 'deception (*vel sim.*)' with the full grade anchored before -*u*-. According to Koch, the cause behind this reanalysis was the disappearance of the *u*-adjective, i.e., the ultimate derivational base, from the language or languages concerned. However, it is questionable whether this is truly the case and, therefore, it is the goal of the last section of this paper to test Koch's scenario against a new set of Anatolian data.

The first root examined here is *h_1u̯es- 'good': A stative verb can be posited on the basis of the CLuw. (though attested in Hittite context) mediopassive form *waššāri* 'to be good, be dear' (< *h_1u̯(e)s-ór + -*i*) and the HLuw. active verb *wa/i-sa-* /was(s)-^{ti}/ 'id.'.[36] A *u*-stem adjective is found in CLuw. *wāšu-* 'good', HLuw. *wa/i-su* (adv.) 'id.' and Pal. *wāšu-* 'id.', all going back to a generalized *o*-grade of an acrostatic paradigm *h_1u̯ós-u-/h_1u̯és-u-. The crucial *nu*-formation is attested in HLuw. *u-sa-nú-wa/i-* /usnuwa-^{i}/ 'to bless'.[37] In the end, an apparent "*u*-extended" verb is attested in HLuw. *wa/i*$_5$*-sú-ha* 'I was good, successful' and probably also in CLuw. *wašuddu*[(3sg.imp.act.?).[38] Unfortunately, due to the lack of attestations, it is not entirely clear what type of conjugation the latter verb has, although, assuming that the Cuneiform Luwian word is complete, it would definitely belong to the non-lenited *mi*-conjugation (just like Lyc. B *welpu-^{ti}*, not ***welpuwe-^{ti}* or ***welpu-^{di}*). The formation in -*nuwa-^{i}*, despite being primarily analyzed as a deadjectival formation (see Section 2.1), may have been reanalyzed at any time as a causative formation of the root verb. The next question is what lies behind the stem formation of the "*u*-verb". Sasseville's explanation as a denominal verb derived from *wāšu-* 'good' falls through (see Section 2.2). A synchronic verbal suffix -*u*- is not possible considering the various conjugations shown by the other data (cf. below). What remains is the explanation as a root extension, which would have been triggered by the reanalysis of the causative formation as an *n*-infixed stem based on a root ending in -*u*. Moreover, it is striking that the meaning of the verb is basically the same as that of the two radical verbs, which also points towards a simple *u*-extension, i.e., back-projected *h_1u̯esu- or *$h_1$useu̯-, backformed

[36] For a summary of the research on the PIE "stative", see recently Grestenberger 2019. In the case of *waššāri*, it is difficult to reconstruct a stative of the type "zero-grade of the root with accent on the suffix" due to the expected effects of Čop's Law in the root, i.e., *éCV > áCCV. Nonetheless, it is always possible that the shape of the root would have been analogically introduced here after another co-existing stem formation.

[37] What appears to be a zero-grade might just be a contraction of /wa-/ > /u-/, which sporadically occurs in Iron Age Luwian. Nonetheless, attestations from the earlier stages of the language are required to verify this.

[38] The Cuneiform Luwian piece of evidence is found in a highly fragmentary context which calls for caution; see Sasseville 2021b: 179.

from the *nu*-formation. If this reasoning is correct, this example alone would contradict Koch's "requirement" for the reanalysis, because both the adjectival base and the root verb without -*u*- are still present in the Luwian language.

The second root adduced here is $^*h_3er$- 'to arise': This root has a stative verb attested in Hitt. *ār-tari* 'he stands' (< $^*h_3$*ér-tor* + -*i*). There are also a few (de)radical formations found in CLuw. *āri-^{ti}* and Pal. *ariye/a-^{ti}* 'to raise' (< $^*h_3$*r̥-i̯é/ó-*) as well as Hitt. *arāi-/-i(ya)-^{hi}* and Lyc. *erije-^{ti}* 'to arise' (< $^*h_3$*rói̯*h_1-/h_3*rih*$_1$-´ << $^*h_3$*ór*-/h_3*r*-´). A *u*-adjective is attested in CLuw. and Pal. *arū-/-aw-* 'standing (straight), tall, high (*vel sim.*)' (< $^*h_3$*(é)r-u-/-éu̯-*).[39] The well-known causative formation in -*nu*- is attested both as Hitt. *arnu-^{mi}* 'to set in motion, carry off' and HLuw. CRUS+*RA/I-nu-wa/i-* /arnuwa-^{i}/ '(lit.) to let stand, abandon (?)' and is commonly analyzed as a deverbal formation, although a deadjectival formation derived from the *u*-adjective is also formally possible.[40] Finally, a "*u*-verb" is attested in CLuw. *aruwarūwa-* 'to lift, erect (?)', HLuw. *261.PUGNUS-*ru*- /aruwaruwa-/ 'to erect (?)', both reduplicated, and in Lyc. *eruwe-^{ti}*, with a highly debated meaning, but which is interestingly intransitive, thus making the Luwian reduplicated formation a causative thereof.[41] From an etymological point of view, the Lycian verb should simply mean 'to stand' (and the CLuw. verb 'to cause to stand'), although its use with the preverb *hri* 'up' in a context of funerary instructions and prohibitions leaves space for a more specialized meaning. Here again the "*u*-verb" of the *ḫi*-conjugation is morphologically unexplained so far, although the Greek formation ὀρούω 'to rise quickly, rush away' built on the same root (EDG 1107) might correspond to the *u*-verb found in Luwian and Lycian, i.e., $^*h_3$*róu̯*-/h_3*ru*-´, with the ablaut pattern of the *ḫi*-conjugation.[42] Taken together, this would suggest that the root $^*h_3er$- was reinterpreted as $^*h_3$*reu̯*- in both Greek and Luwian with the root vocalism placed between the last consonant and the -*u*-, as in the Avestan example *dəbaoman-* mentioned above. Following Koch's reasoning, this extension would have been caused by the reinterpretation of $^*h_3$*r̥-néu̯-/-nu-*´ or $^*h_3$*r̥-né-u̯-/-n-u-*´ (⟵ $^*h_3$*(é)r-u-/-éu̯-*) as an *n*-infixed $^*h_3$*r̥-n-u̯-* to a verbal root $^*h_3$*reu̯*-.

[39] The exact semantics of this adjective are not entirely clear, since it may refer to the king as a favorable attribute (KUB 35.165 obv. 23), to crops (KUB 48.69, 1; here the crops are to grow *arawa* in the manner of a man), and to a tree(?) or better to a door (KUB 35.136+ iv 4', 8', 11', 16', 19', 20'; where the CLuw. $^{\text{GIŠ}}$*ḫīluwa*, which is modified by *arūwa*, may contextually match Hitt. $^{\text{GIŠ}}$IG 'door' in line 24').

[40] The HLuw. verb must be derived from CRUS+*RA/I* 'to stand', as it is made obvious by the logogram (cf. eDiAna ID 2317 on other unpersuasive derivational analyses). The exact meaning of the causative/factitive verb is debatable in the two contexts, especially since it is used each time with a different preverb.

[41] Cf. eDiAna ID 2231.

[42] R. Garnier (pers. comm.).

As a third example, there is the root **serk-* with semantics surrounding the concept of lifting and sublimity found in Hitt. *šarkiške/a-*^{mi} 'to be sublime' (< **sr̥k-sk̂é/ó-*), an imperfective formation derived from an unattested root verb **šark-*. Hittite also has a *u*-adjective *šarku-/-aw-* 'sublime' (< **s(é)rk-u-/-éu̯-*) and a *nu*-formation thereof, i.e., *šarg(a)nu-*^{mi} 'to make sublime (?)' (< **sr̥k-né-u̯-/-n-u-´*), although it is attested in a fragmentary context.[43] Finally, Palaic shows a "*u*-verb" built on the same root, i.e., *šarku-* 'to lift', whose conjugation is however obscured by the lack of at least a 3sg. pres. attestation.[44] As opposed to the last two examples, the "*u*-verb" is transitive, whereas the root verb, relying solely on its imperfective formation, appears to have been a stative verb. It is not impossible that the Palaic verb *šarku-* was originally intransitive and secondarily acquired transitivity, but without further evidence, this claim is unverifiable. Again, it is worth asking how exactly the *u*-verb was created and if the *-u-* could have been reanalyzed as part of the root on account of the *nu*-formation, provided that it existed also in Palaic or in its prehistory. Furthermore, the *nu*-formation is, with respect to its derivational base, ambiguous, just like in other examples. Both the *u*-adjective (via infixation) and the unattested root verb (via suffixation) are morphologically fine as derivational bases, which again illustrates the constant possibility of morphological reanalysis.

More examples can be adduced, even though not all types of formations are attested, which can also be due to chance. The sole example to be discussed here starts with Hitt. *nāḫ-/naḫḫ-*^{ḫi} 'to fear' and HLuw. */nah(h)a-^{i}/ 'id.' found as the base of the imperfective *na-ha-sa-* (< **noh$_2$-(h$_1$)s-*) going back to the root verb of the *ḫi*-conjugation **nóh$_2$-/n$_ə$h$_2$-´*. Furthermore, it is attested next to the mediopassive Hitt. *naḫḫantat* 'they were an object of fear (?)'.[45] A *u*-stem occurs as the base of CLuw. *naḫḫuwašša/i-* 'frightening', which, if it were adjectival, must have been at least substantivized before the genitival adjective in **-osi̯o-* was derived (cf. CLuw. *wāšuwašša/i-* 'of the goods'). Further evidence for a nominal *u*-stem comes from HLuw. *na-hu-ti-* 'fright', which belongs to the nominal type in **-u-ti-* as in Hitt. *-uzzi-*. Unfortunately, a formation in **-nu-* is so far missing from the data, but a CLuw. "*u*-verb" *naḫḫūwa-*^{i} 'to be an object of concern' going back to

[43] Thus also Luraghi 1992: 156.

[44] See eDiAna ID 2736.

[45] Neu (1968: 120) translates it instead as 'they stood in awe', thus as showing no semantic distinction to the active root verb. However, since the context is about the capture of Kaskean men, a translation 'they were an object of fear' for the *naḫḫantat* is even more probable. However, this cannot be proven beyond doubt, since the context is fragmentary and the restoration of the preceding word [*nu-u*]*n*$^{?}$-[*n*]*a*$^{?}$-*aš*$^{!}$ '(to) us' suggested by Neu does not correspond to the readings offered by the autography KBo 50.16.

a *ḫi*-conjugated **n*$_{ə}$*h*$_{2}$*óu̯-/n*$_{ə}$*h*$_{2}$*u-´* (most likely with a restored full grade between *n* and *h*$_{2}$) is attested.[46] This "*u*-verb", just like the others, cannot be satisfactorily explained as a denominative via a suffix *-u-* or even via conversion from an adjectival *u*-stem, because the *-u-* does not act like a synchronic verbal suffix, which would have to be restricted to a specific conjugation and the same applies *a priori* for the process of conversion. The data discussed above, however, demonstrate that these "*u*-verbs" can belong either to the *mi-* or to the *ḫi*-conjugation. In his discussion of PIE *u*-presents, Jasanoff (2003: 141–143) considered the Anatolian evidence to be inconclusive in this respect, while still pleading for the *ḫi*-conjugation as the original one because of the frequent thematization of these verbs in Indo-European and of Hitt. *lāḫū-*^{ḫi} 'to pour', despite Hitt. *tarḫu-*^{mi}.[47] In a recent article, Jasanoff (2023: 67, fn. 24) withdraws this view and argues instead that *u*-presents were originally proto-middles of the **e* : zero-ablauting type, which could secondarily inflect as actives with subsequent thematization. From his perspective, the actual conjugation that is attested in a Hittite *u*-verb need not reflect the original situation of the PIE "*u*-present", which is seen here as a flawed approach. Adding more examples to the list here, e.g., CLuw. *talku-*$^{ti?}$ 'to flatten (?)' (cf. Hitt. *ištalk-*^{mi} 'to flatten'), CLuw. *malḫu-*$^{ti?}$ 'to crush' (cf. Hitt. *malla-/mall-*^{ḫi} 'to grind') and Pal. *āp(p)ū-*^{i} '?' (eDiAna ID 1566), simply confirms the reality of the mixed situation, which should be taken seriously.[48]

The synchronic use of both the *mi-* and the *ḫi*-conjugation is comparable not only to Anatolian root verbs in general, but also to the *n*-infixed verbs in Hittite, which may either take the *mi*-conjugation, e.g., *ḫarni(n)k-*^{mi} 'to make disappear', *šarni(n)k-*^{mi} 'to compensate', *ištarni(n)k-*^{mi} 'to afflict', or the *ḫi*-conjugation, e.g.,

[46] Unfortunately, it cannot be synchronically proven that the ablaut between **-h*$_{2}$*-* and **-u-* was marked by means of the recurring plene spellings with the sign <u>, i.e., *na-aḫ-ḫu-u-wa-* … (eDiAna ID 3109), since the use of this sign following a laryngeal is a strong tendency in Hittite and Luwian (see Kimball 1999: 67 and Kloekhorst 2008: 51–52). Nonetheless, it is worth noticing the same *plene* spelling in *a-ru-u-ru-u-wa-am-mi-iš* (eDiAna ID 2232), but cf. also the adjective *a-ru-u-* 'high'.

[47] One should take note that a late *nu*-formation built on the verb matching Hitt. *tarḫu-*^{mi} has now surfaced in HLuw. TONITRUS-*wa/i-nú-wa/i-tu* /tarhuwanuwa-/ (eDiAna ID 2855). Here, however, the "*u*-extension" is preserved in the base of the *nu*-formation, showing that the *-u-* is understood to be part of the root in Iron Age Luwian.

[48] For a deradical formation in **-i̯e/o-* derived from both the *u*-extended and the bare root, cf. Hitt. *šarḫiye/a-*^{mi} 'to eviscerate, maul' and Lyc. B *zrqqi-*^{ti} 'to maul' going back to **sr̥h*$_{3}$*-i̯é/ó-* and **sr̥h*$_{3}$*u̯-i̯é/ó-* respectively (cf. also CLuw. *šarḫuntalla/i-*, *šarḫuwant(i)-* vs. *šarḫadu*, *šarḫama/i-*, respectively; eDiAna ID 3046). It is also noteworthy that *ḫuišuwe/a-*^{mi} 'to live, stay alive' (next to *ḫuišu-/-aw-* 'alive, raw', *ḫuiš-*^{mi} 'to live' and *ḫuišnu-*^{mi} 'to keep alive, let live'), which was addressed earlier in this paper, may also reflect a formation in **-i̯e/o-* derived from a *u*-extended root **h*$_{2}$*u̯esu-* in the same manner as it happened to **h*$_{1}$*u̯esu-*.

šunna-/šunn-^{ḫi} 'to fill', *tarna-/tarn-*^{ḫi} 'to let (go)'.[49] Furthermore, as shown in Sasseville 2021a: 481–487, the two conjugational types can be reconstructed for Proto-Anatolian formations in **-nu-* based on the Anatolian comparative evidence.[50] Taken at face value, this situation brings further evidence to the working hypothesis that the *u*-extensions and the *n*-infixations are ultimately related through a process of reanalysis involving root verbs, *u*-adjectives and *nu*-formations. Moreover, there would be no need to attempt to find out which conjugation is the original one among "*u*-verbs", since its determination would be parallel to the one among root verbs, i.e., both conjugational types or, in other words, both types of formations can co-exist in the same root.[51]

As it could be deduced from the Anatolian data, the requirement that the *u*-adjective must first disappear in the language for the root to be extended with *-u-* in a given example cannot be maintained. At least the first two examples discussed above clearly demonstrate that Luwian can have the reanalysis of the root with *-u-* while preserving the *u*-adjective. This, however, does not falsify Koch's hypothesis on the origin of *u*-extensions, but rather indicates that the reason behind it must be sought elsewhere. Looking again at the Anatolian examples shown above, the semantics of the "*u*-verbs" seem to match the ones of the stative (or mediopassive) verbs, i.e., 1) CLuw. *waššāri* 'to be good, be dear' = HLuw. *wa/i*$_5$*-sú-ha* 'I was good, successful', 2) Hitt. *ār-tari* 'he stands' = Lyc. *eruwe-*^{ti} '(intr.)' with its causative reduplicated stem CLuw. *aruwarūwa-* 'to lift, erect (?)', HLuw. *261.PUGNUS-*ru-* /aruwaruwa-/ 'to erect (?)' and possibly also 3) Hitt. *naḫḫantat* 'they were an object of fear (?)' = *naḫḫūwa-*^{i} 'to be an object of fear/concern'. This observation implies that there is a strong affinity in these examples between the "*u*-verb" and the stative verb, which in turn has a close affinity to the *u*-adjective.[52] In the end, the missing link would be the *n*-infix/*nu*-formation.

[49] For different opinions on whether this situation is adoptable for PIE or whether it is secondary, see recently Melchert 2022: 107–108 with further literature and Villanueva Svensson 2025.

[50] On the problem regarding the conjugation of the suffix **-sk̑e/o-* in Anatolian, see Oettinger 2023: 176–177.

[51] Jasanoff (2023: 67, fn. 24) sees the mixed situation attested in Hittite as secondary, analyzing, e.g., Hitt. *lāḫū-*^{ḫi} 'to pour' as an originally h_2e-conjugated aorist instead of a *u*-present. This approach, however, cannot be synchronically verified. Another idea put forth is that the *ḫi*-conjugation of **-nu*-verbs comes from the denominal type (just like for the conversion process for stems in **-eh*$_2$*-*), whereas the deverbal type was conjugated according to the *mi*-conjugation (Sasseville 2021a: 485–486; Villanueva Svensson 2024, 2025). It is worth asking whether one could extend this pattern to root verbs derived from root nouns by conversion.

[52] On this type of relationship between *u*-adjectives and stative verbs in state-oriented verbal roots in Indo-Iranian, see Rau 2013.

4 Conclusion

In conclusion, this paper has provided support for the relationship between *u*-adjectives, *nu*-factitives/causatives and *u*-extensions by demonstrating that the phenomenon of reanalysis is responsible for the creation or extension of certain derivational processes. A difficulty encountered in our data concerns the direction of the reanalysis as well as the motivation behind it. To avoid circularity, attention was paid to the rules of word formation to limit the derivational scope. In Anatolian, it could be observed that there is a weak barrier between *nu*-factitives and *nu*-causatives. A given example can be interpreted either as an *n*-infix derived from a *u*-adjective (i.e., *nu*-factitive) or as a *nu*-suffixed formation from a root verb (*nu*-causative) and an objective choice is not always possible. Furthermore, it has become clear that the firmly established denominal *n*-infix in the Anatolian languages began to be reanalyzed as a *nu*-suffix and could thus more easily expand its derivational scope.

Building on the PIE "prototype" *d^h(é)b^h-u-/-éu̯-* 'small' ⟶ *d^h(e)b^h-n-éu̯-/-u-´* ⟶ (root reanalyzed as) *db^heu̯-*, the phenomenon of *u*-extended verbs in Anatolian was explored with a selected set of data, in order to understand the origin of the stem formation. The data presented can be understood as exemplifying the same derivational chain as our proto-type. As a consequence, the "*u*-verbs" HLuw. *wa/i$_5$-sú-ha*, Lyc. *eruwe-^{ti}* (with CLuw. *aruwarūwa-*), Pal. *šarku-* and CLuw. *naḫḫūwa-^{i}* were explained as reanalyzed *u*-extended root verbs based on the reanalysis of the derivational base of the *nu-/n*-infix formation. This Anatolian verbal stem class and its complexity is therefore effectively comparable to the Old Indic 8th class.

Abbreviations

CLuw.	Cuneiform Luwian
Gk.	Greek
Hitt.	Hittite
HLuw.	Hieroglyphic Luwian
intr.	intransitive
KBo	Keilschrifttexte aus Boghazköi
KUB	Keilschrifturkunden aus Boghazköi
Luw.	Luwian
Lyc.	Lycian
Lyd.	Lydian
OInd.	Old Indic
Pal.	Palaic
PIE	Proto-Indo-European
Ved.	Vedic

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Chapter 5

Balto-Slavic denominatives from an Indo-European perspective: An update

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This paper presents an updated and critical overview of the Balto-Slavic denominative formations from an Indo-European perspective. Some important points: 1) denominatives in *-*ō-i̯e/o-* and *-*ā-i̯e/o-* were still directly linked to *o-* and *ā-*stem nouns, respectively; 2) Balto-Slavic has no "*ā*-factitives"; 3) we have a far better understanding of the development of PIE *-*ei̯e/o*-denominatives (viz. causative-iteratives) in Balto-Slavic than just a few years ago; 4) we have new accounts of some suffixes; 5) Balto-Slavic deadjectives included not only essives and fientives, but also specialized factitives (most likely continuing PIE *nu*-factitives); 6) after the breakup of Balto-Slavic, Baltic and Slavic evolved in different directions, in part due to different areal stimuli; 7) issues of directionality in the interplay between denominatives and deverbatives are also discussed.

1 Introduction

The purpose of this article is to present an overview of Balto-Slavic denominatives from an Indo-European perspective, stressing recent findings and pending issues. It should be noted from the outset that the focus is on the Balto-Slavic system, not on PIE reconstruction or on the historical Baltic and Slavic languages (though these are, needless to say, treated as well). The article is divided into two parts: survey of Balto-Slavic denominative formations (Section 2); discussion of their relevance for Indo-European studies (Section 3).

Before proceeding further, a note on terminology. *Deverbative* designates a verb derived from a verb, not a *primary verb*. *Denominative* is used as a cover term

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for both *desubstantive* and *deadjective*. The latter is regularly so specified; the former only occasionally. I distinguish between *stative*, *inchoative* and *causative* for deverbatives and *essive*, *fientive*[1] and *factitive* for denominatives. *Iterative* is sometimes used as a cover term for several semantically differentiated deverbative subtypes. As for language labels, most of them require no comment. I use *Core Indo-European* for the parent language after the separation of Anatolian, *North-Western Indo-European* for the area of Indo-European from which Italo-Celtic, Germanic and Balto-Slavic derive, and *Northern Indo-European* for the common ancestor of Germanic and Balto-Slavic. None of these entities is uncontroversial, but a detailed treatment would be out of place here. *Balto-Slavic* unity is simply taken for granted.

The term "update" in the title should be understood in very vague terms. Denominatives are not a popular research topic and are often absent from short surveys.[2] Handbooks like Birnbaum & Schaeken (1997: 86–93) or Schmalstieg (2000: *passim*) are more informative, but are rarely up to date from an Indo-European perspective. Thus, there is no real *terminus a quo* for the update, though as a rule I do not quote comparative literature published before the classical grammars of Stang (1966) and Vaillant (1966). Finally, any "update" of this sort is per force going to be personal and mine is no exception. I have nevertheless made a conscious effort not to miss alternative views.

2 Balto-Slavic denominative formations

2.1 Introduction

I first present desubstantives (2.2-2.5), then deadjectives (2.6-2.8). Needless to say, a sharp distinction between the two is impossible. Ideally, each entry includes the following information:

1. Balto-Slavic suffix;
2. Baltic and Slavic evidence;[3]

[1]The terms "essive" and "fientive" as used here should not be confused with their use for the alleged suffixes "*-*éh₁-/-h₁-*", "*-*h₁-i̯e/o-*" in part of the Indo-European literature (e.g., LIV[2] 25).

[2]Hardly anything could be grasped from Hock (2005) or the relevant chapters of Kapović (2017) or Klein et al. (2018).

[3]I regularly give the infinitive, present and aorist/preterit. Lithuanian inflected forms are given in the 3rd person, Latvian and OCS forms in the 1st singular (or, in the case of OCS, sometimes just the stem). Quotation of Old Prussian forms naturally depends on what is actually attested.

3. comments on the Balto-Slavic prototype;

4. PIE background;

5. Balto-Slavic contribution.

For considerations of space, quotation of evidence is reduced to a minimum. The same holds true for references to the literature. In particular, I usually omit references to the standard grammars and other reference works.[4] Note that while all three Baltic languages are represented, Slavic is usually limited to Old Church Slavonic as an informal representation of Late Common Slavic. Common Slavic reconstructions are only provided when Old Church Slavonic is uninformative (most saliently, for accentology).[5] The modern Slavic languages stand outside the scope of this paper. Finally, note that for Balto-Slavic I distinguish between (e.g.) "*ī*-presents" and "*ī*-verbs". This is unnecessary for Indo-European, where suffixes like **-ei̯e/o-* were limited to the present stem.

2.2 Bl.-Sl. **-ā̲-i̯e/a-* (inf. **-ā̲-tī*, aor. **-ā̲-s-* [*vel sim.*])

This suffix is well represented in the three Baltic languages: Lith. *núom-oti*, *-oja*, *-ojo* 'rent' (← *núoma* 'rent'); Latv. *nuõm-ât*, *-ãju*, *-ãju* 'id.' (← *nuõma* 'id.'); OPr. *dwigub-ūt*, *dwigubb-ū* 'doubt' (← fem. gen. sg. *dwigubbus* 'twofold'). It regularly builds verbs from nouns, only occasionally from adjectives. The derivational dependency from *ā*-stems is still well preserved in Lithuanian (71,53% of the denominatives in *-oti* are derived from *ā*-stem nouns, according to Pakerys 2011: 281). I do not have exact numbers for Latvian, but the forms given in the grammars point in the same direction. This suffix is also productive for deverbative iteratives: Lith. *šaũkti*, *-ia* 'shout' → *šaũk-oti*, *-oja* 'id. (iter.)', Latv. *sàukt* → *saũkât*.

[4]Descriptive surveys of the East Baltic languages are not hard to find. The best ones for the present topic probably remain the academic grammars produced in the mid-20th century (MLLVG 330–343; LKG 247–268). My own presentation of the facts is mostly based on these sources (and, for Lithuanian, on Jakaitienė 1973). I regret to say that I have found more recent grammars far less informative. On the other hand, older sources like **Skardzius1943** and, especially, Endzelin (1923: 618–651) remain invaluable. For Old Prussian, I mostly rely on Endzelin (1943: 107–114). See also **Smoczynski2005** for the data. Stang (1966: 360–374) remains unsurpassed as a *global* comparative treatment. The case of Slavic is different. For comparative work, only the very securely reconstructed Common Slavic matters, which can often be informally represented with attested Old Church Slavonic. Vaillant (1966) not only remains unsurpassed; all later general treatments I have consulted are less informative. As a result, my presentation of the Slavic facts closely follows that of Vaillant.

[5]Acuteness is marked as *Ē̲, with underline, and non-acute as simple *Ē. Accentology is otherwise not treated in this survey.

According to Jakaitienė (1973: 34), in Lithuanian, deverbatives are twice as common as the denominatives, though usually in extended forms like *-čioti*, *-soti*, *-n(i)oti*, *-l(i)oti* or, above all, *-ioti* (*bė́gti*, *-a* 'run' → *bėgióti*, *-iója* 'id. (iter.)'; see 2.5, 3.5).

Slavic essentially concords with Baltic, with the important difference that *-ati*, *-ajǫ* is immensely productive for derived imperfectives (OCS *čisti*, *čьtǫ* 'read' → *po-čitati*, *-čitajǫ* 'id. (impf.)'). As a result, denominatives are very well represented, but are not productive anymore in most Slavic languages: OCS *igra* 'play' → *igr-ati*, *-ajǫ* 'play', *dělo* 'work' → *děl-ati*, *-ajǫ* 'do'. I do not have numbers indicating whether the connection with *ā*-nouns is as clear as in Baltic. Deadjectives occur, but are very rare.

For Balto-Slavic we can safely reconstruct a denominative suffix *-*ā̠-tī*, *-*ā̠-i̯e/a-* that was still derivationally attached to *ā*-stem nouns. It is impossible to say how productive derivation from other stems was. It was probably not used for deadjectives, but deverbative iteratives were very productive.

The origin of this type is perfectly clear: It goes back to PIE *i̯e/o*-denominatives from *ah₂*-stem nouns. Whether *-*ah₂-i̯e/o*-denominatives expanded beyond their original core within PIE is a matter I will leave open.[6] In any case, it is interesting to note that the original connection with the *ā*-stems is still apparent in Lithuanian. Another observation to make is that, although iteratives in *-*ah₂-i̯e/o-* are found in several languages and may even go back to the parent language, they are nowhere as productive as in Germanic and Balto-Slavic.

Before moving on, a note on the *ā-presents*. As is well known, Baltic is one of the few languages that has an *-*ā*-present distinct from *-*ā-i̯e/o-* (Lith. *laĩko* 'holds' vs. *galvója* 'thinks'). The other branches in which this class is attested are Anatolian (Hitt. *šuppiyaḫḫ-^{hhi}* 'purify'; cf. Jasanoff 2003: 140–141) and Greek (cf. Rau 2009b). This type has recently been put in a new light by Sasseville's (2015) demonstration that in Luvian *-*ah₂*-verbs built denominatives from *-*ah₂*-nouns through conversion. It follows that the now standard assumption of a class of "*ah₂*-factitives of the type *renouāre* derived from *o*-stems" (*vel sim.*) is in need of revision. Here I will limit myself to stress that, against what is often assumed, *ā*-presents are exclusively deverbative in Balto-Slavic; cf. Villanueva Svensson (2020: 392–395).

2.3 Bl.-Sl. *-*ō̠-i̯e/a-* (inf. *-*ō̠-tī*, aor. *-*ō̠-s-* [*vel sim.*])

This suffix is only unambiguously attested in East Baltic, where it is very productive: Lith. *mel-úoti*, *-úoja*, *-ãvo* 'lie' (← *mẽlas* 'lie'), Latv. *mel-uôt*, *-uõju*, *-uõju*

[6] See Sasseville (2021: 126–127) with references for recent discussion.

'id.' (← *mẹ̄li* [pl.t.]). In Lithuanian, verbs in *-úoti*, *-úoja* are normally made from *o*-stem nouns (70,95% of the denominatives, according to Pakerys 2011: 286). I do not have numbers for Latvian, but the impression one gets from the grammars is that the connection with *o*-stem nouns is not as clear as in Lithuanian. Deadjectives are rarer, but not exceptional: Lith. *báltas* 'white' → *balt-úoti*, *-úoja* 'show white', Latv. *skaĩdrs* 'clear' → *skaĩdr-uôt*, *-uõju* 'make clear', refl. 'become clear'. Their meaning is broadly the same as that of the deadjectives in *-*ē̱ti* (fientive or essive, occasionally also factitive). Deverbatives are limited to Lithuanian and clearly secondary (*al̃pti*, *-sta* 'faint' → *alp-úoti*, *-úoja* 'id. (iter.)').

Due to the merger of Bl.-Sl. **ō* and **ā* in West Baltic and Slavic, it is impossible to know whether these branches had verbs in *-*ō̱-tī*, *-*ō̱-i̯e/a-* beside the familiar *-*ā̱-tī*, *-*ā̱-i̯e/a-*. If *-*ō̱-tī*, *-*ō̱-i̯e/a-* is projected into Balto-Slavic, it certainly built denominatives from *o*-stem nouns. It is impossible to know whether it also built desubstantives from other stems and deadjectives.

The only potential *comparandum* of Bl.-Sl. *-*ō̱-tī*, *-*ō̱-i̯e/a-* is the Greek type δουλόω 'enslave' (← δοῦλος 'slave').[7] There is, however, no *communis opinio* concerning the connection between the two formations (if any). During the last decades several scholars have proposed reconstructing for PIE a denominative suffix *-*oh_1-i̯e/o-* (Peters 1999; Malzahn 2010: 401–402) or *-*o-i̯e/o-* (Oettinger 1979: 357–358; Kloekhorst 2008: 132–133). The evidence supporting such a view, however, is either inherently problematic (e.g., Gaulish *marcosior* 'I will ride/I will be ridden') or is better explained in a different way (the Hittite *ḫatrā(i)*-type, for instance, is usually derived from *-*ah_2-i̯e/o-*; see Sasseville 2021: 116–123, with references). The dominant position has always been that Lith. *-úoti*, *-úoja* and Gk. -όω are parallel, but independent innovations; see especially Tucker (1981: 15–22) for the latter. Within this approach Lith. *-úoti*, *-úoja* is usually derived from possessive adjectives in "*-*ō-to-*" of the type Gk. χολωτός 'angry', Lith. *ragúotas* 'horned' (e.g., Endzelin 1923: 627–628; Stang 1966: 364) and this remains the best option today.

I add a few observations on Balto-Slavic:

- it is impossible to know the chronology of the innovation and whether *-*ō̱-tī*, *-*ō̱-i̯e/a-* is really Balto-Slavic in date;
- the now generally accepted derivation of Gk. χολ-ωτός, Lat. *aegr-ōtus* etc. from 'deinstrumental' *-*oh_1-to-* is open to criticism, cf. Fortson (2020: 82–90). For Balto-Slavic, (post-)PIE *-*ō-to-* would work as well;

[7]This statement obviously implies that the short stem vowel of Gk. δουλ-ό-ω (and of τῑμάω 'honor' or ἀνθέω 'be in bloom') is an inner-Greek development (contrast aor. ἐδούλωσα, ἐτῑ́μησα, ἤνθησα), as is the *communis opinio*. The issue cannot be discussed at length here.

- while some Baltic denominatives are susceptible of a deinstrumental gloss ‘be with X’ (e.g., *dūmúoti* ‘smoke; get covered with smoke’), this would be forced for most of them;
- an alternative view considers Lith. *-úoti*, *-úoja* an East Baltic offshoot of the Balto-Slavic denominative type Lith. *-áuti*, *-áuja* (2.4), e.g., Vaillant (1966: 352–353), Kortlandt (1995). See Villanueva Svensson (2014, 2015) for criticism of different aspects of this approach.

2.4 Bl.-Sl. *-au̯-i̯e/a- (inf. *-au̯-tī, aor. *-au̯-ā- [!])

This suffix is productive in Lithuanian and Old Prussian: Lith. *draug-áuti*, *-áuja*, *-ãvo* ‘be friends’ (← *draũgas* ‘friend’), OPr. *dīnk-aut*, pres. 1pl. *dīnk-au(i)mai*, pret. *dīnk-auts* (*dink-owats* I) ‘thank’ (← Pol. *dziękować*). In Lithuanian it derives verbs from nouns of all stems with a basic meaning ‘be X’ (whence ‘work as X’, ‘behave as X’, ‘be engaged in X’, ‘show up as X’, etc.). Deadjectives are rarer, but not exceptional (*šykštùs* ‘stingy’ → *šýkštauti*, *-auja* ‘be stingy’). Deverbative iteratives also occur, but are generally regarded as secondary (*šaũkti*, *-ia* ‘shout’ → *šū́kauti*, *-auja* ‘whoop’). This type has been lost in Latvian, merging with the verbs in *-uôt*, *-uõju* (2.3). The types *-*au̯-tī*, *-*au̯-i̯e/a-* and *-*ō-tī*, *-*ō-i̯e/a-* often overlap in Lithuanian as well.

In Slavic, this type is very productive for both denominatives and deadjectives: OCS *dar-ovati*, *-ujǫ*, *-ovaxъ* ‘make a present’ (← *darъ* ‘present’), *mil-ovati*, *-ujǫ* ‘have mercy’ (← *milъ* ‘pitiful; loved’). In addition, it is abundantly used for borrowings (OCS *kup-ovati*, *-ujǫ* ‘buy’, cf. German *kaufen*). Its use for derived imperfectives can be tracked in the recorded history of Old Church Slavonic, cf. Schuyt (1990: 29–33).

For Balto-Slavic we can confidently reconstruct a denominative suffix *-*au̯-tī*, *-*au̯-i̯e/a-*. It is uncertain whether it was also widely used for deadjectives. From a morphological point of view, it is the only denominative suffix with an *ā*-aorist *-*au̯-ā-*, as borne out by the second stem in *-a-* of Slavic *-ov-a-ti*, *-ov-a-xъ*. This is potentially important from a historical perspective, as the Balto-Slavic *ā*-aorist was otherwise limited to primary verbs, cf. Villanueva Svensson (2020: 382–389). To judge by Slavic (the Baltic *ā*-preterit is fully ambiguous) all other denominative suffixes had a sigmatic aorist in Balto-Slavic (*-*ā-s-*, *-*ē-s-* etc.).

The origin of Bl.-Sl. *-*au̯-tī*, *-*au̯-i̯e/a-* is a traditional *locus desperatus* of Balto-Slavic historical grammar, the only certain thing being that it must be an innovation of this branch. I refer to Villanueva Svensson (2014) for a full treatment, including criticism of the widespread idea that it originated in denominatives from *u*-stem nouns. In this article, I have proposed deriving this suffix from original compounds with the root **h₂eu̯h₁-* 'help' (Ved. *ávati* 'id.' etc.). Parallels for such a reanalysis include Lat. *-ĭ̄gāre*, *-cināre* and, especially, OIr. *-(a)igithir*.

2.5 Bl.-Sl. *-ī-* (inf. *-ī̠-tī*, aor. *-ī̠-s-* [*vel sim.*])

As a result of some recent insights this suffix is now better understood than just a decade ago. I will therefore treat it in more detail and, more importantly, in a different way than the other denominative formations.

The origin of Slavic *i*-verbs like OCS *prositi*, *prosi-* 'ask for' is perfectly clear: they continue PIE *o*-grade causatives and iteratives in **-éi̯e/o-* (PIE **mon-éi̯e/o-* 'remind' > Ved. *mānáyati*, Lat. *moneō*, *-ēre*) and denominatives of *o*-stem nominals in **-e-i̯é/ó-* (Ved. *devá-* 'god' → *devayáti* 'worship the gods'). The PIE functions are preserved nearly intact in Slavic, where *i*-verbs build iteratives (OCS *nesti*, *nesǫ* 'carry' → *nositi*, *nošǫ* 'id. (iter.)'), causatives (OCS *tešti*, *tekǫ* 'run, flow' → *točiti*, *točǫ* 'let run, let flow'), and denominatives from both nouns and adjectives: OCS *měr-iti*, *-jǫ* 'measure' (← *měra* 'measure'), *běl-iti*, *-jǫ* 'whiten' (← *bělъ* 'white'). *i*-denominatives are immensely productive. Deadjectives often (but not exclusively) display factitive value. From an accentological point of view, this is the only Slavic denominative suffix with an old contrast of tone between present (with non-acute *-*ī*-, CSl. 2pl. **nòsite* < **nosîte* < Bl.-Sl. **náśīte*) and aorist-infinitive (with acute *-*ī̠*-, CSl. inf. **nosíti* < Bl.-Sl. **náśī̠tī*).

From a historical perspective, the Slavic *i*-verbs pose two major problems:

1. Can Sl. *-i-* regularly derive from PIE **-ei̯e/o-*? If not, how is Sl. *-i-* to be explained?
2. Is the acute *-*í*- of the aorist-infinitive stem the same as the non-acute *-*î*- of the present stem? If not, how is Sl. *-*í*- to be explained? Note that Baltic shows aor.-inf. *-*í*- to be old with certainty (Lith. inf. *prašýti* = Sl. **prosíti*).

The second question is rarely posited in this way and, accordingly, has received relatively little attention in the literature.[8] As for the first one, it is probably safe to say that the matter has always been seen with a certain degree of perplexity. Accounts not operating with a regular development PIE **-ei̯e-* > Sl. *-i-* have usually taken one of the following four options:

[8] See Klingenschmitt (1978: 3), Jasanoff (2017: 207–208) for scenarios not deriving Bl.-Sl. aor.-inf. *-*ī̠*- from PIE **-ei̯e-*.

1. to start from a PIE semithematic or athematic "ĭ̆-present";[9]
2. to assume that the -*i*- of *prositi*, *prosi*- was taken from that of the stative type *mъněti*, *mъni*- and/or from *i*-stem denominatives (PIE "*-*ĭ̆-i̯e/o*-");[10]
3. to postulate complex analogical scenarios;[11]
4. to operate with irregular phonological changes.[12]

Viewed against this background, the assumption of a regular development PIE *-*ei̯e*- > Sl. -*i*- becomes, in my view, even more attractive (*all* alternatives are unsatisfactory). The idea is far from new,[13] but has certainly gained momentum in recent years; see especially Hock (1995), Hill (2016: 214–222), Villanueva Svensson (2023b: 74–76, 191, 194–199). A detailed treatment is impossible here, but I would like to stress two insights gained from these works:

1. The amount of positive evidence supporting a *regular* development PIE *-*ei̯e*- > Bl.-Sl. *-*ī*- > Sl. -*i*- has increased (e.g., Sl. inf. -*ti*, Lith. -*ti* < PIE *-*tei̯ei̯*);
2. The development PIE *-*ei̯e*- > Bl.-Sl. *-*ī*- was Balto-Slavic in date, as most probably the analogical replacement of *-*ei̯o*- > *-*ii̯a*- with *-*ī*-.[14] In the aorist-infinitive stem *-*ī*- was secondarily acuted to *-*ī̱*- on the model of all verbs with a dissyllabic second stem (*-*ā̱*-, *-*ē̱*- etc. < *-*ah*$_2$-, *-*eh*$_1$-).

Table 1 gives the development of the PIE *-*ei̯e/o*-presents in Balto-Slavic.[15]

[9]"Semithematic *i*-presents" were popular in older says of our discipline, see especially Stang (1942: 23–29). See Schrijver (2003) for a modern defense of the PIE "athematic *i*-presents" and Weiss (2012: 152–155) for a different explanation of the Italic and Celtic facts adduced by Schrijver.

[10]Both statements are frequent in Slavic handbooks, e.g., Vaillant (1966: 439–440), Birnbaum & Schaeken (1997: 91). As often noted (e.g., Jasanoff 1978: 102–103), if one type imposed its morphology on the other(s), it is likely that this was the immensely productive type *prositi*, *prosi*-.

[11]E.g., **Kurylowicz1973**, Klingenschmitt (1978: 3–5).

[12]E.g., **Matasovic2008**, Andersen (2014: 60–69: Late Common Slavic contraction across *jod*.).

[13]Cf. Specht (1934: 61–62; 78–79), with references to J. Schmidt, Bezzenberger, Sommer and Pedersen.

[14]Against a relatively widespread assumption, the regular reflex of PIE *-*ei̯e/o*- in Baltic was not (Lith.) -*ia* or -*ija* (*pace* Kortlandt 1989: 107, Ostrowski 2006: 28, or Jasanoff 2017: 207, among others). See below in the text.

[15]Adapted from Villanueva Svensson (2023b: 198). Arrows indicate analogical forms. The endings, which are sometimes insecure, cannot be discussed here. Column 'Bl. (1)' is deliberately left unaccented as a way to indicate that no direct connection is postulated between PIE stress position (**proḱ-éi̯e-ti*) and that of late Balto-Slavic (**pråśīt(i)*).

Table 1: PIE *-*ei̯e/o*-presents in Balto-Slavic

	PIE	Bl.-Sl. (1)	Bl.-Sl. (2)	Sl.	Bl. (1)	Bl. (2)
1sg.	*prok̂éi̯oh₂	*praśii̯̱ō̱	→ *pȧ́śi̯̱ō̱	*prošǫ́	*praši̯̱ō̱	*praši̯̱ō̱
2sg.	*prok̂éi̯esi	*praśīsi	*prȧ́śīš(i)	*prosîši	?	?
3sg.	*prok̂éi̯eti	*praśīti	*prȧ́śīt(i)	*prosîtь	*praši	*praši
1pl.	*prok̂éi̯omos	*praśii̯amas	→ *prȧ́śīmas	*prosîmъ	*prašīme	→ *prašime
2pl.	*prok̂éi̯ete	*praśīte	*prȧ́śīte	*prosîte	*prašīte	→ *prašite
3pl.	*prok̂éi̯onti	*praśii̯anti	→ *prȧ́śīnt(i)	*prosę̂tь		
Inf.		*praśītī	→ *prȧ́śī̱tī	*prosi̋ti	*prašī̱ti	*prašī̱ti
Aor.		*praśī-s-	→ *prȧ́śī̱-š-	*prosi̋xъ	→ *prašii̯ā	*prašē

The early Baltic development is the product of a series of important developments:

- generalization of the apocopated PIE 3sg. *-*ti* > Bl.-Sl. *-*t* > *-Ø in non-athematic paradigms, yielding **praš-ī(-t)*;
- shortening of unstressed non-acute word-final *-*ī*(C), yielding **praš-i* under 'Bl. (1)' in Table 1 (Hill 2016: 214–222, Villanueva Svensson 2023b: 74–76);
- loss of number distinctions in the 3rd person, with concomitant restructuring of verbal paradigms as an endingless 3rd person to which 1st and 2nd singular, plural and dual endings are attached (e.g., Lith. pres. 3sg. *vẽda* : 1pl. *vẽda-me*, 2pl. *vẽda-te*). This allows us to expect the analogical reshuffling given under 'Bl. (2)' in Table 1.

The most surprising feature of the development of the *ī*-verbs in Baltic, however, is that the expected paradigm does not exist. Instead, we find up to five (!) analogical or mixed types:

1. Iterative and causative type Lith. *laik-ýti*, *laĩk-o*, *laĩk-ė*, Latv. *laîc-ît*, *laîk-u*, *laîc-ĩju*; OPr. *laik-ūt*,[16] *lāik-u* 'hold', with *ā*-present (2.2).
2. Type OLith. *tar-ýti*, *tãr-ia*, *tãr-ė* 'say', attested in relics in Old and dialectal Lithuanian (Stang 1966: 327–328).

[16] Old Prussian has secondarily generalized *-*ā*- to the infinitive stem.

3. Iteratives with 'extra *-i-*' like Lith. *lándž-ioti*, *-ioja*, *-iojo* 'creep, crawl about', Latv. *luõžât*, *-ãju* 'id.', analyzable as **-i-* (pres. **praš-i*) + **-ā̠ti*, **-ā̠i̯a*.

4. Causative and factitive type Lith. *dẽg-inti*, *-ina*, *-ino*, Latv. *dedz-inât*, *-ina*, *-inãja* 'burn (tr.)', OPr. *mukint*, pres. *mukinna* 'teach', analyzable as **-i-* (pres. **praš-i*) + **-na-* (see below 2.8).[17]

5. Denominative type Lith. *dal-ýti*, *-ìja*, *-ìjo* 'divide', Latv. *medît*, *-ĩju* 'hunt', OPr. *madlīt*, *madli*, 1pl. *madlimai* 'pray'.[18]

To disentangle the extraordinarily complex development of the Balto-Slavic *ī*-verbs in Baltic is a major task for the future. It is no less important to stress that the Balto-Slavic *ī*-verbs are better understood today than just a decade ago, to a large degree thanks to recent advances in historical phonology. As for their position in the Balto-Slavic denominative system, probably the main question is whether they were productively used to build deadjectival factitives in addition to plain (transitive) denominatives; see below (2.8).

2.6 Bl.-Sl. **-ē̠-i̯e/a-* (inf. **-ē̠-tī*, aor. **-ē̠-s-* [*vel sim.*])

This suffix is productive in all three Baltic languages, e.g., Lith. *stor-ė́ti*, *-ė́ja*, *-ė́jo* 'grow fat' (← *stóras* 'fat'); Latv. *mel̃n-êt*, *-ẽju*, *-ẽju* 'become black' (← *mel̦̃ns* 'black'); OPr. *druw-īt*, *druw-ē*, 1pl. *druw-ēmai* 'believe' (← *druwis* 'faith'). It builds fientives from adjectives and, less often, nouns. Essives occur, but only as relics: Lith. *garsė́ti*, *-ė́ja* 'be famed' (← *garsùs* 'famous'), Latv. *klusêt*, *-ẽju* 'be silent' (← *kluss* 'silent'); cf. Pakerys (2006), Ostrowski (2006: 109–115). The decay of the essive value is probably related to the decay of nasal fientives (2.7) and the expansion of essives in *-áuti* (2.4) and *-úoti* (2.3). Fientive value may have begun with preverbs, as in Slavic.[19] From a morphological point of view, it is well known that Baltic is one of the few branches distinguishing between *ē*-denominatives (with *-ėja*-presents) and *ē*-deverbatives (with presents in *-ti*, *-i*, *-a* and, rarely, *-ia*). There is, however, a certain degree of contamination between the different

[17] Cf. Villanueva Svensson (2025). See Gorbachov (2007: 175–176) for an alternative account that, however, also derives Bl. **-in-* from the Balto-Slavic *ī*-verbs. The idea is not new, cf. **Otrebski1965**, Vaillant (1966: 227–228), Schmalstieg (2000: 173). See Kortlandt (1989), Hajnal (1996) for different views.

[18] Cf. Villanueva Svensson (2023a), with arguments against the widespread derivation from *i*-stem denominatives (e.g., Stang 1966: 367, Ostrowski 2006: 128, Pakerys 2011: 279–280, among others).

[19] Similarly Ostrowski (2006: 115).

present types. In particular, original denominatives with stative value tend to acquire deverbative morphology (e.g., Lith. *mylė́ti*, *mýli*, Latv. *mīlêt*, *-u* 'love', cf. Lith. dial. *mýlas*, Sl. **mílъ* 'loved').

Slavic essentially agrees with Baltic. The suffix *-*ě̋ti*, *-ě̋jǫ* is productive and derives fientives and, more rarely, essives from adjectives and, less often, nouns: *cěl-ěti*, *-ějǫ* 'recover' (← *cělъ* 'healthy'), *um-ěti*, *-ějǫ* 'be able' (← *umъ* 'mind').[20] According to Vaillant (1966: 369) the fientive value started in preverbed perfectives: OCS *bogatěti* 'be rich' : *raz-bogatěti* 'become rich' (← *bogatъ* 'rich'; cf. factitive *bogatiti* 'make rich'). It clearly predominates already in Old Church Slavonic.

For Balto-Slavic, we can safely reconstruct a deadjective suffix *-*ē̱-tī*, *-*ē̱-i̯e/a-*. It is hard to say whether it also built desubstantives. I assume that it regularly built essives, even though this is mostly based on internal reconstruction. Fientive value is far better attested in historical Baltic and Slavic, but is easily explained as parallel innovations.

An important topic that can only be tangentially mentioned here is the role of the second stem *-*ē̱-* in deverbatives. This is of course not exclusive of Balto-Slavic, but this branch, especially Baltic, stands out by the richness of deverbative formations:

1. With stative *i*-presents: Lith. *budė́ti*, *bùdi*, OCS *bъděti*, *bъdi-* 'be awake'. This is the best-known type.[21]

2. With athematic present: OLith. *gélbėti*, *-mi* 'save', OCS *věděti*, *vědě*/*věmь* 'know'. This type is only well represented in Old Lithuanian.

3. With thematic present: Lith. *tekė́ti*, *tẽka* 'run, flow'. Only in Baltic, but with potential relics in Slavic.[22]

4. With *ā*-present: OCS *iměti*, *imamь* 'have' (only certain example).[23]

5. Several derived verbal classes in East Baltic:
    a) duratives with acute metatony (*stéipėti*, *-ėja* 'lie dying' ← *stìpti* 'die (of animals)');

[20] Vaillant (1966: 366–373).

[21] See Jasanoff (2004) and, more recently, Yakubovich (2014), Grestenberger (2019) on the origin of the Balto-Slavic *i*-presents. See Kortlandt (1989: 109–110), Harðarson (1998), Hill (2016: 216–219) for alternative views.

[22] Cf. **Koelln1977**, Villanueva Svensson (2017: 470–473). The Lithuanian evidence is collected in Jakulis (2004).

[23] Old Prussian verbs like *billītwei*, *billē* ~ *billā* 'speak', with chaotic variation between present *-ā* and *-ē*, could belong here as well (cf. Kaukienė 2004: 187–204).

b) ‘intensives’, also with acute metatony (*svérdėti*, *-i* ‘stagger’ ← *svìrti* ‘bend down’);
c) diminutives with non-acute lengthened grade (*pa-ėjė́ti*, *-ẽji*/*-ėjė́ja* ‘go a little’ ← *eĩti* ‘go’);
d) causatives (only in Latvian, *aûdzêt*, *-ẽju* ‘cultivate, grow (tr.)’ ← *aûgt* ‘grow (intr.)’);
e) several subtypes with ‘extended’ suffix like *-alėti*, *-elėti*, *-erėti*; *-tel(ė)ti*, *-ter(ė)ti*; *-inėti*; *-sėti*.[24]

The precise prehistory of some of these formations is still unclear, just as their interplay with the *ē*-denominatives. As for the latter, Balto-Slavic deadjectives in **-ē̯i̯e/a-* are generally regarded as an archaism (PIE **-eh$_1$-i̯e/o-*). It is important to note that reflexes of **-eh$_1$(-i̯e/o)-* are exclusively denominative in Anatolian and Indo-Iranian (cf. Yakubovich 2014). This strongly suggests that the presence of **-eh$_1$(-i̯e/o)-* among deverbatives is post-Indo-European. With all due caution, I suggest viewing the traditional deverbative *ē*-statives as an innovation of the North-Western language area (the path taken by the Greek η-aorist was completely different). The idea, needless to say, requires more work.

2.7 Bl.-Sl. type ****gru-n-b-e/a-*** (inf. ****grub-tī***, aor. ****grub-e/a-***)

As is well known, nasal presents with anticausative and inchoative value constitute one of the defining features of Balto-Slavic and Germanic, e.g., Lith. *lìpti*, *limpa*, *lìpo* ‘stick to’, OCS *pri-lь(p)nǫti*, *-lь(p)nǫ*, *-lьpъ* ‘id.’, Go. *af-lifnan*, *-lifniþ*, *-lifnoda* ‘be left over’. See Gorbachov (2007) for the reconstruction of the Northern Indo-European prototype **b^{h}u-n-d^h-é-ti* ‘awakes’ (aor. **b^{h}ud^h-é-t*) and Villanueva Svensson (2011) for my views on their origin. The same formation is productive for deadjectival fientives in all three branches, more rarely for desubstantives: Go. *fulls* ‘full’ → *(ga-)fullnan* ‘become filled’, Lith. *šlùbas* ‘lame’ → *šlùbti*, *šlumba* ‘become lame’, OCS *slěpъ* ‘blind’ → *o-slьpnǫti* ‘go blind’.

Fientives are very well represented in East Baltic: Lith. *šlàpti*, *šlampa*, *šlãpo* ‘become wet’ (← *šlãpias* ‘wet’), Latv. *slapt*, *slùopu*, *slapu* ‘id.’ (← *slapjš*).[25] The type, however, is not productive anymore, its place being taken by *-ėti*, *-ėja* (2.6). The absence of denominatives in Old Prussian is surely due to chance.

[24] See Ostrowski (2006: 11–32), **Serzant2014**, Rothstein-Dowden (2022: 100–114) for different issues related to this material.
[25] The evidence is collected in **Pakalniskiene2000**.

Slavic essentially agrees with Baltic: OCS *slěpъ* 'blind' → *o-slьpnǫti* 'go blind'. Denominatives are less well represented than in Baltic, no doubt because the inchoative type *bъ(d)nǫti* itself ceased to be productive and because verbs in *-ěti*, *-ějǫ* became productive for fientives. An obvious archaism is that zero grade is dominant in the older period: OCS *gluxъ* 'deaf' → *o-glъxnǫti* 'become deaf', *xromъ* 'lame' → *o-xrъmnǫti* 'grow lame' (cf. Vaillant 1966: 241–242). This feature recurs in Germanic (Gmc. **starka-* 'strong' → Go. *ga-staurknan* 'become rigid', ON *storkna* 'coagulate') and can be safely projected back into Northern Indo-European. Baltic has some relics as well: Lith. *bjùrti*, *bjū̃ra*/*bjùrsta* 'become ugly' (← *bjaurùs* 'ugly').[26]

For Balto-Slavic, we can confidently reconstruct a class of deadjectival fientives with zero grade of the root. The type was identical with the inchoative *n*-presents and must have involved, among other features, a thematic aorist. From a historical point of view, it is clear that the Northern Indo-European fientives are an offshoot of the deverbative anticausative-inchoatives, obviously via secondary association of some deverbatives to adjectives from the same root.

2.8 Bl.-Sl. ***plit-ne-ti** (inf. ***plit-neu̯-tī**, aor. ***plit-neu̯-s-** [*vel sim.*])

The last denominative formation is a recent addition that is likely to remain controversial, at least for some years. I refer to Villanueva Svensson (2025) for full argumentation of the ideas presented here.

Balto-Slavic clearly had an elaborated system of deadjectives including morphologically differentiated essives (2.6) and fientives (2.7). As for the factitives, probably most scholars would project into Balto-Slavic the Slavic *i*-factitive type OCS *cělъ* 'healthy' → *cěliti* 'heal' (2.5). This is clearly an extension of transitive desubstantives, no doubt supported by the causative value of PIE *-*ei̯e/o-*. The process may go back to Balto-Slavic, but may equally well be exclusively Slavic.[27] From a functional point of view, the Slavic *i*-verbs have a clear pendant in the Baltic verbs in *-*i̲n̲ti*, *-*ina*, which are productive for factitives (Lith. *saldùs* 'sweet' → *sáldinti* 'sweeten') and causatives (Lith. *mir̃ti* 'die' → *marìnti* 'kill'). As noted above (2.5), the *-*i*- of the present *-*ina* can now be identified with the 3sg. *-*i* of the Balto-Slavic *-*ī*-verbs in Baltic. The obvious analysis *-*i*+*na* that this leads to leaves us with a *-*na* that has no obvious source within Baltic (the subsequent reanalysis of *-*i-na* as *-*in-a* is essentially unproblematic).

[26] Cf. Villanueva Svensson (2016).

[27] The same development took place in Germanic (Go. *hails* 'healthy' → *hailjan* 'heal') and Sanskrit (*taruṇa-* 'young' → *taruṇaya-* 'make young').

Let us now turn to PIE. In the 1970s, Koch (1973), Nussbaum (*apud* Tucker 1981: 26) and Tucker (1981) argued that PIE possessed a class of *nu*-factitives derived from *u*-stem adjectives. The type is faithfully continued in the Hittite factitives and causatives in -*nu*-^{mi} (Hitt. *daššu*- 'strong' → *dašš(a)nu*-^{mi} 'strengthen'), in the Greek denominatives in -ῡ́νω (βαθύς 'deep' → βαθῡ́νω 'deepen') and, probably, -αίνω (κῡδρός 'glorious' → κῡδαίνω 'glorify'),[28] and, finally, in scattered relics elsewhere in the family (e.g., Ved. *dabhnóti* 'deceive', cf. Hitt. *tepnu*-^{mi} 'diminish; despise' ← *tēpu*- 'small'). The idea has clearly gained momentum in recent years.[29] In addition, recent research has refined some of the initial assumptions:

- PIE *nu*-factitives were part of the Caland system (Rau 2009a: 112–120, Oettinger 2018);
- new evidence from Luvian suggests that the *nu*-factitives followed the *h₂e*-conjugation, whereas deverbatives followed the *mi*-conjugation (Sasseville 2021: 485–486). I explicitly highlight that Greek factitives in -ῡ́νω and -αίνω continue PIE *nu*-factitives, as this implies that this type was preserved as a distinct class in Core Indo-European.

In Balto-Slavic (as in other branches), a h_2e-conjugation suffix is expected to be thematized at an early date, yielding *-*nu̯e/o*- or *-*neu̯e/o*-. *Qua* secondary verb formation *nu*-factitives lacked an aorist. In Balto-Slavic, this slot is expected to be filled with the present stem minus *-*(i̯)e/o*-, yielding aor.-inf. *-*nu*- or *-*neu̯*-. The major claim of Villanueva Svensson (2025) is that the Slavic type pres. *ri-ne*-, inf. *ri-nǫ-ti*, aor. *ri-nǫ-xъ*, past pass. ptcp. *ri-nov-enъ* 'cast, push' goes back entirely to the Balto-Slavic descendant of the PIE *nu*-factitives. The aorist-infinitive stem extended full-grade *-*neu̯*-. In the present, zero-grade *-*nu̯e/o*- was simplified to *-*ne/o*- by regular sound change. *nu*-factitives were lost in the prehistory of Slavic, being replaced by the *ī*-verbs (2.5). When the Slavic aspect system arose, remaining *nu*-factitives were pressed into new service as secondary perfectives. In Baltic, on the other hand, *nu*-factitives were contaminated with *ī*-verbs (*-*i*+*na*), finally leading to the productive factitive and causative type *-*i̲n̲ti*, *-*ina*.

If this is correct, it implies that Balto-Slavic had a distinct formation for deadjective factitives that directly continues the PIE *nu*-factitives.

[28] The connection of Gk. -ῡ́νω and Hitt. -*nu*-^{mi} was an important aspect of Koch's (1973) argument and has been upheld several times. See Villanueva Svensson (2024), building on Tucker (1981: 22–29), for the idea that Gk. -αίνω continues PIE *nu*-factitives as well.

[29] Cf. Steer (2014: 206–210), Oettinger (2018), Sasseville (2021: 484–486 and this volume), Villanueva Svensson (2024, 2025).

3 Balto-Slavic denominatives: some general issues

3.1 Introduction

The dossier of Balto-Slavic denominatives included the suffixes *-*ā̠-i̯e/a*-, *-*ō̠-i̯e/a*-, *-*a̠u̯̠-i̯e/a*-, *-*ī*- (< PIE *-*ei̯e/o*-) and a clearly delimited deadjectival system involving essives (**grau̯b-ē̠-tī*, **grau̯b-ē̠-i̯e/a*- 'be coarse'), fientives (**grub-tī*, **gru-m-be/a*- 'become coarse') and, probably, factitives (**grau̯b-na̠u̯̠-tī*, **grau̯b-ne/a*- 'make coarse'). The factitive is the only (potentially) controversial slot (other authors would reconstruct **grau̯b-ī̠-tī*, **grau̯b-ī*-). Needless to say, the boundaries between deadjectives and denominatives are unlikely to have been very strict.

This survey does not answer all questions related to the reconstruction of the Balto-Slavic denominative system. Thus, the distribution of the different denominative suffixes is not fully clear (apart from the connection of *-*ā̠-i̯e/a*- and *-*ō̠-i̯e/a*- to *ā*- and *o*-stems, respectively), and the same holds true for their semantics. Problems of this sort are part and parcel of the reconstruction of denominatives, but further research could bring in some clarity (one must note, at this point, that study of Baltic and Slavic denominatives is almost never carried out within a proper Balto-Slavic perspective). Other issues constitute global problems of the Balto-Slavic verb. Thus, although the formation of the Balto-Slavic second stem is generally clear, the way it came into being is not well understood. The caveat with which I accompany aorists like *-*ā̠-s*- ('*vel sim.*'), for instance, indicates that our reconstruction of the Balto-Slavic aorist relies almost exclusively on Slavic and is, accordingly, less certain than other forms. Again, future research may bring in some clarity on this and related issues. In what follows I address some general issues of particular interest for Indo-European studies.

3.2 PIE background 1: The contribution of Balto-Slavic

From a historical perspective, the Balto-Slavic denominative suffix inventory is, as expected, a mix of preservation, loss, and innovation. Clearly innovated are the fientive nasal presents and the denominative suffixes *-*a̠u̯̠-i̯e/a*- and, probably, *-*ō̠-i̯e/a*-. These innovations took place at different stages and are not without interest from a broader perspective. If my account of *-*a̠u̯̠-i̯e/a*- is correct (2.4), it shows that the creation of denominative suffixes out of second compound members is more widespread than one would otherwise expect. From a PIE perspective *-*ō̠-i̯e/a*- is the most interesting item. Following the *communis opinio*, in Section 2.3 I have assumed that it is a Balto-Slavic innovation. The issue, however, certainly deserves more work. In a more pessimistic vein, the contrast with Gk. -όω illustrates a recurrent problem with the Balto-Slavic evidence. Careful attention to the inflectional, derivational and semantic profile of the -όω-denominatives in oldest Greek allowed Tucker (1981: 15–22) to clearly distinguish two different subtypes with, perhaps, two different prehistories. This type of philological work is rarely possible for Balto-Slavic.

As for preservation, the suffixes *-*a̱-i̯e/a*-, *-*e̱-i̯e/a*-, and *-*ī*- (PIE *-*ah₂-i̯e/o*-, *-*eh₁-i̯e/o*-, *-*ei̯e/o*-) are well-known and little needs to be added here. Far more interesting are the *nu*-presents (2.8) and the *ā*-presents (2.2), if correctly interpreted. The matter can be briefly formulated as follows. The Vedic denominative system is extraordinarily simple (nominal stem + -*ya*-).[30] The Hittite system is more complex and includes no less than four more suffixes in addition to -*iya*- (and -*āi*- < *-*ah₂-i̯e/o*-): -*aḫḫ*-, -*e*-, -*ešš*-, and -*nu*- (Hoffner & Melchert 2008: 175–179). The minor Anatolian languages essentially confirm this picture, with some added complications (cf. Sasseville 2021: 547–552). Most Core Indo-European branches offer little or no unambiguous evidence supporting the antiquity of these formations (except for Hitt. -*e*- < PIE *-*eh₁-i̯e/o*-, as generally acknowledged since Watkins 1971). The major exceptions appear to be Greek and, precisely, Balto-Slavic. Put otherwise, the interest of this branch for the reconstruction of PIE denominatives can hardly be overstated.

Finally, a note on dialectology. Little needs to be said about inherited formations (*-*a̱-i̯e/a*-, *-*e̱-i̯e/a*-, *-*ī*-, '*nu*-verbs'). Deverbatives in *-*eh₁(-i̯e/o)*-, as noted above (2.6), display a highly specific profile in the North-Western area. It remains to be seen whether the concept of 'North-Western Indo-European', traditionally limited to studies of vocabulary, may be extended to morphology as well. A clear Northern Indo-European innovation is the anticausative-inchoative type *b^h*u-n-*d^h*-é-ti*, including its extension to deadjective fientives. In this area deverbative iteratives in *-*ah₂-i̯e/o*- display a productivity unmatched elsewhere in the family. The issue clearly deserves more work. Probable or certain Balto-Slavic innovations include the suffixes *-*a̱u̯(i̯e/a)*- and *-*o̱(i̯e/a)*- and the specialization of the *ā*-presents (< post-PIE *-*ah₂-e/o*-) as deverbatives.

3.3 PIE background 2: Formations lost in Balto-Slavic

In my view, no more (or less) denominative formations than those surveyed in 2 need to be reconstructed for Balto-Slavic. Slavic and, especially, Baltic have a

[30] This is of course an oversimplification. See Yakubovich (2014), with references, for a more nuanced discussion of the Vedic facts.

wealth of extended suffixes like Lith. *-dė́ti*, *-dýti*, Sl. *-kati* etc.; see below in Section 3.5. They usually look secondary and, as far as I can see, there is no compelling reason to project them back into Balto-Slavic.

At least one verb appears to entail conversion: Sl. **pȇ́lva* 'chaff' → Sl. **pȇ́lti* (< **pȇ́lv-ti*; Koch 1990: 647–648), **pêlvǫ* 'weed' (OCS *plěti*, *plěvǫ*). A conversion analysis is, in my view, a distinct (and hitherto unexplored) possibility for other etymologically isolated primary verbs, e.g., OCS *prěti*, *perǫ* 'fly' (cf. Sl. **peró* 'feather'), *zobati*, *zobljǫ* 'peck' (cf. Sl. **zȍbъ*, *-ь* 'crop'). At any rate, conversion was purely sporadic in Balto-Slavic.

From a PIE perspective, the most interesting absence is that of some *i̯e/o*-subtypes. The suffix **-i̯e/o-* is of course ubiquitous in Balto-Slavic denominatives, but always as part of longer suffixes (**-ah₂-i̯e/o-* etc.). What is absent are *i̯e/o*-denominatives from consonant, **-i-*, and **-u*-stems (types Ved. *ápas-* 'work' → *apas-yáti* 'works', *jáni-* 'woman, wife' → *janī-yáti* 'desires a wife', *śatru-* 'enemy' → *śatrū-yáti* 'is hostile'). This is not by itself noteworthy (denominative suffixes may easily be lost), but each "missing type" calls for some comments.

Let us begin with consonant stems. Baltic does have denominative *ia*-presents, e.g., *juõkas* 'laughter, joke' → *juõktis*, *-iasi* 'laugh', *šventas* 'holy' → *šv̨ęsti*, *šveñčia* 'celebrate'. These, however, almost certainly are regularizations of the obsolete type *tarýti*, *-ia* and continue, accordingly, PIE **-ei̯e/o-* (2.5). Note, incidentally, that this implies that Baltic has no denominatives of the type Gk. ἄγγελος 'messenger' → ἀγγέλλω 'bear a message' (< **°el-i̯e/o-*), with deletion of the thematic vowel. Slavic has a large class of "expressive" primary verbs such as OCS *glagolati*, *glagolje-* 'speak'. Some of them could be interpreted as denominatives (cf. *glagolъ* 'speech'), but most certainly not, cf. Vaillant (1966: 328–346). Note, finally, that Balto-Slavic has no historically complex suffixes like Gk. -ιζω (< **-id-i̯e/o-*) or Toch. **-ññä/æ-* (< **-n-i̯e/o-*).

The lack of a special suffix for *i*-stem denominatives is fully expected. The development PIE **-ei̯e/o-* > Bl.-Sl. **-ī-* must have started with an early raising **-ei̯e/o-* > *-ii̯e/o-*, which entailed merger with *i*-stem denominatives in **-i-i̯e/o-* (see above Table 1). This is confirmed by Slavic, where *i*-verbs build denominatives from *-o-* and *-i*-stems alike. The Baltic type *dalýti*, *-ìja* does not go back to *i*-stem denominatives (cf. Villanueva Svensson 2023a).

The *u*-stems are more interesting. We do not only lack reflexes of **-u-i̯e/o-*; the stem vowel *-u-* is absent in denominatives: Lith. *saldùs* 'sweet' (OCS *sladъkъ*) → *sald-ė́ti* 'be, become sweet', *sáld-inti* 'sweeten'. In recent years the matter has received two different interpretations. Ostrowski (2006: 117–118) suggests that we are dealing with Caland system relics; see also Majer (2021: 261–262, and *passim*) for discussion. Hill (2012: 6–13), on the other hand, has argued for a Northern

Indo-European rule of *u*-deletion before *jod*. At this point it should be noted that *-*u*- is also absent from the feminine of *u*-stem adjectives: Lith. *saldì*, *-džios* (contrast Ved. *ur-ú-ḥ* : *ur-v-ī́*, Gk. γλυκύς : γλυκεῖα < *-eu̯-i̯a*). As a third option I would suggest analogy with *o*-stem adjectives, where the thematic vowel is (synchronically) deleted: Lith. *plónas* 'thin' → *plon-éti* 'become thin', *plón-inti* 'make thin'. The issue clearly deserves more work.

3.4 Balto-Slavic denominatives from a typological perspective

In this section I will briefly survey what Balto-Slavic tells us about the way denominatives arise and develop. The inference of what we have seen seems to be that (almost) anything goes. Denominatives may be inherited (e.g., *-*ei̯e/o*-, *-*eh*$_1$-*i̯e/o*-), created anew (*-*au̯-i̯e/a*-), or transformed in different ways (e.g., Lith. *-inti*, *-ina* and *-ýti*, *-ìja*). The Balto-Slavic picture is perhaps more interesting from the viewpoint of semantics and, especially, the interplay of denominatives and deverbatives.

I am not aware of detailed studies on denominative semantics from a historical perspective. PIE *-*ei̯e/o*- (Bl.-Sl. *-*ī*-) was often (though not exclusively) transitive, but the semantics of *-*ā-i̯e/a*-, *-*ō-i̯e/a*-, and *-*au̯-i̯e/a*- would definitely benefit from more work. Deadjectives in *-*ē-i̯e/a*- developed from essives to fientives in both Baltic and Slavic, but the interplay between both values is well known. As for the interplay between denominatives and deverbatives, it has been rampant. This may be a consequence, in part, of the architecture of the Balto-Slavic verb, which can be seen as a network of interrelated verbs, each of them filling the slot of a given Aktionsart. See Table 2.

It is also important to note that Balto-Slavic, unlike other branches, has preserved the autonomy of the root up to very recently (in Baltic to some degree up to this day). At any rate, it is evident that such a system provided abundant grounds for mutual contamination between denominatives and deverbatives. Interestingly, not all theoretical configurations took place with the same intensity:

1. "Denominative → secondary verbal formation (iterative, causative, etc.)" is the most common development. It happened in the case of PIE *-*ah*$_2$-*i̯e/o*-, *-*eh*$_1$-*i̯e/o*-, Bl.-Sl. *-*ō-i̯e/a*-, *-*au̯-i̯e/a*-, and Bl. *-*inti*, *-*ina*, if at very different dates. If my account is correct, in the case of the Balto-Slavic *ā*-presents and the Slavic type *rinǫti*, *rine*- an original denominative type survives as deverbative alone. Another generalization is that once deverbatives are established in the system, they easily evolve towards more complicated morphology, whereas denominatives tend to keep simpler morphology.

Table 2: Lithuanian verb system

	I Unmarked (transitive)	II Stative- durative	III Inchoative- anticausative	IV Causative	V Iterative (etc.)
	a-, *ia-* presents	2nd stem *-*ē*-, *-*ā*-	-*n*-, *sta*- presents	-*ýti*, -*ìnti*	-*ýti*, -*óti*, *inė́ti*, etc.
strew (**bʰer*-)	*bérti, -ia*	*byrė́ti, bỹra*	*bìrti, bỹra* *(< *biñra)*	*bìrinti*	*barstýti,* *birénti,* *birsnóti*
lift (**kelh₃*-)	*kélti, kẽlia*	*pa-kylė́ti*	*kìlti, kỹla* *(< *kiñla)*	*kéldinti,* *kìldyti,* *kìldinti*	*kilnóti,* *kilóti,* *kilsúoti,* *kilstelė́ti*
peck (**les*-)	*lèsti, lẽsa*			*lèsinti*	*lesióti*
sleep (**mei̯gʰ*-)		*miegóti, -a*	*už-mìgti,* *-miñga*	*migdýti*	

2. "Denominative → (unmarked) primary verb" certainly occurs in isolated verbs. What seems to be genuinely rare is the creation of primary verbal classes out of denominatives. Potential cases (primary *ē*-statives, *ā*-aorist) almost certainly passed through a stage involving secondary verbal derivation.

3. "Secondary verbal formation → denominative" appears to be very rare. The only certain case is the Northern Indo-European fientive type **gʰru-m-bʰ-e/o*-, which probably responded to a highly specific situation. Slavic *i*-factitives are easily explained as an extension of transitive desubstantives, with deverbal causatives playing at best a supporting role. If my account is correct causatives like Lith. *dẽg-inti*, *-ina* are a secondary extension of factitive deadjectives.

4. Finally, "(unmarked) primary verb → denominative" is unattested. It is noteworthy that secondary deverbal formations appear to go back to denominatives much more frequently than to reanalysis of primary verb morphology.

If these observations are accurate, they are potentially interesting for work on other branches, especially in the case of formations that, in theory, could be of both denominative and deverbative origin (e.g., factitive-causatives). It is evident, however, that they should be tested against a more extensive and clearly delimited dataset before they can be regarded as general tendencies of language change.

3.5 After Balto-Slavic

The Balto-Slavic verbal system sketched above (3.4) provided abundant grounds for the creation of new complex suffixes, semantic specialization, etc. Also, of course, for contamination with denominatives. Here we are only interested in the development of the system after the breakup of Balto-Slavic, with some notes on possible areal phenomena:

1. The system was not only preserved, but actually reinforced in Baltic (note especially, but not only the development of the *ī*-verbs). In addition, Baltic is rich in semantically and formally specialized minor subtypes ("intensives", "diminutives", etc.). It is interesting to note that on this score Baltic contrasts with the major Indo-European branches of Europe. From an areal perspective, it is tempting to attribute the divergent development of Baltic, in part, to Uralic influence (as suggested by Pakerys 2018 for the causative).

2. The system was eliminated in Slavic. Types like the *nǫ*-inchoatives, the *ě*-statives, or the *i*-causatives are still abundantly represented, but are not productive anymore. The *a*-iteratives, of course, became immensely productive, but as part of a new system of grammatical aspect. From a typological point of view late Slavic came to be aligned with the modern Germanic and Romance languages in disfavoring secondary verbal derivation altogether. The Slavic development may indeed have been scattered by contacts with these languages.

If these observations are correct, the recent development of Baltic and Slavic awaits further work. The matter, in addition, has an obvious interest for the areal dimension of language change. Be it as it may, the divergent tendencies towards complication in Baltic and simplification in Slavic are clearly seen in the denominative system as well.

To conclude, I hope to have shown that Baltic and Slavic denominatives are not only interesting for their own sake; they have a genuine contribution to make to the reconstruction of the PIE system.

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Abbreviations

Aor.	Aorist
Bl.	Baltic
Bl.-Sl.	Balto-Slavic
CSl.	Common Slavic
Fem.	Feminine
Gen.	Genitive
Gk.	Greek
Gmc.	Germanic
Go.	Gothic
Hitt.	Hittite
Impf.	Imperfective
Inf.	Infinitive
Intr.	Intransitive
Iter.	Iterative
Lat.	Latin
Latv.	Latvian
Lith.	Lithuanian
OCS	Old Church Slavonic
OIr.	Old Irish
OLith.	Old Lithuanian
ON	Old Norse
OPr.	Old Prussian
Pass.	Passive
PIE	Proto-Indo-European
Pl.	Plural
Pl.t.	Pluralia tantum
Pol.	Polish
Pret.	Preterite
Ptcp.	Participle
Refl.	Reflexive
Sg.	Singular
Sl.	Slavic
Toch.	Tocharian
Tr.	Transitive
Ved.	Vedic

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Part II

Derivational patterns and comparative reconstruction

Chapter 6

Delocatival adjectives surfacing as participles

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This paper proposes to reassess several active participles of the type **CC-en-i̯ónt-* and **CC-ént-* as decasuatives built on a “petrified” locative or oblique **CC-én*. At its face value, Ved. *bhuraṇ-yánt-* ‘quivering, active’ (of Agni) is the participle of the intransitive present stem *bhuraṇ-yá-*^{ti} ‘to be active or restless’, alongside its deverbative doublet *bhuraṇ-yú-* [adj.] ‘quivering, active’. In fact, the indicative *bhuraṇ-yá-*^{ti} is likely to have been backformed after *bhuraṇ-yánt-*, which may be accounted for as a decasuative adjective (reflecting PIE **b*^{h}*r̥h*$_{2}$*-en-i̯ó-nt-*), derived from a locative **b*^{h}*r̥h*$_{2}$*-én* ‘in quick motion’. The suffix **-nt-* was added as a characterizing morpheme to a former **-i̯ó*-stem, with the same pattern as PIE **ĝerh*$_{2}$*-ó-nt-es* ‘the old ones’ (vs. simplex **ĝerh*$_{2}$*-ó-*) or **g*$^{u̯}$*ih*$_{3}$*-u̯ó-nt-es* ‘the living ones’ (vs. simplex **g*$^{u̯}$*ih*$_{3}$*-u̯ó-*). Several other cases will be dealt with, involving PIE 1. **b*^{h}*eh*$_{2}$*-* ‘to shine’, PIE 2. **b*^{h}*eh*$_{2}$*-* ‘to speak, to say’, and PIE **g*$^{u̯}$*eh*$_{2}$*-* ‘to make a step’.

1 Vedic *-anyá*-presents: state of the art

In Vedic, there is a small class of *-anyá*-presents such as *tur-aṇ-yá-*^{ti} ‘to rush’, *bhur-aṇ-yá-*^{ti} ‘to be active or restless’ (transitive ‘to agitate, to stir [a liquid]’), *riṣ-aṇ-yá-*^{ti} ‘to suffer injury’, *kr̥p-an-yá-*^{ti} ‘to desire’, *pr̥t-an-yá-*^{ti} ‘to fight’. YAv. *zar-an-iia-* ‘to be irritated’ reflects the Iranian counterpart of this verbal suffix. One may note that most of these verbs are intransitive (Ved. *bhur-aṇ-yá-*^{ti} only being labile). Jasanoff (2003: 122) suggests identifying this verbal derivation with Hittite duratives in *-anna/i-*, by assuming that the geminates reflect PIE **-nh*$_{2}$*-i-*. To trigger the assimilation **-nH-* > *-nn-*, the suffix requires an initial vowel, reconstructed as PIE **-énh*$_{2}$*-i-* by Melchert (1992: 40–43) or **-ótn-i-* by Kloekhorst

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(2008: 175). According to Jasanoff (2003: 125), Ved. -*anyá*-presents cannot be denominatives to nominal formations in -*an*- (or -*ana*-) because the underlying nasal stems are never attested – a claim which needs to be reassessed. Besides – as he duly points out – thematic nouns in -*ana*- would rather produce denominatives in **-*anā̆-yá-^{ti}*. Jasanoff instead compares Ved. -*anyá*-presents to the type of Ved. *gr̥bh-ā-yá-^{ti}* 'to seize' (< PIE *-*n̥h$_2$-i̯é/ó*-), by assuming an unrecognized Indo-Iranian sound change triggering the shortening of a long vocalic sonorant when following another vocalic sonorant (Jasanoff 2003: 126). He claims that Ved. *dam-an-yá-^{ti}* 'to subdue' would be the regular reflex of PIE **dm̥h$_2$-n̥h$_2$-i̯é/ó*-, whereas the doublet *dam-ā-yá-^{ti}* 'id.' would be analogically remade after the pattern of *gr̥bh-ā-yá-^{ti}* 'to seize'. This seems to me an *ad hoc* rule, which is not supported by any independent lexeme. Furthermore, this implies that forms such as Ved. *riṣ-aṇ-yá-^{ti}* 'to suffer injury', *kr̥p-an-yá-^{ti}* 'to desire', or *pr̥t-an-yá-^{ti}* 'to fight' would be analogical. This phonological account should be rejected, since Jasanoff does not explain why Vedic -*anyá*-presents (always inflected in the active) are often intransitive or labile. By contrast, -*nā́-^{ti}* or -*ā-yá-^{ti}* presents are exclusively transitive (for instance Ved. *iṣ-ṇā́-^{ti}* 'to send'). The alleged lack of evidence for -*an*-stems does not withstand scrutiny: alongside *iṣ-aṇ-yá-^{ti}* 'to cause to make haste, excite, drive', there is *iṣ-áṇi* [loc.] (RV 2.2.9d) 'at [her] impulsion'. Likewise, alongside *tur-aṇ-yá-^{ti}* 'to be quick or swift', there is *tur-áṇa*- [adj.] 'swift' and *bhur-aṇa*- [adj.] 'quick, active' (= stressed **bhur-áṇa*-), akin to *bhur-aṇ-yá-^{ti}* 'to be active or restless' and *bhur-aṇ-yú*- [adj.] 'quivering, active'.

Independently, the comparison of Ved. -*anyá*-presents with Hitt. -*anni*- must now be considered outdated, since this Anatolian iterative suffix cannot reflect PIE *-*n̥h$_2$-i*- (or *-*énh$_2$-i*-), but rather PA *-*ā́d-n-i*-, as *per* Sasseville (2021: 526–528). In my opinion, one has to start with an archaic endingless locative such as Ved. **iṣ-án* 'with drive, impetus', **bhur-án* 'in quick motion', Av. **zar-an* 'with anger'. The lability of the derived verbs vindicates their nominal origin (viz. *bhur-aṇ-yá-^{ti}* 'to be/to set in quick motion'), since **iṣ-án* 'with drive, impetus' could be either agent- or patient-oriented. As a result, several "active" participles of the type **CC-en-i̯ónt*- could in fact be reanalyzed as old decasuative adjectives built on a "petrified" locative or oblique stem **CC-én*. Another strategy of coining a delocative adjective was to add a thematic vowel: PIE *-*én-o*- (cf. Ved. *tur-áṇ-a*- [adj.] 'swift' and **bhur-áṇ-a*- [adj.] 'quick, active' < PIE **bʰr̥h$_2$-én-o*-). I would tentatively account for Ved. *dam-an-yá-^{ti}* 'to subdue' by reconstructing a PIE **démh$_2$-mn̥* [nt.] 'taming', which was prone to undergo the so-called "*ašnō*-rule", viz. the reduction of PIE *-*mn*- to *-*n*- after obstruent, cf. Nussbaum (2010) and Melchert (2010: 186–188). The genitive singular **démh$_2$-(m)n-os* would thus surface as Ved. **dám-n-as* after which an endingless locative **dám-an* 'in subduing' was formed. This very form may have been the base of the denominative verb _dam-an-yá-^{ti}_ ‘to subdue’ (with denominal causativity, viz. *‘to put into subduing, to cause sbdy. or sth. to be in the state of being subdued’). One may reasonably speculate that _dam-an-yá-^{ti}_ ‘to subdue’ is basically an indicative and was built within Vedic, by contrast with _bhur-aṇ-yá-^{ti}_, which is likely to have been inherited, since there is comparative evidence for PIE *_b^{h}r̥h_2-én_ [loc.] ‘in quick motion’ (see Section 2).

As already mentioned, Ved. _bhur-aṇ-yú-_ [adj.] ‘quivering, active’ is the functional equivalent of the participle _bhur-aṇ-yánt-_ ‘id.’, like _tur-aṇ-yú-_ [adj.] ‘rushing’ vs. _tur-aṇ-yánt-_ ‘id.’. One has perhaps to assume that the intransitive or labile _-anyá-_presents were backformed after their participles _-an-yánt-_, which were used as adjectives (‘rushing, quivering’). The starting point may have been an endingless locative *_bhur-án_ ‘in quick motion’ (< PIE *_b^{h}r̥h_2-én_) from which was derived (i) a decasuative _-yánt-_participle, and (ii) a decasuative _-yú-_adjective. I consider that in such cases the _-yánt-_participle is to be segmented as *_-yá-nt-_, with the well-known “characterizing” morpheme *_-nt-_[1] added to a decasuative *_-yá-_adjective (< PIE *_-i̯ó-_). To sum up, one has to start with PIE *_b^{h}r̥h_2-en-i̯ó-_ [adj.] ‘quivering’,[2] a decasuative stem which was recharacterized as PIE *_b^{h}r̥h_2-en-i̯ó-nt-_ ‘the swift one’ (Ved. _bhur-aṇ-yánt-_ ‘quivering, active’ [of Agni]). Independently, there was also PIE *_b^{h}r̥h_2-en-i̯ú-_ [adj.] (Ved. _bhur-aṇ-yú-_) and also PIE *_b^{h}r̥h_2-én-o-_ [adj.] (Ved. *_bhuráṇa-_).

The derivational history of Ved. _bhur-aṇ-yá-^{ti}_ proposed above is not totally unparalleled: There are good Anatolian parallels. Hitt. *_išḫan-iya-_ ‘to bind’ reconstructed by Rieken (1999: 286–287) based on Hitt. _išḫanittar_ [c.] ‘relative by marriage’ (< PA *_sH-an-y-é-tros_) requires a base *_išḫan_ (< PIE *_sh_2-én_ [loc./obl.] ‘with bond’), which she further connects with Hitt. _šaḫḫan-_ [nt.] ‘(feudal) obligation’.[3] Interestingly, “Luwoid” derivatives in _-att-allā-_ (with _-allā-_ from PA *_-élā-_)

[1] See for instance PIE *_ĝerh_2-ó-nt-es_ ‘the elders, the old ones’ (vs. simplex *_ĝerh_2-ó-_ [adj.] ‘wrinkly, decayed’) or PIE *_g^{u}ih_3-u̯ó-nt-es_ ‘the living ones, the living world’ (vs. simplex *_g^{u}ih_3-u̯ó-_ [adj.] ‘quick, alive’). The indicative *_g^{u}ih_3-u̯-é/ó-_ ‘to live’ (Ved. _jī́va-^{ti}_, Lat. _vīvō_) is likely to have been backformed after its (pseudo-)participle stem. That the *_-ont-_participles were formerly independent adjectival stems is assumed by Melchert (2017). For the suffix, one may compare CLuw. _wayant(i)-_ [c.] ‘animal’ from PIE *_g^{u}oi̯h_3-ó-nt-es_ ‘the living ones; creatures’ which is a recharacterization of PIE *_g^{u}oi̯h_3-ó-_ [adj.] ‘living’. PIE *_g^{u}ih_3-u̯ó-nt-es_ is the _Scharnierform_ yielding a present *_g^{u}ih_3-u̯-é/ó-_, which can hardly be accounted for by conversion *_g^{u}ih_3-u̯ó-_ ⟶ *_g^{u}íh_3-u̯-e/o-_ or by deletion (viz. PIE *_g^{u}íh_3-u̯(e/o-)e/o-_), _pace_ Villanueva Svensson (2021: 273–274).

[2] The same pattern is seen in PIE *_ĝ^{h}l̥h_3-en-i̯ó-_ [color adj.] ‘yellow’ ⟶ Ved. _híraṇya-_ [nt.] ‘gold’ (lit. ‘being in the state of shining’).

[3] One should note that, from now on, due to the existence of the so-called “_ašnō_-rule”, Hitt. _šaḫḫan-_ shall no longer be directly assigned to PIE †_séh$_2$-n̥_ ‘obligation’ _pace_ Rieken (1999: 287) and Kloekhorst (2008: 691), but to PIE *_séh$_2$-mn̥_ [nt.] with nasal-deletion after the gen.sg. *_séh$_2$-(m)n-os._

and *-alla/i-* (< PA **-élo-* [with the so-called "Luwian *i*-mutation"]) are built directly on the stem of the base: Hitt. *išḫan-attallā-* [c.] (?) and *išḫan-alla/i-* [c.] (?), both of unclear meaning. As I have recently suggested (Garnier 2022b: 175), Hitt. *kuššan-iye/a-*^{zzi} 'to hire, to employ', which is akin to Proto-Gmc. **χūz-jan-* 'to hire' (Kloekhorst 2008: 498; cf. OE *hȳr* [str.f.] 'wages, interest, usury' < **χūz-ṓ-*), may be accounted for by a base **kuššan* [loc.] 'into [temporary] possession', reflecting PIE **k̂uh*$_1$*-s-én* which was a doublet of PIE **k̂uh*$_1$*-és(i)* [loc.], possibly associated to PIE **k̂éu̯h*$_1$*-ōs* [f.] 'possession' (of holokinetic inflection), or perhaps PIE **k̂éu̯h*$_1$*-s-* [nt.] 'id.' (of proterokinetic inflection). The noun *kuššan-* [nt.] 'pay, salary, fee, hire' is likely to have been backformed after *kuššan-iye/a-*^{zzi} 'to hire, to employ'. For the meaning, one may compare Gk. κῡριεία [f.] 'possession' vs. κῡριος [m.] 'lord, ruler' (< PIE **k̂uh*$_1$*-r-i-i̯ó-* [adj.] 'endowed with strength', cf. PIE **k̂uh*$_1$*-r-í-* [f.] 'strength' derived from PIE **k̂uh*$_1$*-r-ó-* [adj.] 'strong'). Interestingly, the "locative-1" (PA **k̂uʔ-és*) is indirectly reflected by Lyd. *qašaas* [c.] 'hire; rental' (< PA **k̂uʔ-es-yā́-*) and Lyc. *qehñni-te* 'he rented' (< Pre-Lyc. **qes-Vni-*^{ti}), with unlenited endings pointing to PA *-*yé/ó-* (Sasseville 2021: 166–167).

2 A case study: PIE ****b***^{h}***r̥h***$_2$***-én*** 'in quick motion'

2.1 PIE ****b***^{h}***r̥h***$_2$***-én*** 'in quick motion'

I have reconstructed a PIE etymon **b*^{h}*r̥h*$_2$*-én* [loc.] 'in quick motion' (Garnier 2022a: 9), which would surface as Ved. **bhur-án* (for which see above). There is additional evidence for assuming such an "adverbial" stem: (i) PIE **b*^{h}*r̥h*$_2$*-ḗn* [HK f.] (< PIE *-*én-s*) 'vehemence' (Section 2.2); (ii) PIE **b*^{h}*r̥h*$_2$*-én-o-* [adj.] 'being in quick motion' (Section 2.3); (iii) PIE **b*^{h}*r̥h*$_2$*-én-t-* [adj.] 'being in quick motion' (Section 2.4); (iv) PIE **b*^{h}*r̥h*$_2$*-en-i̯ó-* [adj.] 'quivering, active' → "characterized" PIE **b*^{h}*r̥h*$_2$*-en-i̯ó-nt-* 'the swift one' and PIE **b*^{h}*r̥h*$_2$*-en-i̯ú-* [decasuative adj.] 'quivering, active' (Section 2.5). PIE *-*én*-locatives from root nouns are assumed by Neri (2017: 326–327).

2.2 PIE ****b***^{h}***r̥h***$_2$***-ḗn*** [HK f.] 'rush, fury, vehemence'

OIr. *barae* [f.] 'fury, anger' reflects Proto-Celt. **bar-ēs* (with *-*ē* < *-*en-s*), acc.sg. **bar-en-an*,[4] which in turn reflects a remodeling of an old hysterokinetic stem:

[4]Within Celtic a zero-graded secondary derivative **bar-n-iko-* was coined, cf. OIr. *barnech*/*bairnech* [adj.] 'irritated'.

PIE *b^h*r̥h*$_2$*-ḗn* [f.] 'vehemence' (with *-*ḗn* < PIE *-*én-s*).[5] This inherited form (such nasal stems are not productive in Celtic) has been assigned by Stüber *apud* Balles (2002: 1–2) to PIE *b^h*erh*$_x$- 'mit einem scharfen Werkzeug arbeiten' (LIV^2 64), exemplified by Lat. *feriō* 'to strike' and *forāre* 'to bore through, to pierce'. Balles (2002: 7) further compares OIr. *barae* with Hom. φρήν* [f.] 'midriff, diaphragm' (in fact a *plurale tantum* φρένες), which has almost nothing to do with 'anger'.[6] The PIE root *b^h*erh*$_2$- 'sich schnell bewegen' (LIV^2 81)[7] is perhaps a better candidate for a noun 'fury, anger'.[8] As I have previously suggested (Garnier 2021: 53, fn. 6), the PIE root *b^h*erh*$_2$-/*b^h*reh*$_2$- 'to hurry, to rush' (with *Schwebeablaut*) could be explained as a reanalyzed nominal stem *b^h*ér-h*$_2$-/*b^h*r-éh*$_2$- 'hurry, haste, fury' [of proterokinetic inflection], built on the PIE primary root *b^h*er-* (cf. Gk. φέρομαι [mid.] 'to move rapidly').

2.3 PIE *b^hr̥h$_2$-én-o- [adj.] 'being in quick motion'

Ved. **bhur-áṇ-a-* [adj.] 'quick, active', similar to Ved. *tur-áṇa-* [adj.] 'swift' for its formation, may represent something inherited, if one compares W. *bar-an* [m.] 'fury, anger' (< Proto-Celt. **barano-*), which shows a trivial conversion from *'angry' to 'anger' (perhaps via an old neuter stem *baranon*). For the semantic discrepancy between 'active' and 'angry', one may mention Ved. *bhū́r-ṇi-* [adj.] 'restless, active, angry' (< PIE *b^h*r̥h*$_2$*-n-í-* [f.] 'vehemence').

2.4 PIE *b^hr̥h$_2$-én-t- [adj.] 'being in quick motion'

From PIE *b^h*r̥h*$_2$*-én* [loc.] could be derived an adjective *b^h*r̥h*$_2$*-én-t-* 'being in quick motion', with the "characterizing" or "individualizing" morpheme *-*t*-, a delocative combination producing *-*én-t-* which could eventually be reanalyzed

[5] There was no loss of the final nasal after a long vowel due to the presence of the accent, as *per* Harðarson (2005: 218–219). For the internal derivation yielding a hysterokinetic stem from a PIE *-*én*-locative, see Neri (2017: 324, fn. 181).

[6] I have explained Hom. ἄφρων [adj.] 'senseless, foolish, heedless' as the reflex of a PIE etymon *n̥-g^{uh}*r(h$_1$)-on-* 'without sense of smell, not able of scenting' (Garnier 2021), eventually reanalyzed as a privative *bahuvrīhi* (viz. 'lacking [sound] φρένες'), which implies that Hom. φρένες would be a backformation, coined after other names of body parts showing the suffix -ήν. In Homer, the φρένες (this organ is considered double) are rather the seat of the intelligence, memory, and perception. When someone is angry, his φρένες turn black, because they are *filled* with anger, as in A 103, μένεος δὲ μέγα φρένες ἀμφὶ μέλαιναι # πίμπλαντ(ο) "and with rage was his black heart wholly filled" (transl. Loeb).

[7] cf. Ved. *bhar^i*- 'to move rapidly, to rush', Hitt. *parḫ-* 'to chase, pursue, to hunt', and the newly identified HLuw. */par.xa-/ 'to expel' (Melchert 2016: 204–206).

[8] Ved. *bhū́r-ṇi-* [adj.] 'restless, active, angry'.

as a participle *-*ént*- to the zero grade-stem of a root present.[9] This very form is reflected by YAv. *barəṇt-* [ptcp.] 'stürmend' (= Ved. **bhur-ánt-* 'quivering' replaced by the thematic mid. ptcp. *bhur-á-māṇa-*) and Hitt. *parḫ-ant-* [ptcp.] (< PA **barH-ánt-*) meaning both: (i) '*being in quick motion, galloping' [intransitive]; (ii) 'set in quick motion' [transitive passive]. The intransitive meaning (i) is secure in: *pár-ḫa-an-d*[*a-an* [acc.sg.c.] 'galloping' (KUB 35.145 rev. 13), by contrast with (ii) *arḫa pár-ḫa-an-za* [nom.sg.c.] 'expelled/banished' (KUB 8.1 ii 7–8). The scalar process of grammaticalization leading to a "pure" transitive passive participle ('expelled') developed by increasing the verb's valency by turning it into a causative (note the use of *arḫa* [adv.] 'off, away' here with the lexical value of a preverb). As a result, one may speculate that the transitive present *parḫ-zi* 'to chase' is in fact backformed after its passive participle *parḫ-ant-** 'chased'.

2.5 PIE **b^{h}r̥h$_2$-en-i̯ó-* [adj.] 'being in quick motion' and related forms

As already mentioned in Section 1, one has to posit a delocatival adjective **b^{h}r̥h$_2$-en-i̯ó-* [adj.] 'being in quick motion', indirectly reflected by its "characterized" doublet **b^{h}r̥h$_2$-en-i̯ó-nt-* [adj.] 'the swift one'. This very form was eventually reinterpreted as a present "active" participle (Ved. *bhur-aṇ-yánt-* 'quivering, active' [always of Agni]) of an intransitive verb *bhur-aṇ-yá-^{ti}* 'to be active or restless'. Independently, there was also a renewed adjective (Ved. *bhur-aṇ-yú-* [adj.] 'quivering, active' < PIE **b^{h}r̥h$_2$-en-i̯ú-*). The main bulk of the attestations consists of intransitive forms: by contrast with the labile transitive present *bhur-aṇ-yá-^{ti}* 'to agitate, stir (a liquid)' [RV 2x], there is *bhur-aṇ-yá-^{ti}* 'to be active or restless' [RV 3x] (surely backformed after *bhuraṇ-yánt-*), *bhur-aṇ-yánt-* 'quivering, active' [RV 2x], and *bhur-aṇ-yú-* [adj.] 'quivering, active' [RV 5x]. In other words, 10 out of 12 attestations are intransitive.

3 PIE 1.**b^{h}eh$_2$-* 'to shine'

One may also reconstruct PIE **b^{h}h$_2$-én* [loc.] 'into light; showing; appearing', which was an adverb, perhaps compositional only (viz. PIE **°b^{h}h$_2$-én*). This adverbial locative was presumably built on a root noun ([virtual] PIE **b^{h}óh$_2$-* [f.] 'light'), which is well-attested in composition, as is clear from Ved. *vi-bhā́-* [adj.] 'shining' (< PIE **h$_1$u̯i-b^{h}éh$_2$-*). This *-én*-locative would be the derivational base

[9]This derivational pattern is exemplified by PIE **h$_1$d-én* [loc.] 'beim Beißen befindlich' ⟶ hypostatic **h$_1$d-én-t-* 'beißend' (Neri 2017: 150–151, fn. 211). One may also add several Vedic "aorist" participles not referring to a previous action, such as Ved. *dhr̥ṣ-ánt-* 'bold' (Lowe 2017: 163), which could reflect PIE **d^{h}r̥s-én* [loc.] 'with boldness' ⟶ **d^{h}r̥s-én-t-* 'bold'.

of an adjective **-b^h h_2-én-t-* 'shining' with the aforementioned "characterizing" or "individualizing" morpheme *-*t*- (cf. Section 2.4), for instance PIE **pro-b^h h_2-én-t-* [adj.] 'that appears before, manifest' (derived from an adverb **pro-b^h h_2-én*), which was eventually reanalyzed as the "participle" of an amphikinetic present (PIE **b^h éh_2-ti*, **b^h h_2-énti* 'to shine'). One may assume that PIE **pro-b^h h_2-én-t-* [adj.] 'shining forth' ⟶ **pro-b^h h_2-ént-* [ptcp.] would have given rise to a simplex **b^h h_2-ént-* [ptcp.] 'shining'. This form is reflected in the Hesychian gloss φάντα· λάμποντα [acc.sg.m.] 'shining' (Gk. φάντα < PIE **b^h h_2-ént-m̥*). It is noteworthy that this isolated form does not belong to any synchronic verbal stem: The etymologically related verb is φαίνω 'to make appear, to show' (φαίνομαι [mid.] 'to appear'). Within Greek, the privative adjective ἄ-φαν-τος 'invisible' and its sigmatic counterpart ἀ-φαν-ής [adj.] 'id.' are deverbative formations derived from the present stem φαίνω (< **p^h án-yō*). Schirmer (LIV^2 69), after Klingenschmitt (1982: 113), assumes here a nasal-infixed present **p^h án-e/o-* (< PIE **b^h-n̥-h_2-*) with thematization, recharacterized with the suffix *-*ye/o*-. This assumption would imply that Proto-Gk. **p^h án-e/o-* 'to make visible' would be closely related for its formation to Arm. *banam* 'to open' (allegedly from PIE **b^h-n-éh_2-ti*, **b^h-n̥-h_2-énti*), and Albanian (Tosk *bëj*, Gegh *bãj*, Arbëresh, Arvanite *bënj*) 'to make appear; to do, to make'[10] which points to Proto-Alb. **ban-ja-*.[11] However, Arm. *banam* could be a productive -*na*-present of the type seen in Arm. *baṙnam* 'to lift' (< **barj-nam*) and *daṙnam* 'to come back' (< **darj-nam*), whereas Proto-Alb. **ban-ja-* 'to bring to light; to produce' could be derived from its participle **bana-* (T. *bërë*, G. *bënë*), reflecting PIE **b^h h_2-nó-* [adj.] 'shining, appearing' or – alternatively – PIE **b^h h_2-én-o-* [adj.] 'id.'). Besides, *-*nja*-presents are productive in Albanian and should therefore be rejected in comparisons.

As a result, I will tentatively suggest another derivational chain for Gk. φαίνω 'to make appear'. In my opinion, one may start with an adverb **pró-p^h a* 'appearing before' (< PIE **pro-b^h h_2-én*)[12] as the base stem of a denominative verb προφαίνω (< **pro-p^h án-yō*) 'to make appear before', displaying the same pattern as

[10]For a summary of the attested forms see Demiraj & Neri (2021).

[11]By contrast with Vedic -*anyá*-presents, the majority of Proto-Alb. *-*anja*-stems are transitive, as ascertained by Genesin (2005: 169–170) and Schumacher & Matzinger (2013: 50–51).

[12]Except for Hom. *αἰϝέν [adv.] 'for ever', which was oxytone, Greek has restructured the PIE locatives in *-*en* on zero grade *-*n̥*: see for instance Gk. χεῖμα [adv.] 'in winter' vs. Ved. *héman* 'id.' (< PIE **ĝ^h éi̯-m-en*), as *per* Nussbaum (1986: 189). As a result, Gk. **pró-p^h a* [adv.] 'appearing before' (for expected ****pró-p^h an*) should be the reflex of PIE **pro-b^h h_2-én*. Interestingly, alongside *-*én*, there is also evidence for *-*ér* used to build adverbs. Gk. ἄφαρ [adv.] 'quickly' could be explained by PIE **sm̥-b^h h_2-ér* [adv.] 'in a glimpse', an adverbial formation built on PIE **sm̥-b^h éh_2-* [f.] 'glimpse', rather than by PIE †**h_2éb^h-r̥* 'rapid stream' assumed by Willi (2004: 327).

Hom. λίπα [adv.] ‘richly, in a fat way’ (of anointing) ⟶ λιπαίνω ‘to anoint’. One may go a step further and assume that the “complex” suffix *-*nye/o*- is due to resegmentation of the present stem **pro-p^{h}án-ye/o*- as **pro-p^{h}á-nye/o*-, after which were coined such presents as Gk. κλῑ́νω (< **klí-nyō*) ‘to cause to lean’ (with sigmatic deverbatives in *°*klĭn-ḗs*) and πλῡ́νω (< **plú-nyō*) ‘to wash, to clean’ (with sigmatic deverbatives in *°*plŭn-ḗs*).[13]

A possible reflex of PIE **(pro-)b^{h}h$_2$-én* [adv.] may be attested in Germanic: Go. *bandwjan* ‘to give a sign’ and *bandwa** [str.f.] ‘sign’ could be derived from a Proto-Germanic action noun **bandu*- [m.] ‘bright appearance’ (< PIE **b^{h}h$_2$-en-tú*- [abstract] ⟵ **b^{h}h$_2$-en-tó*- [adj.] ‘endowed with shine’). *Pace* EDPG 51–52, Go. *bandwjan* and *bandwa** can hardly be related to Proto-Gmc. **bann-an*- [str.v.] ‘to command, to summon’ (for which see Section 4). The original meaning was obviously ‘apparition, vision’, as is clear from the Langobardic cognate of Go. *bandwa** borrowed in Late Latin (*vexillum quod bandum appellant* in Paulus Diaconus). The concrete meaning ‘flag, banner’ is also supported by Byz. Gk. βάνδον [nt.] (in Procopius). For the meaning, one may compare Ved. *ketú*- [m.] ‘bright appearance’ and ‘flag, banner’ (< PIE **k$^u̯$oi̯t-ú*-).

4 PIE 2. **b^{h}eh$_2$*- ‘to speak, to say’

From PIE 2. **b^{h}eh$_2$*- ‘to speak, to say’, we may assume an adverbial oblique **b^{h}h$_2$-én* as the derivational base of a PIE adjective **b^{h}h$_2$-én-o*- ‘speaking’ whose characterized form **b^{h}h$_2$-én-o-nt*- ‘the speaking one; the one endowed with speaking’[14] is surely reflected by Ved. *bhán-ant-** [ptcp.] which would thus antedate the indicative *bhán-a-ti* ‘to speak, to say’. This stem has been accounted for by Schirmer (LIV^2 69–70 & fn. 6) as the restructuring of an athematic nasal-infixed present **bhanā́-ti*, **bhan-ánti* (< PIE 2. **b^h-n-éh$_2$-ti*, **b^h-n̥-h$_2$-énti* ‘to speak, to say’). Kroonen (EDPG 52) instead assumes a thematic root present **b^{h}énh$_2$-e/o*-, which he compares to Proto-Gmc. **bann-an*- ‘to summon’, allegedly reflecting IE †*b^{h}ónh$_2$-e/o*-, with the final laryngeal triggering the gemination of the nasal. He does not mention the reconstruction of this strong verb as PIE **b^{h}h$_2$-nu̯-é/ó*- by Seebold (1970: 89), which was perhaps less anachronistic than its rewriting

[13]In the final chapter of his thesis, Praust (1998: 121–135) reconstructs an athematic -*ni*-present class: for instance, PIE **ḱli-néi̯-ti*, **ḱli-ni̯-énti* to account for Gk. κλῑ́νω (< **klí-nyō*) ‘to cause to lean’ (Aeol. κλίννω). The adduced comparanda are YAv. *siri-nu*- (or *sri-nu*-) and Lat. °*clīnāre* (with a long -*ī*-), which he considers to be a durative present to a simplex **cliniō* (Praust 1998: 132). This is simply an *ad hoc* assumption.

[14]Likewise, after Hitt. *ḫappina*- [adj.] ‘rich’ (< PIE **h$_3$ép-en-o*-) has been coined *ḫappinant*- [c.] ‘rich person’ (< [virtual] PIE **h$_3$ép-en-o-nt*- ‘the rich one’).

as IE †*b*^{h}*e-nu̯-é/ó-* by Kroonen (EDPG 52), who tacitly suggests a verb **binn-an-* 'to speak' endowed with an intensive doublet **bann-an-* 'to summon'. As already mentioned (see Section 3), the etymological connection with Go. *bandwo*, *bandwa** [f.] 'sign' has little to recommend itself, not to mention the etymon †*b*^{h}*onh*$_{2}$*-tú-h*$_{2}$*-* offered by the author.

I would rather assume here PIE **b*^{h}*h*$_{2}$*-en-u̯ó-* 'endowed with speaking', a decasuative adjective of the pattern **CC-en-u̯ó-* which is required by Proto-Gk. *ξένϝος [m.] 'guest' < PIE **g*^{h}*s-en-u̯ó-* [adj.] 'being at the feast' (Garnier 2013: 65).[15] One may therefore assume a "characterized" **b*^{h}*h*$_{2}$*-en-u̯ó-nt-* 'the one who speaks with authority; the one who gives orders' surfacing as Post-IE **b*^{h}*a-nu̯-ónt-* [m.] 'commanding [adj.], commander [m.]'. This form was normalized as Proto-Gmc. **bánn-and-* [ptcp.] with secondary root accent. This was the starting point for a strong verb **bann-an-* 'to summon'. Such a verbal reinterpretation of an adjectival stem could have been fostered by other genuine **and*-participles used as agent nouns in Germanic: see for instance OE *būend* [m.] 'dweller, inhabitant'.

Incidentally, one may speculate that the Attic infinitive φάναι (vs. Hom. mid. inf. φάσθαι) may be the recharacterization of Proto-Gk. **p*^{h}*án* 'in speaking' (< PIE **b*^{h}*h*$_{2}$*-én*), like the infinitive ἰέναι 'to go' is the reflex of **ién* 'on the way' enlarged with the Greek allative adverbial suffix -αι (cf. Hom. ἴμεν vs. ἴμεναι), as suggested by Garnier & Pinault (2020: 323).

5 PIE *g$^{u̯}$eh$_{2}$- 'to make a step; to be stable'

One may reconstruct PIE **g*$^{u̯}$*h*$_{2}$*-én* [adv.] 'firmly standing' as the base of **g*$^{u̯}$*h*$_{2}$*-én-t-* [adj.] 'stable' (reanalyzed as an "aorist" participle **g*$^{u̯}$*h*$_{2}$*-ént-*). This form is reflected by Hom. εὖ δια-βᾱ́ς (M 458) 'setting his feet well apart', with a clear

[15] PIE **g*^{h}*s-én* [loc.] 'at the feast' may be assigned to PIE **g*^{h}*és-ōr* [AK collective] 'rich meal' derived from **g*^{h}*és-r̥* [PK nt.] 'action of eating' (Garnier 2013: 65). Note that Ir. 2. **xšnaw-* 'to welcome a guest' (Cheung 2007: 457–458) is possibly backformed after its ptcp. **xšnu-tá-* 'invited, welcome' (< [virtual] PIE **g*^{h}*s-n-u-tó-*). Perhaps one has to assume here PIE **g*^{h}*s-en-u̯-ó-nt-* [characterized adj.] secondarily inflected according to the amphikinetic inflection (viz. gen.sg. **g*^{h}*s-n̥-u̯-n̥t-ós* resyllabified in **g*^{h}*s-nu-u̯-n̥t-ós* >> Ir. **xšnuw-ánt-* 'invited, welcome' normalized as as passive ptcp. **xšnu-tá-* 'id.'). More recently, Steer (2019: 338) has rejected the derivation **g*^{h}*s-en-u̯ó-* [adj.] 'being at the feast', suggesting instead a PIE deverbative **g*^{h}*s-nu̯-ó-* [adj.] 'eating' to a *-nu*-present assumed by Lipp (2009: 284, 286) ⟶ *vr̥ddhi*-derivative (viz. PIE **g*^{h}*s-énu̯-o-* 'eater'), which seems impossible to me: One would expect PIE **g*^{h}*s-neu̯-ó-*. See the discussion of Steer's proposal by Schmitt (2020: 160–161), who considers that a base meaning like *'eater' is perhaps too trivial to account for the religious aspects of Homeric hospitality. Schmitt duly mentions the figure of Ζεὺς ξένιος. I had addressed this issue by assuming that the base is PIE **g*^{h}*ós-* [f.] 'rich meal ceremoniously offered to the guests'.

static value, and not referring to a prior event (Létoublon 1986: 133–135; de Lamberterie 1990: 935). For the semantics, one may compare Gk. βέβαιος [adj.] 'firm, stable'. There was possibly also a decasuative **$g^{u̯}h_2$-en-i̯ó-* [adj.] 'firmly standing' endowed with a "characterizing" doublet **$g^{u̯}h_2$-en-i̯ó-nt-* 'the one firmly standing'. This adjectival formation could account for Gk. (εἰσ-, ἐμ-, ἐπι-)βαίνοντ- [ptcp.] 'stepping; setting his feet' (Hom. ποσὶ βαίνων). One must not confuse Gk. 1. συμβαίνειν 'to stand with the feet together' (of a statue) with 2. συμβαίνειν 'to happen' (ultimately from PIE **$g^{u̯}$m̥-i̯é/ó-* 'to come [in contact with]'). Similarly, Gk. σύμβασις [f.] 'agreement' indirectly reflects PIE **kóm-$g^{u̯}$m̥-ti-* [f.] 'coming together; agreement', semantically close to Osc. **kúmbened** 'it is agreed' (Lat. *convēnit*), but is not related to βάσις [f.] 'stepping, step, steps', which rather reflects PIE **$g^{u̯}h_2$-tí-* [f.] 'action of stepping'. Both roots have merged in Greek, due to the so-called "satellite-framing" (for instance προ-βαίνω 'to step forward, to advance' is *dynamic*). As a result, the routinely adduced equation between Gk. °βαίνω 'to step, to walk' and Lat. *veniō* 'to come' (< PIE **$g^{u̯}$m̥-i̯é/ó-*), as *per* Kümmel (LIV^2 209) is only partly true.

6 Concluding remarks

The above study highlights the process of grammaticalization of "petrified" decasuative formations derived from **CC-én* locatives of root nouns (especially in composition). By internal or external derivation, such forms could produce various delocatives:

(1)
a. PIE **CC-ḗn* [HK f.] (< PIE **CC-én-s*)
b. PIE **CC-én-t-* [adj.]
   (→ pseudo-ptcp. **CC-ént-* [root present or root aorist])
c. PIE **CC-én-o-* [adj.]
   → **CC-én-o-nt-* [characterized adjective]
      (→ pseudo-ptcp. **CCén-ont-*)
d. PIE **CC-en-i̯ó-* [adj.]
   → **CC-en-i̯ó-nt-* [characterized adjective]
      (→ pseudo-ptcp. **CCen-i̯ónt-*)
e. PIE **CC-en-u̯ó-* [adj.]
   → **CC-en-u̯ó-nt-* [characterized adjective]
      (→ pseudo-ptcp. **CCénu̯-ont-*)
f. PIE **CC-en-tó-* [adj.]
   → abstract **CC-en-tú-* [m.]

Only the patterns (1a–1e) could produce *-nt*-stems which were eventually reanalyzed as participles of finite verbs. This new morphological analysis may account for Ved. *bhur-aṇ-yánt-* 'quivering, active', Gk. φαίνω 'to make appear', °βαίνω 'to step, to walk', and Proto-Gmc. **bann-an-* 'to summon', which were previously lacking a clear derivational analysis.

Abbreviations

adj.	adjective	Lyc.	Lycian
adv.	adverb	m.	masculine
Alb.	Albanian	nt.	neuter
Arm.	Armenian	Osc.	Oscan
Av.	Avestan	OE	Old English
Byz. Gk.	Byzantine Greek	OIr.	Old Irish
c.	common gender	PA	Proto-Anatolian
CLuw.	Cuneiform Luwian	PIE	Proto-Indo-European
f.	feminine	Pre-Lyc.	Pre-Lycian
Gk.	Greek	Proto-Alb.	Proto-Albanian
Go.	Gothic	Proto-Celt.	Proto-Celtic
Hitt.	Hittite	Proto-Gk.	Proto-Greek
HK	hysterokinetic	Proto-Gmc.	Proto-Germanic
Hom.	Homeric	Ved.	Vedic
Lat.	Latin	YAv.	Young Avestan
loc.	locative	W.	Welsh

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Chapter 7

De-adjectival and de-“adjectival”: Greek κρύπτω ‘cover’ and Proto-Germanic **χreuđaną* ‘deck, equip, adorn’

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This article clarifies the relationships of three secondary roots reflected in, e.g., 1) Greek κρύπτω ‘cover’ and Tocharian B *kraup-* ‘gather, amass’ (: **kreu̯b^h-*), 2) Old Church Slavonic *kryti* ‘cover’ and Lithuanian *kráuti*, 1sg. prs. *kráuju* ‘pile up’ (: **kreu̯h$_x$-*), and 3) Old English *hréodan** ‘deck, adorn’ (: **kreu̯d^h-*), in light of our current understanding of the derivational behavior of nominal compounds formed on the basis of the predicative instrumental periphrasis. After identifying **kreh$_2$-* ‘heap, strew, cover’ (Latvian *krãju*, *krât* ‘put aside, accumulate, amass’) as the primary root, the author proposes that its *u*-stem derivative **kró/éh$_2$-u-* ‘heap, covering’ was the base of an *-*i̯e/o*-present **kreh$_2$-u-i̯e/o-* ‘provide with covering’, whose reanalysis as a primary verb gave rise to **kreu̯h$_2$-*. Since the original denominal stem could only appear in the present system, there was need elsewhere for the use of its base’s instrumental with the light verbs **d^{h}eh$_1$-* ‘put, make’ and **b^{h}uh$_x$-* ‘become’. On the basis of the anticausative periphrasis **kréh$_2$-u-h$_1$ b^{h}uh$_x$-* ‘become covered’ (cf. Ved. *gúhā bhū-* ‘become hidden’) was formed an adjective **kr̥h$_2$-u-b^h(u̯)-ó-* ‘heaped, covered’ >> **kruh$_2$(-)b^h-ó-* basic to a **kruh$_2$b^h-i̯e/o-* ‘render covered, heap (upon)’, whose reflex **krub^h-i̯e/o-* was the preform of κρύπτω and the model for the **kreu̯b^h-* ‘heap’ ancestral to the Tocharian root. On the basis of the causative periphrasis **kréh$_2$-u-h$_1$ d^{h}eh$_1$-* was formed a **kr̥h$_2$-u-d^{h}h$_1$-ó-* >> **kruh$_2$(-)d^{h}h$_1$-ó-*, whose resultative patientive interpretation ‘covered’ paved the way for the quasi-Property Conceptual meaning ‘equipped, adorned’. When this adjective formed an s-stem substantive **kréu̯d$_2$(-)d^{h}h$_1$-o/es-* ‘equipment, adornment’ > **kréu̯d^{h}h$_1$-o/es-*, this quasi-PC meaning and the frequency of *s*-stems beside simple thematic presents, both prototypically radical *é*-grade categories, contributed to the formation of a causative **kréu̯d^{h}h$_1$-e/o-* ‘render equipped, adorned’ derived by suffixal substitution from a new effectively adjectival root.

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1 Introduction: from specific to general

The purpose of this contribution is, on the one hand, to apply a set of theories to explain precisely how the members of a specific set of Indo-European data are historically related and, on the other, to highlight how such explanations offer theoretical implications for our general understanding of various processes whereby verbal stems are formed on the basis of adnominals,[1] whether nominal stems or semantically adjectival[2] roots. It will be argued that this crucial distinction between the two types of de-adjectival verbs, namely, stem-derived versus root-derived, is reflected in the prehistory of, most notably, Greek κρύπτω 'cover, hide, bury' and Old English *hréodan** 'deck, adorn', which have been assumed to be related in terms of root-etymology (IEW 616–617), but whose exact derivational prehistory is unclear. The goal of this paper is to clarify that history by (1) identifying the primitive root as **$kreh_2$*- (LIV^2 367), which seems to have meant 'cover, heap'; (2) proposing that a *u*-stem nominal derivative of that root was used in the predicative instrumental periphrasis with the light verbs **d^heh_1*- 'put, make' (causative) and **b^huh_x*- 'become' (anticausative); and (3) arguing for the derivation, on basis of the respective periphrases, of adjectival nominal compounds whose own derivatives illustrate the interplay of morphology and lexical semantics in de-adjectival derivation.

Our set of data, derived largely from the material under Pokorny's **krā[u]-* : *krəu-* : *krū̆-* 'aufeinander, auf einen Haufen legen, zudecken, verbergen' (IEW 616–617), includes Greek κρύπτω 'cover, hide, bury' (: κρύφα 'secretly' : κρυφ-), Tocharian B *krauptär* 'gathers' (TA *krop*- 'id.'), OCS *kryjǫ*, *kryti* 'cover', Lithuanian *kráuju*, *kráuti* 'pile up' (Latv. *kŗaũju*, *kŗaũt* 'id.'), and Old English *hréodan** 'deck, adorn' (pret. *hréad* : pret. part. *gehroden*, ON *hrjóða** 'id.' : *hroðinn* 'painted', *gull-roðinn* 'gilt'). Given that all the above-listed data bear segments reflecting the sequence **kru*- and convey lexical meanings involving the notion of covering or placement together, it remains appropriate to maintain that Pokorny is basically correct in supposing that these Greek, Tocharian, Slavic, Baltic, and Germanic forms ultimately share a root reflected in a series of roots reconstructible as secondary. The question that confronts us if we accept this supposition is the origin of the material following the reflexes of **kru*- in each subset.

[1] I use "adnominal" in a general sense to refer (with the exception of finite verbs) to any element attributed to or predicated of a substantive.

[2] For the derivational behavior of such roots, see Nussbaum 2022.

2 The existence of *$kre\underset{\frown}{u}b^h$-, *$kre\underset{\frown}{u}h_x$-, and *$kre\underset{\frown}{u}d^h$-

The first consists of Greek κρύπτω and Tocharian B *kraup-* (: *kraupe* m. ‘group’), the latter of which, since it forms a class II present, a class II subjunctive, and a class I preterite (Adams 2013: 236–238), is likely, in light of the oddity of that pattern, to be inherited (cf. athematic Tocharian A *kroptär* class I pres., 3pl. *kropäntär*, Malzahn 2010: 614–616). What this comparison allows us to infer is, therefore, that κρύπτω and *krauptär* reflect derivatives of a secondary root at least as old as Proto-Nuclear Indo-European, namely a *$kre\underset{\frown}{u}b^h$- ‘heap, strew, cover’ which formed, in descriptive terms, a radical zero-grade *-$\underset{\frown}{i}e/o$-present *$krub^h$-$\underset{\frown}{i}e/o$- ancestral to κρύπτω as well as what appears to be a radical *ó*-grade uncharacterized present ancestral to *krauptär* (< Transponat *$kró\underset{\frown}{u}b^h$-e-tor*). Even if we attribute this root to PIE, it is not, in accordance with our hypothesis, a primary product of reconstruction, as one would assume for a lemma in LIV^2. To deal with our second subset, which includes OCS *kryti* ‘cover’ and Lithuanian *kráuti* ‘pile up’, LIV^2 reconstructs a *$kre\underset{\frown}{u}h_x$- ‘aufhäufen, bedecken’ (LIV^2 371), from which were formed, in descriptive terms, the *$kruh_x$-$\underset{\frown}{i}e/o$- ancestral to *kryjǫ* and the *$kre\underset{\frown}{u}h_x$-$\underset{\frown}{i}e/o$- indirectly ancestral to *kráuju*[3] (cf. dial. Lith. *kriáuju* : Latv. *kŗaũju*).

While LIV^2 ignores κρυφ- and *kraup-* encountered in Pokorny, it does take account of the Germanic data via the assumption that OE *hréodan** ‘bedecken’ reflects a *-d^he/o-present *$kré\underset{\frown}{u}h_x$-$d^h$e/o-*. Though we have good reason to believe that there was indeed a morphological boundary before that dental segment at some historical stage, it appears premature to characterize *hréodan** as the reflex of a present in *-d^he/o-*. Unlike, for example, βαρύθω ‘get weighed down (intr.)’, which is formed, with the reflex of a descriptive *-d^he/o-*, from the actual stem of an adjective (βαρύς ‘heavy’), the Proto-Germanic **χreuðanq* ‘deck, equip, adorn’ ancestral to *hréodan** behaves as a root-based formation parallel to other derivatives of its root (: **χrauðō-* [ON *hrauð* f. ‘chainmail shirt’, OE *earm-hréad* ‘arm-ornament’] : **χrusti-* [OHG *hrust* ‘armor’, OE *hyrst* ‘decoration, armor’]). In this formal respect, **χreuðanq* rather resembles βρῑ́θω ‘become weighty’, which, though historically related to βαρ- ‘heavy’ (v. Rasmussen 1989: 94–95), is formed from its own root (: βρῑθ-ύς ‘heavy’, βρῖθ-ος ‘weight’). While it is possible to claim that the *-*ði/a-* of **χreuðanq* reflects an erstwhile presential suffix that formed a stem eventually reanalyzed as a (secondary) *$kre\underset{\frown}{u}d^h$- ‘cover’, such a claim is of no profit in the absence of an attempt to explain the *$kre\underset{\frown}{u}d^h$- in question, which must have existed at some stage, in light of any explanations of the origins of the *$kre\underset{\frown}{u}b^h$- and *$kre\underset{\frown}{u}h_x$- to which it is assuredly related.

[3]On the acute intonation, see Villanueva Svensson 2014.

3 Evidence for primary ***kreh₂-** 'heap, strew, cover'

The initial clue serving in an explanation of how the three secondary roots, **kreu̯bʰ*-, **kreu̯hₓ*-, and **kreu̯dʰ*-, are related is in fact contained in Pokorny's entry. The brackets of his **krā[u]*-, the first-listed variant of the root, betray the inkling that the truly primary form of the root did not include the *-*u*-. Especially in light of the Latvian *krãju*, *krât* 'put aside, accumulate, amass' listed under **krā[u]*-, it is deeply appealing to reconstruct the truly primary form as a **kreh₂*- 'heap, strew, cover', which LIV² 367 quizzically represents as a ?**kreh₂*- 'aufhäufen, sammeln' supposedly reflected solely in Balto-Slavic. According to Pokorny, one may group *krãju*, *krât* together with OCS *kradǫ*, *krasti* 'steal' "mit präsensbildendem d" and, with an apparent *p*-extension, Lithuanian *krópti* 'id.' and Latvian *krâpju*, *krâpu*, *krâpt* 'cheat, deceive'. Though the origin of such formants or extensions remains to be determined, what is clear is the likelihood that these verbs, given their senses 'steal' or 'deceive', should be related to the one meaning 'amass'. Since accumulation generally involves the placement or position of a substance or set of objects over and upon an entity, and since the entity positioned over and upon another entity may serve to effect the latter's non-stimulus status (e.g., soil of a mound covering a dead body and/or treasure), it is likely that the senses of theft and deception proper to *krasti*, *krópti*, and *krâpt* developed via 'hide, conceal' from 'heap, strew, cover' (cf. PDE *bury*; OIr. *arcelim* 'steal' : *ceilid* 'hides', Buck 1949: 791).

Some evidence for this original sense of accumulation proper to **kreh₂*- may also be adduced from Greek. Beside evidence of a κλῶμαξ, -ακος, m. 'heap of stones, rocky place' and a κλωμακόεις 'rocky' (B 729), Hesychius attests to a κρώμαξ· σωρὸς λίθων and a κρωμακόεν· κρημνῶδες. While Beekes (EDG 720), rejecting the association of the former set with κλάω 'break', considers the variation between -λ- and -ρ- to be evidence that both sets are of non-Indo-European origin, an evaluable hypothesis would take them back to two PIE roots. Before considering the question of the roots themselves, we should first observe that the sequence -μακ- points to derivation from stems in -μα (cf. ἕρμαξ, -ακος f. 'heap of stones, cairn' ← ἕρμα 'prop, support' [Hom.]). By comparison to Latin *rūpēs* f. 'rock' (: *rumpō* 'break'), it is probable that, in the case of κλῶμαξ, the μα-stem in question was indeed akin to κλάω. If, however, the earlier meaning of that μα-stem was 'heap', it is not inconceivable that its preform was derived from **kleh₂*- 'heap up, put' (**k⁽ᵘ̯⁾leh₂*- 'hinbreiten, hinlegen', LIV² 362), evidence of whose verbal derivatives, namely Lithuanian *klóju*, *klóti* 'spread, lay down' (Latv. *klâju*, *klât* 'id.') and OCS *kladǫ*, *klasti* 'load, lay', incidentally resembles that of **kreh₂*- 'heap, strew'. In the case of the base of κρώμαξ, it is fairly certain that its meaning was 'heap'.

Two additional data deserve mention: Κρῶμνα, -ης (· πόλις Παφλαγονίας B 855 [Hsch.]; a toponym) and the reportedly Paphlagonian κρωμακωτός 'rocky and steep'. Given the identity of place as well as toponomastic practice (cf. Αἴπεια 'Hightown' : αἰπύς, αἰπός 'high, lofty'), it is likely that the base of the *devī́*-derivative ancestral to Κρῶμνα was reflected as the μα-stem basic to the κρῶμαξ from which κρωμακωτός was derived. In all likelihood, this μα-stem would go back to a **-men-* stem derived from **kreh₂-* 'heap, strew'.

Comparable in form to the **póh₂-mn̥* 'protection' (: **peh₂-* 'cover, protect, care for') ancestral to πῶμα 'lid, cover' (Hom.),[4] this **-men-* stem would have been an **ó/é* acrostatic. In addition to the association between κρῶμαξ and **kreh₂-* effected by the attribution of the **-men-* stem to this type, the reconstruction of a **króh₂-mn̥/kréh₂-mn̥-* 'heaping, heap' has the potential advantage of providing a way to account for the origin of κρημνός 'overhanging bank, cliff'. Since the appearance of its -η- in Pindar appears to preclude an **-ā-* in the preform (EDG 777), it is conceivable that κρημνός goes back to a genitival **krēh₂-mn-ó-* 'of the heaping, heaped' derived from **króh₂-mn̥/kréh₂-mn̥-* 'heaping, heap'. Though this scenario has some problems,[5] it does provide κρημνός with a semantically straightforward etymology (cf. Hitt. *wappu-* 'riverbank' < **(h₂)u̯óp-u-/h₂u̯ép-u-* 'heap' : **h₂u̯ep-* 'cast, strew' [Hitt. *ḫuwapp-/ḫupp-* 'cast, throw' : Ved. *vápati* 'strews'], Catsanicos 1985: 125; Melchert 2013: 176) whose formal features are mostly in line with a simple thematic item conceivably derived from a **-men-* stem to **kreh₂-* 'heap, strew'.

By comparison to the form of Latin *fās* 'morally right' (< **bʰéh₂-o/es-* 'saying, utterance', EDL 203), we may hypothesize that **kreh₂-* also formed an *s*-stem substantive **kréh₂-s-* 'heaping, strewing, spreading, covering' for which there is some indirect evidence in Baltic. Beside Lithuanian *krū́snis*, *-ies* f. 'pile of stones in a field' (v. **Smoczynski2018**), identifiable as the reflex of a **kruh₂-s-n-i-* derived from a possessive in **-nó-* to an *s*-stem **kréu̯h₂-o/es-* 'heap, covering' to be discussed below (: **króu̯h₂-o-* 'covering' [> OCS *krovъ* 'roof']), Lithuanian and Latvian respectively attest to a *krósnis*, *-ies* f. '(stone-)stove; pile of stones' and *krâsns* 'stove', which **Smoczynski2018** and Karulis (1992: 419) both connect with Latvian *krãju*, *krât* 'amass'. While Smoczyński's connection with *krât* and thus **kreh₂-* is entirely justified, what this scholar analyzes as a suffix *-sni-* is in need of further clarification. If we compare the dialectal Latvian *krāsts* m. 'oven' (: *krāstît* 'collect, hoard'), we may suppose a formation similar to Latin *fāstus* 'right' (←

[4]Analysis of πῶμα 'lid, cover' as the reflex of an **ó/é* acrostatic derives from Nussbaum (p.c.).

[5]One would have to assume that this derivation took place at a very early date in the protolanguage and that one of the nasals failed to be deleted as expected or was restored.

fās 'id.'), that is, a possessive in **-tó-* derived from an *s*-stem basic to a **-nó-* stem from which the ancestor of this *-sni-* stem was formed. Since that *s*-stem had the sense 'heaping, heap', its derivative, namely **kr̥h₂-s-nó-* (cf. λύχνος 'lamp' < **lúk-s-no-* 'the lighted' ← **luk-s-nó-* 'lighted' ← **léu̯k-o/es-* 'light'), would have borne a meaning 'heaped' subsequently determined by means of the acrostatic *i*-stem derivative **kréh₂-s-n-i-* 'the heaped, heap' ancestral to the Baltic cognates. Since the above nominal and verbal evidence supports the reconstruction of **kreh₂-* as a primary root, the immediate question is the nature of the **-u-* in the secondary roots. As the old extensions invoked by Pokorny have increasingly ceded to the discipline's growing powers of morphosemantic analysis, we can assume that the **-u-* is of suffixal origin.

4 The nature and derivational role of **kró/éh₂-u-* 'heap, covering'

4.1 Background

This assumption opens up two paths of hypothesis: (1) it is a verbal stem formant, in which case it would be the suffix of a *u*-present, or (2) it is the suffix of a nominal stem. The preferability of the latter hypothesis becomes clear in light of parallelism with the derivational history assigned in previous research to καλύπτω 'cover' (Merritt 2021). This particular history derives its existence from the understanding, originating in and enriched by, respectively, comparative-historical (Jasanoff 1978: 118–126; 2004; Schindler 1980; Balles 2006; Rau 2009) and theoretical-typological work (Dixon 1982, 2004; Francez & Koontz-Garboden 2015), that PIE was a language in which certain stative concepts, such as Property Conceptual (PC) states, were disposed to realization via substantives in the instrumental case. In prelinguistic terms, a stative concept could be accessed by conceiving of a substance, whether concrete or abstract, as an entity possessed by the being of which the state is predicated. While much attention has been paid to the role of adjectival (i.e., PC) abstracts in this means of stative predication, we should not forget that the specific stative/anticausative/causative[6] construction which inaugurated much of this research tradition, namely Vedic *gúhā as-/bhū-/dhā-* 'be/become/make hidden', includes the reflex of a substantive derived ultimately from a root that basically meant 'cover' (Lith. *gũžti* '[of hens] sit,

[6]I use such terms in a general sense (cf. introduction to this volume: 14–16). Accordingly, a causative and anticausative derived from an adjective or adjectival root are specifically factitive and fientive respectively.

squat, keep warm'), corresponding to an Activity Concept, rather than a Property Concept. Though *gúhā* in the sense 'hidden, secret' does express (approximately) a PC state, its preform, in light of the metonymy COVERED FOR HIDDEN underlying that sense, must have resembled a verbal resultative state, namely 'covered' (lit. 'with covering'), predicated of a patient/theme. In like manner, there is, at the heart of the history of καλύπτω, an acrostatic *u*-stem substantive **ḱó/él-u-* 'covering', whose instrumental was used to predicate just such an Activity Concept state. Correspondingly, it seems feasible to suppose that our **kreh₂-* was ultimately the base of a **kró/éh₂-u-* 'heap, covering'.

4.2 **kreh₂-u-i̯e/o-* 'provide with a covering'

In accordance with this hypothesis, it is conceivable that the instrumental of **kró/éh₂-u-*, namely **kréh₂-u-h₁*, was used as an adnominal in nominal sentences or (anti)causative periphrases with forms of **bʰuhₓ-* 'become' and **dʰeh₁-* 'put, make'. While the instrumental form would participate in periphrasis, it is also conceivable that the meaning of the instrumental could be expressed without the morphology of that case by means, for instance, of the suffix **-i̯e/o-*. As a denominal verbal stem formant, **-i̯e/o-* lets the constructional and lexical semantic idiosyncrasies of the base shape the meaning of what it derives. In principle, it does not strictly specify the quantity or quality of the arguments. In practice, however, there seems to be a tendency for transitivity and for the agentivity of the argument serving as subject. For example, the derivational semantic history of Latin *arguō* 'demonstrate' is discernible by supposing that it reflects a transitive **-i̯e/o-*present **h₂erĝ-u-i̯e/o-*[7] '[*x*] provide [[*y*] [with clarity]]' derived from **h₂ó/érĝ-u-* 'clarity' in the sense corresponding to its instrumental **h₂érĝ-u-h₁* 'with clarity', from which in theory was formed a *t*-stem (abstract)[8] substantive basic to the possessive adjective ancestral to the "presuffixally lengthened" perfect passive participle *argūtus* 'demonstrated' (< Transponat **h₂erĝ-u-h₁-t-o-*; Nussbaum 1996; cf. EDL 53). Given the frugality of this analysis, it is appealing to suppose that **kró/éh₂-u-* 'heap, a covering' was the base of an instrumentative PIE **-i̯e/o-*present **kreh₂-u-i̯e/o-* 'provide with a covering'.

[7]The cognation of Hittite *arkuwae-ᶻⁱ* 'make a plea', which Melchert 1998 associated with *arguō* (< **argu-yé/ó-*), would be difficult to maintain, unless one assumes on the basis of Kloekhorst 2006 and 2008: 206 that, since the root **h₂erĝ-* of the form ancestral to *arkuwae-ᶻⁱ* was in *o*-grade (rather than the *e*-grade found elsewhere), the initial *h₂* was lost.

[8]Cf. Hitt. *ḫappina-* 'rich' → abstract *ḫappinatt-* (c.) 'wealth', Nussbaum 2004.

4.3 The genesis of ***kreu̯h$_2$-** 'heap, cover'

Since this verb's meaning was equivalent to that of the verb originally derived from **kreh*$_2$- 'heap, strew, cover', it is conceivable that it was reanalyzed as a primary present derived from a "long-diphthongal" **kreh*$_2$*u*- 'heap, strew, cover'. Accordingly, the root of this present would have appeared eligible as a base to a *-*tó*- verbal adjective, namely a **kr̥h*$_2$*u-tó*- 'heaped, covered'. Though not of the canonical structure reflected in Vedic *pītá*- 'drunk' (< **pih*$_3$*-tó*- < **ph*$_3$*(-)i-tó*-), this **kr̥h*$_2$*u-tó*- may be assumed to have undergone laryngeal metathesis in the same way as the preform of Latin (← Sabellic) *brūtus* 'heavy, stupid', which via **g*$^{u̯}$*ruh*$_2$*tó*- would reflect, whatever its exact analysis, a **g*$^{u̯}$*r̥h*$_2$*-u-t(-)ó*- 'heavy' related to the preform of βαρύς 'id.'. The resulting **kruh*$_2$*-tó*- 'heaped, covered', arguably basic to the preform of Lithuanian *krū̃tis* f. 'pile of stones, stone-oven' (< **kruh*$_2$*-t-i*-) and directly ancestral to the past passive participle of OCS *kryti* 'cover' (cf. Russian *krýtyj* 'covered'), appears to have been pivotal in the creation of what we may now rewrite as our secondary **kreu̯h*$_2$- 'heap, cover' (→ τόμος-substantive **króu̯h*$_2$*-o*- 'covering' [> OCS *krovъ* 'roof'] : *s*-stem **kréu̯h*$_2$*-o/es*- 'heap, covering' → **kruh*$_2$*-s-ó*- 'heaped, cover(ed)' [→ **króu̯(h*$_2$*)-s-o*- → **króu̯(h*$_2$*)-s-i̯o*- > ON *hreysi* 'stone heap'] → **kruh*$_2$*-s-i̯o*- → **krúh*$_2$*-s-i̯e-h*$_2$- > Russian *krýša* 'roof'). While Slavic points to a radical zero-grade **kruh*$_2$*-i̯e/o*- derived from **kreu̯h*$_2$- on the model of **kruh*$_2$*-tó*-, Baltic inherited a radical *e*-grade *-*i̯e/o*-present **kreu̯h*$_2$*-i̯e/o*- 'heap' (>> dial. Lith. *kriáuju* : Latv. *kr̦aũju*) that arose conceivably by analogy to families involving (as usual) radical zero-grade *-*tó*- verbal adjectives beside radical *e*-grade *-*i̯e/o*-presents (e.g., **ten-i̯e/o*- 'stretch' [τείνω, Alb. *n-den* 'stretches out', LIV^2 627] : **tn̥-tó*- 'stretched' [Ved. *tatá*- 'id.', τατός 'extensible']).

5 The origin of κρύπτω

5.1 Background

To understand the emergence of **kreu̯b*^{h}- beside the **kreu̯h*$_2$- whose origin we have just considered, we should return to the original instrumentative **kreh*$_2$*-u-i̯e/o*- 'provide with a covering'. Since, by its very nature, this denominal was a creature of the present system, the non-present system had to be served by periphrases consisting of the instrumental of the denominal's base and the light verbs **d*^{h}*eh*$_1$- and **b*^{h}*uh*$_x$-. Given the necessity of periphrasis at some level, it is conceivable that the option to form compounds based on that periphrasis was realized. We may thus suppose that the anticausative phrase **kréh*$_2$*-u-h*$_1$ *b*^{h}*uh*$_x$-

‘become covered’ inspired a resultative compound adjective *$k\textsubring{r}h_2$-u-$b^h uh_x$-ó- ‘(having) become heaped, covered’, whose νεογνός-reflex was a *$k\textsubring{r}h_2$-u-b^h(u̯)-ó-. This form was ancestral to an eventually simple thematic adjective *$kruh_2b^h$-ó-[9] ‘heaped, covered’ via laryngeal metathesis and morphological reanalysis. As a thematic adjective, this *$kruh_2b^h$-ó- would in theory have been eligible as a base to a *nēwaḫḫi*-type factitive. What we find instead is the reflex of a *-i̯e/o-present which, like χαλέπτω ‘treat harshly’ beside χαλεπός ‘harsh’, was derived by substitution of *-i̯e/o- for the thematic vowel of the adnominal base. The derivative would thus have been *$kruh_2b^h$-i̯e/o- ‘render covered, heap (upon)’.

5.2 The parallel of πλάσσω ‘form, mould’

The morphological and phonological nature and fate of this denominal may be compared to the history of πλάσσω ‘form, mould’, which, given the sequence πλαθ- (: κοροπλάθος ‘figure-modeler’), Pokorny (IEW 805–807) characterizes as a *-i̯e/o-present formed from a base containing the remnant of a d^h-formant of a present stem derived ultimately from what would now be represented as *$pleh_2$- ‘broad, flat’ (Hitt. *palḫi-/palḫai-* ‘wide, broad’, Lat. *plānus* ‘flat’, OE *flōr* f. ‘floor’). Since, however, πλάσσω does reflect a present in *-i̯e/o- (PGk. *$pl\text{ə}t^h$-i̯e/o-), it is preferable to analyze the preform of this verb as denominal, specifically de-adnominal from a *-d^hh_1-o- compound. In maintenance of the connection with the descriptive *$pleh_2$- ‘broad, flat’, such an analysis would begin with the derivation of what may be provisionally represented as a root noun *$pl\acute{e}h_2$-/$p\textsubring{l}h_2$-´ ‘the flat, flatness’, whose capacity to modify substantives rendered it eligible for use in the familiar predicative instrumental periphrasis (i.e., the *cale-faciō*-type). Accordingly, on the basis of a causative phrase *$p\textsubring{l}h_2$-$\acute{e}h_1$ d^heh_1- ‘provide with the flat, flatten’ was formed *$p\textsubring{l}h_2$-d^hh_1-ó- ‘flattened, pressed, kneaded, moulded’, apparently basic to the substantive *$p\textsubring{l}h_2$(-)d^hh_1-e-h_2- ‘the moulded, mould’ ancestral, via the παλάμη- or *palma*-rule (Rix 1996: 157–158, Meiser 1998: 108–109, Höfler 2017, Weiss 2020: 119, Hackstein 2021: 193–194), to the παλάθη ‘cake of preserved fruit’ adduced by Pokorny (‘flacher Fruchtkuchen’; cf. παλάθαι· σύκων μαζία [Hsch.] : μάσσω ‘knead, mould’).

The sequence παλαθ- brings up the question of the origin of the radical base πλᾰθ- underlying πλάσσω and apparent in its nominal relatives (e.g., πλάθανον ‘baking-mould’, πλαθά f. ‘modelled figure’). Given that πλᾰθ- does not display a laryngeal reflex, it is necessary to suppose that speakers posited the root on the basis of a form in which the laryngeal was deleted. Since the substitution

[9] For the PIE status of the development *-b^hu̯- to *-b^h-, see Merritt 2021: 312–313.

of *-$\underset{\frown}{i}$e/o- for the thematic vowel of our *$p\d{l}h_2$(-)d^hh_1-ó- would have produced a sequence subject to Pinault's Law and, subsequently, the *Weather* Rule (cf. Ion. γλᾱ́σση- 'tongue' < *$g\d{l}h_3g^h$-$\underset{\frown}{i}eh_2$- 'pointed' [cf. γλωχῖ́ς 'point, barb'] : *gl(e)h_3g^h-ih_2- [: γλῶττα 'tongue']; Neri 2017; Weiss 2020: 123), it is appropriate to suppose that *$p\d{l}h_2$(-)d^hh_1-$\underset{\frown}{i}$e/o- 'get x moulded' was reflected as an eventual *$p\d{l}d^h$-$\underset{\frown}{i}$e/o- directly ancestral to πλάσσω, whose meaning 'form' resulted from a metonymic extension on the part of *$p\d{l}h_2$(-)d^hh_1-ó- from 'flattened, pressed' to 'kneaded, moulded' via the notion of applying pressure to change or create a surface. Morphologically, the result is clear: the *-$\underset{\frown}{i}$e/o-present *$p\d{l}d^h$-$\underset{\frown}{i}$e/o- was reanalyzed as primary from a *$pled^h$- 'form, shape' that could now form derivatives of its own, such as the τόμος-type *$plód^h$-o- 'formation' arguably ancestral to OCS *plodъ* 'fruit, fetus' (cf. *kre$\underset{\frown}{u}h_2$- 'heap, cover' → *kró$\underset{\frown}{u}h_2$-o- 'covering' > OCS *krovъ* 'roof').

5.3 The origin of *kre$\underset{\frown}{u}b^h$-

If we apply the comparandum that is *$p\d{l}h_2$(-)d^hh_1-$\underset{\frown}{i}$e/o- to our *$kruh_2b^h$-$\underset{\frown}{i}$e/o- 'render covered, heap (upon)', the result is equally clear. Once the *Weather* Rule deleted the laryngeal, the sequence consisted of a suffix *-$\underset{\frown}{i}$e/o- and a base *$krub^h$- interpretable as a root in its own right, namely *kre$\underset{\frown}{u}b^h$- 'heap, cover'. At this point, we can revisit the thematic Tocharian B *krauptär* 'gathers' in light of its unambiguously athematic cognate *kroptär* in Tocharian A (class I pres., 3 pl. *kropäntär*, Malzahn 2010: 614–616). Given this discrepancy, it appears probable that the athematicity of the Tocharian A form is a retention (Jasanoff p.c.). Given, moreover, the likelihood of this retention and the possibility that *kraup-* and *krop-* reflect an *o*-grade, it is probable that Proto-Tocharian inherited the reflex of a *molō*-present (Jasanoff 2003: 70–79) *kró$\underset{\frown}{u}b^h$-/kré$\underset{\frown}{u}b^h$- 'heap up together' derivationally parallel to the τόμος-type *kró$\underset{\frown}{u}b^h$-o- 'accumulation' apparently ancestral to TB *kraupe* 'group'. In other words, once a *kre$\underset{\frown}{u}b^h$- 'heap, cover' was posited on the basis of a historically secondary formation synchronically interpretable as primary, namely a *-$\underset{\frown}{i}$e/o-present, the formation of primary stems, namely a *molō*-present and τόμος-type nominal, became possible.[10]

[10] A similar account may be assigned to the preform of Proto-Germanic **walð-i/a-* : Proto-Slavic **vold-e/o-* : Proto-Baltic **véld-* 'be powerful, rule', which Rothstein-Dowden (2022: 129–135) has analyzed as a *molō*-present formed with suffixal *-d^h- (*(h_2)$\underset{\frown}{u}$ólh_x-d^h-/*(h_2)$\underset{\frown}{u}$élh_x-d^h- 'be powerful').

6 The origin of *χreuðaną

6.1 Introduction

A similar kind of root-based account is crucial to explaining the origin of secondary *kreu̯d^h- and Proto-Germanic *χreuðaną ‘deck, equip, adorn’. In light of our account of the origin of the secondary *kreu̯b^h- in the anticausative phrase *kréh$_2$-u-h$_1$ b^{h}uh$_x$- ‘become covered’, it is reasonable to suppose that *kreu̯d^h-, provisionally definable as ‘cover’, originated ultimately in the causative variant of the same construction, namely *kréh$_2$-u-h$_1$ d^{h}eh$_1$- ‘provide with covering’, on the basis of which a compound adjective *kr̥h$_2$-u-d^{h}h$_1$-ó- was formed. Though this adjective could certainly convey the meaning ‘providing with covering’ as a function of the agentivity assignable by the causative light verb, the necessary existence of a patient/theme in the structure of *d^{h}eh$_1$- would always render possible the interpretation of *kr̥h$_2$-u-d^{h}h$_1$-ó- as an adjective of patientive and resultative sense equivalent to its sister *kr̥h$_2$-u-b^{h}uh$_x$-ó- ‘(having) become heaped, covered’. We may thus suppose that *kr̥h$_2$-u-d^{h}h$_1$-ó- also meant ‘provided with cover, rendered covered’.

6.2 COVERED FOR *X*

This patientive and resultative sense ‘covered’ is a nucleus disposed, via the basic notion of positioning one entity over and upon another entity, to generate by metonymy a variety of senses, some of which correspond approximately to Property Concepts. For example, the sense of Vedic *gúhā* ‘secret’ is, as discussed above, the product of COVERED FOR HIDDEN. The structurally and semantically interesting feature of this metonymy is its introduction of the end-oriented notion that the action relates to a third entity, which in this case is a non-experiencer, beside the patient/theme and implied agent. In other words, when a verb that means ‘cover’ is used in the sense ‘hide’, there arises the implication that the end or purpose of the agent’s action is to prevent the patient/theme from becoming a stimulus to some additional person (third entity), whose existence is implied, and which may even surface as an participant, as in (agent) x’s hiding of y (non-stimulus) from z (non-experiencer). The metonymy COVERED FOR HIDDEN that generates the implication of this third entity is structurally comparable to two other metonymies, COVERED FOR EQUIPPED and COVERED FOR ADORNED, which both imply ends, respectively, something for which the patient/theme is equipped and someone for whose experience the patient/theme is adorned. If we assume that *kr̥h$_2$-u-d^{h}h$_1$-ó- ‘provided with cover, covered’ and its reflex *kruh$_2$(-)d^{h}h$_1$-ó- could develop senses of generic end-orientation and even end-orientation in relation to an experiencer considered capable of assigning positive value to the stimulus (= patient/theme), it would be conceivable that *$kruh_2(\text{-})d^hh_1$-ó-* expressed, on the one hand, the senses 'equipped, prepared, ready', and, on the other, 'adorned, embellished, beautiful'. In this connection, it is appropriate to observe that Latin *parātus* 'ready' (: *parāre* 'prepare') and *ōrnātus* 'adorned' (: *ōrnāre* 'fit out'), the resultative adjectives corresponding to these two sense-domains, have been lexicalized as adjectives bestriding the Activity and Property Conceptual domains. Correspondingly, we may suppose that *$kruh_2(\text{-})d^hh_1$-ó-* meant something like 'equipped, adorned'.

6.3 ***$kruh_2$-d^hh_1-ó-* → ***kréu̯$h_2(\text{-})d^hh_1$-o/es-***

The interaction of this quasi-Property Conceptual meaning and the behavior of the proto-language with respect to word-formation is now of great importance. Like its sister, the merely resultative *$kruh_2b^h$-ó-* 'heaped, covered', *$kruh_2(\text{-})d^hh_1$-ó-* may be assumed to have been at some point analyzed as a simple thematic adjective *$kruh_2d^hh_1$-ó-*. Given, however, its developed and thus distinctive PC meaning, it is conceivable that speakers felt a greater need for that meaning to be expressed by means of an associated substantive. For the substantival expression of both Activity Concepts and Property Concepts, the readily available nominal stem formant was of course *-*o/es*-, the addition of which was prototypically accompanied by the installation of radical *é*-grade for both primary and secondary derivation (e.g., **men*- 'think, feel' → **mén-o/es*- 'mind, feeling' > Ved. *mánas*- 'spirit'; **sru-tó*- 'flowing' → **sréu̯-t-o/es*- [i.e., **sréu̯-to/e-s*-] 'flow' > Ved. *srótas*- n. 'stream'). It is therefore reasonable to suppose that our *$kruh_2d^hh_1$-ó-* 'decked, equipped, adorned' was at some point the base of an *s*-stem substantive *kréu̯$h_2(\text{-})d^hh_1$-o/es-* that expressed the resultative state of being equipped or adorned as a *nomen qualitatis* (cf. PDE *prepared-ness* 'readiness').

6.4 ****χrusti*- < ****krud^h-s-ti*-**

Before considering the influential role of this *s*-stem in our account, we should of course adduce evidence that such a stem existed in the first place. That evidence, while indirect, is all the same fairly clear. Old High German *hrust* 'armor' and OE *hyrst* 'decoration, armor', being exact cognates, point to a **χrusti*- derived from **χreuð*- by means of a suffix which Seebold (1970: 275) represents as *-*sti*- (**hrud-sti*- → **hrust-i-z* [f.] 'Rüstung'). At face value, such a sequence is necessary to account for the -*st*- cluster actually attested, as we would expect an underlying Pre-Proto-Germanic *krud$^h(h_1)$-ti-* to be reflected as **χrussi*- (cf. OE *ġewiss* 'sure, certain’ : *$\underset{\frown}{u}$id-tó- ‘seen, known’). Moreover, though it is imaginable that *χrussi- was refashioned as *χrusti- or that the reflex of an original *$krud^h$-ti- developed an excrescent -s- resulting in *χrusti- as well, such claims would be ad hoc. It is therefore preferable to suppose that the preform was a *$krud^h$-s-ti- of some description.

6.5 ***χrausta- < *kro$\underset{\frown}{u}$$d^h$-s-to- : *$krud^h$-s-tó-***

Since *-ti- in this non-primary context points to derivation from an adnominal in *-tó- (Schindler 1980: 390), a safe assumption would be that *$krud^h$-s-ti- was derived from a *$krud^h$-s-tó- (cf. Pre-Hitt. *dalugaš- ‘length’ → *dalugašta- ‘lengthy, long’ → dalugašti- ‘lengthiness, length’, Jasanoff 2004: 148). Indirect evidence for this *$krud^h$-s-tó- may be derived from Old Norse *hraustr* ‘strong, valiant, doughty’, which certainly reflects a *χrausta- related to *χreuðanę (Hill 2003: 174), the lexical semantic basis being the connection between strength or vitality and preparedness, as evident in the relation between *rüstig* ‘healthy’ and *rüsten* ‘equip’, the reflex of a denominal to *χrusti- (cf. OE *hyrstan* ‘adorn’). Formally, *χrausta- ‘ready’ would thus reflect a *kro$\underset{\frown}{u}$$d^h$-s-to- interpretable as an adjective formed by insertion of *o*-grade into the root of *$krud^h$-s-tó- (cf. *b^ho$\underset{\frown}{i}$d-ro- ‘cutting’ [Goth. *baitrs* ‘bitter’] : *b^hid-ró- ‘id.’ [OE *bitter*, etc.]).

6.6 ***kré$\underset{\frown}{u}$$d^hh_1$-o/es-* ‘equipment, adornment’**

If by comparison we consider, for example, the relation between (1) the thematic adjective basic to the preform of OE *āst* f. ‘kiln’ (< PGmc. *aistō-) and (2) the *s*-stem that is ancestral to Vedic *édhas-* ‘fuel’ and basic to that thematic adjective (*h_2e$\underset{\frown}{i}$$d^h$-s-to- ‘hot’ ← *$h_2e\underset{\frown}{i}$$d^h$-es- ‘heat’, Nussbaum 1998: 527), it is appropriate to assert that the absolute zero-grade *$krud^h$-s-tó- ‘equipped, adorned’ (< *$krud^hh_1$-s-tó-)[11] was derived from an *s*-stem *kré$\underset{\frown}{u}$$d^hh_1$-o/es- ‘state of being equipped/adorned; equipment, adornment’, whose preform *kré$\underset{\frown}{u}$$h_2d^hh_1$-o/es- reconstructed above had presumably undergone Hackstein’s (2002) rule of laryngeal deletion (*CH.CC > *C.CC). The question that now confronts us is the relation between this *kré$\underset{\frown}{u}$$d^hh_1$-o/es- and the preform of Proto-Germanic *χreuðanę.

6.7 πλήθω (re-)revisited

This question is best addressed by considering the history of πλήθω ‘be/become full’, which LIV[2] 482–483 classifies as the reflex of a present in *-d^he/o-, namely

[11]For loss of *h_1 between obstruents, see Jasanoff 2003: 77.

**pléh$_1$-d^{h}e/o-*, also reflected in the diathetically divergent *frāda-* 'increase (tr. act./ intr. mid.)' of Young and Old Avestan. What needs to be fundamentally explained in an account of the origin of this suffixal material is not this divergence in relation to some original semantic value of that material, but how the preform of πληθ- and *frād-* came to behave derivationally as a root when we know it is not a root at the deepest reconstructible stratum. The primary root was of course **pleh$_1$-* 'fill', which, like **k̂el-* 'cover' or **kreh$_2$-* 'cover, heap', corresponded to an Activity Concept, rather than a Property Concept (cf. Nussbaum 2022: 208–209). As such, **pleh$_1$-* was eligible to form a root noun **pléh$_1$-/pl̥h$_1$-´* with the value of a *nomen actionis* 'filling'. Since, moreover, the specific lexical meaning of **pleh$_1$-* was of such a kind that the predication of **pl̥h$_1$-éh$_1$* 'with filling' could express a state of being filled and thus full (cf. 'with covering' ≈ 'covered'), the stative expression could be used in the causative periphrasis **pl̥h$_1$-éh$_1$ d^{h}eh$_1$-* 'provide with filling, get *x* full', on the basis of which was formed a compound **pl̥h$_1$-d^{h}h$_1$-ó-* 'filling; filled, full'. Like the resultative adjective **pl̥h$_1$-nó-* 'filled, full' (Ved. *pūrṇá-* 'full') beside **pélh$_1$-n-o/es-* 'fullness' (: *parīṇas-* 'fullness', Av. *parənah-vant-* 'copious', Rau 2009: 132, 174), **pl̥h$_1$-d^{h}h$_1$-ó-* 'filled, full', indirect evidence for which may be derived from the reflexes of nominals belonging to clearly de-thematic categories (e.g., *ā*-stem Locrian πληθα 'assembly, the commons' < **pléh$_1$-d^{h}h$_1$-e-h$_2$-* 'the full, fullness', *u*-stem Lat. *plēbēs* 'the commons' << **pléh$_1$-d^{h}h$_1$-u-*; v. Nussbaum 2014), appears to have been the base of an *s*-stem **pléh$_1$-d^{h}h$_1$-o/es-* 'fullness' ancestral to πλῆθος (cf. **frādah-* in Av. personal name *Daŋ́hu-frādah-* 'whose increase is of the land', Yt.13.116, Bartholomae 1904: 682).

6.8 **pléh$_1$-d^{h}h$_1$-o/es-* 'fullness' : **pléh$_1$-d^{h}h$_1$-e/o-* 'get full'

Since that *s*-stem was the substantival exponent of a Property Concept FULL, it could serve as a means of derivational inspiration on the model of (primary) adjectival roots interpretable as basic to *s*-stems and to prototypically radical *é*-grade simple thematic presents (e.g., **tep-* 'hot' → **tép-o/es-* 'heat' [> Ved. *tápas-* 'id.'] : **tép-e/o-* 'be/become hot' [Ved. *tápa-ti/te* 'is/becomes hot'], **g^{uh}er-* 'warm' → **g^{uh}ér-o/es-* 'warmth' [θέρος 'summer'] : **g^{uh}ér-e/o-* '(be) warm' [θέρομαι 'become warm'], etc.). While the ultimate origin of the latter category is mysterious,[12] Rau (2013: 263–265; this volume) has observed what appears to be an archaic correlation of active morphology and intransitive semantics, which may point to the frequent – though not necessarily exclusive – anticausative use of the proto-middle. There may have been a degree of lability comparable to the

[12] On the thematization of the proto-middle, see Jasanoff 2023.

(anti)causative use of PDE *get* (e.g., *get* (x) *warm*; cf. PDE *become* : NHG *bekommen* ‘get’). In any case, given the cooccurrence of such Calandish derivatives, namely radical *é*-grade *s*-stem substantives and largely anticausative radical *é*-grade simple thematic presents, it is highly likely that **pléh₁-dʰh₁-o/es-* ‘fullness’ invited the suffixally substitutive formation of a **pléh₁-dʰh₁-e/o-* ‘get full’, whose Greek reflex is active and intransitive and whose Iranian one either happened to be active and transitive or was an original anticausative active remodeled as a middle corresponding to a causative active backformed from that middle in a manner suggested by Rau (2009: 155). However it might have been, the fundamental idea is that, for the purpose of derivation, the sequence **pleh₁dʰh₁-* was treated as a root with the meaning ‘filled, full’.

6.9 **-dʰh₁e/o-* and **-éh₁i̯e/o-* compared

Such an account is not designed to deny the existence of *-*dʰe/o*-presents with intransitive semantics. Indeed, Greek evidence makes it very clear that such a formant existed (φλεγέθω ‘burn’, φθινύθω ‘perish’, βαρύθω ‘am heavy’, φαέθω ‘shine’, μινύθω ‘diminish’; see Magni 2008: 268; Rothstein-Dowden 2022: 18–24). What may be suggested concerning the origin of the category of *-*dʰe/o*-presents is an inference based on cases like **pléh₁-dʰh₁-e/o-* ‘get full’, where the “root-based” formation from a base ending in *-*dʰh₁*- was still analyzable with a morphological boundary before *-*dʰh₁*-. Thus, a reanalyzed *-*dʰh₁e/o*- could even be used to form primary verbal stems with intransitive meaning (e.g., **i̯eh₂*- : OCS *jadǫ* ‘go’ : Lith. *jóju*, *jóti* ‘id.’, **pelh₂*- : πελάθω/πλᾱ́θω ‘approach’ : πελάζω ‘id.’). In this respect, a largely anticausative *-*dʰh₁e/o*- would, in terms of the history of the possessive adnominal system, intimately resemble the stative *-*éh₁i̯e/o*-, which originated in the addition of *-*i̯e/o*- to the instrumental of root noun abstracts used in nominal sentences (Jasanoff 2004: 147). Since these instrumental root nouns were usually predicated of themes or non-agents generally (e.g., **pédom tr̥séh₁* ‘the ground is dry’, **h₃rḗĝōn h₂n̥réh₁* ‘the king is strong’, etc.),[13] the historically complex suffix has a stative or anticausative value. Though in principle a **h₁rudʰ-éh₁-i̯e/o-* might have been a causative (i.e., factitive) ‘redden (*x*)’ like other adnominally based *-*i̯e/o*-presents, such as what are arguably the preforms of κρύπτω and πλάσσω, the fact that it meant ‘be/become red’ is plausibly attributed to the use of **h₁rudʰ-éh₁* ‘with redness’ in stative predication of themes.[14] Like hypostatic *-*ó*- and *-*i̯i̯o*- in nominals, *-*i̯e/o*- could, as one of its functions,

[13] I use “theme” to refer an argument of which a property is predicated.

[14] For a formal analysis of the development of *-*eh₁*- into a verbal stem formant, see Grestenberger 2023: 15–23.

merely render an already inflected substantive – and thus uninflectable adnominal – inflectable. Once a unitary **-éh₁i̯e/o-* ‘be/become (x)’ emerged, it could be used in primary derivation (e.g., Lat. *iaceō* ‘lie’ : *iaciō* ‘throw’). A comparable development would be the case for **-d^h h_1 e/o-*, whose denominal origin from compounds based on the causative **d^h eh_1-* periphrasis would follow from the greater importance and thus frequency of mentioning agents in causation, and whose largely intransitive semantics would be a consequence of the tendency for verbal adjectives to group patiency and resultativity together for the sake of characterization (Haspelmath 1994). Because verbal adjectives in **-d^h h_1-ó-* were largely statives of patient/theme-oriented and resultative sense (**pl̥h_1-d^h h_1-ó-* ‘filled’), the adjectival abstracts formed from them on the basis of their resultative stative meaning (or secondary simple stative sense) could inspire the creation of largely intransitive, radical *é*-grade “simple” thematic presents formed from roots (or derivationally root-like bases) basic to Caland systems. Accordingly, the largely intransitive value of the suffix **-d^h h_1 e/o-* would be an inheritance from the formation of which its immediate ancestor **R(é)(-)d^h h_1-e/o-* was a subtype: radical *é*-grade simple thematic presents (**R(é)-e/o-*).

6.10 **kréu̯d^h h_1-e/o-* ‘render equipped, adorned’

For our purpose, the question is how the **pléh_1(-)d^h h_1-e/o-* ancestral to **pléh_1-d^h h_1 e/o-* was formed. By appeal to an *s*-stem **pléh_1(-)d^h h_1-o/es-* ‘fullness’ interpretable as derivative of a root-like base **pleh_1(-)d^h h_1-* of Property Conceptual sense, the *s*-stem could easily have inspired a verbal formation in **-e/o-* (e.g., **g^{u̯h}er-* ‘warm’ → **g^{u̯h}ér-o/es-* ‘warmth’ : **g^{u̯h}ér-e/o-* ‘(be) warm’ :: **pléh_1(-)d^h h_1-o/es-* ‘fullness’ : x = **pléh_1(-)d^h h_1-e/o-*; thus **pleh_1(-)d^h h_1-* ‘full’ → **pléh_1(-)d^h h_1-e/o-* ‘get full’). In the same way, I argue, our **kréu̯d^h h_1-o/es-* ‘state of being equipped, adorned, equipment, adornment’, interpretable as derivative of a quasi-Property Conceptual **kreu̯d^h h_1-* ‘equipped, adorned’, inspired the formation of a **kréu̯d^h h_1-e/o-* ‘render equipped, adorned’ whose transitivity would be the result of whatever underlies the transitivity of the Avestan reflex of **pléh_1(-)d^h h_1-e/o-* ‘get full’ (> *frāda-* ‘increase [tr. act./intr. mid.]’) and whose Proto-Germanic reflex **χreuðanq̨* (> OE *hréodan** ‘deck, adorn’) consequently came to mean ‘equip, adorn’.

7 Conclusion

The main purpose of this contribution has been the argument that, most notably among a group of formally and semantically similar verbal derivatives, Greek κρύπτω ‘cover, hide, bury’ and Old English *hréodan** ‘deck, adorn’ are not only related in terms of root-etymology, but also paradigmatic of the two ways in which a verb can be de-adjectival: stem-derived (κρύπτω) vs. root-derived (*hréodan**). While I have in general focused on the morphological and derivational semantic foundations underlying the formation of the etyma of both verbs and their relatives, it has been my particular intention to describe the conceptual and lexical semantic basis for a type of root-based derivation that in IE terms may be designated as “de-adjectival”, namely Caland system verbal derivation (see the introduction to this volume). The formal prerequisite to this description has been the identification of the origin of the roots ancestral to those of κρύπτω and *hréodan** in nominal derivation and the predicative instrumental (anti)causative periphrasis – a type of construction closely, but not exclusively, associated with the Caland system and the actional expression of Property Concepts.

Despite the subsequent importance of Property Conceptuality in this contribution, I have argued that the primary root underlying all the forms compared was in lexical semantic terms a verbal one **kreh₂*- ‘heap, strew, cover’ (Latvian *krãju*, *krât* ‘put aside, accumulate, amass’) corresponding to an Activity Concept and basic to a *u*-stem verbal noun **kró/éh₂-u*- ‘heap, covering’ whose instrumental was employed with anticausative **bʰuhₓ*- ‘become’ and causative **dʰeh₁*- ‘put, make’. While the expression of an action involving this substantive as a possessum was in the present system handled by means of a derivative in *-*i̯e/o*- (**kreh₂-u-i̯e/o*- ‘provide with covering’), whose reanalysis as a primary verb gave rise to a historically secondary verbal **kreu̯h₂*- (OCS *kryti* ‘cover’), constructions with the light verbs **dʰeh₁*- and **bʰuhₓ*- were required elsewhere. I have argued that **kr̥h₂-u-bʰ(u̯)-ó*- ‘heaped, covered’ (>> **kruh₂(-)bʰ-ó*-), a thematic compound formed on the basis of the latter construction, gave rise to the secondary verbal root **kreu̯bʰ*- (Tocharian B *kraup*- ‘gather, amass’) and that the origin of this root lies in stem-based de-adjectival derivation. In a manner comparable to the use of *-*i̯e/o*- to represent the functionally adnominal **kréh₂-u-h₁* ‘with covering’ in the derivative **kreh₂-u-i̯e/o*- ‘provide with covering’, the inflectable adnominal **kruh₂(-)bʰ-ó*- ‘covered’ was the base of a **kruh₂bʰ-i̯e/o*- ‘render covered, heap (upon)’ whose reflex **krubʰ-i̯e/o*- was the preform κρύπτω.

The crucial contrast in terms of de-adjectival derivation is with the preform of *hréodan** ‘deck, adorn’, which arguably originated in a similar way, namely, from a compound **kr̥h₂-u-dʰh₁-ó*- ‘covered’ (>> **kruh₂(-)dʰh₁-ó*-) formed on the basis of the causative periphrasis **kréh₂-u-h₁ dʰeh₁*- ‘provide with covering’, but whose ancestral root’s meaning emerged as a result of a lexical semantic development that endowed the resultative verbal adjective **kruh₂(-)dʰh₁-ó*- ‘covered’ with the capacity to express an idea in the vicinity of a Property Concept, i.e., 'equipped, ready' or 'adorned'. This adjacency to Property Conceptuality was the precondition for the emergence of Calandish derivational behavior. Once **kruh₂(-)dʰh₁-ó-* 'equipped, adorned' became the base of an *s*-stem substantive **kréu̯h₂(-)dʰh₁-o/es-* 'state of being equipped/adorned', the resulting **kréu̯dʰh₁-o/es-* 'equipment, adornment', now analyzed as a primary derivative, implied the existence of a derivative belonging to a verbal stem type closely associated with the Caland system: the radical *é*-grade simple thematic present (**kréu̯dʰh₁-e/o-* 'render equipped/adorned; provide with equipment/adornment'). In sum, the preforms of both Greek κρύπτω 'cover' and Proto-Germanic **χreuðaną* 'deck, adorn' not only exemplify the complexity of de-adjectival derivation, but also offer some insight into the ways in which morphology and derivational semantics interact with our conceptual world.

Abbreviations

act.	active	OE	Old English
Alb.	Albanian	OCS	Old Church Slavonic
Av.	Avestan	OE	Old English
f.	feminine	OHG	Old High German
Hitt.	Hittite	OIr.	Old Irish
Hom.	Homeric	OLith.	Old Lithuanian
Hsch.	Hesychius	ON	Old Norse
IE	Indo-European	PC	Property Concept
intr.	intransitive	PDE	Present-day English
Lat.	Latin	PGmc.	Proto-Germanic
Latv.	Latvian	PGk.	Proto-Greek
Lith.	Lithuanian	PIE	Proto-Indo-European
m.	masculine	TA	Tocharian A
mid.	middle	TB	Tocharian B
n.	neuter	tr.	transitive
NHG	New High German	Ved.	Vedic
OCS	Old Church Slavonic	Yt.	Yašt

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Chapter 8

Reassessing the emergence of the PIE *-*eh*$_2$-factitive present type

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The PIE *-*eh*$_2$-factitive present type is currently known under the canonical instance of ***néu̯-eh*$_2$- 'to make new' (Hitt. *nēwaḫḫ-*, Lat. *(re)nouāre*, etc.), as based on the thematic adjective ***néu̯-o-* 'new'. The type has reflexes in Latin, Greek, Germanic, Balto-Slavic, Celtic, and possibly Tocharian, mostly in the thematic form *-*eh*$_2$*-i̯e/o-* (like denominatives from abstract *-*eh*$_2$-stems). The Anatolian facts have confirmed that this formation was originally athematic and inflected according to the **h*$_2$*e*-conjugation, reflected by the *ḫi*-conjugation of the Hitt. *nēwaḫḫ*-type, while the other Anatolian languages (Luwian, Lycian, Lydian) adopted the *mi*-conjugation of the comparable types. According to the revision of the data by Sasseville (2021), the original function of this type in Anatolian was the derivation of denominal verbs from substantives, which was then extended to adjectives of various sorts. Several instances conclusively point to -*ah*$_2$-abstracts as the basis, which raises the issue of the so-called "conversion" of a noun into an athematic present as the original PIE derivational process. As a preliminary investigation, the paper reviews other alleged reflexes of this process in other present types, which seem to point to denominative formations without apparent suffix, alternatively with zero suffix, for instance PIE **g*$^{u̯}$*ih*$_3$*u̯é/ó-* 'to live', vis-à-vis the PIE adjective **g*$^{u̯}$*ih*$_3$*u̯ó-* 'living'. It will be shown that these apparent cases of conversion of a noun into a verb are illusory and result from analogical developments, which presuppose the PIE existence of the "classical" full grade thematic present type, to wit **b*^{h}*ér-e/o-*.

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1 Vedic *rāj-* 'to rule, be king'

1.1 Verb and corresponding nouns with a lengthened grade

The Vedic verb *rāj-* (RV +) is currently glossed by an array of notions which leaves as somewhat uncertain the basic meaning, either 'to rule, reign', or 'to shine, beam', compare 'herrschen, König sein, strahlen, glänzen' (EWAia II 444). This semantic ambiguity is actually linked to derivational and etymological issues. The corresponding root noun *rā́j-* clearly means 'king, ruler'. It is rare as independent noun (RV 3x), while being mostly found as second compound member, in *sam-rā́j-*, masc. (RV 38x, fem. *sam-rā́jñī-* 4x), *vi-rā́j-* (7x), *sva-rā́j-* (20x), *jyeṣṭha-rā́j-* (2x), *eka-rā́j-* (1x), *apna-rā́j-* (1x), *vane-rā́j-* (1x), likewise in Old Iranian, see Av. *bərəzi-rāz-*.[1] Those compounds belong to the type of verbal governing compound with root nouns as second member. The normal word for 'king, ruler', is *rā́jan-*, masc. (RV 256x), nom.sg. *rā́jā*, acc. *rā́jānam*, gen. *rā́jñas*, dat. *rā́jñe*, nom.pl. *rā́jānas*, etc. If one adopts the current reconstruction of the root, this item can be traced back to a noun **h_3rḗĝ-on-*, with the suffix suggesting AK inflection.[2] At face value, it looks like an individualizing derivative (*-*on-*/-*n-*) besides **h_3rḗĝ-*, which would be an abstract meaning 'ruling, rulership'. This root noun is effectively attested as animate, and masculine, nom.sg. **h_3rḗĝ-s*, which is reflected by RV *rā́ṭ*, matching Lat. *rēx*, OIr. *rí*, Gaul. *-rix*, but as animate, it could also be feminine, without specific marking of the gender, as testified by RV 5.46.8b. According to Weiss (2017), PIE **h_3rḗĝ-* may have originally been a neuter root noun at the time when PIE had no feminine, meaning 'ruling, directing'. The basis of the nasal stem was the locative sg. of this root noun: **h_3rēĝ-én* 'in the power' (> Ved. *rāján-i*, RV 10.49.4c), from which the heteroclitic neuter stem found in Avestan, *rāzarə*, gen.sg. *rāzəṇg*, instr.sg. *rašnā*, gen.pl. *rāšnąm* was backformed. The delocatival animate yielded the nasal stem **h_3rḗĝ-on-* masc. 'being in power' > 'ruler', hence Ved. *rā́jan-*. The *-n-*stem abstract 'rulership' is also the basis of **h_3rḗĝ-n-ih_2* fem. possessive 'having rulership' (female) > 'queen', cf. Ved. *rā́jñī-*, OIr. *rigain*, Lat. *rēgīna* < **rēgnīnā* (as per Pinault 2014: 293–295).[3]

This reconstruction leaves intact the fact of the apparent coexistence of the root noun Ved. *rā́j-* and of an athematic root present of the verb *rāj-*, as reflected by Ved. (RV) 3sg.act. *rā́ṣṭi*, which is unexpected from a synchronic point of view.

[1] Survey of the RVic material by Scarlata (1999: 446–451).

[2] Henceforth, I use the following abbreviations for the standardly assumed PIE inflectional types: AS = acrostatic, PK = proterokinetic, HK = hysterokinetic, AK = amphikinetic, see Widmer (2004), Fortson (2010).

[3] Adopted with some minor differences by Weiss (2017: 796–797, 799).

Projected back into PIE, the verb *_h_3rḗĝ-ti_ has been interpreted as a denominative verb formed without suffix from the noun *_h_3rḗĝ-_ because of the meaning of the former: ‘to rule’ = ‘to be king’ (s. KEWA III: 56 and EWAia II 444 with refs.). This is of course linked to the etymology of ‘king’ itself, which connects this noun to the PIE root *_$h_3$reĝ-_ (LIV^2 304–305) ‘gerade richten, ausstrecken’.[4]

In Indo-Iranian, only Ancient Vedic has the root athematic present _rā́ṣṭi_. The root thematic present is much better attested: RV _rā́jati_, matched by Av. _°rāzaiti_, cf. _vī-rāzaiti_ ‘rules’ Yt.14.47 < Indo-Iranian *_rā́ja-ti_. In Iranian the long vowel of the root has been extended to the verbal adjective, Av. _rāšta-_ ‘directed, arranged’, OP _rāšta-_ ‘straight’, MP _rāšt_ ‘true’, etc.[5] In the RV, the root thematic present _rā́jati_ is the prevalent verbal stem (49x), subjunctive (2x), participle active _rā́jant-_ (8x), beside inf. _rājáse_ (2x). There are only a few forms of other stems, aor.act.3pl. _arājiṣuḥ_ (1x),[6] intensive pres.ptcp.act. _rā́rajat_ (1x). Projected back into PIE, this would point to a present *_h_3rḗĝ-e-ti_, besides *_$h_3$réĝ-e-ti_, the normal root thematic present, reflected by Gk. ὀρέγω, Lat. _regō_, _-ere_, OIr. _rigid_, Go. _rikan_. The whole issue has been treated by Strunk (1987), with further literature. Strunk’s own solution (1987: 389–390) is as follows: The root *_$h_3$reĝ-_ had an acrostatic (Narten) root present with a 3sg.act. indicative *_h_3rḗĝ-ti_, subjunctive *_$h_3$réĝ-e-t(i)_. The thematic root present found in most languages reflects this subjunctive. Indo-Iranian *_rā́ja-ti_ is explainable by generalization of the length of the root vowel to the subjunctive, and then to the injunctive and indicative. However, the RV subjunctive 1sg.act. _vi-rā́jāni_ (10.159.6d = 10.174.5d, in relatively late hymns) constitutes poor evidence for a subjunctive of the athematic root present because it is most probably based on the thematic stem _rā́ja-_ which is widespread already in the RV. As far as the verb is concerned, this doctrine has been endorsed by LIV^2 305. As a minor and feeble alternative, the long grade of the verb would have its source in the nominal basis. Apparently, Strunk (1987) took the long vowel grade of the Narten present and of the root noun ‘king’ as a coincidence, even though they are from the same root. Schindler’s oral communication, mentioned in a footnote (1987: 390, fn. 3), would point, with question mark, to a root noun with acrostatic root ablaut, *_rḗĝ-_/*_réĝ-_ (according to the notation used by

[4] See Adams in Mallory & Adams 1997: 329–331, with previous literature. I leave aside the speculative and unsuccessful attempts to derive *_h_3rḗĝ-_ ‘king’ from a different root such as *_reh_1ĝ-_ or *_$h_x$reh_1ĝ-_ (more specifically *_$h_2$reh_1ĝ-_ through the connection with Gk. ἀρήγειν ‘to help’, ἀρήγων ‘helper’) by several scholars in the past; see the literature in KEWA III 50 and EWAia II 445–446. In the Indian tradition itself, Skt. _rā́jan-_ has also been connected to roots other than _rāj-_, see Hara 1969.

[5] See further Cheung 2007: 196–198, under *_Hraz-_ ‘to draw a line, to direct’.

[6] See Narten 1964: 221–222, under _rāj-_ ‘glänzen’.

Strunk), with further generalization of the lengthened grade, while the expected alternation is found in the heteroclitic neuter noun, OAv. *rāzarə* 'statute, order', besides instr.sg. *rašnā* (already quoted above) to wit **rḗĝ-r̥/*réĝ-n-*, comparing (implicitly) the word for 'liver'. This idea was later adopted in the framework of "Narten systems" favored by Schindler in the late period of his research and teaching in Vienna. Schindler assumed that some PIE roots displayed "Narten" behavior (i.e., lengthened grade of the root in place of normal full grade **e* in the strong stem, besides full grade in place of expected zero grade), both in their verbal and in their nominal forms. This is reflected, for instance, in the treatment of the derivatives of the root **(h₃)reĝ-* 'lenken, herrschen' by Widmer (1997: 29–31). This included the reconstruction of the root noun as an original abstract noun and of the heteroclitic neuter as **(h₃)rḗĝ-r̥/*(h₃)réĝ-n-*. In this discussion, the formation of the root present Ved. *rā́ṣṭi* was not addressed, however, possibly because it was considered as marginal. The issue is not substantially modified by the reconstruction of the root as **h₃reĝ-*, which imposed itself in the meantime.

1.2 Explanation of Ved. *rā́ṣṭi* 'reigns'

The coexistence of Ved. *rā́ṣṭi*, *rā́jati* 'rules, reigns' and of the noun (°)*rā́j-* 'ruler' 'ruler' is puzzling because these verbs cannot be considered denominative presents in the normal sense. One would not deny that the prevalent use of RV *rā́ṣṭi* and *rā́jati* 'to rule', i.e., 'to be king' has been influenced by the meaning of (°)*rā́j-* 'ruler', independent of whatever the original meaning of the root *rāj-* was. An alternative strategy, which was favored in the early stages of Sanskrit lexicography, is to set up two roots on the basis of the translations of the occurrences in the most ancient text, the collection of hymns of the RV. There is however no need, *pace* Grassmann (1996: 1156), to set up two homophonous verbal roots: 1. glänzen, strahlen ('shine, be bright') 2. herrschen, gebieten über ('rule (over)', governing the genitive). Both verbs would share the same present formation *rā́jati*, ptcp. *rā́jant-*, so that this solution seems uneconomic. In most cases, the meaning 'to shine, be bright' is only inferred from the context, because the agent is a shining god (Agni, fire) or object (sun, liquid). The separation of these two meanings is actually superfluous, as can be shown by checking the majority of standard translations of the RV, which consistently situate the understanding of *rāj-* under the notions of ruling and kingship of the agent of this verb.[7] In the most recent translation of the RV by Jamison & Brereton (2014), one finds 'rule (over)', 'reign' in most occurrences. Alternative possibilities are considered in

[7]See Geldner 1951 and Renou, EVP.

a minority of instances, and mostly between brackets: 'rule [/shine]' in 3.56.8b, 8.7.1c, 8.19.31d, 9.75.3d, 10.170.1d, 'reign [shine forth]' in 5.55.2b, 'shine [/rule]' in 8.60.15d, 'shine forth' in 8.14.10c (for the drinks of *soma,* subject of the aorist *arājiṣuḥ*, hapax, with preverb *ví*), 9.5.1b, 9.5.3b, 9.61.18b, 10.189.3a, 'radiates' in 9.71.7d. This survey shows that one cannot absolutely exclude the interpretation of several occurrences of *rāj-* as connected to the notion of shining (up) or beaming. This should not be taken to entail that 'king' itself (*rā́j-*, *rā́jan-*) originally meant 'the bright one', following a by now obsolete interpretation. The meaning 'king, ruler' is anchored in the inherited lexicon of Indo-Aryan and is supported without any doubt by the external comparanda. However, the semantic link between PIE *h_3*rḗĝ-* and the root *h_3*reĝ-*, for which there have been somewhat different approaches, poses yet another issue which would lead us beyond the limits of the present paper. For the purposes of my argument, it is sufficient to recall that this etymological connection is accepted by a wide majority of scholars.[8] The semantic link from 'stretch, direct' to 'rule' is certainly inherited. This was probably based on the concept of the ruler whose power and control covers a vast domain, his realm, and extends in several directions. In Ancient Vedic, the transfer 'ruling in every direction' > 'shining forth' is due to the metaphorical association of the glory and supreme power of the king with the brightness of the sun and the radiance of the weapons of the warriors. In this area of expressions of symbolic notions, one may also note the favored use of the preverbs *ví* 'apart, asunder, in different directions' and *sám* 'together' with the verb *rāj-* and with its nominal derivatives. For the occurrences in the RV where the reference to 'shining' seems to be more convenient and more exact than 'ruling over', one may also invoke some rhyming effect with the verb *bhrāj-* 'to shine, beam, glitter'[9] in some cases, which also forms a thematic root present, RV middle *bhrā́jate*, which is par excellence the verb used to describe the shining of the fire god Agni. In a somewhat conservative way, Gotō (1987: 267–271) sets up two verbs (i.e., two thematic root presents) *rā́jati* (1. strahlen, 2. herrschen), basically following Grassmann, but admits that the two meanings are difficult to discriminate in most instances. Moreover, there is no link between these two meanings and different verbal formations nor etymologies. The whole picture is murky.[10]

Returning now to the problem of derivation, the issue of *rā́ṣṭi* and *rā́jati* as anomalous denominatives is actually a pseudo-problem. The basis for the athe-

[8]As an aside, it should be recalled that the Anatolian data show that this etymon is limited to the languages of Core IE, so that *h_3*rḗĝ-* cannot be deemed any longer as the unique and prototypical noun for 'king' in PIE.

[9]EWAia II 279–280; PIE root *b^h*re*h_1*ĝ-* in LIV[2] 92.

[10]See also the discussion by Scarlata (1999: 445–446).

matic root present is small indeed, since it its only found in two occurrences in the 3sg. active, once in the indicative, and once in the injunctive:

(1) RV 1.104.4b
rā́ṣṭi _śū́raḥ_
reign.3SG.PRS.IND.ACT champion.NOM.SG.M
'The champion reigns.'

(2) RV 6.12.5cd
sadyó _yáḥ_ _syandró_ _víṣito_
immediately who.NOM.SG.M hasty.NOM.SG.M loosened.NOM.SG.M
dhávīyān | _ṛṇó_ _ná_ _tāyúr_ _áti_
hurrying.COMP.NOM.SG.M debtor.NOM.SG.M like thief.NOM.SG.M across
dhán(u)vā _rāṭ_ ||
wasteland.ACC.PL.N reign.3SG.PRS.INJ.ACT
'He [Agni] who immediately, when unloosened, streams ever faster, like a debtor (turned) thief he has headed straight across the waste places.' (transl. Jamison & Brereton 2014: 786).

By chance (or by a single poet's choice?), the same hymn features one of the few (three in total) occurrences of the nom.sg. _rā́ṭ_, 6.12.1a, at the same place in the pāda. Besides _rā́ṭ_, the simplex noun _rā́j-_ has no other forms in the RV. Therefore, it seems better to explain this root athematic present as an innovation within Ancient Vedic, which remained unsuccessful compared to _rā́jati_, inherited from Indo-Iranian *_rā́ja-ti_. One should find a way to explain both presents, without resorting to the notion of denominative formations. My scenario is as follows. The Indo-Iranian thematic present was inherited parallel to the formations attested in other IE languages and originally with normal full grade of the root: *_rája-ti_ < *_$h_3$réĝ-e-ti_, ptcp. *_rájant-_ < *_$h_3$réĝ-o-nt-_. Those forms were remade as *_rā́ja-ti_, ptcp. *_rā́jant-_ under the influence of the nasal stem *_rā́jan-_ < *_h_3rḗĝ-on-_, and of the second compound member °_rā́j-_ < *°_h_3rḗĝ-_. The two latter items had been independently inherited. The nasal stem was derived via a delocatival derivation as outlined above and became firmly entrenched as the designation of the ruler at some stage. The process of reshaping the verbal stem hinges on two crucial points. The first point is the equivalence of the second member of a verbal governing compound with a participle and an agent noun, to wit 'ruling over'. The second point is the crucial role of the present participle as indicative of the structure of the corresponding present stem. This will also be relevant in the interpretation of the thematic present *_$g^{u̯}ih_3u̯é$-ti_, which we will discuss in the second section of this paper.

The king's function was correlated with the territory where his power was in force and where the clans (Ved. *víś-*) under his authority reside, live and prosper in peace, a notion covered in Ancient Vedic by the root *kṣay-/kṣi-* 'to dwell' and its derivatives, see especially *kṣéma-* masc. 'peace, peaceful dwelling'. The Vedic term *rāṣṭrá-* nt. (RV 10x) 'realm, kingdom' (basis of *rā́ṣṭrī-* 'ruler' 3x) has been formed after the model of *kṣétra-* nt. 'field, ground, country' (RV 20x, plus in several compounds), the latter being inherited, cf. OAv. *šōiθra-* 'district, country'. The phrase RV *kṣétrasya páti-* is matched by Av. *šōiθrahe paiti-* 'lord, master of the field' = 'king'. This lexical configuration yielded the causal force for the association between noun and verb. The verb *kṣay-/kṣi-* has a root athematic present inherited from Indo-Iranian and from PIE, namely RV 3sg.act. *kṣéti*, 3pl. *kṣiyánti*, etc. (LIV^2 643, root **tk̂ei̯-*). Based on these inherited lexemes, a proportion took shape: noun *kṣétra-* : root present *kṣéti* :: *rāṣṭrá-* : X = *rā́ṣṭi*.

A proportional analogy such as this suffices to account for the root athematic present and de facto eliminates one of the favorite instances of an alleged denominative verb without suffix, i.e., of conversion of a noun into a verb. One further advantage of the whole solution is to dispense with the otherwise interesting idea of a "Narten" root in the case of the root **h$_3$reĝ-*. I may recall from personal recollection that Schindler was forcefully opposed to the idea of denominative verbs without marking of derivation by a suffix. The preceding paragraphs should be understood as a first modest answer to this recurring problem.

2 The PIE thematic present **g$^{u̯}$ih$_3$u̯é-ti*

2.1 A PIE *-*u*-present

A PIE thematic present can be safely reconstructed on the basis of several IE languages: Ved. *jī́vati*, YAv. *juuaiti*, Lat. *uīuit*, OPr. *giwa*, Latv. *dzîvu*, OCS *živǫ*. As for the root grade, Toch. B *śāw-*, stem CToch. **śāw'ä-* > Toch. B **śāyä-*, pres. act. 3sg. *śaiṃ* (besides CToch. **śāwæ-* in Toch. A act. 3pl. *śāweñc*) is ambiguous, since **śāw-* may reflect **g$^{u̯}$i̯oh$_3$u̯-* as well as **g$^{u̯}$ih$_3$u̯-*.[11] Gk. ζώω has probably been reshaped with full grade on the basis of the root athematic aorist 3sg.act. **g$^{u̯}$(i)i̯éh$_3$-t* (cf. Gk. ἐβίων). In Vedic, the retraction of the accent to the root was due to the restructuring of the neo-root *jīv-* as a plain thematic present.

The mainstream account (e.g., LIV^2 215–216, Jasanoff 2003: 142) assumes a PIE thematic present which originated as a thematization of an athematic *-*u*-present **g$^{u̯}$i̯éh$_3$-u-*/**g$^{u̯}$ih$_3$-u-´*. This type followed the **h$_2$e*-conjugation and hence had a

[11] See Malzahn 2010: 916–919 with previous literature.

3sg. *$g^{u̯}$i̯éh$_3$-u̯-e that was the nucleus of the later thematic paradigm (Jasanoff 2003, 2023). To this scenario one can object that there are no reflexes of an athematic *-*u*-present from the root *$g^{u̯}$i̯eh$_3$-, contrary to the reflexes of other verbs: **terh$_2$-u-* 'overcome, cross over' (Hitt. *tarḫu-*, 3sg.act. *taruḫzi*, *tarḫuzzi*, Ved. mid. *tarute*, thematic Ved. *tū́rvati*, YAv. *tauruua-*), **leh$_2$-u-* 'pour' (or **leh$_3$-u-*, as per Melchert 2011), Hitt. *laḫu-* (3sg.mid. OH *laḫuwāri*, ptcp. *laḫu(w)ānt-*), **u̯er-u-* 'ward off', **u̯el-u-* 'turn', etc.[12]

2.2 Questioning the theory of conversion

An alternative theory that was entertained for a long time is that *$g^{u̯}$*ih$_3$u̯é-ti* was a denominative present based on the adjective *$g^{u̯}$*ih$_3$u̯ó-s* (nom.sg.masc.) reconstructed for Core IE as well: Ved. *jīvá-*, Av. *juua-*, Lat. *uīuus*, OIr. *biu*, Go. *qius*, Lith. *gývas*, Latv. *dzîvs*, OCS *živŭ*.[13] The infinitive Lat. *uīuere*, Ved. *jīváse* reflects a stem *$g^{u̯}$*ih$_3$-u̯e/os-* (underlying also Toch. B *śaiṣṣe* 'world', A *śoṣi* 'people')[14] of the abstract based on this adjective.

This pair of adjective and verb has been put into service by Villanueva Svensson (2021: 273–277), claiming that the thematic present issued from the thematic adjective by a process of "conversion", which may have been a regular derivational process at an early PIE stage:

(3) adj. *$g^{u̯}$*ih$_3$-u̯é/ó-* ⟶ verb 3sg. *$g^{u̯}$*ih$_3$-u̯e-t(i)*, 3pl. $g^{u̯}$*ih$_3$-u̯o-nt(i)*.

According to Villanueva Svensson, this pair is but a peculiar instance, which has been exceptionally preserved in Core IE, of the derivation by conversion of a thematic present from a thematic adjective (2021: 273–277), which was a purely nominal formation, not based on an alleged, but actually non-existent *-*u*-present (with strong stem *$g^{u̯}$*i̯éh$_3$-u-*). This pair reflected a standard process which is hypothetically assumed for early Core IE, through which the simple thematic present of the type **b^{h}ér-e/o-* 'to carry' was derived from a reconstructed verbal adjective **b^{h}er-ó-* 'carrying' (Villanueva Svensson 2021: 277–284). Consequently, the whole root thematic present type emerged *after* the separation of Anatolian from PIE and became widely productive in Core IE. The latter point is the major claim of his paper, because Villanueva Svensson dismisses previous theories

[12] See Jasanoff 2003: 141–143; 2023: 63–66 for these examples and a few others. *Pace* Jasanoff 2023: 66, Toch. B *śaumo* 'man, human' (A *śom* 'boy') cannot be projected back to PIE *$g^{u̯}$*i̯éh$_3$u-mō(n)* based on the suffixed root. This item ought to be explained as created at the CToch. stage, as per Pinault 2021: 119–120.

[13] See Tichy 2004: 55, with part of the previous literature.

[14] As per Pinault 2008: 234, 2021: 117.

about the place of the deverbative type *$b^hér$-e/o- in the PIE verbal system (2021: 264–273).

The same process is then found in other pairs consisting of a thematic adjective and a thematic root present, e.g. Gk. λευκός 'white' (*leu̯k-ó-): Ved. *rócate* 'shines' (*léu̯k-e-to+i*), yielding the thematic root present through additional adjustments of accentuation to obtain an intermediate *léu̯k-o- 'shining' that could serve as the basis for conversion. However, Villanueva Svensson fails to account for the full grade and the root accentuation of the *$b^hér$-e/o- present type as well as the typical *-e/o-alternation of the thematic vowel in the inflection, except through circular and *ad hoc* hypotheses (such as assuming that the original accentuation of the adjective *leu̯k-ó- > Gk. λευκός was **léu̯k-o-, just for complying with the accent of the present stem *léu̯k-e/o-, cf. Villanueva Svensson 2021: 281–282, and fn. 43).

An earlier variant of the conversion account for the verb 'living' has been formulated by Tichy (2004: 55) plainly as follows: "3.Sg. Ind. Präs. uridg. *$g^u ih_3$-u̯e-ti 'ist *$g^u ih_3$u̯ó-, lebt'." This is however a mere pedagogical paraphrase instead of an explanation, which leaves out the real nominal form that could be used as a predicate in a nominal sentence, the nom.sg. *$g^u ih_3$u̯ó-s, in favor of the transposition of the thematic stem. Tichy quotes Rix (1994: 79), who pushes back the phenomenon into PIE: "hocharchaische Denominativbildung mit Nullsuffix".

The basic problem is: What does "conversion" really mean? In other words, how would it work in syntactic and morphological terms? Does it imply that PIE was not a fully-fledged inflectional language? Under this account, there is no clearcut link through reinterpretation of a predicate thematic stem, with nominal endings, nom.sg. *°o-s (animate) or *°o-m (neuter) as verbal 3sg.act. *-e (active or proto-middle; active *°e-t, middle *°e-to (injunctive). How could this gap be bridged? This cannot be understood remotely in the spirit of "Watkins' Law" (Watkins 1969), according to which a form *$b^hér$-e with "proto-middle" 3sg. ending was reinterpreted as a thematic stem with zero ending, triggering the generalization of the stem *$b^hér$-e-.[15]

The *°e form of the thematic vowel did not surface in the nominal inflection except for the vocative and possibly in some adverbial forms. Nobody has claimed that the thematic inflection started from the imperative, while taking for granted that the imperative with zero ending in the second singular active could be akin to a vocative (and vice versa). Nor is there any basis for setting up a predicative adverb **b^her-é 'while carrying', which is mentioned here only for the sake of the

[15]See Jasanoff 2003: 224–227. Watkins' "law" was used as powerful tool for interpreting various verbal formations, see Joseph 1980 and Janse 2009 with further references.

argument. Therefore, it appears that the theory of conversion cannot be pushed very far, since it entails many additional speculations.

Jasanoff (2023: 74–77), on the other hand, argues that PIE did possess the full grade thematic present type, which did not arise through conversion but through a process of reanalysis of the "proto-thematic" 3sg. active ending *-*e* of the **h*$_2$*e*-conjugation, coexisting with active (transitive) 1sg. *-*e*-*h*$_2$*e*, 2sg. *-*e*-*th*$_2$*e*, later recharacterized as injunctive 3sg.act. *-*e*-*t*, hence indicative *-*e*-*ti*. There are instances of standard root thematic presents already in Anatolian, not to speak of the numerous complex suffixed thematic presents, in *-*i̯e/o*-, *-*sk̂e/o*-, *-*ei̯e/o*- with active thematic inflection. As for the above pair itself, the scenario developed by Jasanoff (2023: 62–71) would take the two items as independent formations from the *-*u*-present which followed the **h*$_2$*e*-conjugation (type **térh*$_2$-*u̯*-*e*, with -*i* added to the ending in the active, as per Jasanoff 2003: 141–143): The present **g*ᵘ*ih*$_3$-*u̯e/o*- arose through thematization based on the 3sg.act. **g*ᵘ*ih*$_3$-*u̯*-*e(+i)* (for expected ***g*ᵘ*i̯éh*$_3$-*u̯*-*e*, presupposing levelling of the root ablaut) and was characteristically active, while the adjective was presumably a kind of thematic verbal adjective based on the regular weak stem **g*ᵘ*ih*$_3$-*u̯*-. The latter point remains somewhat in the dark, if not taken for granted.

2.3 An alternative derivation with a verbal adjective as intermediate item

In the following, I will try a different course, based on well-known facts of nominal derivation and semantics. PIE **g*ᵘ*ih*$_3$-*u̯ó*- 'living' was a derivative from the root **g*ᵘ*i̯eh*$_3$- by means of the archaic verbal adjective-forming suffix *-*u̯o*-,[16] which can be compared to its semantic opposite **mr̥*-*u̯ó*- 'dead' > OIr. *marb* from the root **mer*- 'to die', replaced by **mr̥*-*tó*- > Ved. *mr̥tá*-, Av. *mərəta*-, Arm. *mard*, etc., with (parallel) contamination into **mr̥*-*tu̯o*- > Lat. *mortuus*, OCS *mĭrtvŭ*. The development of **g*ᵘ*ih*$_3$-*u̯ó*- should therefore be viewed in the context of the semantic field which covers life, old age and death. The root **ĝerh*$_2$- 'to wear out, age, make look older' (LIV^2 165) formed a plain thematic verbal adjective PIE **ĝerh*$_2$-*ó*- 'decayed' (Arm. *cer*) and a parallel athematic synonymous **ĝérh*$_2$-*ont*-/**ĝr̥h*$_2$-*n̥t*-´ > Ved. *járant*- and *jurat*- (dat.sg. *juraté*, gen.pl. *juratā́m*), fem. *járatī*-, Gk. γέρων, γέροντ-, Oss. *zærond*) 'old, decayed', featuring AK inflection, and belonging to a Caland system (cf. the neuter abstract **ĝérh*$_2$-*s* > Gk. γέρας shifted to 'honor, perk' and remade as γῆρας under the influence of the

[16] This is more cogent than resorting to the oppositional suffix found in **dek̂s(i)*-*u̯o*- 'right' (Gk. δεξι(ϝ)ός, Go. *taihswo*), since the basis of this term was not a verbal root.

aorist ἔγηρᾱ, as per Stüber 2002: 84).[17] The idea of *_ĝérh$_2$-ont-_ as the participle of a Narten verb, present or aorist, as per Widmer 1997: 21 (see also LIV^2 165, n. 2) does not withstand scrutiny. The Ancient Vedic verbal system included the transitive-factitive present _járati_ 'to make old', besides intransitive stems such as the present RV _jū́ryati_/AV _jī́ryati_ 'to grow old, become frail' and the stative perfect RV participle act. _jujurvā́n_, _jujurúṣ-_, AV _jajā́ra_ 'ist gealtert'. One cannot reconstruct any PIE aorist of this verb. An *_-(o)n-_stem is reflected by the Toch. B stem _śrānä-_ masc. 'adult man',[18] nom.pl. _śrāy_, obl.pl. _śrānäṃ_, gen.pl. _śrānäṃts_; _śrāy_ < *_śärāyä_ < *_śärāñä_ < *_ĝérh$_2$-ōn-es_ (alternatively, but less likely, *_ĝérh$_2$-n-es_), presupposing as stem *_ĝérh$_2$-ōn-_ < *_ĝérh$_2$-o-on-_ 'the old one', with an individualizing suffix based on the thematic adjective mentioned above.

After the model of *_ĝérh$_2$-ont-_, reanalyzed as *_ĝérh$_2$-o-nt-_, parallel to the synonymous *_ĝerh$_2$-ó-_, *_ĝérh$_2$-ōn-_, a remodelled form *_g$^{u̯}$ih$_3$u̯ó-nt-_ meaning 'the living one' was created, opposed to *_mér-to-_ 'the dead one' (Ved. _márta-_ masc.) 'mortal, human' and *_mertōn-_ < *_merto-on-_ (Av. _marətān-_, masc. 'human', nom.sg. _marətā_, nom.pl. _marətānō_). At a later stage, when the verbal adjective formation in *_-u̯ó-_ had been replaced by *_-tó-_ and *_-(e/o)nt-_, _g$^{u̯}$ih$_3$u̯ónt-_ could be reanalyzed as *_g$^{u̯}$ih$_3$u̯ó-nt-_, the participle of a thematic verbal stem, whence the present injunctive act. 3pl. *_g$^{u̯}$ih$_3$u̯o-nt_, 3sg. *_g$^{u̯}$ih$_3$u̯e-t_. The final stage of this process presupposes the PIE existence of root thematic presents of the "classical" type, to wit *_b^{h}ér-e/o-_, injunctive act. 3sg. *_b^{h}ér-e-t_, 3pl. *_b^{h}ér-o-nt_, participle *_b^{h}ér-o-nt-_. If not for this precise root, one may surmise that this type existed in PIE, see, e.g., the *_-i̯e/o-_ and *_-sk̑e/o-_presents with "complex" thematic suffixes. The PIE existence of full grade thematic presents of the type 3sg.act. *_b^{h}ér-e-ti_ has recently been supported by new independent arguments,[19] even though the presents of this type were not as abundant at the early stage as they were later in Core IE.

To conclude, the thematic present *_g$^{u̯}$ih$_3$u̯é/ó-_ was wholly analogical, while being based ultimately on an adjectival stem. This derivation offers an additional instance of creating the inflectional forms of a finite present stem based on a verbal adjective reinterpreted as participle. My scenario thus differs markedly from the recent accounts of the pair *_g$^{u̯}$ih$_3$-u̯ó-_ 'alive' : *_g$^{u̯}$ih$_3$-u̯e/o-_ 'to live' by Villanueva Svensson (2021) and Jasanoff (2023).

[17] Consequently, the routine reconstruction of an acrostatic neuter *-_s_-stem, as per Widmer (1997: 20, 2004: 50) and other scholars, is not necessary.

[18] Pinault 2008: 441, 484, differently Adams 2013: 705.

[19] See Jasanoff 2023: 72–78.

3 PIE *$n\acute{e}\underset{\frown}{u}$-$ah_2$- 'to make new'

3.1 Brief review of the Anatolian evidence

The type Hitt. *nēwaḫḫ-* 'to make new' has been much discussed in the literature.[20] The Anatolian evidence has recently been reassessed by Sasseville (2015 and 2021: 16–77). The type was productive in Hittite in forming deadjectival factitives from thematic and other kinds of stems: *nēwaḫḫ-* 'to make new' : *nēwa-* 'new', *maršaḫḫ-* : *marša-* 'unholy', *dannattaḫḫ-* : *dannatta-* 'empty', *ḫappinaḫḫ-* : *ḫappina-*, *ḫappinant-* 'rich', *šuppiyaḫḫ-* : *šuppi-* 'pure', *tepawaḫḫ-* 'to make little' : *tēpu-* 'little'; from substantives, e.g. *armaḫḫ-* 'to make pregnant' < 'make round like the moon' : *arma-* c. 'moon', *luriyaḫḫ-* 'to humiliate' : *luri-* c. 'humiliation', *šagiyaḫḫ-* 'to give a sign' : *šagāi-* c. 'sign'. This type belongs to the derivational set of deadjectival verbs together with fientive *-ēšš-* and factitive *-nu-* verbs.

The *°aḫḫ-* type followed the *ḫi*-conjugation in Hittite (OH and MH), which was later replaced by the *mi*-conjugation. By contrast, in the other Anatolian languages, the *mi*-conjugation prevails, with non-lenited endings in the corresponding presents in Luwian /*-a-*/, Lycian *-a-* and Lydian *-a-*. Those verbs are mostly derived from substantives, rarely from adjectives: HLuw. *tabariya-* 'to command' : *tabariya-* c. 'command', Lyc. A *prñnawa-* 'to build (a grave-house)' : *prñnawa-* c. 'grave-house', Lyc. B *muwa-* 'to strengthen' : coll. *muwa* 'might, power', Lyc. B *qla-* 'to build a precinct' : *qala-*/*qla-* c. 'precinct', Lyc. A *sm̃ma-* 'to bind, oblige' : *sm̃ma-* c. 'binding, obligation', Lyd. *pita-* 'to make a donation' ('to make pay' according to Garnier, p.c.) : **pita-* c. 'donation', cf. Lyc. A *pijata-* c. 'gift', Hitt. *piyetta-, pitta-*, coll. 'allotment, payment'. They could also be derived from adjectives or agent nouns: CLuw. *tannatta-* 'to make empty, devastate' : *tannatta/i-* 'empty', CLuw. *arkammanalla-* 'to make someone into a tributary' : *arkammanalla/i-* 'tribute-bearing', Lyc. A *kumaza-* 'to perform a sacrifice' < 'to act as a priest' : *kumaza/i-* 'sacrificing priest', HLuw. *hantawatta-* 'to act as a king' : *hantawatt(i)-* c. 'king'. Those agent nouns actually reflect formations in *°ā* < *-eh_2-, the individualizing suffix, e.g., Lyc. *-*za-* < *-$t\underset{\frown}{i}ah_2$-, derived from secondary adjectives in *-*t̯io-* (see Sasseville 2017, 2018).[21]

[20] Cf. Oettinger 1979: 155, 240, 247, 253, 455–458, Melchert 1997: 132, Jasanoff 2003: 139–141. These references are not intended to be exhaustive. See also the introduction to this volume for an overview.

[21] Following Melchert 2014, see further below.

3.2 Considering the issue of conversion

The usual comparanda for the type are the *-_ā_-factitives in Latin, Greek, Germanic, Balto-Slavic, and Celtic. This is often summarized by the equation Hitt. _nēwaḫḫ-_, Lat. _(re)nouāre_, Gk. νεάω 'to plough up anew', OHG _niuwōn_, despite some secondary restructurings. Thus, Gk. νεάω is actually associated with νειός fem. 'new-ploughed land' (< *_neu̯ii̯ó-_), not with the adjective νέος 'new' and OHG _niuwōn_ is based on the alternative Germanic adjective 'new', OHG _niuwi_ (< *_neu̯(i)i̯o-_, cf. Ved. _náv(i)ya-_, doublet of _náva-_). Nonetheless, the derivational process can be summarized by the following scheme in PIE:

(4) thematic adjective *_néu̯-o-_ 'new' ⟶
verb *_néu̯-e-h$_2$-_ 'to make new', 3sg.act. *_néu̯-e-h$_2$-e(i)_, 3pl.act. *_néu̯-e-h$_2$-n̥ti_

This entails that the thematic vowel of the base took on the *-_e_-grade, assuming that the verbal suffix is reconstructed as plain *-_h$_2$-_. Alternatively, the suffix *-_eh$_2$-_ would be substituted for the final *-_o-_ of the base. This moot point is ordinarily not considered in the scholarship, even though it has significant consequences for ascertaining the origin of the formation.

This present formation was subsequently thematized to *-_eh$_2$-i̯e/o-_, which is found in Latin, Greek, Balto-Slavic, Vedic, and possibly Tocharian,[22] as denominatives from reflexes of PIE *-_ah$_2$-_stems. This type became quite productive in several languages. One may leave aside henceforth the case of the alternative thematization in *-_eh$_2$-e/o-_, which became productive in Balto-Slavic.[23] This type has left reflexes in Aeolic and Mycenaean Greek contract verbs, pointing to an originally athematic 3sg.act. *_neu̯ah$_2$ei̯_, cf. Lesb. -αι as 3sg.act. in the athematic paradigm of contract verbs (with 1sg.act. -ᾱμι instead of -άω), and Myc. _te-re-ja_, 3sg.act. present < *_telehi̯ai̯_, from the verb *_telehi̯ā̆-_ 'to accomplish, fulfill one's obligations' (Rau 2009). The basis can be either the adjective *_telehi̯o-_, a syncopated form of *_telehii̯o-_ > Gk. τέλειος 'perfect, accomplished' (derived itself from the neuter _s_-stem reflected by τέλος 'service, obligation'), or an abstract *_telehi̯ā_ < *_teles-i̯ah$_2$_ (Rau 2009: 181, fn. 4). The Myc. infinitive _te-re-ja-e_ < *°_āhen_ (cf. Lesb. -ᾱν) has the expected form for an athematic stem, since the ending may reflect ultimately a virtual locative sg. *°_ah$_2$-sen_, similar to other enlargements of abstracts, such as the well-known types *-_ah$_2$-ter/n-_, *-_ah$_2$-u̯er/n-_.[24]

[22] Pinault 2008: 579, further discussion in Malzahn 2010: 393–402.

[23] See Jasanoff 2003: 140–141; 2017: 200–201; Villanueva Svensson 2020: 393–395.

[24] Nussbaum 1986: 32–34, 135, 198, fn. 4; Rieken 1999: 347–354, 380–382, 414–417.

The verbal suffix *-h_2- (after thematic stems) with factitive value has not been accounted for from the etymological point of view. There is no comparable suffix in the verbal derivational system of PIE. The segmentation itself (as *-*e*h_2- or *-*e*-h_2- if the basis was a thematic adjective) remains elusive. If one assumes the "conversion" of a nominal stem in *-*(e)*h_2- as the basis of a verbal stem, it would be required to ascertain what the underlying nominal derivative was, e.g., an abstract/collective neuter in *-*e*h_2- (*-*e*-h_2-) or an individualizing animate in *-*e*h_2-.[25] This issue would not matter so much for scholars (such as one anonymous reviewer of the present paper) who consider "that at an early stage of Indo-European nominal and verbal inflection differed only in the endings, but not in the stem formation", so that the suffix *-*e*h_2- could form both nominal and verbal stems. This hypothesis is based on a very simplistic and glottogonic view of the morphological reconstruction. In any case, it would not account for the *h_2*e*-conjugation of the PIE **néu̯-a*h_2- present type, nor for its factitive meaning. The present stems in *-*i*- and *-*u*- cannot be straightforwardly identified with suffixed nominal stems either, which feature several inflectional patterns which are not all found in the verbs. In addition, this approach ignores the existence of denominative presents in PIE and in the IE daughter languages, as well as of verbal adjectives provided with specific and different morphemes. I will therefore not consider it further.

As shown by Sasseville (2017: 287–292; 2021: 69–70), the original Anatolian function of verbal -*aḫḫ*- was the derivation of verbs from substantives, which was then extended to adjectives, as in the Hitt. *nēwaḫḫ*-type. Several instances conclusively point to *-*a*h_2-abstracts as the base, for instance Lyd. *pita*- 'to make a donation' from a *-*ta*h_2-stem, Hitt. *waštaḫḫ*- 'to sin, to offend' vs. *waštai*- c. 'sin', CLuw. *wašta*- nt. 'sin', Hitt. *armaḫḫ*- 'to make round like the moon, to impregnate' vs. *arma*- c., CLuw., HLuw. *arma*- c. 'moon', Lyc. A *arm̃ma*- c. 'moon-god' < stem in *-*ma*h_2-. Concerning the latter example, I follow the derivation proposed by Sasseville (2017: 282–284; 2021: 69–71), which complies with the expected semantic evolution of a verb of the factitive Hitt. °*aḫḫ*-type. Therefore, this account is certainly to be preferred to the one suggested by one anonymous reviewer for the original meaning of this verb: 'to bring to the months', or perhaps 'to bring to the month (of birth)'. This remains arbitrary, and at variance with the factitive meaning warranted by other instances of °*aḫḫ*-verbs which are clearly based on substantives. There is no support for the alleged directive (or allative) meaning which would be taken by the basis of this denominative formation.

[25] Cf. Melchert 2014, Nussbaum 2014: 274–275, 306, and Fellner & Grestenberger 2016 on *-*(e)*h_2- as marker of substantivization.

In addition to these factitive verbs which were apparently derived without any overt suffix, Anatolian had normal denominatives formed with the suffix *-*i̯e/o-*, featuring essive value: e.g. Hitt. *armā(i)-* < *°*ah₂-i̯e/o-* 'to be round like the moon' > 'be pregnant', contrasting with *armaḫḫ-* 'to make round like the moon', quoted above.[26]

Sasseville (2015: 292–293) assumes the "conversion" of *-*ah₂*-abstracts into factitive *-*ah₂*-verbs, challenging the standard view of the *nēwaḫḫ*-type as derived from thematic adjectives, for Anatolian as well as for PIE. The derivation from thematic adjectives would be a development specific to Hittite within Anatolian and independently of Core IE. The process of restructuring of a nominal suffix into a verbal suffix is taken for granted, without setting up any intermediate step. This approach is endorsed by Villanueva Svensson (2020: 391–392, 2021: 275–276) because it provides welcome support to his own account which argues for the derivation of the full grade thematic present type from thematic verbal adjectives. This entails that there was originally no proper verbal suffix in verbs with a 3sg.act. *-*ah₂-e*, which followed the **h₂e*-conjugation. One may surmise that the thematic renewal as *-*ah₂-e/o-* found in Balto-Slavic issued originally from the reinterpretation of this 3sg. active form. The missing link could be a further individualizing derivative in *-*on-* or *-*(e/o)nt-*, the latter being reinterpreted as a participle. This would not explain the original *-*h₂e* conjugation, however. Moreover, such secondary derivatives of abstracts or animate substantives in *-*ah₂*- did not leave any reflex in unmistakable form. The case of adj. **gᵘ̯ih₃-u̯ó-* 'living' vs. verb *gᵘ̯ih₃-u̯é-ti* '(s)he lives, is alive' does not add support to the "conversion" account because this present *does* feature the normal thematic conjugation of the **bʰér-e-ti* type; see Section 2.3 above.

3.3 The Vedic evidence

In previous discussions of possible comparanda of the Hitt. *nēwaḫḫ*-type, Indo-Iranian has not been mentioned so far. However, some intriguing evidence in Ancient Vedic can be detected: The present RV *vr̥ṣāyáti/-te* 'to make rain' (3x), which must be distinguished from *vr̥ṣāyáte* (RV 7x, ptcp. *vr̥ṣāyámāṇa-* 2x) 'to behave like a bull, act the bull', denominative from *vŕ̥ṣan-* masc. (see also the parallel adjective *vr̥ṣayú-* 'acting the bull' in the herd among the cows, RV 9.77.5d). The structure of the former item has not yet been cogently accounted for. The reading of the occurrences does not leave any doubt:

[26] As argued cogently by Sasseville (2015: 287–288).

(5) RV 10.98.1d [to Bṛhaspati]
sá *parjányaṃ* *śáṃtanave* *vṛṣāya* ||
DEM.PRON.NOM.SG.M P.ACC.SG S.DAT.SG rain.make.2SG.IMP.ACT
'Make Parjanya rain for Śaṃtanu.' (in a plea for rain by Devāpi, priest and poet. Transl. Jamison & Brereton 2014: 1555)

(6) RV 9.71.3ab
ádribhiḥ *sutáḥ* *pavate* *gábhast(i)yor*,
stone.INS.PL pressed.NOM.SG.M purify.3SG.PRS.IND.MID hand.LOC.DU
vṛṣāyáte *nábhasā* *vépate*
rain.make.3SG.PRS.IND.MID cloud.INS.SG tremble.3SG.PRS.IND.MID
matī́ |
thought.INS.SG
'Pressed by the stones, he (Soma) purifies himself between the two hands. With his cloud he makes rain (the rain = his juice); he trembles (in poetic inspiration) with his thought.' (differently from Jamison & Brereton 2014: 1304, who translate 'he acts the bull [/rains]').[27]

(7) RV 10.44.4ab
evā́ *pátiṃ* *droṇa-sā́caṃ* *sácetasam*,
thus lord.ACC.SG cup-companion.ACC.SG like.minded.ACC.SG
ūrjá *skambháṃ* *dharúṇa* *ā́*
nourishment.GEN.SG prop.ACC.SG support.LOC.SG PRVB
vṛṣāyase |
rain.make.2SG.PRS.IND.MID
'Just in this way you [Indra] drench yourself in the lord [= soma], the companion of the cup, the like-minded one, the prop of nourishment upon its support.' (transl. Jamison & Brereton 2014: 1449).

This present stem provided the basis for the secondary -*iṣ*-aorist *ā́vṛṣāyiṣata* (VS II.31), following the pres. imperative *ā́vṛṣāyadhvam* (cf. Narten 1964: 62, 292), both with preverb *ā́*, as in RV 10.44.4b. The formation of the present is not discussed by Narten.

This present *vṛṣāyáti/-te* can hardly represent a formation of the *gṛbhāyáti* type, based originally on nasal infixed present, to wit *gṛbhṇā́ti* from *grabh*^{i}- 'to

[27]The first option corresponds to Geldner's choice: "durch die Regenwolke wird er wie ein Bulle, durch die Dichtung wird er beredt" (Geldner 1951: III, 63). See further discussion by Renou (EVP VIII 83 and IX 81), who notes that the reference to the rain is implied by *nábhas-* as complement, evoking the cloud.

seize', with *-āyá-* < **-aH-yá-* < **-n̥H-i̯é/o-*,[28] since *varṣ-/vr̥ṣ-* (*aniṭ* root) did not form a *-nā*-present. Nor can it be a plain essive or fientive denominative from a noun **vr̥ṣā-* (or **vr̥ṣa-*) meaning 'rain', because it has factitive meaning.[29] This verb did not survive long, since it was in competition with the regular causative *varṣáyati* 'to make rain' (RV 3x, participle *varṣáyant-* 1x, with complement *dyā́m* 'sky' or *vr̥ṣṭím* 'rain'), besides the intransitive pres. *várṣati* 'to rain'.[30]

Instead, I propose that *vr̥ṣā-yá-ti* replaced a factitive athematic present **u̯r̥šā-* < **h₂u̯r̥seh₂-*. The latter cannot be based on an adjective **h₂u̯r̥s-ó-* 'rainy, raining', which is unattested. The basic nominal derivatives of the PIE root **h₂u̯ers-* are well known (see Nussbaum 2017: 255–256): agentives which have been variously substantivized, namely **h₂u̯órs-o-* or **h₂u̯ors-ó-* (Hitt. *warša-* c.), **h₂u̯ors-ó-* or **h₂u̯ers-ó-* (Ved. *varṣá-* nt.), collective **h₂u̯ersé-h₂* (Ved. *varṣā́* 'the rains', 'rainy season'), abstract **h₂u̯érse-h₂-* (Gk. ἑέρση, ἀέρση 'dew'). The potential stem **h₂u̯r̥s-éh₂-* could be a primary derivative of the type Gk. φυγή, Lat. *fuga* 'flight' or a replacement of the expected root noun **h₂u̯érs-/*h₂u̯r̥s-* 'raining' reflected by RV *prāvr̥ṣ-* fem. (2x) 'beginning of the rainy season', derived adj. *prāvr̥ṣī́ṇa-* (1x).[31]

In order to explain the factitive present from an abstract, the best way lies in the syntactic reconstruction of an intermediary nominal sentence:

(8) dative sg. in predicative function **h₂u̯r̥séh₂-ei* '(s)he/it (is) for the raining' > '(s)he/it makes rain'.

Next, the sequence could be resegmented as a 3sg. present **h₂u̯r̥séh₂-e+i* with a primary ending, hence as a verbal stem **h₂u̯r̥séh₂-*, which was later thematized as Indo-Iranian **u̯r̥šā-i̯á-ti*, featuring the accent on the thematic suffix as in regular deverbative or denominative presents. As for the syntax, one can compare the well-known construction of the dative sg. of verbal nouns in predicative function, such as Vedic RV *ná dábhāya* 'not to be deceived' (5.44.2c, 7.91.2a, 9.73.8a) or *ná (...) ādábhe* (RV 8.21.16d) with the same meaning. These negative phrases in nominal sentences are equivalent in the RV to the privative adjectives *ádabdha-*

[28] Gotō 2013: 107. The type is reconstructed as **-n̥h₂-i̯e/o-* by Jasanoff (2003: 123–124). The root *grabhⁱ-* is currently traced back to **gʰrebh₂-* (LIV² 201), but the precise laryngeal in this present type does not matter in the present context.

[29] *Pace* Gotō 2013: 131. In any case, one should renounce searching the origin of the factitive in the denominative of 'bull', through the metaphorical assimilation of the jet of sperm by the male to the flow of rain. This cannot be supported by the problematic etymological interpretation of PIE 'bull' itself, from a root that would mean 'to impregnate', *vel sim.*

[30] Jamison 1983: 117–118.

[31] Scarlata 1999: 525.

(48x), *ádābh(i)ya-* (34x) 'undeceivable', *adábha-* (5.86.5b) 'undeceptive'.[32] Compare also the reading of the final dative of a root noun used as quasi-infinitive with factitive meaning in RV 1.89.5c *védasām ásat vr̥dhé* 'he will be for increasing/growing of possessions' = 'for letting grow the possessions'.[33]

3.4 Consequences for Anatolian and PIE

The derivational scheme devised for explaining Ved. *vr̥ṣāyáti* applies in principle to any *-*eh₂*- abstract dative sg. **X-eh₂-ei̯*. This was the precise basis of the so-called conversion. The predicative **[X-eh₂]-ei̯* in nominal sentences was reinterpreted as *X-eh₂-ei̯*, thus as 3sg. active present in *-*e+i* of the **h₂e*-conjugation, with **X-eh₂*- reanalyzed as a denominative verbal stem. This happened after the 3sg. ending of the **h₂e*-conjugation was reanalyzed as an active ending, which yielded the Anatolian *ḫi*-conjugation. The further step to be made consists in transposing this process into PIE to generate the *-*eh₂*-factitive present type. This can be supported by a concrete example found in Anatolian: **armaH-ei̯* '(in)to the moon' ≈ 'to become/make like the moon' > Hitt. 3sg.act. *armaḫḫ-i* 'makes N like the moon' = 'makes N pregnant', see Section 3.1 above. This is consistent with the prevalent derivation of this factitive present type from abstract substantives, and by extension from other *-*ah₂*- substantives, as shown by Sasseville (2021, see again Section 3.1). The form **armaH-ei̯* '(in)to the moon' could in principle generate a labile verb, 'to become/make (N) [round] like the moon'. But it was used transitively, in contrast to the essive/fientive suffixed denominative Hitt. *armā(i)*- 'to be pregnant' < *°*ah₂-i̯e/o*- 'to be(come) [round] like the moon'. Hence, the former predicative was verbalized as Hitt. 3sg.act. *armaḫḫ-i* 'makes N [round] like the moon' > 'makes N pregnant'. One should assume that the transitive reading prevailed for *-*ah₂*-abstracts used predicatively in the dative singular with an overt nominal complement.

On the other hand, one cannot deny that the productivity of the type **neu̯eh₂*- 'to make new' in Core IE as based on thematic adjectives had a primeval source in PIE, although its prevalence and extension to other types of adjectives is a phenomenon proper to Hittite in the Anatolian language family. One may surmise that this development was only embryonic in PIE. Nonetheless, it is likely that some *-*eh₂*- abstracts could be derived from thematic adjectives. Therefore, the chain of derivation was extended with respect to the base, as in ex. (9).

[32] On the relationship of infinitives and quasi-infinitives with verbal adjectives, see Renou 1937.
[33] See Gippert 1978: 95, 96 and Hettrich & Stüber 2018: 23–25 for further instances.

(9) adjective **néu̯-o-* ⟶
abstract **néu̯-e-*h_2- 'newness' ⟶
dative sg. **néu̯-e*h_2*-ei̯* 'for the purpose of newness, into newness' ⟶
verb **néu̯-e*h_2*-e+i* '(s)he makes X into newness' > '(s)he makes X new'.

This derivational chain was then reduced to **néu̯-o-* ⟶ **néu̯e*h_2*-e+i*, resegmented as **néu̯-e-*h_2*-e+i*, in the deadjectival Hitt. *nēwaḫḫ*-type and its cognates in other IE languages. The restructuring of a complex chain by "jumping over" the intermediate link is a well-known phenomenon. The surface effect of the process was finally the apparent substitution of the nominal thematic vowel by the verbal suffix, thus contrasting with the *-*o-i̯é/ó*- or *-*e-i̯é/ó*- denominative presents.

This "de-datival" derivation of *h_2*e*-conjugation presents is to some extent corroborated by the derivation of Hitt. *ḫi*-verbs from adverbs, as described by Melchert (2009): *āppai-* 'to go back, be finished' besides *āppa* 'after, (temporally) behind', *parā-* 'to appear, come/go forth' (cf. HLuw. *ARHA para-* 'go missing, be missing, lack') from *parā* 'forth, out', *šanna-* 'to conceal' from **šanna-* 'isolated, separated' (< **sn̥(H)o(-)*, cf. *šannapi* 'in an isolated place'). An example of the latter is given in (10).[34]

(10) KBo 5.3. I 28
n-at-mu-kan *mān šannatti*
CONJ-3SG.ACC.N-1SG.ACC/DAT-PTCL if conceal.2SG.PRS.ACT
n-at-mu *ŪL mematti*
CONJ-3SG.ACC.N-1SG.ACC/DAT NEG tell.2SG.PRS.ACT
'if you keep it secret from me, and do not tell me'

The secondary verbal inflection of local adverbs resides ultimately in the use of adverbs as predicate in nominal sentences.

The process underlying the formation of the *-*e*h_2-factitive is strikingly parallel to the derivation of the stative in *-*e*h_1- from a predicative adverb based on the instrumental sg. of a root noun, e.g. *h_1*rud*^{h}*-é*h_1 'with redness' ⟶ stative *h_1*rud*^{h}*-é*h_1- 'to be red', verbalized through addition of the present suffix *-*i̯é/ó*-[35] (see the introduction to this volume). In some languages, the stative itself coexisted with the productive formations of fientives in *-*e*h_1*-s-* and inchoatives in *-*e*h_1*-sk̂e/o-*, or of periphrastic factitives, type Lat. *ārē-faciō* 'to make dry', beside

[34]Further instances in Puhvel 2017: 111–113. The derivation proposed by Melchert (2009: 336–337) is more straightforward than through a verbal stem with a nasal infix (Oettinger 1979: 159–160; Kloekhorst 2008: 719) or with a nasal suffix (Puhvel 2017: 116).

[35]See Jasanoff 2004: 145–149; also Merritt, this volume.

ārē-scō 'to grow dry'. In addition, the stative present stem became the basis of nominal forms in *-eh_1-to-*, *-eh_1-ti-*, etc. The thematic renewal of the *-eh_2-* factitive type into *-eh_2-i̯e/o-* would be similar to the replacement of the *-eh_1-stative type by *-eh_1-i̯e/o-*, e.g. in Lat. *rubeō*, *-ēre*, etc. Parallel to the formations issued from the stative, secondary adjectives in *-to-* and *-(e/o)nt-* based directly on the abstract probably also existed, which were later grammaticalized as participles of the newer *-ah_2-factitives, to wit Lat. *(re)nouātus*, Hitt. *nēwaḫḫant-*.

4 Concluding remarks

In the first two sections of this paper, two alleged instances of denominative presents without an overt suffix have been falsified. This was made possible by resorting to more economic analogical processes. As an exemplary point, the theory of conversion of an adjective into a verb has been challenged in the case of *$g^{u̯}ih_3$-u̯ó-* 'living' and *$g^{u̯}ih_3$-u̯é-ti* 'lives'. This case suggests that the analogical process took place already in PIE. In the third section, the intriguing case of conversion apparently featured by the factitive *-eh_2-present type has been treated in some detail and was resolved by resorting to a further analogical process in PIE. The crucial fact lies in the prevalent derivation of this factitive type from abstracts and individualizing derivatives, as still reflected by the Anatolian data. Furthermore, the later productive development of *-eh_2-abstracts from thematic adnominal bases provides the missing link for deriving the *néu̯-eh_2-type from thematic adjectives. The reconstructed process also explains without further ado the original *h_2e-conjugation of the *-eh_2-factitive type, which had been never accounted for. As an alternative approach, which would effectively be the admission of failure, one would be forced to assume that the verbal suffix *-eh_2-* of the present type and the nominal suffix *-eh_2-* had nothing in common and were simply homophonous. It does not seem commendable to multiply homophonous PIE morphemes only by projecting back the data on a purely formal basis without paying attention to the functions of the derivatives. This amounts to giving up on any perspective for a solution with the excuse of ignorance. With respect to the semantic side, the main issue concerned explaining the factitive meaning of the denominative *-eh_2-present type, which follows from the syntactic reconstruction of predicatively used verbal abstracts with nominal complements.

Abbreviations

acc.	accusative
act.	active
adj.	adjective
aor.	aorist
Arm.	Armenian
AV	Saṃhitās of the Atharvaveda
Av.	Avestan
CLuw.	Cuneiform Luwian
CToch.	Common Tocharian
dat.	dative
fem.	feminine
Gaul.	Gaulish
gen.	genitive
Gk.	Greek
Go.	Gothic
HLuw.	Hieroglyphic Luwian
Hitt.	Hittite
IE	Indo-European
instr.	instrumental
Lat.	Latin
Latv.	Latvian
Lesb.	Lesbian
Lith.	Lithuanian
Lyc.	Lycian
Lyd.	Lydian
MH	Middle Hittite
MP	Middle Persian
masc.	masculine
mid.	middle
Myc.	Mycenaean
nom.	nominative
nt.	neuter
OAv.	Old Avestan
OCS	Old Church Slavonic
OH	Old Hittite
OHG	Old High German
OIr.	Old Irish
OP	Old Persian
OPr.	Old Prussian
Oss.	Ossetic
PIE	Proto-Indo-European
pl.	plural
pres.	present
ptcp.	participle
RV	Saṃhitā of the R̥gveda
sg.	singular
Toch.	Tocharian
Ved.	Vedic
VS	Vājasaneyi-Saṃhitā
YAv.	Young Avestan

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Chapter 9

A Late Nuclear Proto-Indo-European verbal type: Intransitive thematic nasal-infix present actives

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This paper examines primary verbal formations in the ancient Indo-European languages that also play a role in deadjectival derivation. It builds on recent work on the historical morphology of the intransitive change of state nasal-affix thematic active presents in the "Northern Indo-European" languages – Baltic, Slavic, and Germanic – which continue an anticausative to the well-known athematic nasal-infix transitive presents. The paper argues that this formation has exact cognates in Ancient Greek, Vedic Sanskrit, and Indo-Iranian, more generally, where we find morphologically identical presents that as in the Northern Indo-European languages make semantically intransitive change of state formations to both primary verbs and root-derived property concept adjectives. The agreement across these branches supports the reconstruction of this formation for late Nuclear Indo-European, as a continuation of a deeper Proto-Indo-European type.

1 Introduction

One of the more remarkable features of Indo-European (IE) morphology is that deadjectival verbal formations are partially handled with dedicated denominative morphology and partially with the root-derived – "primary" – morphology that otherwise holds across Proto-Indo-European (PIE) verbal classes. While the deadjectival denominative formations have been thoroughly studied and are well

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understood, the insight that primary formations also played a role in deadjectival derivation – at least in the sphere of property concept adjectives – is a recent discovery whose full parameters have yet to be elucidated.[1]

This paper aims to further advance this discussion and takes its start from recent work on the historical morphology of the intransitive change of state nasal-affix thematic active presents in the "Northern Indo-European" languages (Baltic, Slavic, and Germanic), whose origins have been suggested to continue a morphologically peculiar anticausative to the well-known athematic nasal-infix transitive presents. The paper argues that this type has exact cognates in Ancient Greek and Vedic Sanskrit (and Indo-Iranian, more generally), where we find morphologically identical presents that as in the Northern Indo-European languages make semantically intransitive change of state formations to both primary verbs and subsets of root-derived property concept adjectives. Based on the agreement of these branches, it is possible to reconstruct the formation for late Nuclear Indo-European[2] and to project its origins deep into PIE.

2 Northern Indo-European

The starting point for this investigation is the magisterial 2007 Harvard PhD thesis of Yaroslav Gorbachov (Gorbachov 2007; see further Villanueva Svensson 2006, 2011) on the inchoative nasal-affix presents in Baltic, Slavic, and Germanic, the Northern Indo-European languages. This nasal-affix verbal type is well attested in all three Northern Indo-European branches, where it generally serves to make 1) intransitive change of state/location formations – often categorized as "inchoatives" – to intransitive statives, less frequently activity verbs, 2) anticausatives to causative formations, and 3) denominatives to property concept adjectives and, more rarely, substantives. For a survey of forms across all representative semantic categories, cf. the examples in (1).

(1) Inchoative nasal-affix presents in Slavic, Baltic, and Germanic
   a. **Old Church Slavonic (OCS)**
      - *sędǫ, sesti, sědŭ* 'sit down'
      - *vŭzlęgǫ, -lešti, -legŭ* 'lie down'

[1] See Rau 2009 with lit., 2013, 2016, 2017, and further Nussbaum 2022.

[2] "Nuclear Indo-European" is used to refer to the subgroup of Indo-European languages that shared innovations after the departure of Anatolian from the PIE speech community. I use "late Nuclear PIE" to refer to the subgroup of Nuclear PIE languages that remain after the departure of Tocharian and then later Proto-Italo-Celtic.

- *stanǫ, stati, staxŭ* 'stand up'
- *pri/vŭzniknǫ, -niknǫti, -nikŭ* 'spring up'
- *promŭknǫ, -mŭknǫti sę, -mŭkŭ* 'go out, spread'
- *prilĭ(p)nǫ, -lĭ(p)nǫti, -lĭpŭ* 'cleave, cling to'
- *uglĭbnǫ, -glĭbnǫti, -glĭbŭ* 'get stuck'
- *izběgnǫ, -běgnǫti, -běgŭ* 'flee, avoid'
- *postignǫ, -stignǫti, -stigŭ* 'come upon, reach'
- *vŭzbŭ(d)nǫ, -bŭ(d)nǫti, -bŭdŭ* 'wake up'
- *usŭ(p)nǫ, -sŭ(p)nǫti, -sŭpŭ* 'fall asleep'
- *navyknǫ, navyknǫti, navykŭ* 'get used to'
- *oslĭpnǫ, -slĭpnǫti, -slĭpŭ* 'go blind' (: *slěpŭ*)
- *ugasnǫ, -gasnǫti, -gasŭ* 'go out, be extinguished'
- *prisvędnǫ, -svędnǫti, -svędŭ* 'wilt'
- *omrĭknǫ, -mrĭknǫti, -mrĭkŭ* 'grow dark' (: *mĭrkŭ**)
- ORuss. *svĭ(t)nuti* 'grow light'
- *sŭ/pomrŭznǫ, mrŭznǫti, -mrŭzŭ* 'freeze, congeal' (: *mrazŭ*)
- *sŭxnǫ, sŭxnǫti, sŭxŭ* 'dry up' (: *suxŭ*)

b. **Baltic**

- OPr. *sīnda-* 'sit down'
- OPr. *polīnka, polāikt* 'stays' (: Lith. dial. *-liñka* 'remains')
- Lith. *tam̃pa, tàpti, tàpo* 'become'
- Lith. *knim̃ba, knìbti, knìbo* 'sink, fall'
- Lith. *kañka, kàkti, kàko* 'start out, go'
- Lith. *muñka, mùkti, mùko* 'break free, escape'
- Lith. *lim̃pa, lìpti, lìpo* 'cling, stick to, climb'
- Lith. *pliñta, plìsti, plìto* 'spread out'
- Lith. *skiñda, skìsti, skìdo* 'fall apart, split asunder'
- Lith. *atbuñda, atbùsti, atbùdo* 'wake up'
- Lith. *miñga, mìgti, mìgo* 'fall asleep'
- Lith. *suprañta, supràsti, supràto* 'understand'
- Lith. *patiñka, patìkti, patìko* 'suits, fits'
- Lith. *šviñta, švìsti, švìto* 'dawn'
- Lith. *sniñga, snìgti, snìgo* 'snow'
- Lith. *tuñka, tùkti, tùko* 'grow fat' (: *taukaĩ*)
- Lith. *pų̃va, pùti, pùvo* 'rot, decay'
- Lith. *sų̃sa, sùsti, sùso* 'dry up' (: *saũsas*)
- Lith. *dum̃ba, dùbti, dùbo* 'grow hollow' (: *dubùs*)

c. **Germanic**
- Go. *standan* ‘stand’, ON *standa* ‘id.’
- Go. *aflifnan* ‘remain’, ON *lifna* ‘id.’
- OE *climban* ‘climb’, OHG *klimban* ‘id.’
- Go. *rinnan* ‘run’, ON *renna* ‘id.’
- Go. *aflinnan* ‘leave off, desert’, ON *linna* ‘cease, stop’
- Go. *wiknan* ‘start to give in, yield’, ON *vikna* ‘id.’
- Go. *usgutnan* ‘spill out’
- Go. *usbriknan* ‘break off’
- Go. *gawaknan* ‘wake up’, ON *vakna* ‘id.’
- ON *sofna* ‘fall asleep’
- Go. *ga(h)nipnan* ‘grow sad’, ON *hnipna* ‘become downcast, droop’
- Go. *gawairþnan* ‘become reconciled’
- Go. *gathaursnan* ‘get dry’, ON *þorna* ‘id.’
- Go. *gastaurknan* ‘grow stiff, rigid’, ON *storkna* ‘id.’ (: ON *starkr*)
- Go. *bulgnan* ‘swell up’, ON *bolgna* ‘id.’
- ON *fúna* ‘rot’
- Go. *gafullnan* ‘grow full’, ON *fullna* ‘id.’ (: *fulls*)
- ON *hvítna* ‘goes pale’ (: *hvítr*)
- ON *róðna* ‘turn red’ (: *rauðr*)

This Northern Indo-European verbal type also shows a large number of two- and three-way cognate sets among the languages – cf. the forms in Table 1.

In his thesis, Gorbachov makes three important observations about this type and its prehistory. First, he notes that these verbs inflect as *active* and *thematic* presents with a nasal affix across all three branches – so in Gothic and residually elsewhere in Germanic –, and that this active thematic inflectional pattern should be reconstructed for the ancestor of the type in all three languages. See Gorbachov 2007 *passim*, with discussion of alternative analyses. Second, he points out that the type systematically corresponds to transitive, athematically inflected nasal-infix presents in other IE languages, and is ultimately to be aligned with this transitive type – see, e.g., the forms in Table 2.

Third, he argues that based on the function of the formation, its active thematic inflection, and its correlation with transitive athematic nasal-infix verbs in the other IE languages, it is possible to set up a descriptive derivational process for the ancestor of these branches, whereby an anticausative is made to the transitive athematic nasal-infix present by putting the stem in the zero-grade, suffixing the

Table 1: Cognate inchoative nasal-affix presents in Slavic, Baltic, and Germanic

OCS	**Baltic**	**Germanic**
sędǫ, sesti, sědŭ 'sit down'	OPr. *sīnda-* 'sit down'	
prilĭ(p)nǫ, -lĭ(p)nǫti, -lĭpu 'cleave, cling to'	Lith. *lim̃pa, lìpti, lìpo* 'cling, stick to, climb'	Go. *aflifnan* 'remain'
promŭknǫ, -mŭknǫti sę, -mŭkŭ 'go out, spread'	Lith. *muñka, mùkti, mùko* 'break free'	
uglĭbnǫ, -glĭbnǫti, -glĭbŭ 'get stuck'		OE *climban* 'climb', OHG *klimban* 'id.'
vŭzbŭ(d)nǫ, -bŭ(d)nǫti, -bŭdŭ 'wake up'	Lith. *atbuñda, atbùsti, atbùdo* 'id.'	
usŭ(p)nǫ, -sŭ(p)nǫti, -sŭpŭ 'fall asleep'		ON *sofna* 'fall asleep'
ORuss. *svĭ(t)nuti* 'grow light'	Lith. *šviñta, švìsti, švìto* 'dawn'	ON *hvítna* 'goes pale'

accented thematic vowel, and inflecting the formation as an active, as outlined in Table 3.

In terms of its deeper PIE origins, Gorbachov (2007: 200–214) argues that the formation is to be traced to a **h_2e*-conjugated version of the athematic nasal-infix present, which ultimately served to make anticausatives and which as all **h_2e*-conjugation active present types underwent analogical thematization on the way to NPIE.[3] In my view, Gorbachov is correct in his analysis and reconstruction of this type for late PIE. The intention of this paper is to further support his analysis by outlining additional reflexes of the type in Ancient Greek and Indo-Iranian.[4]

[3]See Jasanoff 2003 for the PIE **h_2e*-conjugation and its systematic thematization in the Nuclear IE languages.

[4]A further important feature of these presents is that in Slavic – the only Northern Indo-European branch to preserve the inherited IE aorist – they systematically correlate with the thematic aorist.

Table 2: Correspondences between inchoative nasal-affix presents in Slavic, Baltic, and Germanic and transitive athematic nasal-infix presents in other IE languages

OCS	Baltic	Germanic	Nasal-infix pres.
prilĭ(p)nǫ, -lĭ(p)nǫti, -lĭpu 'cleave, cling to'	Lith. *limp̃a, lìpti, lìpo* 'cling, stick to, climb'	Go. *aflifnan* 'remain'	Ved. *limpáti* AV+, ?Gk. λιπαίνω 'anoint' Att. Ion.
promŭknǫ, -mŭknǫti sę, mŭkŭ 'go out, spread'	Lith. *muñka, mùkti, mùko* 'break free, escape'		Ved. *muñcáti* 'release, let go', Lat. *ēmungō, -ere* 'blow the nose'
vŭzbŭ(d)nǫ, -bŭ(d)nǫti, -bŭdŭ 'wake up'	Lith. *atbuñda, atbùsti, atbùdo* 'id.'		OIr. *ad·boind* 'give notice' (Gk. πυνθάνομαι Hom.+)
	Lith. *skiñda, skìsti, skìdo* 'fall apart, split asunder'		Ved. *chinátti* 'split off, break off', Lat. *scindō* 'id.'
		OHG *scrintan* 'split open, burst'	Ved. *kr̥ntáti* 'cut, split', YAv. *kərəṇta-^{ti}* 'id.'
ORuss. *rínut'* 'stream, flow'		Go. *rinnan*, ON *renna* 'run'	Ved. *riṇā́ti* 'set in motion, swirl', Gk. ὀρῑ́νω 'stir, move excite' Hom.+

Table 3: Derivational relationship between athematic nasal-infix transitives and thematic nasal-infix intransitives

Athematic Nasal-Infix Tr.		Thematic Nasal-Infix Intr.
**munék-^{ti}/munk-^{tói}* 'release, set free' Ved. *muñcáti*, Lat. *ēmungō, -ere*	→	**munk-é-^{ti}* 'go free, escape' OCS *promŭknǫ, -mŭknǫti sę, -mŭkŭ* 'go out, spread', Lith. *muñka, mùkti, mùko* 'break free'
**linép-^{ti}/linp-^{tói}* 'make adhere to' Ved. *limpáti* AV+,?Gk. λιπαίνω 'anoint' Att. Ion.	→	**linp-é-^{ti}* 'cling, stick to' OCS *prilĭ(p)nǫ, -lĭ(p)nǫti, -lĭpu* 'id.', Lith. *limp̃a, lìpti, lìpo* 'cling, stick to, climb', Go. *aflifnan* 'remain'

3 Ancient Greek

In Ancient Greek (and further Classical Armenian[5]), it is possible to identify several sets of thematic nasal-affix presents that can be aligned with the Northern Indo-European type outlined by Gorbachov – cf. the five types listed in examples (2)–(6).[6]

(2) *Nu*-presents,[7] e.g., φθῑ́νω/φθῐ́νω 'decay, waste away, disappear' Hom.+: φθινύθω Hom.+ (tr., intr.), ἔφθιτο Hom.+, ἔφθιον Hom.+, ἔφθ(ε)ισα Hom.+, ἔφθιται Hom.+ (: Ved. *kṣiṇā́ti*, *kṣiṇóti* 'destroy' AV, YAv. *jinā-*[ti] 'id.'; ON *dvena* 'disappear, decrease', ON *dvína* 'id.')[8]

(3) Nasal-infix to sonorant-plus-laryngeal-final roots

a. θάλλω 'blossom, flourish' Hes. *hHom.*+: ἔθαλον *hHom.*+, τέθηλα Hom.+, θαλέθω Hom.+ (: θαλερός Hom.+, ἡ θάλεια Hom.+, τὸ θάλος Hom.+)[C] (: Alb. 1sg. *dal*, *del* 'go out' < **dalnō*, *dalnet* and MW *deillyaw** << **daln-*, see Schumacher 2004: 257–259 and Schumacher & Matzinger 2013: 968–969)

b. δῡ́νω 'go into, enter; go down, sink' Hom.+[A]: δύω, -ομαι Hom.+, ἔδῡν Hom.+ (intr.), ἔδῡσα, -αμην Hom.+

[5]The Greek type has a close match in Classical Armenian, where, e.g., presents in *-ane/im* regularly correlate with nasal-infix presents elsewhere, are made to thematic preterites, both inherited thematic aorists and recategorized thematic imperfects, and show up in both transitive and intransitive formations. For the facts and discussion, see Kocharov 2019: 162ff. with lit. The details of this type, whose core likely depends on a prehistoric Greek and Armenian genetic or convergence-related innovation, I leave for discussion elsewhere. For a useful orientational assessment of the relationship between Greek and Armenian, see most recently Kim 2018 with lit.

[6]Superscript [A] marks verbs that at least optionally take an acc. object. Verbs that show strong Caland system associations, which are frequent across these categories, are marked with superscript [C]. Note that all Caland-system-associative verbs can in principle be understood as deadjectival provided they continue to stand in a synchronically transparent relationship with property concept adjectives derived from the same roots. For primary verbs as a foundational element of the PIE and IE Caland system, see Rau 2009, 2013, 2016, 2017, and further Nussbaum 2022.

[7]The poetic verb ἄνω, -ομαι 'accomplish, finish' Hom.+ (: ἀνύω 'id.' Hom.+) may also belong in this class if it is a relatively late innovation of the sort noted below. Another possibility is that the form is a recategorized thematic aorist, either an inherited formation – viz. *ásanat* RV+ (: *asāniṣam* RV): *sanóti* RV, OAv. subj. *hanāṯ* – or an inner-Greek remodeling of a root aorist, assuming the root is **senh₃-*. This latter analysis may have support in the peculiar aorist ἤνεσα *IG* 7.3226, Orchomenos and elsewhere, if it represents the contamination product of this thematic aorist and the *s*-aorist otherwise well attested for this verb.

[8]As comparanda, I provide forms that are directly cognate and those that are closely parallel.

c. θῦνω ‘rush, dart along’ (: analogical θῡνέω Hes. *Sc.*; θῦω/θυίω ‘rage, seethe’ Hom.+, ἔθῡσα Call.) (: Ved. *dhūnóti* ‘shake’ to *d^heu̯H- [plus θῦω/θυίω ‘rage, seethe’ Hom.+, so LIV^2], but LIV^2 *s.r.* to *d^heu̯- because of θῡνέω Hes. *Sc.*[!])[9]

d. κάμνω ‘(tr.) produce/acquire through labor/toil (Hom.[+], but rare); (intr.) labor, toil, grow weary toiling’ Hom.+^{A}: ἔκαμον Hom+, κέκμηκα Hom.+ (: ἀκάμας , -αντος Hom.+; *śamnīṣe* ‘labor’ YV, *śamāyáte* ‘id.’ RV, *aśamīt* ‘ist ruhig geworden’ AVP, *áśamanta* ‘labored’ YV)

e. ὀφείλω Att. Ion., ὀφέλλω Hom. Lesb. Arc., ὀφήλω Cret. Arc. (Myc.) ‘owe, be obliged to pay for/render; be bound, obliged to (+ inf.)’^{A}: ὦφλον Hom.+ (Myc.), ὤφληκα Hom.+ Dial. (: ὀφλισκάνω Att., etc.; τὸ ὄφελος Hom.+ [Myc.], ὀφλητής [gloss]; cf. LIV^2 Addenda, **$h_3b^helh_1$-*)

(4) Nasal-infix to obstruent-plus-laryngeal-final roots

a. aor. (⇐ impf.) ἔχανον ‘yawn, gape’ Hom.+: κέχηνα Hom.+, χάσκω Att. Ion., χαίνω late^C (: OCS *zinǫti* ‘gape’, OHG *ginēn* ‘yawn’, ON *gína* ‘id.’; LIV^2 Addenda, **$\hat{g}^heh_2$-* and *i*-present **$\hat{g}^heh_2i$-*)[10]

b. φθάνω/φθᾱ́νω ‘(tr.) be beforehand, anticipate; (intr.) come/act first (±part.)’ Hom.+^{A}: ἔφθην Hom.+ Att., φθάμενος Hom. Hes., ἔφθασα Att. Ion. (: Ved. opt. *daghnuyāt* ‘würde knapp verfehlen’ KS, *mā́ dhak* ‘soll nicht verfehlen’ RV. See LIV^2 **$d^heg^{u̯h}h_2$-* for further possible comparanda.)

(5) Nasal-infix to stop-final root

a. λάμπω ‘shine’ Hom.+ (: λαμπρός Hom.+; Hitt. *lap-^{zi}*, pret. *lāp-^{ta}* [cf. Lith. *lópė*, etc.])^C

b. [ὁ θρόμβος ‘lump, clot’ Att. Ion.: τρέφω, -ομαι ‘make firm, congeal’, ἔτραφον Hom. (intr.), ἔτρεψα, τέτροφα Hom.+^{C} (: ταρφύς Hom.+, τάρφα Hom.+, τὸ τάρφος Hom.+; Lith. *drim̃ba, drìbti* ‘fall in flakes, drop’)][11]

[9] Given the structure of the root, the form is best assigned, *pace* LIV^2, who align it with the *nu*-class above, to the category of nasal-infix formations to laryngeal final roots, with Ved. *dhūnóti* an analogical innovation.

[10] Here I assume, with LIV^2, that aor. ἔχανον is an aspectually recategorized imperfect. A parallel shift is responsible for perfective aspect of OCS *zinǫti*.

[11] This and the following two nouns obviously depend on an underlying nasal-infix present, either an athematic transitive or a thematic intransitive of the sort under discussion. I assign them to the latter type, first because of the semantic profile of the roots and their derivational *averbo*, which – at least in the case of ὁ θρόμβος and τὸ θάμβος – is exactly parallel to the

c. [τὸ θάμβος 'astonishment' Hom.+: θαμβέω 'be astonished' (: ἐθάμβησε Hom.+)[A], ἔταφον Hom.+, τέθηπα Hom.+ (: θώψ, θωπός Att. Ion.[C]; deverbal[?] PGmc. ***dumba*-). Non-nasal root, *pace* Hackstein 2002: 237 with lit.[12]]

d. [ὁ στρόμβος 'top; spinning object' Hom.+: στρέφω, -ομαι 'twist, turn' Hom.+, ἔστρεψα, -αμην Hom.+] Cf. YAv. nom. sg. *θrąfš°* 'satisfaction (*vel sim.*)' Yt.5.26, YAv. *θrąfəδra*- 'rich in; satisfied': Ved. *tr̥mpáti* 'take delight in, enjoy (+gen.)' RV. See Section 4.

(6) -άνω to obstruent final roots[13]

a. ἁνδάνω 'be pleasing' Hom.+ (: ἥδομαι Att. Ion. Dial.): ἕαδον Hom.+, ἕαδα Hom.+ (: ἡδύς Hom.+, τὸ ἦδος Hom.+, etc.[C]; Ved. *s(u)vádati* 'make sweet, sweeten' RV, *svā́date* 'become sweet' RV)

b. ἐκφλυνδάνει 'break out (of sores)' Hp. (: φλυδαρός 'soft, flabby' Hp. *ap.* Gal.19.152, φλυδάω 'have excess of moisture, be soft/flabby' Hp.)[C]

c. ἀργάνω 'be white' E.+ (: ἀργός Hom.+, τὸ ἄργος Hom.+, etc.)[$C]

d. οἰδάνω 'swell (intr.)' I646 (οἰδάνεται), Ar.*Pax*.116 (οἰδάνων); 'make swell' I554 (≃ A.R.1.478): οἰδαίνω 'swell (intr.)' A.R.+ (tr., late and rare), οἰδέω 'id.' Hom.+ (: οἰδαλέος Hom.+, τὸ οἶδος Hom.+, etc.[C]; ClArm. *aytnowm* 'swell [intr. *tantum*]')[14]

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other verbs that can be assigned to this class, and second, because the retention of the nasal-infix formant in a nominal derivative is more easily explained as the result of a synchronically isolated thematic nasal-infix intransitive – replaced prehistorically in Greek by the -ανω type below – than an otherwise reasonably well-paralleled athematic nasal-infix transitive.

[12]The long vowel in the perfect τέθηπα Hom.+ and the root noun θώψ, θωπός Att. Ion., whose secondary origins are difficult to motivate, make the reconstruction of an underlying non-nasal root the default assumption.

[13]This is a productive present-stem formant in Greek, which is regularly used to generate presents from thematic aorist forms. I mostly omit obviously – e.g., ἀλφάνω Att. (: ἄλφω Hsch.): ἦλφον Hom.+ – and likely – e.g., λιμπάνω Hom. (*v.l.*) Sa. Th. Hp. (+) (: λείπω Hom.+): ἔλιπον Hom.+ – secondary creations. I include some obviously or likely secondary, inner-Greek formations here to illustrate more fully the semantic subcategories and coherence of this subtype. I use [$] to mark such formations as secondary, Greek-internal innovations. Forms are arranged according to semantic subtype, from property concept to perception/cognition and possession, *inter alia*.

[14]Intransitive οἰδάνων at Ar.*Pax*.116 is an obvious *forma difficilior* and points to an original intransitive meaning for the active. The Homeric opposition transitive active: intransitive middle has been analogically created according to a well-attested Greek morphological restructuring process – cf., e.g., αἴθω 'burn (intr.)' Pi.S., αἴθουσα 'portico' Hom.+: αἴθω 'burn (tr.)', αἴθομαι 'burn (intr.)' Hom.+. The intransitive active is also supported by largely intransitive and synchronically/semantically isolated οἰδαίνω 'swell (intr.)' A.R.+ (tr., late and rare), which is likely

e. κῡδάνω ‘be triumphant (intr., Y.42); glorify (tr., ξ73)’: κῡδαίνω ‘glorify, honor (tr. *tantum*)’ Hom.+, ἐκῡ́δηνα Hom.+ (: κῡδρός Hom.+, κῡδάλιμος Hom.+, τὸ κῦδος Hom.+, etc.)$^{\mathrm{C}}$

f. λανθάνω, -ομαι ‘escape notice (±acc.); forget’ (: λήθω, -ομαι Hom.+)$^{\mathrm{A}}$: ἔλαθον, -ομην Hom.+: λέληθα Hom.+ (: λάθρηι Hom.+, etc.)$^{\$\mathrm{C}}$

g. ὀλισθάνω ‘slip’ Att. (: ὀλισθαίνω ‘id.’ Arist.+): ὤλισθε Hom.+ (: Go. *slindan* ‘schlingen’; Ved. *srédhati* ‘gleitet aus, handelt falsch’ RV, *sridhat* RV)$^{\mathrm{C}}$

h. ἁμαρτάνω ‘miss the mark, go astray’ Hom.+: ἤμαρτον Hom.+

i. τυγχάνω ‘chance on, happen, hit the mark’ Hom.+: ἔτυχον Hom.+, τετύχηκα Hom.+ (: τεύχω ‘make ready, produce’ Hom.+, ἔτευξα Hom.+, τετευχώς Hom. [Myc.])

j. πυνθάνομαι ‘learn, ascertain’ Hom.+$^{\mathrm{A}}$ (: πεύθομαι Hom.+): ἐπυθόμην Hom.+: πέπυσμαι Hom.+ (: OCS *vŭzbŭnǫti, vŭzbŭde* ‘wake up’, Lith. *atbuñda, atbùsti, atbùdo* ‘id.’; OIr. *ad·boind*)$^{\mathrm{C}}$

k. αἰσθάνομαι ‘perceive, take notice of’ Att. Ion.$^{(\mathrm{A})}$: ᾐσθόμην Att. Ion., ᾔσθημαι Att. Ion.$^{\$}$

l. μανθάνω ‘learn, notice, understand’ Pi+$^{\mathrm{A}}$: ἔμαθον Hom.+, μεμάθηκα Att. Ion.$^{\$}$

m. δαρθάνω ‘fall asleep’ Hierocl.: ἔδαρθον Hom.+$^{\$}$

n. θιγγάνω ‘touch, handle’ Att. Ion.+: ἔθιγον Archil.+[15]

o. χανδάνω ‘take hold; be capable’ Hom.+$^{\mathrm{A}}$: ἔχαδον Hom.+, κέχα/ονδα Hom. (: Lat. *prehendō*, *-ere* ‘seize, grasp’, OIr. *ro·geinn* ‘occupies’, Go. *duginnan* ‘begin’[16])

p. λαγχάνω ‘obtain by lot, come to possess’ Hom.+$^{(\mathrm{A})}$, intr. ‘fall to one’s lot’ Hom.+: ἔλαχον Hom.+, λέλογχα Hom.

q. λαμβάνω ‘take hold, grasp, seize’ Att. Ion.$^{\mathrm{A}}$ (: λάζομαι Hom., Dial.): ἔλαβον Hom.+, εἴληφα Att./λελάβηκα Ion., Dial.$^{\$}$

a backformation from an innovative aorist to present οἰδάνω. Similar considerations hold for immediately following transitive and intransitive active κῡδάνω ‘be triumphant (intr., Y.42); glorify (tr., ξ73).’

[15] On this verb, see the recent discussion of Nikolaev (2023).

[16] See Fortson (2009), and further Jasanoff (2022), who proposes a special origin for this set of verbs and at least this subset of -ανω presents in Greek (and Armenian). In my view, this set is best understood as an instance of the general type under discussion, with Lat. *prehendō* as the continuation of $^{*}g^{h}\underset{\circ}{n}d^{h}$-*e/o*-, Go. *duginnan* as a re-characterization of this inherited stem according to the productive Germanic type, and OIr. *ro·geinn* ‘occupies’ as the phonologically regular development of the same stem *à la* Fortson (2009).

It is possible to divide this material into two basic sets of forms. The first set, which includes everything but the final άνω-class in ex. (6), consists of verbs that are phonologically directly derivable from a thematic nasal-infix formation of the Gorbachov type made to the PIE root in question, with a subtype (a) that pairs with a root aorist or for which the aorist is not attested – viz. φθῐ́νω/φθῑ́νω Hom.+: ἔφθιτο Hom.+, ἔφθιον Hom.+, ἔφθ(ε)ισα Hom.+, δῡ́νω Hom.+[A]: ἔδῡν Hom.+ (intr.), ἔδυσα, -αμην Hom.+, etc. – and a subtype (b) that pairs with a thematic aorist – viz. θάλλω Hes. *hHom.*+: ἔθαλον *hHom.*+, etc. The second set consists of verbs in -άνω, a highly productive and expansive category in both prehistoric and historical Greek. As far as can be demonstrated, the verbs found in this second set are not made to roots in a final laryngeal and so cannot at first sight be explained as the direct phonological results of the thematic nasal-infix formation in question – many if not all must be of analogical origin. Nevertheless, it is important to note that these verbs largely show the same semantics as set 1 (see further below), systematically pair with the thematic aorist (viz. ἁνδάνω Hom.+: ἕαδον Hom.+, etc.), as set 1 (and further as in Slavic), have a substantial number of forms that correlate with the Caland system, as set 1, and of course share a similar thematic active nasal stem formant, exactly as set 1. There is thus every reason to assume that both sets continue the same underlying type.

Across both sets of forms, these verbs regularly correlate with nasal presents in other IE languages. Some have exact or close formal and semantic matches of the type that Gorbachov outlined, while others have athematic nasal-infix cognates where they pattern semantically with the intransitive middle of the athematic nasal-infix cognate but not the transitive active, cf. (7)–(9).

(7) Close semantic and morphological equation or parallel[17]
   a. φθῐ́νω/φθῑ́νω 'disappear, decrease' Hom.+: ON *dvena* 'disappear, decrease', ON *dvína* 'id.' (: Ved. *kṣiṇā́ti*, *kṣiṇóti* AV, YAv. *jinā-^{ti}*)
   b. θάλλω 'blossom, flourish' Hes. *hHom.*+: MW *deillyaw**, Alb. *dal*, *del* 'go out'
   c. ἔχανον 'gape, yawn' Hom.+: OCS *zinǫti* 'gape', OHG *ginēn* 'yawn', ON *gína* 'id.'
   d. πυνθάνομαι 'learn, ascertain' Hom.+: OCS *vŭzbŭnǫti, vŭzbŭde* 'wake up', Lith. *atbuñda, atbùsti, atbùdo* 'id.' (: OIr. *ad·boind*)
   e. [?]χανδάνω 'take hold' Hom.+: Lat. *prehendō*, *-ere*, OIr. *ro·geinn* 'occupies', Go. *duginnan* 'begin' (cf. Fortson 2009; Jasanoff 2022)

[17]A likely further instance of this type underlies ὁ θρόμβος 'lump, clot' Att. Ion.: Lith. *drim̃ba, drìbti* 'fall in flakes, drop'.

(8) Semantic correspondence with athematic nasal-infix middle

a. κάμνω '(tr.) produce/acquire through labor/toil (Hom.[+], but rare); (intr.) labor, toil, grow weary laboring/toiling' Hom.+: *śamnīṣe* 'labor' YV, *śamāyáte* 'id.' RV, *aśamīt* 'ist ruhig geworden' AVP, *áśamanta* 'labored' YV

b. ἁνδάνω 'be/become pleasing' Hom.+: ?Ved. *s₍ᵤ₎vádate* 'wird schmackhaft' RV[18] (: Ved. *s₍ᵤ₎vádati* 'macht schmackhaft' RV)

c. οἰδάνω 'swell (intr.)' I646 (οἰδάνεται), Ar.*Pax*.116 (οἰδάνων); 'make swell' (tr.) I554 (≃ A.R.1.478): ClArm. *aytnowm* 'swell [intr. *tantum*]'

(9) Semantic mismatch with athematic nasal-infix active[19]

a. φθῖνω/φθῐ́νω 'disappear, decrease' Hom.+: Ved. *kṣiṇā́ti*, *kṣiṇóti* 'destroy' AV, YAv. *jinā-ti* 'id.' (: ON *dvena* 'disappear', decrease', ON *dvína* 'id.')

b. θῡ́νω 'rush, dart along': Ved. *dhūnóti* 'shake'

c. ἁνδάνω 'be/become pleasing' Hom.+: ?Ved. *s₍ᵤ₎vádati* 'macht schmackhaft' RV, *svā̆́date* 'wird schmackhaft' RV

d. ὀλισθάνω 'slip' Att. (: ὀλισθαίνω 'id.' Arist.+): ὤλισθε Hom.+: Go. *slindan* 'schlingen' (: Ved. *srédhati* RV, *sridhat* RV)

e. πυνθάνομαι 'learn, ascertain' Hom.+ (: πεύθομαι Hom.+): OIr. *ad·boind* (: OCS *vŭzbŭnǫti, vŭzbŭde* 'wake up', Lith. *atbuñda, atbùsti, atbùdo* 'id.')

The semantic patterning here aligns closely with the fact that nearly two-thirds of the forms in Greek are intransitive and that transitive verbs are largely limited to cases where an original accusative of respect or goal is likely, such as λανθάνω, -ομαι 'escape notice (±acc.); (mid.) forget' (: λήθω, -ομαι Hom.+) or φθάνω/φθᾱ́νω '(tr.) be beforehand, outstrip, anticipate; (intr.) come/act first (±part.)' Hom.+, or where the root denotes a cognitive-perceptual or possessional state, state-oriented semantic categories that are regularly transitive cross-linguistically – e.g., πυνθάνομαι 'learn, ascertain' Hom.+ (: πεύθομαι Hom.+) or χανδάνω 'take hold of' Hom.+.[20]

[18] Assuming this form continues a nasal-infix present.

[19] For the pair φθάνω/φθᾱ́νω '(tr.) be beforehand, outstrip, anticipate; (intr.) come/act first (±part.)' Hom.+: Ved. opt. *daghnuyāt* 'würde knapp verfehlen', it is important to note that active *nu*-presents are often intransitive in Vedic and in some cases replace the thematic nasal-infix intransitive actives discussed here – viz. Ved. *tr̥mpáti* 'take delight in' RV: *tr̥pṇóti* RV 'id.'

[20] On state-oriented semantic subclasses, see Rau 2013, 2016, and 2017.

The traditional approach to these verbs, which stretches back to the 19th c. and is followed by all modern scholars, including Klingenschmitt 1982, Rix 1992, LIV² *passim*, and others, explains them as the result of an inner-Greek functionless thematization. This is problematic on all fronts and can be excluded.[21]

[21]The traditional approach, which explains the whole type this way and especially forms with suffixal -άνω ⇐ *-Cn̥hₓont(i) bzw. -Cn̥u̯ont(i)*, has four fatal issues:

1. As noted in the survey, the Greek forms show a fundamental and unmotivated semantic mismatch with active athematic nasal-infix presents, which are in effect exclusively transitive in the languages and mostly make causative change of state/location (±activity) formations – apart from late NPIE **dʰr̥snéu̯-ti* (: Ved. *dhr̥ṣṇóti* 'dare', etc.) and some *nu*-presents in Indic and Iranian, which are obviously secondary. On the general semantic profile of nasal-infix presents, see Meiser 1993.
2. There is no reason to assume that Greek generalized the thematic 3rd pl. in **-ont(i)* to athematic stems and positive evidence against this assumption. See Peters 2004 and further Willi 2018: 286–298.
3. There is no reason to assume that the syllabification of the 3rd pl. of the nasal-infix presents in Greek or the other IE languages was *-Cn̥hₓont(i) bzw. -Cn̥u̯ont(i)*. The nasal-infix formant fails to show syllabic variants in its *inflectional* paradigm (see Praust 2004), at least in the environments in question here – viz. when infixed before a sonorant or laryngeal in a minimally tri-consonantal root – and instead developed an epenthetic vowel, a situation also evidenced by the Greek treatment of the nasal-infix presents to laryngeal final roots. The default assumption is that this also held in the 3rd pl. and that the epenthetic vowel was later deleted as in Indo-Iranian.
4. Even if these assumptions were reasonable, there is no other support for the assumption that nasal-infix presents were regularly or even occasionally subject to thematization in the deeper prehistory of the language. Nasal-infix presents instead follow other well documented developmental pathways, see Sturm 2021.
    a) Some have replaced their athematic forms with *gr̥bhāyáti*-derivatives – e.g., ὑφαίνω 'weave' (: Ved. *unábdhi* 'id.'), φαίνω 'show, make appear' (: ClArm. *banam* 'disclose, discover'), ἰαίνω 'warm, cheer' (possible athematic ἰνάω 'empty') (: Ved. *iṣṇā́ti* 'set in vigorous motion'), κλῑ́νω 'incline' (: YAv. *-sirinao-ti* 'lean on (tr.)'), κρῑ́νω 'judge' (: Lat. *cernō*, *-ere* 'distinguish, discern'). On the type, see Jasanoff 2003, and on the Greek forms, Kölligan 2010.
    b) Nasal-infix stems to roots in a stop were regularly remodeled as *nu*-stems – cf. ζεύγνῡμι 'yoke, join' (: Ved. *yunákti* 'id.'), ὀρέγνῡμι 'stretch out' (: Ved. *r̥ṇákti** 'id.'), etc.
    c) Nasal-infix stems to roots in *-h₂* develop normally and then in Attic Ionic are restructured as *nu*-verbs – cf. πίτνημι 'spread out' Hom.+ (: πετάννῡμι Att.), etc.
    d) Nasal-infix stems to roots in *-h₃* develop normally and then in the history of Greek are restructured as *nu*-stems, with further late remodeling in some cases – cf. στόρνῡμι 'strew' Hom.+ (: στορέννῡμι Eust.), etc.
    e) Nasal-infix stems to roots in *-h₁* are early analogically restructured as thematic stems – cf. βάλλω, τάμνω, etc.

The obvious alternative to this approach is to motivate the form and function of these verbs by aligning them with the active thematic nasal-infix inchoative type that Gorbachov has outlined for Baltic, Slavic, and Germanic. This is straightforward in the case of the forms in set 1) above – that is, everything but the ἀνω-presents. These can all easily be explained as direct phonological and semantic continuants of this type – viz. φθῑ́νω/φθῐ́νω Hom.+ < **d^{h}g$^{u̯h}$inu̯-o/e-* (: φθινύθω Hom.+, Ved. *kṣiṇā́ti*, *kṣiṇóti* 'destroy' AV, YAv. *jinā-^{ti}* 'id.'; ON *dvena* 'disappear, decrease', ON *dvína* 'id.'), θάλλω Hes. *hHom.*+ < **d^{h}l̥nh$_{1}$-o/e-* (: Alb. 1sg. *dal*, *del* 'go out' < **dalnō*, *dalnet* and MW *deillyaw** << **daln-*), etc.

The forms in set 2) – ἀνω-presents to roots in obstruent without a demonstrable laryngeal –, which largely show the same morphology, semantics, and derivational relations as set 1) and cannot be explained as the result of an inner-Greek functionless thematization (see footnote 21), can also be aligned with this type under two straightforward assumptions. First, the suffix -ἀνω was extracted from roots that ended in an obstruent plus laryngeal – viz. *-Cano/e-* < **-Cn̥h$_{x}$e/o-* – and then extended to all obstruent final roots. This development can be understood as an example of the "externalization" of the nasal formant, a process commonly found in all IE languages and well-attested in Greek – cf., e.g., ζεύγνῡμι (: Ved. *yunákti*), ὀρέγνῡμι (: Ved. *r̥ṇákti**), etc. or, with the *gr̥bhāyáti*-type, Gk. ὑφαίνω (: Ved. *unábdhi*), κῡδαίνω 'glorify, honor' Hom.+ (⇐ ἰαίνω 'invigorate, warm' [: Ved. *iṣṇā́ti* 'set in vigorous motion']), etc. – and can be seen to be exactly parallel to what has happened in Slavic and Germanic, where the synchronically normal shape of the formant has been extracted from sonorant-plus-laryngeal-final roots – viz. *-I/R̥ne/o-* < **-I/R̥nh$_{x}$e/o-* – and extended at the expense of all other stem shapes – e.g., OCS *prilĭ(p)nǫ, -lĭ(p)nǫti* 'cleave, cling to', Go. *aflifnan* 'remain', ON *lifna* 'id.' (: Lith. *lim̃pa, lìpti* 'cling, stick to; climb': Ved. *limpáti* AV+). Second, the extension of the formation to transitive verbs denoting cognitive-perceptual and possessional states, which would not a priori be expected from an anticausative formant made to causative change of state verbs of the sort found here, can be understood to result from either 1) the general reinterpretation and extension of the formant as an all-purpose change of state marker – e.g., Lith. *suprañta, supràsti,*

The nasal-infix stems that otherwise show up as thematic in Greek are 1) late backformations from the aorist – cf., e.g., τανύω 'stretch' Hom.+ (: τάνυμαι Hom.), ἀνύω 'seek, strive' Hom.+ (: ἄνυμες Theoc.+), 2) of denominative origin – cf., e.g., δῑνέω 'whirl around', κῑνέω 'set in motion', on which see Nikolaev 2025a,b – or obviously secondary – ἱκνέομαι 'reach, arrive' Hom.+ Att. Ion. (: ἵκω Hom. Dial.) –, 3) likely due to reinterpretation of archaic volitional/desiderative subjunctives as indicatives in performative, speech-act contexts *et sim.* – viz. ὀμνύω 'swear' Hom.+ Dial. (: ὄμνῡμι Hom.+), τῑ́νω/τίνω 'pay' Hom.+ (: <τιννυνα(ι)> Kyme, ca. 700 BCE, Bartoněk & Buchner 1997: 201–214, τίνυμαι Hom. Hes. Hdt. E.) –, or 4) simply morphologically obscure – viz. ἀρνέομαι 'deny' Hom.+, κυνέω 'kiss' Hom.+.

supràto 'understand' (: Go. *fraþjan* 'id.') – and/or 2) the precocious generalization of the suffix as a regular present-stem formant for (state-oriented) thematic aorists (and stative/presential perfects), as normal in later Greek.[22]

4 Indo-Iranian

The type is also unambiguously attested for Indic and indirectly for Iranian – though here less robustly than for Greek.[23] For the nasal presents under discussion, see further Kulikov 2000, Hill 2007, and especially Zasada 2021 *s.vv.*, who provides critical discussion of the relevant forms. The forms can usefully be divided into finite and participial examples.

(10) Intransitive nasal-infix thematic actives in Vedic: Finite Forms

a. *tr̥mpáti* 'enjoy, take delight in, fill up on (+gen.)' RV (12x), *tr̥pṇóti* 'id.' RV (5x), *tr̥pyati* AV+: *átr̥pat* AVŚ 3.13.6 (: adv. neut. *tr̥pát* RV 3x); cf. YAv. n. sg. *θrąfš°* 'satisfaction (*vel sim.*)' Yt.5.26, YAv. *θrąfəδra-* 'rich in; satisfied', which points to the Indo-Iranian age of the present.

b. *śrathnā́s*, *śrathnité* 'loosen, make slack' RV+: *śrathayanta* 'loosen, become slack'.

RV 2.24.3ab:

tád	*devā́nāṃ*	*devátamāya*	*kártuvam*	***áśrathnan***
that.NOM	god.GEN.PL	god.SUPERL.INS.SG	do.GER	loosen.3PL.IPF.ACT

dr̥ḷhā́⁺	*ávradanta*	*vīḷitā* |
firm.NOM.PL.N	soften.3PL.IPF.MID	hard.NOM.PL.

'That had to be done by the foremost god of gods: what was firm became loose, what was hard became pliant.' (Jamison & Brereton 2014)

Cf. TS 6.1.9.7 *ánu śr̥nthati* 'become loose.' On the geographic localization of the forms of the root with nasal, see Insler 1991: 95[3].

c. *r̥ṇóti*, *r̥ṇváti* 'set in motion, impel (tr., intr. mid.)', *íyarti*, *ī́rate* 'id.' RV+: *ārta* (intr.) RV.+.

[22]From this starting point the type will then have been fully morphologized as the regular present stem correspondent to thematic aorists and then further extended. A similar process will have taken place in Armenian.

[23]Possible indirect evidence for the type may also survive in the RV and AV in the tendency noted by Kulikov 2000 for nasal-infix verbs that have relatively well attested athematic and thematic by-forms to show a preference for the thematic variant in intransitive usages.

RV 6.2.6ab:
tveṣás *te* *dhūmá* ***r̥ṇvati***
turbulent.nom.sg you.2sg.gen smoke.nom.sg impel.3sg.prs.act
diví *ṣáñ* *chukrá*
heaven.loc.sg be.prs.ptcp.act.nom.sg gleaming.nom.sg
ā́tataḥ |
stretched.out.nom.sg

‘Your smoke, when it is in heaven, is turbulent in motion, stretched out (there) gleaming, …’ (Jamison & Brereton 2014).

Possibly further RV 1.144.5cd, 3.11.2c.

d. *jinóti/te*, *jínvati/te* ‘revitalize, refresh (tr., intr. mid.)’ RV+.

AVŚ 12.1.46cde:
krímir ***jínvat*** *pr̥thivi* *yádyad*
worm.nom.sg revitalize.3sg.ipf.act Earth.voc.sg whichever
éjati *prāvŕ̥ṣi* *tán* *naḥ*
move.3sg.prs.act rain.season.loc.sg that.nom.sg us.1pl.acc
sárpan *mópa* *sr̥pat*
crawling.prs.ptcp.nom.sg.act neg+towards.prvb crawl.3sg.aor.act

‘Welcher auflebende Wurm auch immer, o Erde, sich in der Regenzeit regt, der möge, wenn er kriecht, nicht zu uns herankriechen.’ (Zasada 2021: 105).

Possibly also AVŚ 11.4.14c, 16d = AVP 16.22.4b, 6d, AVŚ 12.1.3c. Cf. mid. 3rd sg. *jinvate* (intr.) RV 3.2.11ab. See Zasada 2021: 104ff.

e. *śvíndate* ‘shine’ Dhātup. (: *áśvitan*, *áśvait*, *śvitāná-* RV) (: ORuss. *svĭ(t)nuti* ‘grow light’, Lith. *šviñta, švìsti, švìto* ‘dawn’, ON *hvítna* ‘goes pale’).[24]

(11) Intransitive nasal-infix thematic actives in Vedic: Participial forms

a. *siñcáti/te* ‘pour out (tr., intr. mid.)’ RV+ (: YAv. *-hiṇca-*^{ti} ‘id.’): *sécate* ‘pour out (intr.)’ RV 10.96.1 (: OAv. PN *Haēcat̰.aspa-*), *sicyáte* ‘id.’: RV *ásicat/ta* RV+.

[24]Given its predominantly active but semantically intransitive aorist forms, which points to a verbal root primarily lexified as an intransitive, I tentatively include this verb here, despite its likely finite middle inflection, which can easily be explained as analogical *bzw.* pleonastic, as outlined below.

RV 5.85.6cd:

ékaṃ yád udnā́ ná pr̥ṇánti
single.acc.m which water.instr.sg neg fill.3pl.prs.act
*énīr **ā-siñcántīr***
mottled.nom.pl.f prvb-pouring.prs.ptcp.act.nom.pl.f
avánayaḥ samudrám ||
stream.nom.pl.f sea.acc.sg.m

'That the mottled streams, pouring out, do not fill the single sea with water.' (Jamison & Brereton 2014).

Possible also RV 1.121.6cd (*siñcáñ*). See Hill 2007: 92–95 with lit.

b. *prá … r̥ñjánti* 'stretch out (tr.)', 3rd pl. mid. *r̥ñjáte* 'stretch oneself out; move quickly forwards', them. *r̥ñjate* 'id.': *abhí … r̥jyate* RV 1.140.2 'id.'.

RV 1.95.7ab:

úd yaṃyamīti savitéva bāhū́ ubhé
up raise.3sg.prs.int.act Savitar.like arm.acc.du.m both.acc.du.f
sícau yatate bhīmá
seam.acc.du.f fearsome.nom.sg.m moving.towards.3sg.prs.mid
r̥ñján |
charge.on.prs.ptcp.act.nom.sg.m

'Like Savitar, he raises up his arms again. Healing himself along the two seams (of the world?), the fearsome one charging straight on.' (Jamison & Brereton 2014).

RV 1.172.2abc:

āré sā́ vaḥ sudānavo |
distance.loc dem.pron.nom.sg.f you.gen.dat.pl drop.rich.voc.pl
*máruta **r̥ñjatī́** śáruḥ* |
Marut.voc.pl moving.straight.prs.ptcp.nom.sg.f arrow.nom.sg.f
āré áśmā yám ásyatha ||
distance.loc stone.nom.sg which.acc hurl.2pl.prs.act

'In the distance be your straight-aiming arrow, you Maruts rich in drops, in the distance the stone that you hurl.' (Jamison & Brereton 2014).[25]

[25] Given its intransitive semantics, I include this form here; its athematic inflection represents a trivial inner-Vedic innovation.

RV 3.31.1:
śā́sad *váhnir* *duhitúr*
instruct.PRS.PTCP.ACT.NOM.SG.M conveyor.NOM.SG.M daughter.GEN.SG
naptíyaṃ *gād*
(grand)daughter.ACC.SG come.3SG.AOR.INJ.ACT
vidvā́m̐ *ṛtásya* *dī́dhitiṃ*
knowing.PRF.PTCP.ACT.NOM.SG.M truth.GEN.SG contemplation.ACC.SG
saparyán | *pitā́* *yátra* *duhitúḥ*
serving.PRS.PTCP.ACT.NOM.SG.M father.NOM.SG where daughter.GEN.SG
sékam ***ṛñján*** *sáṃ*
outpouring.ACC.SG stretching.out.PRS.PTCP.ACT.NOM.SG together
śagmíyena *mánasā* *dadhanvé* ||
capable.INS.SG mind.INS.SG run.3SG.PRF.MID

'Instructing the (grand)daughter [= Āhavanīya?] of his daughter [= Gārhapatya?], the (offering-)conveyor [= Agni] has come – he the knowing one, ritually serving the visionary power of truth – to where the father [= priest? Agni?], stretching out straight, has run toward the outpouring of his daughter [=butter offering?] with capable mind.' (Jamison & Brereton 2014).

RV 4.38.7,8:
utá *syá* *vājī́* *sáhurir*
and DEM.PRON.NOM.SG.M prizewinner.NOM.SG.M victorious.NOM.SG.M
ṛtā́vā *śúśrūṣamāṇas* *tanúvā*
truthful.NOM.SG.M fame.seeking.DES.PCTP.MID.NOM.SG.M body.INSTR
samaryé | *túraṃ* *yatī́ṣu*
clash.LOC headlong.ADV go.PRS.PTCP.ACT.LOC.PL.F
turáyann *ṛjipyó* *ádhi*
hasten.PRS.ACT.PCTP.NOM.SG.M hurrying.NOM.SG.M towards
bhruvóḥ *kirate* *reṇúm*
eyebrows.LOC.DU.F scatter.PRS.MID.3SG dust.ACC
ṛñján ||
stretch.out.PRS.PTCP.ACT.NOM.SG.M
utá *sma* *asya* *tanyatór* *iva* *dyór*
and EMPH DEM.PRON.GEN.SG.M thunder.ABL like heaven.GEN
ṛghāyató *abhiyújo* *bhayante* | *yadā́*
tremble.PRS.PCTP.ACT.GEN.SG charge.ABL fear.PRS.MID.3PL when
sahásram *abhí* *ṣīm* *áyodhīd*
thousand to PERS.PRON.ACC.SG battle.AOR.ACT.3SG

durvártuḥ *smā* *bhavati*
difficult.to.obstruct.NOM.SG.M EMPH become.PRS.ACT.3SG
bhīmá ***r̥ñján*** ||
fearsome.NOM.SG.M stretch.out.PRS.PTCP.ACT.NOM.SG.M

'And this prizewinner, victorious, truthful, himself seeking fame with his own body in the clash, hastening headlong toward them (fem. [=mares?]) as they go hastily, straight-flying, scatters dust up to the eyebrows as he stretches out straight. And they take fear at his charge as he shows his mettle, as if at the thundering of heaven. When a thousand have battled him, the fearsome one becomes difficult to obstruct, as he stretches out straight.' (Jamison & Brereton 2014).

c. *r̥ṇádhat* 'accomplish' RV AV, *r̥dhnóti* 'id.' RV 1x, AV 1x: *r̥dhyate*/*r̥dhyáte* 'succeed' RV+.

RV 1.173.11:
yajñó *hí* *ṣma* *índaraṃ*+ *káś cid*
sacrifice.NOM.SG.M for EMPH Indra.ACC whichever.NOM.SG.M
r̥ndháñ *juhurāṇáś*
suceeding.PRS.PTCP.ACT.NOM.SG.M swerving.AOR.PTCP.MID.NOM.SG.M
cin *mánasā* *pariyán* | *tīrthé* *ná*
EMPH mind.INS.SG go.around.PRS.PTCP.ACT.NOM.SG.M ford.LOC.SG as
áchā *tātr̥ṣāṇám* *óko* *dīrghó* *ná*
to thirst.PRF.PTCP.MID.ACC.SG.M home.ACC.SG.N long.NOM.SG.M as
sidhrám *ā́* *kr̥ṇoti* *ádhvā* ||
goal.ACC.SG PRVB do.3SG.PRS.ACT road.NOM.SG.M

'For any sacrifice that reaches fulfillment, even though it swerves along, meandering in mind, brings Indra to the house, as if bringing a thirsting man to a ford – as a long road brings home a man who reaches his goal.' (Jamison & Brereton 2014).

d. *minā́ti/te* 'damage, harm; exchange, change (tr., intr. mid.)' RV+ (: *mī́yate* 'disappear, pass away' RV).

RV 5.2.1cd:
ánīkam *asya* *ná* ***mináj***
face.NOM.SG.N his NEG changing.PRS.PTCP.ACT.NOM.SG.N
jánāsaḥ *puráḥ* *paśyanti* *níhitam* *arataú* |
people.NOM.PL front see.3PL.PRS.ACT down.set.ACC.SG.N spokes.LOC

'His face is not one that changes (its face): the peoples see it in front, set down in the circle of spokes.' (Jamison & Brereton 2014).

Cf. intr. them. mid.:

RV 1.79.2a:

ā́ te suparṇā́ ***aminantam̐***
PRVB your.GEN.SG fine.feathered.NOM.PL.M change.3PL.IPF.MID
évaiḥ
way.INS.PL

'Your fine-feathered (lightning flashes) zigzagged along their ways.' (Jamison & Brereton 2014).

e. *muñcáti/te* 'release, set free (tr., intr. mid.)' RV+: *múcyate/mucyáte* 'get free' RV+ (: *ámucat* RV+, *ámugdhvam* RV+; OCS *promŭknǫ, -mŭknǫti sę*, Lith. *muñka, mùkti*).[26]

AVŚ 8.7.10ab:

unmuñcántīr *vivaruṇā́ ugrā́*
PRVB.get.free.PRS.PTCP.NOM.PL.F off.Varuṇa.NOM.PL.F mighty.NOM.PL.F
yā́ viṣadū́ṣaṇīḥ |
who.NOM.PL.F destroying.poison.NOM.PL.F

'Sich befreiend, frei von Varuṇa, mächtig, die Gift zerstörend sind… [= Pflanzen].' (Zasada 2021: 214).

Cf. 8.7.4 (intr.) *prastr̥ṇatīs*, *pratanvatīs*. For full discussion, see Zasada 2021: 214.

f. *ásinvant-* 'insatiable' RV (5x; *asinvá-* 'id.' RV) if as likely to tr. **sinóti* 'satiate' (: TA *sim̐señc* 'satiate', mid. *sim̐santär* 'become satiated', TB *sinastar* 'id.').

RV 10.79.1d:

ásinvatī *bápsatī bhū́ri*
insatiable.NOM.DU.F gnawing.PRS.PTCP.NOM.DU.F much.ACC.SG
attaḥ
eat.3DU.PRS.ACT

'Insatiable, gnawing, they eat amply.' (Jamison & Brereton 2014).

There are two interesting observations that can be made about this material.[27] First, there is a fundamental difference in the distributional attestation of this

[26]Note further OP <a-m^u-u-θ> /amunθa/ D I 01, ii 2, 71, iii 71 'fled, escaped', which as generously pointed out by a reviewer is most easily taken as an intransitive thematic imperfect.

[27]It is also important to note that this type regularly correlates with the thematic aorist in Indic and Iranian – cf., e.g., *tr̥mpáti* 'enjoy, take delight in, fill up on (+gen.)' RV: *átr̥pat* AVŚ 3.13.6 (: adv. neut. *tr̥pát* RV) or *siñcáti/te* 'pour out (tr., intr. mid.)' RV+ (: YAv. *-hiṇca-^{ti}* 'id.'): RV *ásicat/ta* RV+ – as regularly in Greek (and Armenian) and systematically in OCS. This is obviously an inherited feature of a subset of the verbs in this class.

type in Greek and Indo-Iranian. In Greek, nearly all the forms we find are independent lexical items that synchronically lack the corresponding transitive nasal-infix form from which they have been derived. In Indo-Iranian, while there are such cases, we mostly find that the forms are independent, one-off active thematic intransitive uses of an otherwise well-attested nasal stem that normally signals causativity alternations with active vs. middle voice. This distribution suggests that the derivational type survived as a living process into Indo-Iranian and then early Indic at least in some verbs and/or as a dialectal phenomenon. What we have in our texts should thus likely be considered a mix of archaic and/or dialect forms and poetic innovations based on them.[28]

Table 4: Derivational & chronological relationship between athematic and thematic nasal-infix stems

Nasal-Infix Causative		**Nasal-Infix Anticausative**
PIIr.$_1$ **sinákti, sinktái* 'pour out (tr., intr.)'	⟶	**sinčáti* 'pour out (intr.)'
		⇓
⇓		**sinčáti*, *sinčátai* 'id.'
		⇓
PPIr.$_3$ **sinčáti*, *sinčátai* 'pour out (tr., intr.)' (> Ved. *siñcáti/te*, YAv. *hiṇca-*^{ti})	⇐	**sinčátai* 'id.' (residual **sinčánt-* > Ved. *siñcánt-* 'id.')

The second thing to note is that the preservation of this type into Indo-Iranian provides a straightforward explanation of another remarkable feature of some of the verbs that show this usage – the fact that already in the RV and in some cases in Avestan they are fully thematized, without any trace of their original athematic inflection – viz. *siñcáti/te* RV+: YAv. *-hiṇca-*^{ti}, *muñcáti/te* RV+ (: OCS *promŭknǫ, -mŭknǫti sę*, Lith. *muñka, mùkti*). For the attestation of these forms and others like it, see Hill 2007. The highly precocious thematization we find here and in similar cases – so, e.g., *kr̥ntáti/te* RV+, YAv. *kərəṇta-*$^{ti/te}$ (: OHG *scrintan*) and other stems with obstruent final roots liable to the causative alteration – makes perfect sense under the assumption that in some cases the thematic intransitive active was subject to an early pleonastic medialization, which then led to the backformation of a thematic transitive active and the displacement of the original athematic stem. The full derivational and chronological relationship between these forms is given in Table 4.

[28]The poetic character of the type can be seen clearly in the formulaic-template-type usage of, e.g., *pāda*-final *r̥ñján*.

Both analogical developments are well paralleled in Indic and Iranian – cf., e.g., Ved. *édhate* 'flourish, increase' (: αἴθω 'burn (intr.)' Pi.S., αἴθουσα 'portico' Hom.+; αἴθομαι Hom.+, ClArm. *ayrem* 'burn'), for "pleonastic" medialization, and Ved. *śráyati* 'lean on (tr.)' (: YAv. *-sirinao-^{ti}*; Ved. *śráyate*), for the backformation of a transitive active from an intransitive middle stem.[29]

5 Conclusion

Indo-Iranian and Ancient Greek[30] thus provide a very close parallel to the type outlined by Gorbachov for the Northern Indo-European languages – one that matches it morphologically, semantically, and derivationally. Based on the close agreement of these languages, it is possible to confirm Gorbachov's analysis of the Northern Indo-European material and to reconstruct the formation for late Nuclear Proto-Indo-European as an important deverbal and deadjectival intransitive change of state type.[31]

Abbreviations

A.R.	Apollonius Rhodius	E.	Euripides
Alb.	Albanian	Eust.	Eustathius
Ar.	Aristophanes	Gal.	Galen
Arc.	Arcadian	Gk.	Greek
Archil.	Archilochus	Go.	Gothic
Arist.	Aristotle	Hdt.	Herodotus
Att. Ion.	Attic Ionic	Hes.	Hesiod
AV	Atharvaveda	Hes. Sc.	Hesiod Scutum
AVP	Atharvaveda Paippalada	hHom.	Homeric Hymn
		Hitt.	Hittite
AVS	Atharvaveda Saunakiya	Hom.	Homer
		Hp.	Hippocrates
Call.	Callimachus	Hsch.	Hesychius
ClArm.	Classical Armenian	Intr.	Intransitive
Dhātup.	Dhātupātha	KS	Katha Samhita
Dial.	Dialects	Lat.	Latin

[29]See further footnote 14.

[30]Further Classical Armenian. See footnote 5.

[31]The type is also attested for Italo-Celtic as Gorbachov (2007: 217), following others, rightly notes – cf., e.g., Lat. °*cumbō*, *-ere* Pl.+, *ning(u)it* Lucr.+, and possibly °*hendō*, *-ere* Pl.+. A task for the future is to identify further instances in these branches.

Lesb.	Lesbian	Pi.	Pindar
Lith.	Lithuanian	Pl.	Plautus
Lucr.	Lucretius	PN	Proper name
MW	Middle Welsh	RV	Rigveda
Myc.	Mycenaean	S.	Sophocles
OAv.	Old Avestan	Sa.	Sappho
OCS	Old Church Slavonic	TA	Tocharian A
OE	Old English	TB	Tocharian B
OHG	Old High German	Th.	Thucydides
OIr.	Old Irish	Theocr.	Theocritus
ON	Old Norse	Tr.	Transitive
OP	Old Persian	TS	Taittiriya Samhita
OPr.	Old Prussian	Ved.	Vedic Sanskrit
ORuss.	Old Russian	YAv.	Young Avestan
PGmc.	Proto-Germanic	YV	Yajurveda

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Part III

Beyond Caland: Typological and theoretical perspectives

Chapter 10

Diachronic trends in the formation of Italian deadjectival verbs

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This chapter provides an overview of deadjectival verb formation in the history of the Italian language by outlining the productivity and comparing the semantics of the three available verbalising processes, namely, suffixation (*chiaro* 'light, clear' > *chiar-eggiare* 'clear up'), conversion (> *chiar-ire* 'clarify') and parasynthesis (> *s-chiar-ire* 'lighten'). While in the early periods of the language history parasynthesis is the best attested process, from the mid-19th century on suffixation becomes predominant. Significantly, it is most commonly through suffixation that complex adjectives (and thus most of those of more recent formation) are verbalised. Like parasynthesis, conversion, despite being well attested in the early stages, has seen a marked decline in productivity, and is overall the least productive of the three processes. Variation in productivity over time is also observed when comparing different derivational patterns (e.g., different suffixes) within a single process. The semantic analysis of the derived verbs confirms that when adjectives are turned into verbs, this is mostly to express a change of state ('make/become Adj'), which is the semantic space for which all three processes potentially compete. States and activities ('be/appear Adj', 'behave in a Adj'), on the other hand, are only encoded by suffixation and conversion, with a prevalence of the former. Parasynthesis is reserved for expressing change-of-state events, as well as for deriving the largest number of verbs expressing an inchoative meaning ('become Adj') in the active form, i.e., without the anticausative marker *-si*. Overall, this study allowed us to reconstruct the development of Italian deadjectival verbalisation over a wide time frame, while also gaining insights into the factors that might have influenced it and possibly contribute to shaping current trends as well.

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1 Introduction

This contribution explores the derivation of deadjectival verbs in the Italian language over a timeline spanning from the origins (13th century) to the end of the 20th century. As Italian can avail itself of three different derivational processes to form verbs from adjectives – namely suffixation, conversion, and parasynthesis – which exhibit overlapping semantic functions, our goal is to investigate how these processes have interacted and competed over time.

The study, based on a dataset consisting of all the attested verbs derived from the most frequent Italian adjectives throughout history, outlines and compares the productivity of the different verbalising processes by tracking novel coinages, while also taking into account the competition between synonymous verbs, a pivotal factor contributing to the marginalisation and eventual obsolescence of lexemes. Indeed, it is by considering the relationship between new formations and the existing lexicon at a given point in time that more light can be shed on the dynamics determining the productivity of word-formation processes.

The few studies on Romance verb formation that take a diachronic perspective point out an abundance of verbs formed from the same adjective in the first stages of the language and their reduction over time. This trend was first highlighted in the seminal research by Malkiel (1941), as well as in recent contributions on Portuguese and Spanish by, respectively, Rio-Torto (2019) and Espejel (2023). The only diachronic investigation on Italian deadjectival verb formation available is Grossmann & Iacobini (2025), from which the present study differs in methodology, contributing new data to the documentation and analysis of the phenomenon.

This chapter is organised as follows. In Section 2, we begin by describing the methodological choices underlying the data collection (2.1) and the classification (2.2). In Section 3 we present the results. We first outline the diachronic distribution of verbs according to derivational process (3.1) and pattern (3.2). Subsequently (3.3), some insight into the phenomenon of competition between the derivational processes is provided: Sections 3.3.2–3.3.3 focus on the adjectival side and highlight the relationship between the semantic and derivational features of the adjectives and their availability to serve as bases for verbal derivation, while Section 3.3.4 is devoted to describing how the derivational processes have been employed for the expression of the various possible semantic types that are typically yielded in deadjectival verbalisation. Section 4 wraps up the paper with a summary of the main findings.

2 Data

2.1 Sample selection

The collection of verbs departed from a sample of 858 adjectives, i.e., those identified as the most frequent and widespread according to the *Basic Vocabulary of the Italian Language* (De Mauro 1980).[1] These adjectives were selected to obtain a wide and representative list of deadjectival verbs formed over a broad historical range, extending from the 13th century to the end of the 20th century. A total of 917 verbs was gathered by means of a manual search in the *Grande Dizionario della Lingua Italiana* (GDLI) historical dictionary.

This method allowed us to identify the number and types of verbal derivatives for each adjective, as well as the adjectives lacking documented verbal derivatives, thus offering a means to investigate the propensity of adjectives for verbalisation.

The use of the GDLI was a necessary choice due to the lack of large diachronic corpora for Italian and of other electronically searchable resources. Our productivity measures, based on dictionary data, quantify the number of verbs first documented in a given time period. The use of the GDLI made it possible to compile a comprehensive list of verbs belonging to the diasystem of the Italian language. The verbs were subsequently sought in the *Grande Dizionario della lingua Italiana* (GRADIT), the largest dictionary of contemporary Italian and our source for the present-day usage of the verbs. Through this second search we could also identify the subset of verbs exclusively lemmatised in the historical dictionary that did not make it into the GRADIT.

2.2 Annotation criteria

2.2.1 General

The adjectives were classified based on morphological make-up following Thornton et al. (1997), who distinguish between simplex and complex adjectives, and

[1] The adjective collection was carried out by consulting the BDVDB database (Thornton et al. 1997), which contains the lemmas of the *Basic Vocabulary of the Italian Language* annotated with orthographic, phonological, morphological, syntactic and etymological information, and provided with the date of first attestation. Adjectives from the *Basic Vocabulary* were selected from those belonging to the categories *Fundamental Lexicon* and *High frequency*. The adjectives of the *Basic Vocabulary* cover about 90% of the adjectives that can be found in any modern Italian text. Besides, they are to a large extent of ancient attestation: More than 70% are in fact attested as early as the 14th century, and only 5% are attested after the mid-19th century.

identify, with regard to the latter category, the following adjective-forming processes: prefixation (*disonesto* 'dishonest' < *onesto* 'honest'), suffixation (*maschile* 'masculine' < *maschio* 'male'), compounding (*maleducato* 'ill-mannered' < *male* 'bad' and *educato* 'mannered') and, finally, denominal and deverbal conversion (*marrone* 'brown' < *marrone* 'chestnut', *sveglio* 'awake, alert' < *svegliare* 'wake up'). Aiming at the maximal identification of morphological compositionality possible, they include within complex adjectives also those in which a morphological element is identifiable solely through the comparison of words sharing similar derivational characteristics, even if not fully analysable in discrete elements. The choice of adopting this classification is motivated by the diachronic setting of our study, and is further supported by finding that the distinction between synchronically fully-compositional and non-fully-compositional would not significantly impact the distribution of the dataset's adjectives into morphologically complex and non-complex.[2]

In order to follow the evolution of the use of the different derivational processes and patterns over time, as well as to classify the verbs from a semantic and aspectual point of view, the verbs were annotated for a) first occurrence, b) current usage, c) verb-forming process and pattern expressing each process, and d) semantic and aspectual features. We discuss these criteria in more detail in the following sections.

2.2.2 First occurrence

The verbs' date of first attestation was aligned to the temporal scan outlined in the MIDIA corpus (D'Achille & Grossmann 2017), which divides the history of the Italian language into five periods based on significant linguistic, literary, and cultural events:

1. The period from the early 13th century to 1375, spanning from the emergence of literature and vernacular writing in Tuscany to Boccaccio's death and Coluccio Salutati's appointment as Chancellor of Florence.

2. From 1376 to 1532: This period includes the era of Humanism and the Renaissance, characterised by the evolution from "Silver Florentine" to the establishment of "Golden Florentine" as the literary norm, as advocated by Pietro Bembo in his *Prose della volgar lingua* (1525). It concludes with the publication of the third edition of *Orlando Furioso* by Ludovico Ariosto, embodying the realisation of Bembo's linguistic principles in poetry.

[2] On the derivability of verbs from morphologically complex adjectives in Italian, see Grossmann (1999).

3. From 1533 to 1691: The Late Renaissance, Mannerism and Baroque era, culminating in the publication of the third edition of the *Vocabolario degli Accademici della Crusca* (1691), and the foundation of the Arcadia in 1690.

4. From 1692 to 1840: This epoch encompasses the Arcadia, the Enlightenment, and Romanticism. It culminates with the definitive edition of Alessandro Manzoni's *I Promessi Sposi*, which, drawing upon spoken Florentine, served as a linguistic model in post-unification Italy.

5. From 1841 to 1947: The era of the unification of Italy, World Wars, extending to the advent of the Italian Republic and the ratification of its constitution. Our periodisation includes a subsequent period corresponding to the Republican era.

6. From 1948 to the present: An epoch characterised by mass schooling, the spread of the mass media, and the resulting gradual affirmation in speech of the Italian standard language at the expense of the dialects (De Mauro 2014).

2.2.3 Current usage

Verbs are grouped according to the usage labels of the GRADIT dictionary. We distinguished the verbs that remain in present-day usage from those which have become obsolete. The latter group includes the verbs lemmatised in the GDLI but not in the GRADIT (about 17% of the original set) and those marked in the GRADIT as obsolete and/or only attested in literary texts.

2.2.4 Verb-forming process and pattern

The processes available in Italian for the derivation of verbs from adjectival bases are suffixation, conversion and parasynthesis. Each process is characterised by distinct patterns: Suffixation involves the addition of one of a set of suffixes, which all assign the verbs to the -*are* inflectional class. Conversion involves the use of inflectional marks of the -*are* or -*ire* class. Parasynthesis results from the combination of one of a set of prefixes and conversion.

Given the wide diachronic perspective adopted here, only the patterns allowing a comparison over the whole history of the language have been taken into consideration. The verbalising suffixes that have been used productively from the origins until present day are -*izzare*, -*eggiare*, and -*ificare* (e.g., *estremo* 'extreme' > *estremizzare* 'take to the extremes', *verde* 'green' > *verdeggiare* 'be/become green, lush', *intenso* 'intense' > *intensificare* 'intensify'). Verbs suffixed with -*itare* were also included because of the wide use of some of them (e.g., *facile* 'easy' > *facilitare* 'facilitate', *mobile* 'movable, moving' > *mobilitare* 'mobilise') and their morphological and semantic transparency, despite their limited number and the current lack of productivity of the suffix.

The suffixes -*izzare* and -*eggiare* go back to the same Greek etymon, -ιζειν. The form -*izzare* comes to the Romance languages through learned borrowings from Latin, particularly used in religious and philosophical writing, while the suffix -*eggiare* goes back to the vernacular Latin form -*idiare*, manifesting clear Romance phonetic developments.

The suffix -*ificare* results from Latin compound verbs like *certificare* (*certum facere*), *vīvificare*, *mānsuēfacere*, *stupefacere* and from other verbs derived either from nouns ending in -*fex* or from adjectives ending in -*ficus*. Suffixed verbs have only ever belonged to the default inflectional class (-*āre*).

Conversion can be defined, following Valera (2015: 322), as "a word-formation process where the form of the converted item does not change, while its inflectional potential, its syntactic function and its meaning do, such that the item displays inflectional, syntactic and semantic properties of a new word class." While well suited to languages with poor inflectional systems, this definition does not fully apply to derivation by conversion in languages, like Italian, in which the inflectional features of verbs are marked by a rich set of inflectional endings, which clearly distinguish the categories of verbs from those of nouns and adjectives.

Conversion assigns a verb to the -*are*, by large the most productive, or the -*ire* inflectional class, e.g., *libero* 'free' > *liberare* 'set free', *scuro* 'dark' > *scurire* 'darken'.

Parasynthesis can be defined as the simultaneous application of two derivational processes (prefixation and conversion), which results in the coinage of prefixed verbs directly derived from an adjectival (or nominal) base (see Iacobini 2021). In the current synchronic stage neither the non-prefixed verb nor the prefixed adjective is an actual word. However, as already noted in Iacobini (2004), and corroborated by a number of examples from the present dataset, in prior stages of the language, parasynthetic and converted verbs formed from the same adjective were attested in the same time period.

For the data collection, we sought the verbs with an ingressive change-of-state meaning ('make/become Adj') that are derived by prefixes still currently productive, i.e., *ad*-, *in*-, *s*-, *de*-/*di*-/*dis*-, *ri*-/*ra*-/*rin*-, e.g., *dolce* 'sweet' > *addolcire* 'sweeten', *grasso* 'fat' > *ingrassare* 'gain weight; fatten', *bianco* 'white' > *sbiancare* 'whiten', *lucido* 'shiny, polished, lucid' > *delucidare* 'elucidate', *giovane* 'young' > *ringiovanire* 'look younger; make sb./st. look younger'.

2.2.5 Semantic and aspectual features

The semantic and aspectual classification is based on the correlation between the meaning of the adjective and the predicative function expressed by the verb. Verbs can fall, based on telicity and dynamicity, into the categories of change of state, activity and state. Change-of-state (CoS) and activity verbs share the feature dynamic, whereas activity and state verbs have the feature atelic in common. Based on argument structure, a further distinction can be made, within change-of-state verbs, into causative and inchoative. Activity (behaviour) and state (copulative) verbs, on the other hand, are only intransitive. Table 1 shows the interplay between these defining criteria.

Table 1: Semantic and aspectual types of deadjectival verbs

	Dynamic		Static
	Telic	Atelic	Atelic
	Change of state	Activity	State
Tr.	Causative *legalizzare* ‘legalise’ *chiarire* ‘clarify’ *rallegrare* ‘cheer up’		
Intr.	Inchoative *impallidire* ‘turn pale’ *curiosare* ‘snoop around’	Behaviour *americaneggiare* ‘act American’	Copulative *pianeggiare* ‘be flat’

In most cases the verbalisation of an adjective results in a CoS verb, i.e., the situation most commonly lexicalised by deadjectival verbs is the causation and/or the acquisition of the state or property denoted by the adjectival base. Such verbs can occur in causative or inchoative constructions.[3] In causative constructions, the subject referent is the agent that causes the change of state of a patient/theme. With reference to the relationship between the adjective and the verb, the meaning of a causative verb can generally be paraphrased as ‘make (more) Adj’, e.g.,

[3]In line with the terminology generally employed in the Italian word-formation literature, our use of the terms “causative/inchoative” does not presuppose a distinction to “factitive/fientive”: We use the former pair as general terms for transitive and intransitive change-of-state deadjectival verbs.

legale 'legal' > *legalizzare* 'legalise', *chiaro* 'light' > *chiarire* 'clarify', *allegro* 'cheerful' > *rallegrare* 'cheer up'.

In inchoative constructions, the subject referent takes on the role of the patient/theme affected by the CoS, regardless of the cause. The meaning of the verb can thus be paraphrased as 'become (more) Adj', e.g., *pallido* 'pale' > *impallidire* 'turn pale', 'go white'. These verbs may exhibit a causative-inchoative alternation, which can either lack morphological marking (e.g., *gli zuccheri ingrassano Giovanni* 'sugars make John fat'/*Giovanni ingrassa* 'John is getting fat') or, more commonly, be marked by the clitic pronoun *-si*. For some verbs, the clitic pronoun is obligatorily expressed (e.g., *grande* 'big' > *ingrandire* 'make bigger'/*ingrandirsi* 'grow bigger'), while in others, its use is optional (e.g., *zitto* 'silent' > *zittire*/*zittirsi* 'go silent'). In contemporary Italian, the prevailing method of conveying the inchoative sense involves the use of the pronominal form. However, in earlier stages of the language, numerous verbs were employed in inchoative constructions without the need for a clitic pronoun (cf. a.o. Cennamo 2015; Bentley 2024).

The CoS can manifest as either ingressive, emphasising the transition into a state characterised by the possession of the property denoted by the adjective from a preceding state, or egressive, if the departure from the original state is highlighted during the transition from one state to another (cf. Grossmann 1994; Iacobini 2021). Put differently, the CoS may involve either the acquisition of a property not previously held or, conversely, the deprivation or loss of a property. In verbs where the adjective signifies a scalar, gradable attribute, and the scale is unbounded, the CoS occurs as an approximation or deviation towards an indefinite endpoint, representing a transition between relative states. This transition involves either an augmentation or diminution of a property already possessed to some degree.

Unlike denominal verbs, whose interpretation relies on context and on some kind of knowledge regarding the usual or expected use of the noun referent, deadjectival verbs do not normally require any encyclopaedic integration, as their essential function is to relate the property expressed by the adjective with the object referent in causative constructions or with the subject in inchoative constructions.

Deadjectival verbs also lexicalise situations in which no change of state is predicated. Within this category, further distinctions can be drawn, based on dynamicity and argument structure, into copulative and behaviour verbs. Copulative verbs denote an inherent or permanent mode of being of the subject referent, which can be paraphrased as 'be Adj', e.g., *piano* 'flat' > *pianeggiare* 'be flat'.

Behaviour verbs are dynamic verbs paraphrasable as 'behave in the manner denoted by Adj' or 'appear Adj'. The subject referent refers to an entity that demonstrates or possesses a certain property to varying degrees or intermittently over time, e.g., *americano* 'American' > *americaneggiare* 'play the yankee, act American', *curioso* 'curious' > *curiosare* 'nose around, snoop', *folle* 'deranged, lunatic' > *folleggiare* 'frolic, paint the town red'.

The analysis also revealed the presence of a small number of verbs that do not fit into the grid identified by the intersection of the parameters indicated so far. In this case, transitivity does not coincide with a causation in the object referent of the property expressed by the base adjective, e.g., *frequente* 'frequent' > *frequentare* 'attend, frequent, date', *violento* 'violent' > *violentare* 'rape'. These verbs lack the usual semantic transparency characterising the transitive verbs previously mentioned, as the property expressed by the base adjective does not denote the final state or property of the object referent, but may be differently interpreted to refer, for example, to the subject referent, or to the event as a whole (cf. also Fábregas 2023: 108–112). Due to their extremely small number, these verbs are not included in the quantitative evaluations in Section 3.3.3.

3 Results

3.1 The derivational processes

Table 2 shows the percentage distribution of all verbs in the dataset by process and time period. Parasynthetic verbs are the most numerous, accounting for 45.6% of the total. Suffixed verbs account for approximately a third of the total (32.7%), whereas converted verbs make up the smallest group (21.7%). While about half of the verbs are formed during the initial two time periods, all three processes remain productive until present day.

Table 2: Process productivity over time: Percentage of new coinages from the total of verbs

Process	**I: 1200–**	**II: 1376–**	**III: 1533–**	**IV: 1693–**	**V: 1841–**	**VI: 1947–**	**Total**
Suffixation	3.7	2.3	5.2	3.9	8.9	8.7	**32.7**
Conversion	10	4.5	2.8	1.4	2.1	0.9	**21.7**
Parasynthesis	19.2	7.1	8.1	3.8	5.9	1.5	**45.6**
Total	**32.9**	**13.9**	**16.1**	**9.1**	**16.9**	**11.1**	**100**

Table 3 shows the percentage of verbs formed by the different processes for each of the time periods. While parasynthesis is the best attested process in each of the first three periods, from the end of the 17th century it starts being overridden by suffixation, which becomes the most prominent process from the mid-19th century onwards. A remarkable decrease in productivity concerns also conversion, which, while being well attested in the first two periods, shows a significant decrease in use from the middle of the 16th century onwards.

Table 3: Process productivity over time: Percentage of new coinages per time period

Process	**I: 1200–**	**II: 1376–**	**III: 1533–**	**IV: 1693–**	**V: 1841–**	**VI: 1947–**
Suffixation	11.3	16.5	32	42.8	52.9	78.5
Conversion	30.4	32.3	17.7	15.5	12.3	7.8
Parasynthesis	58.3	51.2	50.3	41.7	34.8	13.7
Total	**100**	**100**	**100**	**100**	**100**	**100**

Being aware that selecting high-frequency adjectives might result in an over-representation of the processes employed in the earliest stages, we compared the distribution of the verbs collected with the distribution of all derived verbs lemmatised in the GRADIT (represented, respectively, by the solid and the dotted line in Figure 1), with the aim of checking whether our collection method significantly skewed the proportions of the verbal coinages across the different time periods.

Although the two distributions diverge in the earliest and latest time periods, they show a largely similar trend. As expected, the verbs in our dataset are slightly more numerous than those in the GRADIT during the earliest period, whereas the reverse is true in the most recent one. Nonetheless, our data distribution does not appear significantly skewed, in that even in the two most recent time periods it closely mirrors that resulted from the GRADIT, and also aligns with the findings of quantitative studies on 20th century Italian word formation (Iacobini & Thornton 1992, Peša Matracki 2006, Micheli 2024), which show a decrease in parasynthetic verbs and an increase in suffixed verbs when compared to previous stages of the language.

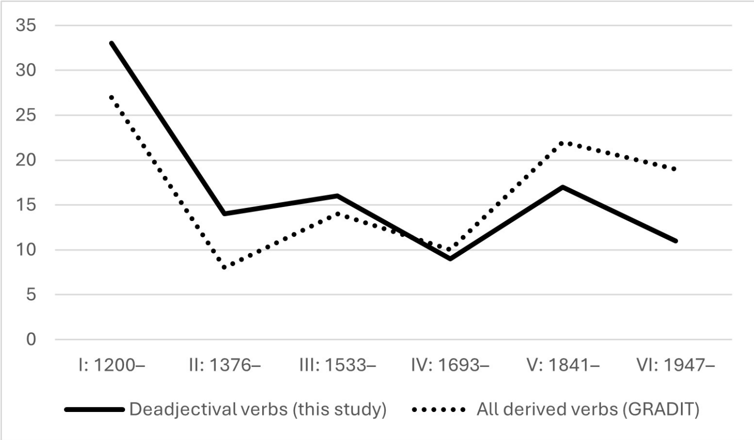

Figure 1: Coinage over time of all complex verbs (deadjectival, denominal, deverbal) vs. deadjectival verbs collected in this study

3.2 The derivational patterns

3.2.1 Suffixation

The formation of suffixed verbs is especially concentrated in the two most recent time periods, as shown in Table 4.

Table 4: Productivity of suffixes over time: Percentage of new coinages on the total of suffixed verbs

Pattern	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–	Total
-izzare	1	1.7	6.3	8.3	19.3	22.7	**59.3**
-eggiare	5.3	3.7	8	2.3	5.3	2.7	**27.3**
-ificare	4.7	1	0.7	1	2.7	1.3	**11.4**
-itare	0.3	0.7	0.7	0.3	0	0	**2**
Total	**11.3**	**7.1**	**15.7**	**11.9**	**27.3**	**26.7**	**100**

Table 5 shows the percentage of verbs formed by different suffixes in each period. In the earliest centuries, -*eggiare* is the most frequently used suffix, accounting for about half of suffixed verbs in each of the first three periods. From the end of the 17$^{\text{th}}$ century, its productivity decreases, becoming very scarce in contemporary Italian: From the second half of the 20$^{\text{th}}$ century onwards, coinages in

Table 5: Productivity of suffixes over time: Percentage of new coinages per time period

Pattern	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–
-*izzare*	8.8	23.8	40.3	69.4	70.7	85
-*eggiare*	47.1	52.4	51.1	19.5	19.5	10
-*ificare*	41.2	14.3	4.3	8.3	9.8	5
-*itare*	2.9	9.5	4.3	2.8	0	0
Total	**100**	**100**	**100**	**100**	**100**	**100**

-*eggiare* constitute just over 10% of suffixed deadjectival verbs. Conversely, from the mid-16th century onwards, there is a significant increase in the productivity of the suffix -*izzare*, which becomes the most employed suffix by the end of the 17th century; its productivity continues to grow steadily, so that from the second half of the 20th century onwards, verbs in -*izzare* constitute about 85% of suffixed formations. The success of this suffix is linked to the influence of Modern Latin, and, more recently, to contact with French and English, cf. Fally & Goryczka (2021); Radtke (1994).

Of the verbalising suffixes, -*ificare* is the least productive, forming 11.4% of total suffixed verbs, and appears to be employed mainly in the first two temporal phases. The recent reduction of productivity that is shown here may be overrepresented due to the characteristics of our dataset, which, being built from the highest-frequency adjectives, cannot fully account for the verbal coinages in the technical-scientific domains, where recourse to -*ificare* (as well as -*izzare*) is frequent.

The data concerning the productivity of the three suffixes agrees with what is reported by Grossmann (2004) for the second half of the 20th century, according to which the vast majority of deadjectival suffixed verbs are in fact formed by means of -*izzare*, while recent productivity of -*ificare* and -*eggiare* is rather low.

In contemporary Italian, the suffix -*eggiare* is predominantly used to lexicalise activities and states, while verbs formed in earlier stages more often also present CoS, causative meanings. At present, these verbs (or, specifically, their causative meanings, in the cases of polysemous verbs) are generally of low frequency or have fallen out of use, e.g., *aspro* ‘sour, harsh’ > *aspreggiare* ‘treat harshly’; *falso* ‘false’ > *falseggiare* ‘falsify’; *stanco* ‘tired’ > *stancheggiare* ‘tire’. Only in rare instances is the causative meaning preserved in high frequency verbs, e.g., *amaro* 'bitter, sour' > *amareggiare* 'embitter, sour'; *pari* 'equal, even' > *pareggiare* 'level', besides 'tie, draw'.

Concerning the suffix -*izzare*, the opposite case is observed, whereby a number of verbs testifies to an earlier use of this suffix for the expression of an activity, i.e., a manner of behaving. This is in fact the function originally exhibited in the early attestations in Latin, later reinterpreted as causative (cf. Gibert-Sotelo & Pujol Payet 2020). Only rarely, however, do -*izzare* verbs exclusively express an activity meaning, e.g., *avaro* 'greedy, miserly' > *avarizzare* 'behave like a miser', or *ostile* 'hostile' > *ostilizzare* 'behave in a hostile way'. More often, both an activity and a causative meaning are documented for the same verb, e.g., *ebreo* 'Jew/Jewish' > *ebraizzare* 'make Jewish or similar to Jews' besides 'imitate the Jews, follow Jewish customs and traditions'; *unico* 'unique' > *unicizzare* 'be unique, behave as unique' besides 'unite, unify'. Such verbs are currently of low frequency or obsolete.

Due to this semantic overlap between the suffixes -*izzare* and -*eggiare*, especially observable in earlier periods (cf. also Tronci 2019), various doublets are attested, most often lexicalising an activity, e.g., the forms *latineggiare*/*latinizzare* (< *latino* 'Latin'), both of which could express the meaning 'make use of Latinisms' in the past. In contemporary Italian, however, said meaning is predominantly (if not exclusively) expressed by the verb in -*eggiare*, while the verb in -*izzare* survives with the causative meanings of 'introduce to a people the Latin or neo-Latin language and culture' and 'give Latin form to words of another language'. The very few verbs in -*izzare* that currently lexicalise an activity do not have an -*eggiare* counterpart, e.g., *solidale* 'supportive, sympathethic' > *solidarizzare* 'sympathise (with sb.)', *sottile* 'subtle' > *sottilizzare* 'split hairs'. In the fewer cases of doublets in which the -*eggiare* verb used to express a causative value, the verb has fallen out of use, with the rival in -*izzare* prevailing, e.g., *ridicolo* 'ridicule' > *ridicoleggiare*/*ridicolizzare* 'ridicule'.

While specialisation over time is observed for -*eggiare* and -*izzare*, -*ificare* expresses from the beginning only a causative value, accordingly to its etymological origin. It is therefore not surprising that in the dataset only one doublet in -*ificare* and -*eggiare* is attested, i.e., *chiareggiare*/*chiarificare* (< *chiaro* 'light, clear'), among whose meanings the causative 'make clear, clarify' is shared. More numerous are doublets in -*ificare* and -*izzare*. Although the suffixes both contribute (primarily or exclusively) a causative meaning, the verbs forming the doublets are never fully interchangeable, as in *elettrizzare*/*elettrificare* (< *elettrico* 'electrical'), where the technical meaning of 'charging a body with electricity' is shared, but only *elettrizzare* is also used figuratively to mean 'put in a state of excitement'.

3.2.2 Conversion

Converted verbs are for the most part attested in the first two time periods, while being scarcely represented in the more recent ones (see Table 6). Table 7 shows that the vast majority (83.4%) of converted verbs fall into the default inflectional class, the *-are* class, and that only few verbs belong to the *-ire* inflectional class: Most of them date back to the earliest periods, while from the 18th century onwards only an extremely scarce number of *-ire* verbs, of low-frequency or limited to literary register, is coined, e.g., *imbecillire* 'become, make imbecile' (1865), *giallire* 'become yellow' (1878). After the mid-20th century no *-ire* verbs are attested in our dataset.

Table 6: Productivity of conversion patterns over time: Percentage of new coinages on the total of verbs

Pattern	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–	Total
-are	39.8	16.1	9	6.5	8	4	**83.4**
-ire	6.6	4.5	4	0	1.5	0	**16.6**
Total	**46.2**	**20.6**	**13.1**	**6.5**	**9.5**	**4**	**100**

Table 7: Productivity of conversion patterns over time: Percentage of new coinages per time period

Pattern	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–
-are	85.9	78	69.2	100	84.2	100
-ire	14.1	22	30.8	0	15.8	0
Total	**100**	**100**	**100**	**100**	**100**	**100**

It is rather uncommon for the same adjective to yield more than one converted verb. When this occurs, the adjective at hand is generally prolific and is the base for a number of verbs derived by other derivational processes, too. A case in point is *chiaro* 'clear, light' > *chiarare*/*chiarire*, both attested with the meanings 'make clear; make intelligible, explain; to clear up'; for *chiarare* the meaning 'illuminate' is also attested. In present-day Italian only *chiarire* survives, with the meaning ‘clarify, explain’. Alongside these, the following suffixed and parenthetic verbs are attested: *chiareggiare* ‘clear up, clarify’, *chiarificare* ‘clarify’, *dichiarare* ‘declare’, *dischiarare* ‘clear up’, *schiarare* ‘illuminate, become clear’, *schiarire* ‘lighten, become clear, clear up’. Conversely, cases of adjectives being verbalised by both conversion patterns and no other are rarely found. Among the very few examples is *oscuro* ‘dark’ > *oscurare* ‘make, become dark’ and *oscurire* ‘become dark’.

3.2.3 Parasynthesis

Recourse to parasynthesis is extensive in the early stages, with more than half (57.7%) of all parasynthetic verbs coined in the first two periods. The contribution of this process to deadjectival verb formation decreases over time, becoming very limited in the most recent period (see Table 8).

Table 8: Productivity of parasynthetic prefixes over time: Percentage of new coinages on the total of parasynthetic verbs

Prefix	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–	Total
in-	12.5	6.2	9.9	3.3	5.5	1.9	**39.3**
ad-	17.9	2.9	3.8	1.5	1.4	0.2	**27.7**
ri-/ra-/rin-	6.7	3.1	2.4	2.7	3.8	0.7	**19.4**
s-	1.9	2.9	1.4	1	1.9	0.5	**9.6**
de-/di-/dis-	3.1	0.5	0.2	0	0.2	0	**4**
Total	**42.1**	**15.6**	**17.7**	**8.5**	**12.8**	**3.3**	**100**

The best attested prefixes are *in-* (39.3%), *ad-* (27.7%), and *ri-/ra-/rin-* (19.4%), while the presence of *s-* and *de-/di-/dis-* is rather limited (respectively 9.6% and 4%). As shown in Table 9, the productivity of the prefix *in-* is consistently high – the highest, in fact, from the second time period onwards. In the earliest period the prefix *ad-* is the most robustly attested, but then undergoes a quite notable decrease in use, becoming the second least productive prefix from the mid-17th century onward. While already being well attested in the earlier stages, from the fourth period onwards the prefixes *ri-/ra-/rin-* emerge as the most employed after *in-*.

As for the overall less productive prefixes, *s-* shows a significant increase in use from the first to the second time period (from 4.5% to 18.5%), and from then on,

Table 9: Productivity of parasynthetic prefixes over time: Percentage of new coinages per time period

Prefix	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–
in-	29.6	39.9	55.4	40.1	42.6	57.2
ad-	42.6	18.5	21.6	17.1	11.1	7.1
ri-/ra-/rin-	15.9	20	13.5	31.4	29.6	21.4
s-	4.5	18.5	8.1	11.4	14.8	14.3
de-/di-/dis-	7.4	3.1	1.4	0	1.9	0
Total	**100**	**100**	**100**	**100**	**100**	**100**

a rather constant usage. On the other hand, verbs prefixed with *de-/di-/dis-* are predominantly attested in the earlier stages, with very scarce recent coinages – an exception being *dimagrire* 'lose weight, slim down', first attested in 1855 and a highly frequent verb. The current widespread use of the prefixes *de-/di-/dis-* mainly concerns the formation of deverbal verbs to which they contribute an egressive (i.e., reversative or privative) meaning.

From Table 10, which illustrates the assignment of parasynthetic verbs into the two inflectional classes, we can observe a significant difference in both the overall proportions and the diachronic development compared to those outlined for converted verbs. While in each time period converted verbs are predominantly assigned to the *-are* class, the distribution of parasynthetic verbs between the two classes is more balanced (although still with a prevalence of *-are* verbs). In particular, the first two time periods are characterized by a predominance of *-are* verbs, the third period sees a shift toward the prevalence of *-ire* verbs, while in the fourth period the productivity of the two classes is comparable. The trend observed from the second half of the 19th century onwards is opposite to that of the earlier periods, with the majority of parasynthetic verbs being assigned to the *-ire* class. In this way, the non-default *-ire* class, which has almost ceased to accommodate verbs formed by conversion since the late 17th century, retains its productivity, e.g., *curioso* 'curious' > *incuriosire* (1887) 'intrigue, arouse curiosity'; *piatto* 'flat' > *appiattire* (before 1920) 'flatten'; *nervoso* 'nervous, tense' > *innervosire* (1950) 'annoy, irritate'; *volgare* 'vulgar, coarse' > *involgarire* (1969) 'vulgarise, coarsen'.

Finally, Table 11 shows the overall occurrence of parasynthetic prefixes over the *-are* and *-ire* inflectional classes. Verbs prefixed with *in-* are more often as-

Table 10: Distribution of parasynthetic patterns

Pattern	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–	Total
prefix-ADJ-*are*	27.9	8.9	7.4	4.3	5	1	**54.5**
prefix-ADJ-*ire*	14.1	6.7	10.3	4.1	7.9	2.4	**45.5**

signed the -*ire* class, but are also numerous in the -*are* class. Two-thirds of verbs prefixed with *ad*- belong to the -*are* class. Verbs prefixed with *ri-/ra-/rin-* do not show a marked preference towards either of the two classes. Verbs prefixed with ingressive *s*- predominantly fall within the -*are* class, but the percentage of -*ire* verbs is also high. Finally, the few verbs prefixed with ingressive *de-/di-/dis-* almost exclusively belong to the -*are* class: In fact, no -*ire* verb is attested with the prefixes *dis*- and *de*-, while by the pattern *di*-ADJ-*ire* only the aforementioned *dimagrire* 'slim down' (< *magro* 'slim') is derived.

Table 11: Distribution of parasynthetic prefixes over the inflectional classes

Infl. class	Prefix				
	in-	*ad-*	*ri-/ra-/rin-*	*s-*	*de-/di-/dis-*
-*are*	41.5	66.4	54.4	57.5	94.1
-*ire*	58.5	33.6	45.6	42.5	5.9
Total	**100**	**100**	**100**	**100**	**100**

Parasynthesis is the process by which the most derivatives from the same adjective are formed. These are largely synonyms, e.g., *caro* 'expensive, pricey' > *incarare*/*incarire* 'become more expensive', both obsolete; *largo* 'broad, large' > *allargare*/*slargare* 'enlarge'; *ruvido* 'rough' > *arruvidare* (obsolete)/*irruvidire* 'roughen'. Possible differences in meaning do not correlate systematically with the different patterns, e.g., *profondo* 'deep' > *sprofondare* 'sink, collapse, make sb./st. collapse'/*approfondire* 'deepen, delve into'; *sicuro* 'sure, safe' > *assicurare* 'assure, secure, fasten'/*rassicurare* 'reassure'; *rosso* 'red' > *arrossare* 'make red'/ *arrossire* 'turn red, blush'.

3.3 Competition in deadjectival verb formation

3.3.1 Background

The notion of competition is at the centre of the current theoretical and methodological debate in morphology.[4] Italian deadjectival verb formation can serve as a useful testbed for the study of this phenomenon, in view of the several strategies available, both at the process and the pattern level, for producing the shift from the adjectival to the verbal lexical category.

Competition between derivational strategies often results in the instantiation of a word from a set of possible candidates, which, despite fulfilling all formal requirements, remain "virtual" (in the terminology of Rainer 2012), as they get blocked by the already existing word. However, blocking is more of a tendency than a strict condition, and synonymous words can indeed be "actualised", especially if we do not exclude partial synonymy as an instance of synonymy. This is the case with many adjectives from which multiple synonymous verbs have been derived. In such cases competition becomes particularly manifest, but the interplay between derivational strategies can also be considered at a more abstract level, where these have overlapping preferences in base selection and can potentially fulfill the same semantic function.

To understand how competition contributes to shape deadjectival verb formation, we examined the adjectives within our dataset, providing a quantitative overview of their propensity to verbalisation (3.3.2) and – when this has taken place – by which derivational strategies (3.3.3). We then explored how the different processes and patterns have overlapped or, conversely, divided up the labour in the expression of the main semantic and aspectual categories (3.3.4).

3.3.2 Adjectives as bases for verbalisation: a quantitative overview

Out of the 858 adjectives examined, half have served as bases for verb derivation. Among the adjectives that have yielded verbal derivatives, 48.8% count only one, 24.2% two, 12.6% three, 14.4% four or more.

An example of a very prolific base is *molle* 'soft, mushy, weak', which has yielded 11 verbs (listed, by period of coinage, in Table 12). All three processes have over time selected this adjective, from which two verbs are derived by conversion, one by suffixation (with -*ificare*), and as many as eight by parasynthesis.

[4] For recent comprehensive reviews on the theoretical issues and the methodological approaches developed for the study of competition among affixes, and, more generally, in word formation, see Huyghe & Varvara (2023); Nagano et al. (2024).

As in many other cases, most verbs are formed in the earlier periods, but recent coinages are also attested. All derivatives lexicalise CoS: Some are attested only with a causative meaning, i.e., 'soften, appease, mollify' (i.e., *mollificare*, *ammollare*, *ammollire*, *immollare*, *smollare*, *mollire*), others with both a causative and an inchoative one, i.e., 'soften' and 'become soft, limp' (*mollare*, *immollire*, *rammollare*, *rammollire*), and, finally, *smollirsi*, attested only in the pronominal form, is necessarily only used inchoatively. Of these derived verbs, six are obsolete or rare (*mollificare*, *mollire*, *immollire*, *immollare*, *rammollare*, *smollirsi*), while five are still in use and highly frequent (*mollare*, *ammollare*, *ammollire*, *rammollire*, *smollare*). While the verbs still used in present-day Italian share the general meaning 'make soft, mollify', they do vary in terms of usage context and collocations: *rammollire*, for example, is mostly used figuratively to mean 'deprive of strength and energy'; *ammollare* is instead generally used to mean 'dunk, soak', i.e., 'soften by immersing in a liquid'; finally, among the meanings of *mollare* (the most frequent and polysemous verb) are 'release, leave, ditch, give, give in', whereas 'soften, weaken' is marked in the GRADIT as regional. No clear relation emerges between the time period of coinage and the survival of a verb. For example, among the most recently attested verbs, *smollare* survives, while *smollirsi* has not established itself in contemporary usage.

Table 12: Timeline of the coinages of derivatives from *molle* 'soft, mushy'

Process	I: 1200–	II: 1376–	III: 1533–	IV: 1693–	V: 1841–	VI: 1947–
Suffixation	*mollificare*					
Conversion	*mollare*		*mollire*			
Parasyn.	*ammollare* *ammollire* *immollare* *rammollare* *rammollire*	*immollire*			*smollare* *smollirsi*	

A general overview of a single case such as this exemplifies the tendency to form multiple derivatives from the same base, often synonyms (whether full or partial), and to reduce their number over time. This trend is shared by other Romance languages and already documented by Grossmann & Iacobini (2025) for Italian. This general tendency can be appreciated with reference to our data if we compare the ratio of the verbs derived from the adjectives first attested in the earliest time period to said adjectives (2,37%) with the ratio of the verbs in present-day use to the total adjectives having ever yielded at least one verb (1,62%). The reduction of the verbal derivatives does not result, however, in a one-to-one verb-meaning correspondence.

In order to consider the complete landscape of competition and the interaction between verbs derived from the same adjective, as well as the specialisation of meaning and the differentiation of semantic nuances in different domains or registers of verb usage, a detailed historical reconstruction of individual verbs is needed, but is beyond the scope and the dimension of this paper. An analysis that is more limited in scope, but also more fine-grained was carried out by Barbato & Necker (2006), who trace the history of the verbal derivatives of the variants of the two adjectives *bianco* 'white' and *chiaro* 'light, clear' within the Italo-Romance varieties.

3.3.3 Adjectives as bases for verbalisation: a morphosemantic analysis

Many factors possibly play a role in the instantiation and the establishment in the lexicon of a word derived by a certain strategy. Here we confine ourselves to observing the relevance, for verbalisation, of the adjectives' morphological and semantic features. The adjectives that yield multiple verbs are, for the most part, morphologically simplex and express the prototypical property concepts (Dixon 1982), as shown in Table 13. Many of these properties can be understood both literally and figuratively.

As stated above, the morphological structure of the adjectives notably impacts their propensity for verbal derivation. As shown in Table 14, morphologically simplex adjectives, which account for just over a third of all adjectives, undergo verb derivation much more frequently than complex adjectives. In fact, 73.2% of simplex adjectives are used to derive at least one verb. The percentage of verbs derived from complex adjectives is significantly lower: Only slightly more than one third (37.8%) of complex adjectives has served as a base for verb derivation.

As shown in Table 15, the three verbalising processes show different propensities with regard to the selection of simplex versus complex adjectives.

The leftmost column recalls the percentage contribution of each process to the total pool of derived verbs. The middle column shows verbs derived from simplex adjectives, while the rightmost column shows the percentage of verbs derived from complex adjectives. We see that parasynthesis and suffixation are the two processes most influenced by the derivational structure of adjectives, with parasynthetic verbs being predominantly derived from simplex adjectives

Table 13: The adjectives with the highest number of verbal derivatives

Colours	*azzurro* 'blue', *bianco* 'white', *bruno* 'dark, dark brown', *nero* 'black', *pallido* 'pale', *rosa* 'pink', *rosso* 'red', *scuro* 'dark', *chiaro* 'light'
Size	*basso* 'short, low', *corto* 'short', *grande* 'big', *largo* 'wide', *lungo* 'long', *magro* 'skinny', *piccolo* 'small', *profondo* 'deep', *sottile* 'thin, subtle'
Physical properties	*biondo* 'blond', *calvo* 'bald', *curvo* 'curved', *denso* 'dense', *dolce* 'sweet', *doppio* 'double', *folto* 'thick', *freddo* 'cold', *lento* 'slow', *liquido* 'liquid', *lucido* 'shiny', *molle* 'soft, mushy', *morbido* 'soft', *nudo* 'bare, naked', *piano* 'flat', *secco* 'dry', *solido* 'solid', *sordo* 'deaf', *tenero* 'tender', *tiepido* 'lukewarm', *umido* 'moist'
Value judgments	*bello* 'nice, beautiful', *brutto* 'ugly', *buono* 'good', *caro* 'dear', *cattivo* 'bad', *cupo* 'gloomy', *grave* 'serious, deep', *misero* 'miserable, meager', *prezioso* 'precious', *puro* 'pure', *vero* 'true', *vile* 'vile', *vano* 'vain', *vivo* 'alive'
Human attitudes	*curioso* 'curious', *gentile* 'kind', *sereno* 'clear, peaceful', *superbo* 'arrogant', *triste* 'sad'

Table 14: Relationship between the morphological complexity of the adjectives and their propensity to be derived into verbs

All adjectives		**Verb-forming**	**Non verb-forming**	**Total**
Simplex	34.8	73.2	26.8	**100**
Complex	65.2	37.8	62.2	**100**
Total	**100**			

Table 15: Relationship between the morphological complexity of the adjectival bases and the verb-forming processes

Verbalising process	Verbs from all Adjs	Verbs from simplex Adjs	Verbs from complex Adjs
Conversion	21.7	23.4	19
Parasynthesis	45.6	57.8	23.8
Suffixation	32.7	18.8	57.2
Total	**100**	**100**	**100**

and suffixed verbs from complex ones. Converted verbs do not strongly relate with either category, and simplex bases are only slightly more numerous.

The morphological make-up of the adjectives gains increased significance on the derivation of verbs when we correlate the adjective-forming process and the verb-forming one(s), as shown in Table 16. The two left-hand columns show the propensity of the different types of complex adjectives to be verbalised, while the subsequent three columns show the specific processes by which the verbs are derived.

Table 16: Relationship between the derivational structure of the adjectives and that of the resulting verbs

Complex Adjs	Non-verb forming	Verb forming	Total	Conv. Verbs	Parasyn. Verbs	Suffixed Verbs
Compound	75	25	**100**	17	–	83
Prefixed	68	32	**100**	46	–	54
Suffixed	58	42	**100**	14	17	69
Converted	72	28	**100**	42	26	32

Parasynthetic verbs are only derived from suffixed and converted adjectives, whereas suffixation and conversion allow for verb derivation from all types of bases.

Verbalisation of adjectives formed by conversion is limited by the fact that many of them are converted from verbs, which strongly reduces the possibility of further derivation into verbs, e.g., *decidere* 'decide' > *deciso* 'decisive' > **decisare*; *distendere* 'stretch, unwind' > *disteso* 'stretched, relaxed' > **distesare*. The increasing use of suffixation for verb derivation (cf. Section 3.1) can be partly explained by the fact that suffixed adjectives lend themselves to further derivation through suffixation. Given that new adjectives are predominantly coined through suffixation, the pool of potential bases available for verb derivation via suffixation is continuously growing.

Compound adjectives tend to have only one derivative, mainly formed by *-izzare* suffixation (e.g., *autonomo* 'autonomous' > *autonomizzare* 'autonomise', *democratico* 'democratic' > *democratizzare* 'democratise'), or by *-are* conversion, e.g., *uniforme* 'uniform' > *uniformare* 'uniform'.

Prefixed adjectives also tend to derive only a single verb. Exceptions with two derivatives also occur, e.g., *impaziente* 'impatient', which gives rise to two converted verbs, *impazientare* and *impazientire*, both 'become or make impatient'. Furthermore, from *infinito* 'infinite', both converted *infinitare* 'make infinite' and suffixed *infinitizzarsi* 'become infinite' are derived. The most commonly used suffix to form verbs from prefixed adjectives is *-izzare*, e.g., *internazionale* 'international' > *internazionalizzare* 'internationalise', *immobile* 'immobile' > *immobilizzare* 'immobilise'.

Approximately half of the adjectives formed by conversion form only one verb, e.g., *tecnico* 'technical' > *tecnicizzare* 'technicalise'. Some adjectives converted from nouns, on the other hand, yield up to four derivatives, e.g., *rosa* 'rose' > *rosa* 'pink' > *arrosare*, *inrosare*, *rosare*, *roseggiare*, the former three meaning 'colour, paint pink', with *rosare* also figuratively meaning 'illuminate with a pink light', and *roseggiare* meaning 'take on a pink hue due to the effect of a diffused pink light (of the sky) or due to the presence of numerous pink-coloured flowers (of a field)'.

Among complex adjectives, suffixed ones are those from which the highest number of verbs are derived, both in general and in terms of the number of derivatives per single adjective: Roughly one third of adjectives yields two or more derivatives: *civile* 'civil' > *civilizzare*, *incivilire* both 'civilise'; *cristiano* 'Christian' > *cristianificare*, *cristianizzare* both 'christianise', *cristianeggiare* 'behave as a Christian'; *curioso* 'curious' > *curiosare*, *scuriosare*, *curioseggiare* all 'look around, snoop', *incuriosire* 'make curious, arouse curiosity'.

The suffixes *-ale*, *-bile*, and *-oso* are among the most productive for adjective formation. The verbs derived from adjectives in *-ale* are predominantly formed by *-izzare* (e.g., *commerciale* 'commercial' > *commercializzare* 'commercialise', *nazionale* 'national' > *nazionalizzare* 'nationalise'), although verbs in *-eggiare* are also documented, e.g., *orientale* 'Eastern' > *orientaleggiare* 'assume attitudes or characteristics typical of the East'. Adjectives in *-bile* also form verbs with *-izzare*, e.g., *responsabile* 'responsible' > *responsabilizzare* 'make responsible', *sensibile* 'sensitive' > *sensibilizzare* 'sensitise, raise awareness'. The numerous adjectives in

-*oso* tend not to be verbalised, with just over 10% forming (mostly parasynthetic) verbs, e.g., *pietoso* 'compassionate' > *impietosire* 'move to pity', *goloso* 'greedy', 'delicious' > *ingolosire* 'make sb's. mouth water', *nervoso* 'nervous' > *innervosire* 'irritate'.

3.3.4 The verbal derivatives: the meanings expressed

Examining the semantic and aspectual properties of attested verbs can help to gain more insight into the competition between processes and patterns for the same meanings. Table 17 displays the percentage distribution of verbs over the main semantic and aspectual classes. Deadjectival verbs predominantly convey a CoS (86.5% of total verbs, equivalent to the cumulative value of the first three columns), whereas verbs denoting states or activities make up 7.5%. Additionally, alongside verbs falling into the two main classes, there exists a subset (6%) of verbs that can express CoS along activities and stative meanings. In their default, active form (i.e., non-pronominal), most of the CoS verbs are causative (53.2%), while about a quarter (24.8%) can convey both causative and inchoative meanings. 8.5% solely express an inchoative value, of which a small subset is exclusively found in the pronominal form, i.e., is not attested in the default, active form.

Table 17: Distribution of verbs over the semantic and aspectual classes

CoS			**State/ activity**	**CoS + state/ activity**	**Total**
Caus.	Inch.	Caus. + Inch.			
53.2	8.5	24.8	7.5	6	**100**

Table 18 shows the percentage contribution of each of the three processes in the expression of the different semantic types.

Regarding the expression of CoS, all three processes contribute in roughly equal proportions to the formation of causative verbs, with conversion slightly less represented than suffixation and parasynthesis. Conversely, verbs that exclusively express inchoative meanings and verbs that occur in both inchoative and causative constructions are predominantly formed by parasynthesis, with suffixation playing a minor role. Parasynthesis is notably scarce among verbs expressing states and activities, as well as among those capable of denoting both CoS and states and activities. Suffixation emerges as the primary process for

Table 18: Contribution of the derivational processes to the expression of the main semantic and aspectual types

	CoS			**State/ activity**	**CoS + state/ activity**
	Caus.	Inch.	Caus. + Inch.		
Suffixation	39.2	6.7	4.1	66.7	64.1
Conversion	23.2	16	16.5	27.3	30.2
Parasynthesis	37.6	77.3	79.4	6	5.7
Total	**100**	**100**	**100**	**100**	**100**

forming verbs denoting states and activities, while conversion accounts for approximately 30% of their derivation.

Tables 19 to 21 detail the expression of the main semantic types by the patterns within each process. Table 19 displays the contribution of each suffix to the formation of CoS and of state and activity verbs, from which a specialisation exhibited by individual suffixes emerges.

Table 19: Contribution of each suffix to the expression of the main semantic and aspectual types

	CoS			**State/ activity**	**CoS + state/ activity**	**Total**
	Caus.	Inch.	Caus. + Inch.			
-*izzare*	48.2	1.3	1.3	1.1	7.4	**59.3**
-*eggiare*	1.3	0	1.1	21	4	**27.4**
-*ificare*	10.4	0.3	0.6	0	0	**11.3**
-*itare*	2	0	0	0	0	**2**
Total	**61.9**	**1.6**	**3**	**22.1**	**11.4**	**100**

The suffix -*ificare* is used exclusively in the formation of CoS verbs, whereas -*izzare*, besides being the most frequently employed suffix in the expression of CoS, is also exhibited by verbs denoting states and activities. The suffix -*eggiare* is unfrequently used in the derivation of CoS verbs, its main function being the expression of states and activities. The suffix -*itare* is found in only a very small number of causative verbs.

Verbs formed by conversion belonging to the *-are* class (83%) significantly outnumber those in the *-ire* class (17%). The contribution of each of the two classes to the expression of the different semantic and aspectual values can be seen in Table 20. We can see that among the verbs of the *-are* class there is a predominance of verbs with only causative value, whereas the verbs of the *-ire* class can be mainly used in both causative and inchoative constructions. States and activity verbs are almost exclusively of the *-are* class.

Table 20: Percentage of converted verbs of the two classes in the expression of the main semantic and aspectual types

	CoS			**State/ activity**	**CoS + state/ activity**	**Total**
	Caus.	Inch.	Caus. + Inch.			
-are	59.7	5.4	11.4	15.7	7.8	**100**
-ire	29.4	8.8	50.1	2.9	8.8	**100**

The data regarding parasynthetic verbs are summarised in Table 21, wherein the various prefixes are grouped together due to their minor individual difference in contribution in this respect. This grouping also helps highlighting the differences in terms of inflectional class, and thus a more straightforward comparison with converted verbs.

Table 21: Percentage of *-are* and *-ire* parasynthetic verbs in the expression of semantic and aspectual classes

	CoS			**State/ state/act.**	**CoS + act.**	**Total**
	Caus.	Inch.	Caus. + Inch.			
Prefix_*are*	32.9	6	13.2	1.4	0.7	**54.2**
Prefix_*ire*	9.4	8	28.4	0	0	**45.8**
Total	**42.3**	**14**	**41.6**	**1.4**	**0.7**	**100**

The data confirm the substantial percentage of CoS verbs, among which a significant proportion exhibits an inchoative value and the near absence of verbs conveying states and activities. A difference that emerged between parasynthesis and conversion consists in a higher propensity of the former to derive inchoative verbs – among which particularly numerous are those belonging to the *-ire* class. Some of the very few (only 6) verbs classified as states or activities may also be interpreted as prefixed verbs, cf. *scuriosare*/*curiosare* ‘snoop’, *svigilare*/*vigilare* ‘watch’. The only example of an unarguably parasynthetic verb is *intristare* ‘look sad’, not in common usage. Conversely, the verb *impazzare* is currently employed with an atelic, activity meaning, ‘infuriate, manifest in a tumultuous way, rage, be all the rage’, while its inchoative meaning, ‘go mad’, has fallen out of contemporary usage.

4 Conclusions

The study aimed to give a comprehensive overview of deadjectival verb formation in Italian, both in terms of the broad time span considered and the comparative approach used in describing the productivity and semantics of the different verbalising strategies.

By taking the most frequent Italian adjectives as a starting point for the verb collection, we could distinguish the verb-forming from the non-verb-forming ones and identify the cases where multiple verbal derivatives have been formed from one and the same base. There emerged that half of the 858 sampled adjectives have served as a base for verbal derivation, of which 48.8% have yielded a single derivative, 24.2% two, 12.6% three, and 14.4% four or more.

Among the 917 verbs collected, parasynthetic derivatives are the highest in number (45.5%), suffixed verbs account for about a third (32.6%), while converted verbs make up the remaining 21.9%. Most parasynthetic and converted verbs are first attested in the earliest periods, while in 20th-century Italian the vast majority of deadjectival verbs is instead derived by suffixation, with a trend reversal beginning in the mid-19th century.

The extensive use of parasynthesis in the early centuries and of suffixation in the later ones may, at least in part, be related to the processes’ preference for, respectively, morphologically simplex and complex bases, as revealed by the analysis of the adjectives’ morphological structure. The processes showing stronger preferences regarding the structure of the bases are parasynthesis and suffixation. Conversion, on the other hand, lies in the middle, showing only a minimal preference for simplex adjectives. Overall, verbalisation involves simplex adjectives twice as often than complex ones. As for the latter, parasynthesis is mostly employed to verbalise suffixed and converted adjectives, while suffixation and conversion show more flexibility in this regard. For compound and suffixed adjectives suffixation emerges as the preferred verbalising process.

In addition to examining base selection preferences, we also considered how the different processes and patterns are distributed with respect to the possible meanings expressed in deadjectival verbalisation. Indeed, an onomasiological perspective allows us to determine the extent to which the different strategies functionally overlap (and potentially compete), or differ.

The vast majority of deadjectival verbs, regardless of the derivational class, denote change-of-state events. The three processes contribute almost equally to the formation of causative verbs, with suffixation and parasynthesis slightly more prevalent than conversion, whereas verbs that exclusively express inchoative meanings and those employed in both inchoative and causative constructions are mainly parasynthetic. While inchoative meanings are normally achieved by adding the anticausative particle -*si* to causative verbs, a number of derivatives can convey such meaning in the active (i.e., non-pronominal) form: These are mostly parasynthetic and converted verbs falling in the non-default, -*ire* inflectional class. While change-of-state events can be expressed by all three processes, the encoding of states and activities is limited to -*eggiare* suffixation and -*are* conversion.

Considering the verbs' first attestation and their persistence or exit from contemporary usage revealed two diachronic phenomena: A decrease in the number of verbs derived from the same adjective and an increasing presence of suffixed coinages. The expansion of suffixation in verb formation is reasonably expected to continue, also in the light of the fact that suffixation is the preferred process for the verbalisation of suffixed and compound adjectives, which are, in turn, the adjectival classes most frequently enriched by new formations.

The original data and analyses presented in this research allowed for a wide-ranging diachronic investigation of the productivity of the strategies involved in deadjectival verb formation and their functional distinctions. The insights provided contribute to a better understanding of the factors underlying the trend towards the decrease in the number of verbal lexemes over time and the current tendencies in the verbalisation of adjectives.

Abbreviations

Adj	Adjective	inch.	inchoative
caus.	causative	intr.	intransitive
CoS	change of state	tr.	transitive

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Chapter 11

Deriving the Akkadian Stative: A deadjectival verb?

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Recent studies on the Akkadian Stative conjugation have defended its verbal nature (Kouwenberg 2000, Carver 2016, Kamil 2023), arguing against approaches that have treated it as a nominal element (Huehnergard 1987, Buccellati 1988). The nominal argument stems from the Stative's diachronic origin as a verbal adjective (VA); synchronically, it is thus often treated as a denominal/deadjectival verb (Kouwenberg 2000). This present contribution argues for two separate trajectories of Stative derivation: Primary Statives, i.e., Statives whose base is a VA, are derived deradically, while secondary Statives, i.e., Statives whose base is a nominal or adjectival stem, are derived denominally or deadjectivally. I motivate these different trajectories based on mismatches observed between Statives and VAs derived from the same LEXICAL ROOT, pertaining to 1) the argument predicated by either Stative or VA, and 2) the resultative readings of Statives vis-à-vis VAs. First, I show that Experiencer-subject transitives feature mismatches in argument predication: While their Statives predicate the Experiencer-subject, their VAs predicate the Theme-object. Secondly, I motivate the resultative reading of Statives as a consequence of their derivation: Primary Statives inherit their change-of-state event directly from the ROOT. Secondary Statives require external syntactic or pragmatic cues to achieve resultative readings.

1 Introduction

The Akkadian (East Semitic†) Stative[1] paradigm is special among the Semitic verbal paradigms for its ability to be derived, in theory, from any verb, noun,

[1] I use an uppercase "Stative" to refer to the conjugation paradigm thematised in the present paper and a lowercase "stative" to describe an aspectual property of predicates.

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or adjective. Both its morphological and its semantic status have been widely debated in the previous literature with neither its verbal status unanimously accepted nor its semantic functions clearly understood.

More recent works on the form by Kouwenberg (2000), Carver (2016), and Kamil (2023) have provided evidence for the form's verbal nature, with theoretical morphosyntactic arguments made in Kamil (2023). Morphologically, the conjugation is made up of two components: a base and a conjugation-specific suffix. Thereby, two types of bases may be differentiated: a verbal adjectival base and a nominal stem base. In Assyriological terminology, a verbal adjective (henceforth VA) is the most common type of adjective. It is understood to be derived directly from the ROOT into the VA template, *XaY(i)Z-*. The VA's "verbal" nature derives from the observation that these VA templates are found across the verbal derivational patterns (D/INTENSIVE and Š/CAUSATIVE), too. It has remained unclear, however, whether these adjectives are deverbal themselves.

Understanding the VA and its syntax is imperative to understanding the Stative conjugation. While their derivational connection has been established before, most notably by Huehnergard (1987), I argue that the primary Stative is *not* a direct projection of the VA and its semantics, but rather a resultative verb. In other words, I argue against the *synchronic* treatment of the primary Stative as a deadjectival verb.[2] Apart from the forms' different phrase-syntactic behaviour, evidence for the dissonance between VA and Stative comes from the different arguments being described by the respective forms when it comes to Experiencer-subject verbs.

Building on the understanding of the conjugation as a verbal form, this paper now sets out to determine the more precise morphosemantic and morphosyntactic configurations of the Stative, by pursuing two main goals: 1) Arguing against a deadjectival treatment of the primary Stative, thereby providing the first syntactic and semantic account of the structure of Akkadian VAs and Statives, and 2) motivating the interpretation of the Stative as a conjugation denoting a resultative state, thereby accounting for the resultative nature of secondary (actually) deadjectival/denominal Statives.

In the context of the present volume on deadjectival verbal derivations in Indo-European, this paper further sets to provide a typological comparandum to a productive conjugation that appears to be morphologically deadjectival, but is (mostly) syntactically independent.

The paper is structured as follows: Section 2 introduces Akkadian morphology and the Stative paradigm. Section 3 then treats the primary Stative, outlining its

[2] I naturally recognise the diachronic origin of the form to be from the VA. How this apparent process of reanalysis came about shall not be thematised in this article.

morphology in Section 3.1 and syntax and semantics in Section 3.2. Section 3.2 is further broken up into a discussion of the relevant ROOT classes in 3.2.1, the suggested derivational analyses for VAs and Statives in 3.2.2, the motivation of the resultative reading in 3.2.3, and finally the placement of the Stative within the verbal system of Akkadian in 3.2.4. Building on the analysis of the primary Stative, I discuss the secondary Stative in Section 4. I again discuss morphology (Section 4.1) and syntax/semantics (Section 4.2) separately. I also present a survey of secondary Statives in Section 4.3 to further back the claim for their resultative readings. Finally, I conclude in Section 5.

2 The Akkadian Stative

As a preface to understanding the Akkadian verbal system, a brief introduction to the general derivational mechanism is necessary. Like all Semitic languages, Akkadian employs the ROOT-and-pattern/template mechanism for lexical word building: Typically triradical lexical/semantic ROOTS are inserted into so-called templates or patterns, which in turn encode word class (i.e., noun, adjective, verb, and in some cases adverb), Tense/Aspect/Mood-features (henceforth TAM), voice-features, φ-features (i.e., person, number, and gender features), etc. An Akkadian example of the variety of words derivable from a single lexical ROOT is given in (1), with verbal derivations given in (1a-1b), nominal derivations given in (1c–1f), and adjectival derivations given in (1g–1h).

(1) Akkadian words derived from the ROOT √*šrq*

a. *šarāqu* ‘steal’
b. *našruqu* ‘be abducted’
c. *šarrāqu* ‘thief’
d. *šarqūtu* ‘theft’
e. *šurqu* ‘theft’
f. *šerqum* ‘stolen goods’
g. *šarqu* ‘stolen’
h. *šarriqu* ‘thieving’

The ROOT consonants *š-r-q* can be seen inserted in the same order in all the lexical items in (1) above. The *templates* into which they are inserted can be depicted as *XaYāZu* ‘infinitive’ (1a), *naXYuZu* ‘passive infinitive’ (1b), or *XaYZu* 'verbal adjective' (1g), whereby *X-Y-Z* refer to the respective ROOT consonants. This will be the notational framework used to describe templates in this paper.[3]

For the verbs, six TAM-encoded templatic conjugations are attested: The perfective (*iXYV*$_R$*Z*) and imperfective (*iXaYYV*$_T$*Z*) encode the basic aspects, the imperative (*XaYaZ*) and precative (*liXYuZ*) encode two moods. The two remaining conjugations, namely the *t*-aspect (*iXtaYV*$_T$*Z*) and the so-called "Stative" (*XaYiZ*) paradigms are more difficult to place within the traditional TAM-notions.[4] I will suggest a placement for the Stative in Section 3.2.4 below.

The conjugation has been treated by numerous Akkadianists (see Buccellati 1968, 1988, Kraus 1984, Huehnergard 1987, Kouwenberg 2000, Carver 2016, Kamil 2023, *inter alia*) in the past, with the main scholarly arguments usually revolving around a) its verbal nature and b) its meaning(s) and function(s). Thereby, most of the arguments for or against its verbal nature were centred around the Stative's morphological make-up.

While in most grammars (e.g., Reiner 1966, von Soden 1995), the form is treated as a verb, morphologically it is made up entirely of nominal elements, namely a VA base and a Stative-specific suffix as summarised in (2), which is itself also derived from nominal elements. The Stative-specific suffixes have further developed the ability to attach to, theoretically, any nominal or adjectival stem, too. This has led various researchers (most notably Huehnergard 1987 and Buccellati 1988) to analyse it as a nominal form.

(2) Basic morphological make-up of the Stative
Base + Stative suffix
   a. Primary Stative: VA + Stative suffix
   b. Secondary Stative: Nominal stem + Stative suffix

More recent accounts (e.g., Kouwenberg 2000, Carver 2016, Kamil 2023) provide stronger arguments for a synchronically verbal interpretation: Syntactically, the Stative forms appear exclusively in the predicate position, they may take verbal suffixes such as the ventive, subjunctive, and *ma*-conjunction, as well as arguments in the forms of accusative and dative objects both in bound (suffixal)

[3] Another important notion of Semitic morphology are the template patterns SIMPLE (G), INTENSIVE (D), and CAUSATIVE (Š). SIMPLE verbs typically mirror the root's semantics, INTENSIVE verbs typically increase transitivity or intensity (i.e., 'cut off' vs. 'chop off'), and CAUSATIVE verbs are, as their name implies, causative in nature. Onto these patterns three voices (active, middle, passive) may be mapped and all verbal conjugations derive forms in all template patterns. While they do not play an important role to the argument of the present paper, they will be referenced on multiple occasions. For the full system, see Kamil (2025).

[4] For literature on the *t*-aspect see Streck (1995, 2003), Hackl (2007), Brose (2019).

or unbound form. Finally, all Statives, i.e., primary ROOT- and secondary stem-derived, may be "middled", i.e., it is possible to insert the middle *t*-morpheme into them to create medio-passive or reflexive/reciprocal derivations.

The latter argument was provided by Kamil (2023) alongside a new morphological categorisation schema summarised in Table 1, dividing Statives into "primary" Statives, and "secondary" Statives. Primary Statives refer to Statives utilising a VA as their base. VAs are derived *exclusively* deradically, through the insertion of the ROOT into a VA template *XaY(i)Z-*. Here, one may generally differentiate between verbal/eventive ROOTS, e.g., $\sqrt{}$*wbl* 'bring', $\sqrt{}$*mqt* 'fall', and adjectival/property concept (PC) ROOTS, e.g., $\sqrt{}$*wrq* '(be) green/yellow', $\sqrt{}$*dmq* '(be) good'.[5] Secondary Statives on the other hand refer to Statives derived from adjectival and nominal *stems* (i.e., pre-derived nominal forms granting no access to root-encoded information). Morphologically, this means that primary Statives will *always* be found in the same template, while secondary Statives do *not* conform to any one template. We will adopt this categorisation schema here, too, and address the question of the Stative semantics in greater detail in the following.

Table 1: Morphological Stative categorisation

Primary Statives		Secondary Statives	
XaY(i)Z- + suffix		Stem + suffix	
PC ROOT	Eventive ROOT	Adjectival stem	Nominal stem

The general semantics of the Stative have been previously described to denote a state of being (von Soden 1995: 126), earning the conjugation its name, though some scholars have described its function as a "passive" to the perfective/imperfective conjugations (Rowton 1962: 237, Huehnergard 1987: 225). This paper will on the one hand challenge and on the other hand refine these functional generalisations and claim that the Stative denotes a *resultative* state,[6] making implicit reference to a CoS event preceding the eventuation of the state.

The following Section 3 will introduce the primary Stative, providing a morphological overview (Section 3.1) and a semantic/functional analysis (Section 3.2).

[5] I follow Beavers & Koontz-Garboden (2020); Beavers et al. (2021) in their distinction between PC ROOTS and what they refer to as "result roots", although this distinction is restricted to change-of-state (CoS) verbs, whereas I will use the term "eventive roots" to refer to verbs patterning with result ROOTS, but with no inherent CoS semantics. See also the introduction to this volume for further discussion.

[6] Loesov (2012) has previously argued against a resultative interpretation of the Stative. I acknowledge this publication, but will not centre my argumentation around disproving Loesov's.

Section 3.2 is further organised as follows: Section 3.2.1 will introduce the issue of different ROOT classes deriving different VA-Stative patterns, Section 3.2.2 will then attempt to account for these patterns, offering a tentative solution. The resultative interpretation of the form will be motivated in Section 3.2.3, and finally Section 3.2.4 places the Stative paradigm within the TAM system of Akkadian.

3 The primary Stative

3.1 Morphology

The "primary Stative" consists of a VA base in the template *XaY(i)Z-* and the Stative-specific suffixes. The paradigm for the SIMPLE (G) template pattern is given in Table 2 below. Note, however, that the Stative conjugation is found in the INTENSIVE (D), CAUSATIVE (Š), and SIMPLE PASSIVE (N)[7] template patterns, too.[8]

Table 2: The Akkadian Stative in the SIMPLE template

	SINGULAR	PLURAL
3. MASC	*XaYiZ*	*XaYZ-ū*
3. FEM.	*XaYZ-at*	*XaYZ-ā*
2. MASC.	*XaYZ-āta*	*XaYZ-ātunu*
2. FEM.	*XaYZ-āti*	*XaYZ-ātina*
1. COM.	*XaYZ-āku*	*XaYZ-ānu*

As mentioned in Section 2, the Stative suffixes that are morphologically added to the VA base are derived from nominal elements. For the second and first persons, the origin can be traced back to the independent personal pronouns (IPP, Kouwenberg 2000), given in Table 3. The third person suffixes, though not traceable to the IPP, are nonetheless nominal in origin, too (Hasselbach-Andee 2022).

[7]Capitals are used to abbreviate template patterns. Especially the INTENSIVE capital D (for *Dopplungsstamm*) will be used in glosses of examples.

[8]For scope reasons, this paper will not consider further derivations of the Stative paradigm, i.e., D, Š, or N Statives, though I am aware that the incorporation of these derivations is crucial to our understanding of the morphosyntax and the therefrom derived semantics of the resultative/Stative. As it stands, I assume that these morphemes are added further up in the derivation and that their relevance to the basic mechanism of resultative/Stative derivation is thus only peripheral at this point, though I may offer no evidence for this within the scope of the present paper. I, for now, refer the reader to Kamil (2025).

Table 3: Independent personal pronouns in Akkadian

	SINGULAR	PLURAL
3. MASC.	*šū*	*šunu*
3. FEM.	*šī*	*šina*
2. MASC.	*attā*	*attunu*
2. FEM.	*attī*	*attina*
1. COM.	*anāku*	*nīnu*

A VA is typically differentiable from a Stative by taking on a case marker, which is also marked for number. Feminine VAs additionally add a feminine affix, which takes on the (lengthening) number feature instead of the case-suffix in masculine forms. The paradigm, given in Table 4, shows that potential overlaps could occur between Stative 3pl. masc. forms with masc. nom.pl. adjectives, as well as Stative 2sg. fem. forms with fem. gen./acc.pl. adjectives. This makes their distinction not always easy. Syntactically, however, clearly identifiable VAs are distinguishable from clearly identifiable Statives through their phrasal syntactic positions: While VAs may be used attributively *and* predicatively, Statives are only found in predicative usage.

Table 4: Akkadian case marking on nouns and adjectives

		NOM	GEN	ACC
MASC	SG	*-u*	*-i*	*-a*
	PL	*-ū*	*-ī*	
FEM	SG	*-atu*	*-ati*	*-ata*
	PL	*-ātu*	*-āti*	

Examples for the morphological forms of primary Statives are given in (3) below, for now without their respective contexts.

(3) Primary Stative examples

a. *gapš-ū*
$\surd$*gpš* 'huge'
huge.VA-STAT.3.M.PL
'They are swollen.' (CT 39 18:101)

b. *amr-āku*
√*ʔmr* 'see'
see.VA-STAT.1SG
'I have seen.' (Lambert BWL 194 rev. 8)

c. *nadn-āku*
√*ndn* 'give'
give.VA-STAT.1SG
'I have been given (sold).' (TIM 2 100:14)

3.2 Syntax and semantics

I argue in this paper that the Stative conjugation primarily denotes a resultative *state*. In other words, it denotes the state that follows a CoS event. Said CoS event, I argue, is implied within the ROOT used to derive the VA serving as a base for the resultative (Kamil 2025).

To illustrate, consider the following examples with the Statives from (3) in their respective contexts in (4). Example (4a) shows a PC ROOT in the Stative used in a conditional clause. The context indicates a situation in which the waters of a river are not typically swollen by a a flood. Following the event of a flood, the waters rise/swell (√*gpš*), and the waters are now in a state after having swollen. The conditional then goes on to prescribe the appropriate reaction to the now-swollen waters. Conditional usages for the Stative are indeed common.

Example (4b) shows a Stative for a non-agentive transitive ROOT. The context here suggests that the speaker is now in a state of knowing what it is like to be under the influence or hold of the addressee. Finally, (4c) gives a Stative for an agentive transitive ROOT. The speaker claims to currently hold a state of having been sold, following a 'giving' (√*ndn*) transaction, which has taken place in the past, thus changing the speaker's state to an unfree person.[9]

(4) Primary Stative examples in usage

a. *šumma magal* ***gapš-ū***
√*gpš* 'huge'
if flood.CSTR huge.VA-STAT.3.M.PL
'If (the waters of a river) are swollen by flood.' (CT 39 18:101)

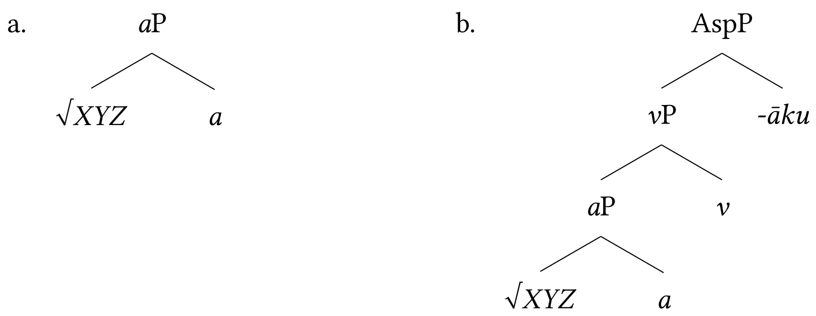
[9] I will elaborate on the aspectual "post-perfective state" nature of the Stative in Section 3.2.4 further below.

b. *ṣibit=ka* **amr-āku**
√*ʔmr* 'see'
hold=your.M.SG see.VA-STAT.1SG
'I have experienced your hold.' (Lambert BWL 194 rev. 8)

c. *aššumi=ka* *ana* *kasp-am* ***nadn-āku***
√*ndn* 'give'
because-your.M.SG DAT silver-ACC give.VA-STAT.1SG
'Because of you, I have been sold for money.' (TIM 2 100:14)

With the primary Stative *morphologically* deriving from the VA, one may be inclined to propose a syntactic structure for Statives as given in (5b) below, assuming that the basic structure for VA is that in (5a).[10] The structure in (5b) would then imply that a Stative is derived through the verbalisation of an adjectival base, to which then Agreement features are added at AspP.

(5) (Preliminary) syntactic structures for VAs and primary Statives



A comparison with the VAs derived from the same ROOTS as the Statives given in (4) quickly complicates this proposition, however. While the VAs in (6a, 6c) appear to describe the same argument as the Statives in (4a, 4c), the VA in (6b) does not describe the argument as the argument predicated in (4b): The VA of √*ʔmr* 'see' selects the Theme, i.e., the entity that is being seen, and the Stative selects the Experiencer, i.e., the entity that is seeing.

[10] I refer to the Agreement phrase as AspP, in accordance with Akkadian marking only aspect and no tense on its verbs. Given that the Stative is a conjugation and may not be combined with other TAM-marked forms, I assume its marking to thus take place on AspP, just like the *perfective* and *imperfective*.

(6) VAs

a. *gapš-u*
√*gpš* 'huge'
huge.VA-NOM
'swollen'

b. *amr-u*
√*ʔmr* 'see'
see.VA-NOM
'seen'

c. *nadn-u*
√*ndn* 'give'
give.VA-NOM
'given'

Experiencer-subject verbs thus demonstrate that a blind VA-to-Stative mapping, either semantically or syntactically, cannot be upheld. It is perhaps due to the disregard of this class of verbs in previous treatments of the Stative that the direct and simple connection between VA and Stative was made.

In his treatment of the Stative, Huehnergard (1987) focusses his analysis on deriving the semantics of the VA-base of the form and treating Stative semantics as the projection of these VA semantics. He categorises three derivational patterns for VAs: "stative" (i.e., PC) ROOTS, "active-intransitive" (i.e., unergative) ROOTS, and "active-transitive" (i.e., transitive ROOTS with Agent-subjects) ROOTS. Huehnergard suggests that stative/PC ROOTS derive "descriptive" VAs (and hence also Statives), active-intransitive/unergative ROOTS derive "resultative" VAs (and Statives), and finally active-transitive ROOTS derive "passive" VAs (and Statives). His generalisations are summarised in Table 5 below.[11]

Table 5: Semantic types of Akkadian VAs

Stative ROOTS	Active-intr. ROOTS	Active-tr. ROOTS
↓	↓	↓
VA is descriptive	VA is resultative	VA is passive
idmiq 'improved'	*ušib* 'sat down'	*iṣbat* 'seized'
→ *damiq* 'is good'	→ *wašib* 'is seated'	→ *ṣabit* 'is seized'

[11]For a comprehensive overview over the literature pertaining to the Stative see Carver (2016).

Four counterarguments can be raised against these generalisations. First of all, as was shown in the, admittedly provisional, structure in (5) above, I assume that the verbal Stative is formed at least through the derivation of a *verbalising head*,[12] and the Stative suffix, which agrees for person, number, and gender. This means that both morphologically as well as syntactically, the Stative *adds* to a VA base and should thus not be blindly assumed to be semantically equivalent to it.

Secondly, Huehnergard's ROOT-classifications are too broad and do not include the full range of derivational patterns found in the corpus. Most importantly, he excludes the crucial class of non-agentive Experiencer-subject transitive predicates (such as *hear*, *give birth*, *know*, *love*, etc.), which we have seen poses a problem for straight syntactic VA-to-Stative mapping.

Thirdly, while the pattern Huehnergard observed for the VA concerning stative/PC ROOTS may remain uncontested, the differentiation between the resultative VAs derived from active-intransitive/unergative ROOTS and passive VAs derived from active-transitive ROOTS seems redundant, if not incorrect. For one, Akkadian derives its passives by *derivation*, adding the *n*-morpheme to its SIMPLE (G) predicates, or the *t*-morpheme to the INTENSIVE (D) and CAUSATIVE (Š) predicates. The Stative, however, is a TAM-encoded conjugation, and thus falls under the category of *inflection*, which falls under a different syntactic domain. Crucially against the passive-claim, one may also add that a few Statives even appear in derivationally passive forms, i.e., the N stems.

Finally, the resultative nature of VAs derived from unergatives may be contested, too. It appears that rather than resultative, these VAs appear more aspectually durative, deriving stage-level predicates, as shown in (7).

(7) Durative VA derived from unergative ROOT

ina šēp-ē=ya	***lasm-āt-e***	*ina pašḫ-i*	*a-dûk*
	√*lsm* 'run'		√*dʔk* 'kill'
in foot-PL.OBL=my	run.VA-F.PL-OBL	in spear-OBL	1.SG-kill.PFV

'On my running/fleet feet (and) with a spear, I killed (lions).' (KAH 2 84:124)

For present purposes, this means that while the overt morphological observation that Statives derive in some form from VAs may still hold, something more complex is going on in the syntax. It will be the goal of this Subsection to determine and motivate what exactly that is.

The answer, I argue, lies in the interaction of the lexical semantics of IDIOSYNCRATIC LEXICAL ROOTS with the syntactic construction of event-templates. This

[12] See Kamil (2023).

means that we must consider both the argument structural encoding of LEXICAL ROOTS, as well as the derivation of event templates for VAs and Statives respectively. Different functional heads, I argue, set different s-selectional requirements for the derivation of their argument structure.

I will begin by providing some observations on the relevant ROOT-classes in Section 3.2.1 and then suggest a derivation-schema of Statives vs. VAs on the interfaces of lexical semantics and syntax in Section 3.2.2. I will then move on to motivate the resultative reading and function of the Stative in Section 3.2.3 by drawing parallels to the resultative construction described by Levin & Rappaport Hovav (1995). To differentiate the Stative from the (Indo-European) concept of a perfect(ive), I will finally make several notes on the aspectual nature of the Stative in Section 3.2.4.

3.2.1 Root classes

The schema in Table 1 above has already introduced the basic split between adjectival/PC ROOTS and verbal/eventive ROOTS for primary Statives. This split is semantically motivated by the different VAs that are derived from these two kinds of ROOTS.

Beginning with PC ROOTS, Dixon (1982: 3f.) differentiates seven types of PC adjectives/roots, as given in (8) below. ROOTS describing these concepts derive VA describing stative properties, which are not implied to have been acquired through any type of change event. Their respective verbs on the other hand are always inchoative, describing a (usually) non-causal CoS process leading up to the state otherwise denoted by the VA, as illustrated in (9), with the G infinitive given in the (i.) examples and the VA given in the (ii.) examples.

(8) Types of property concept adjectives (following Dixon 1982: 3f.)
  a. DIMENSION: big, long, wide, etc.
  b. AGE: old, young, new, etc.
  c. VALUE: good, strange, necessary, etc.
  d. COLOUR: red, black, white, etc.
  e. PHYSICAL PROPERTY: soft, heavy, clean, etc.
  f. HUMAN PROPENSITY: happy, jealous, cruel, etc.
  g. SPEED: fast, quick, slow, etc.

(9) PC ROOT adjectives in Akkadian

a. Dimension: √*gpš*
   i. *gapāšu* 'become huge'
   ii. *gapšu* 'huge'
b. Age: √*šʔb*
   i. *šiābu* 'become old'
   ii. *šîbu* 'old'
c. Value: √*dmq*
   i. *damāqu* 'become good'
   ii. *damqu* 'good'
d. Colour: √*wrq*
   i. *warāqu* 'become green/ yellow'
   ii. *warqu* 'green/yellow'
e. Physical property: √*dnn*
   i. *danānu* 'become strong'
   ii. *dannu* 'strong'
f. Human propensity: √*ḫdʔ*
   i. *ḫadû* 'rejoice, become happy'
   ii. *ḫadû* 'happy'
g. Speed: √*ḫmṭ*
   i. *ḫamāṭu* 'hasten, be fast'
   ii. *ḫamṭu* 'fast'

Eventive ROOTS pattern differently. For one, their verbs do not behave as uniformly as the PC ROOTS verbs do. Verbs derived from eventive ROOTS may be of all transitivities, featuring all sorts of argument structural components, and different (aspectual) event types. A crucial commonality of eventive ROOTS containing a Patient argument, however, is their derivation of adjectives that entail change.[13] Examples are given in (10) below, again with the G infinitive given in the (i.) examples and the VA given in the (ii.) examples. As shown in (10b), unergatives, which lack a Patient (or more generally, an internal argument), fail to derive a result-adjective.

(10) Result ROOT adjectives in Akkadian

a. Unaccusative: √*mqt*
   i. *maqātu* 'fall'
   ii. *maqtu* 'fallen'
b. Unergatives: √*wšb*
   i. *wašābu* 'sit, dwell'
   ii. *wašbu* 'sitting, dwelling'

[13]An exception to this rule is found in ROOTS of non-voluntary stimuli emissions, such as *stink*, *sparkle*, etc. These ROOTS derive adjectives that pattern with PC ROOTS, i.e., denoting adjectives that do not entail change. They do not pattern with PC ROOTS when it comes to their verbs, however, which denote statives, too. One may argue that the argument of these verbs is not a Patient per se, but rather a Theme, which thus prevents result-adjectives.

c. Transitive withAGENT subject: √*mḫṣ*
  i. *maḫāṣu* 'strike, beat'
  ii. *maḫṣu* 'beaten'

d. Transitive with non-AGENT subject: √*šmʔ*
  i. *šemû* 'hear'
  ii. *šemû* 'heard'

This pattern of PC ROOTS deriving adjectives not entailing change and eventive ROOTS deriving adjectives that do entail change is by no means unique to Akkadian. In their large-scale typological study on CoS predicates, Beavers et al. (2021)[14] motivate the distinction of two ROOT types among CoS predicates: PC ROOTS, i.e., in our case adjectival ROOTS, and result ROOTS, i.e., in our case non-adjectival ROOTS. Crucial to our present purposes is their finding that PC ROOTS derive adjectives that do not entail change, while result ROOTS derive adjectives that *do* entail change, much like the eventive ROOTS (i.e., result ROOTS + ROOTS not necessarily entailing a CoS) found in Akkadian.

But so far, this only strengthens our case for separating PC ROOTS from eventive ROOTS in their derivation of VAs. When it comes to the derivation of Statives, however, we have already seen in Section 3.2 above that further sub-classification of eventive ROOTS is necessary. Consider the following Stative examples in (11) of the four ROOTS given in (10) above.

(11) Stative usage examples

a. ***maqit*** *bēl* *mešr-em=ma*
√*mqt* 'fall'
fall.VA.STAT.3SG.M lord.CSTR riches-GEN=CONJ
'(Even) the lord of riches is fallen.' (Lambert BWL 80: 187)

b. UD.3.KAM *ina bīt-i* ***ašib***
√*wšb* 'sit'
3 days in house-OBL sit.VA.STAT.3SG.M
'(You draw a line around a sick person's bed,) he will stay home for three days.' (AMT 88,2: 6)

c. *Ḫumman-ḫaldašu šar* *Elamt-i* *ina muṣlāl-i*
RN king.CSTR Elamites-GEN in afternoon-OBL

[14]See also Beavers (2012), Beavers & Koontz-Garboden (2020).

maḫiṣ=*ma*
√*mḫṣ* 'strike'
strike.VA.STAT.3SG.M=CONJ
'Ḫumman-ḫaldašu, the king of the Elamites was struck this afternoon and (... he died).' (CT 34 50 *iii* 31)

d. *pî* PN *ūl* ***šamê-āku***
√*šmʔ* 'hear'
mouth.CSTR PN NEG hear.VA-STAT.1SG
'I have not heard (~ had orders) from the mouth of PN.' (AbB 1 30:12)

In (11a), the only argument of the unaccusative verb 'fall' is the argument predicated by the Stative, namely the lord of the riches. As the ROOT derives verbs with only one Patient argument, it is not greatly surprising to us to find the same argument described both in the VA and in the Stative derived from this ROOT. In (11b), we see a rare example for an unergative Stative, where again the only argument available in perfective/imperfective verbs is the one surfacing in the VA and the Stative. Notably, the Stative takes on a non-resultative, durative meaning, denoting a stage-level predicate. So far, it thus seems as though there is no particular preference for Patient vs. Agent-subjects for Stative formation, although the event types appear to differ.

(11c) gives an example for the Stative of a ROOT, which otherwise derives transitive verbs with Agent-subjects and internal Patient objects. The Stative, just like the VA, predicates the internal Patient argument of these verbs. This behaviour is not unusual, if one compares with adjectives of result/eventive ROOTS in other languages, e.g., *beat* → 'the beaten *path*', *cook* → 'the cooked *meal*'. What it does show us, is that given the two options, VAs as well as Statives, appear to s-select for the internal Patient object, rather than the Agent-subject.

(11d) presents us with a challenge, however: While the VA of √*šmʔ* 'hear' selects for the Theme object as in (10d) (i.e., *šemû* '(Theme is) heard'), the Stative selects for the Experiencer-subject (i.e., *šamî/ê* '(Experiencer is in the state of having) heard'). The same holds for predicates with Experiencer-subjects and Patient objects, such as √*wld* 'bear', which is given in (12) below. It appears thus that we are dealing with an allosemic s-selectional mismatch between VAs and Statives in Experiencer-subject verbs.

(12) √*wld* 'bear'

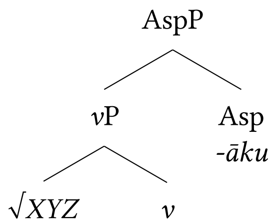
a. VA
waldu 'born'

b. Stative
awilt-um ... *ana mutī=ša* 2 *mār-ī*
woman-NOM ... DAT husband.CSTR=her 2 son-PL.GEN
ald-at
√*wld* 'bear'
bear.VA-STAT.3SG.F
'The woman has borne two sons for her husband.' (AbB 7 106:19–21)

This allosemy further challenges Huehnergard's analysis of Statives being projections of VAs. Despite the morphological similarity, syntactically, there must be two different underlying structures.

As hinted at further above, I suggest that the locus of this alternation happens in the syntax-semantic interface, more specifically in the s-selectional requirements of functional heads: While VAs are formed through the combination of a ROOT with an adjectival head, Statives are formed through a ROOT's combination with a verbal head. Its form is then distinguished from other conjugations such as the perfective and imperfective at AspP. I summarise this in (13).

(13) Revised primitive syntactic structure of a Stative



Before turning to the mechanism of the s-selectional mismatch of verbal and adjectival heads in the next Subsection, we hold that given the different Stative derivations, the primary Stative may be further divided by ROOT class, as summarised in Table 6.

3.2.2 Deriving the Stative – From lexical semantics to syntax

Given the identical morphological makeup of Statives and VAs respectively, regardless of the ROOTS inserted, I currently posit no difference in the "higher" syntax, i.e., in the operations concerning functional heads such as Asp, D, etc. Instead, I suggest that the core of the Experiencer-subject alternation lies in the s-selectional requirements a functional head imposes on the ROOT it is merged with.

Table 6: Basic morphological Stative categorisation (revised)

Primary Statives					Secondary Statives	
XaY(i)Z- + suffix					Stem + suffix	
Verbal ROOT				Adj. ROOT	Adj. stem	Nom. stem
Unerg.	Unacc.	Agent-tr.	Exp-tr.			

Crucial to this argument is the conception that verb meanings consist (at least partly) of *event structures* and that event structures in turn consist of two components: *event templates* and *idiosyncratic lexical semantic roots* (Beavers et al. 2021: 2, but see also Dowty 1979, Levin & Rappaport Hovav 1995, Rappaport Hovav & Levin 1998, i. a.).

Event templates are constructed "from a small number of basic eventive primitives" (Beavers et al. 2021: 2), which define the general temporal and causal shapes of events. The event template is what groups verbs into "types". The IDIOSYNCRATIC LEXICAL SEMANTIC ROOTS then fill in specific actions or states for the eventive templates of verbs. With the eventive template grouping types of verbs into classes, the LEXICAL ROOT is what distinguishes verbs within a class. The separation of these two components is maintained in both syntactic (e.g., bifurcation hypothesis, Marantz 1997, Embick 2004, Harley 2012) and lexicalist (e.g., Levin & Rappaport Hovav 1995, Rappaport Hovav & Levin 1998, Beavers & Koontz-Garboden 2020) theoretical approaches.

By the example of CoS predicates, Beavers et al. (2021) demonstrate that some lexical semantic ROOTS naturally entail change, thereby making a strict distinction of event templates and lexical semantic ROOTS difficult to maintain. In other words, some lexical semantic ROOTS appear to already contain the eventive "operators" that are introduced in the eventive templates of other CoS predicates. This information is contained within the ROOT next to the otherwise recognised θ-grids, i.e., the thematic representations that later on surface or rather dictate the course of derivation in the syntax (Belletti & Rizzi 1988: 291, see also Irwin & Kastner 2020, Kastner 2020).

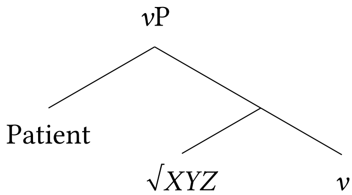
Building on the notion that different ROOTS may have different effects on syntactic derivation, I propose to synchronise two tentative approaches to dealing with the present Experiencer-subject alternation: a "syntactic mapping" and a "syntactic filtering" approach. Regardless of which approach one follows, I maintain that the argument structural encodings of LEXICAL ROOTS play a crucial role in the explanation of the Experiencer-subject alternation.

The ROOT classes discussed so far are given in (14) with their respective minimal encoding of the necessary arguments. Note that the difference between (14a) and (14c) lies in the unaccusative (14a) disallowing the introduction of an Agent.

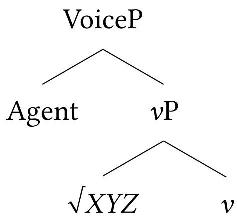
(14) Minimal argument structural encodings in the lexical idiosyncratic ROOTS
  a. √*mqt* 'fall': $[\lambda x.\lambda e.\text{fall}(e) \wedge \text{Patient}(x,e)]$
  b. √*wšb* 'sit': $[\lambda e.\text{sit}(e)]$
  c. √*mḫṣ* 'beat': $[\lambda x.\lambda e.\text{beat}(e) \wedge \text{Patient}(x,e)]$
  d. √*šmʔ* 'hear': $[\lambda y.\lambda x.\lambda e.\text{hear}(e) \wedge \text{Experiencer}(x,e) \wedge \text{Theme}(y,e)]$

Beginning with the syntactic mapping approach, it has been argued before, for instance for Greek (Anagnostopoulou 1999) and Italian (Belletti & Rizzi 1988), that Experiencer arguments are syntactically equivalent to internal arguments with Patient/Theme roles. Belletti & Rizzi (1988) and Anagnostopoulou (1999) show that this is true for Experiencer-subject verbs (e.g., *see*, *love*, *know*) and Experiencer internal *and* external objects (Anagnostopoulou 1999). Building on these cross-linguistic studies, one may postulate θ-grids as in (15). I further follow Kratzer (1996) in introducing Agent-subjects at VoiceP.

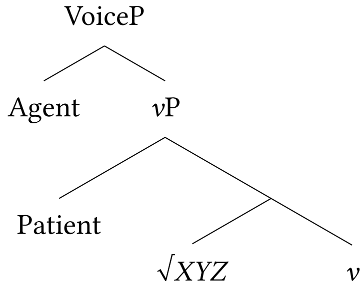
(15) Syntactic mapping of θ-roles
  a. Unaccusative verb



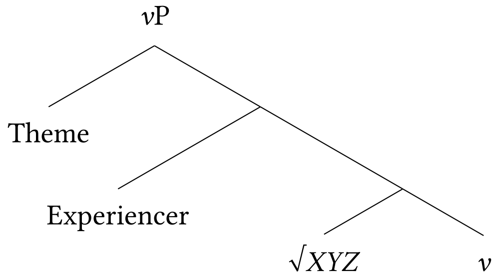
  b. Unergative verb



c. Agentive transitive verb



d. Non-agentive transitive verb



The initial observation one may make based on these θ-mappings is that VAs appear to always predicate the argument in the specifier of *v*P, i.e., the Patient/ Theme in unaccusative and transitive verbs. In the rare cases of unergative VAs, the derivation typically fails, and when it does not, the VA does not express the resultative observed in most other verbal classes. Statives, on the other hand, appear to always predicate the internal argument, i.e., the Patient in unaccusative and agentive transitive verbs, but the Experiencer in non-agentive transitives.[15]

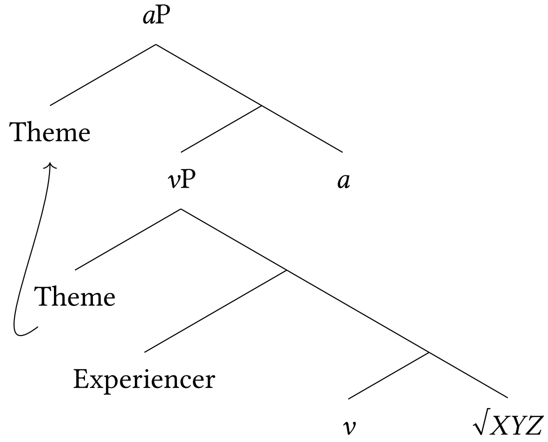
[15]VAs, despite being primarily adjectival, display strong verbal behaviour morphologically and syntactically, earning them their name: Not only may one derive INTENSIVE/D, CAUSATIVE/Š, and medio-passive N/*t*-marked VAs, but Huehnergard (1987), Buccellati (1988), Kouwenberg (2000), i. a., report that VAs may also be used predicatively in copular structures, e.g., (i). Typologically, this is not entirely unexpected: Adjectives have long been studied as an intermediate category between nouns and verbs, with languages differing in terms of where their adjectives place on the noun-verb scale (Berg 2000, Sasse 2001).

(i) *šakn-u* *ša* *ana* *šūtuqūt-i* *ša* *unqāt-i* … *ina* *Nippur* ***paqd-u***
√*pqd* 'entrust'
official-NOM REL DAT forwarding-OBL REL sealed.letter-OBL … in GN entrust.VA-NOM
'The *šaknu*-official who was appointed in Nippur for forwarding sealed (royal) letters.'
(ABL 1 238 rev. 10)

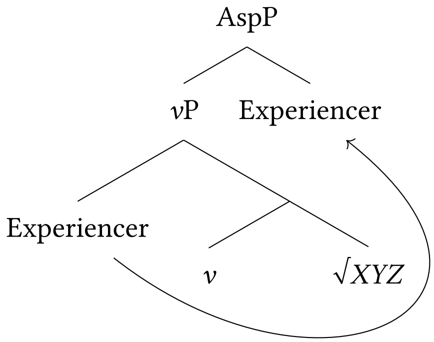
This means that one could potentially assume the VAs and Statives to s-select for their arguments as illustrated in (16).

(16) Syntactic selection of arguments for VAs and Statives

a.



b.



For Statives, this does not seem problematic: Derivation would go as far as the internal argument is derived and assign that argument its inflectional features at AspP. Assuming an inherent *v*P base for VAs is more problematic, however: Non-agentive transitive would derive two arguments within the *v*P with only the Theme predicated and the Experiencer left derived but not predicated. Underlying *v*Ps for VAs are thus difficult if not impossible to maintain.

A modification to the syntactic mapping approach can be found in the filtering approach. I follow Irwin & Kastner (2020: 4) in assuming "that the syntax is free to generate syntactically well-formed structures, but lexical ROOTS can impose semantic constraints at Logical Form [...]. In this way, the interfaces can be said to not only interpret but also 'filter' the output of the syntax".

The core idea here is that functional heads select for the arguments/roles they require through an operation called "event-identification" (Irwin & Kastner 2020: 7). An adjectival head could thus specify an eventive template that deliberately specifies for a Patient/Theme to be filtered into syntax, just as we see. By contrast, a verbal head may set a more complex templatic event structure in which it will, as in the syntactic mapping approach, require the one role to be derived that fills the internal argument position.[16] This of course requires one to follow syntactic approaches such as Belletti & Rizzi (1988), Anagnostopoulou (1999) in positing that Experiencer-arguments behave like internal arguments and transitive Experiencer-subject verbs derive their Patient/Theme objects more like external arguments. I recognise that an approach that assumes syntactic uniformity for Patients/Themes would require a more elaborate explanation for why Experiencers are drawn up to be predicated by Statives in particular.[17]

Thus, by assuming different lexical ROOT decompositions and different event-templatic requirements for different functional heads, we can account for the Experiencer-subject alternation. By contrast, all other ROOT classes appear to overlap in their adjectival and verbal head requirements: Patients/Themes are usually derived as internal arguments.

In summary, this Section set out to show that Statives cannot be directly derived from VAs, basing the argument on Experiencer-subject verbs. I suggest an analysis in which different syntactic heads set different event-templatic requirements for the derivation of their categorial type. Thereby, adjectival heads always s-select for Patients/Themes, while verbal heads rather select for internal arguments. The fact that Experiencers are filtered for the position of internal argument over Patients/Themes might hint at some ROOT-internal structure of thematic roles, though I leave this speculative point open for future discussion.

[16] A similar if not identical approach is attempted in Geist (2010) for Russian adjectives, which distinguish a so-called "long form" (LF) and a "short form" (SF). Thereby, the SF appears to be verbal, as our present Stative is, while the LF is adjectival. In Geist's analysis, different functional heads set different event templatic requirements, too, accounting for the different syntactic and eventive scopes of the respective "adjectives".

[17] An attempt to solve this issue, though for present purposes not further elaborated on here, would be the treatment of Experiencer-arguments as so-called "holders of state" (Hirsch 2018: 204). Statives would then have to consist of a more complex underlying eventive structure than simply being specified at AspP. But it could explain why they are particularly drawn to Experiencers who are "by default" holders of state, and filter them out of the ROOT and into the syntax. This theoretical discussion shall remain subject for future work.

The next Section will deal with said further derivation of (16b), i.e., the primary Stative, and motivate its resultative semantics and respective syntactic representation.

3.2.3 Motivating the resultative

In their monograph on unaccusative verbs, Levin & Rappaport Hovav (1995) dedicate a chapter to resultative constructions, in particular resultative phrases like the phrases in (17), as a diagnostic on whether English syntactically encodes the difference between unaccusative and unergative verbs. Thereby, resultative phrases refer to "an XP that denotes the state achieved by the referent of the NP as a result of the action denoted by the verb in the resultative construction" (Levin & Rappaport Hovav 1995: 34). The resultative construction is thus achieved by *secondary predication*, wherein result state (underlined) and CoS event (bold script) are not expressed by the same syntactic node.

(17) Examples of English resultative phrases *apud* Levin & Rappaport Hovav (1995: 34ff.)

a. *… a 1,147 page novel that* ***bores*** *you* <u>*bandy-legged*</u> … (P. Andrews, *Abandoned in Iran*, 28)

b. *Well, the conclusion was that my mistress* ***grumbled*** *herself* <u>*calm*</u>. (E. Brontë, *Wuthering Heights*, 78)

c. ***Drive*** *your engine* <u>*clean*</u>. (Mobil ad)

The resultative construction underlies the Direct Object Restriction (DOR), given in (18) below. This means that only the direct object of a non-resultative predicate may undergo the CoS said predicate would denote in a resultative construction.

(18) Direct Object Restriction (DOR; Levin & Rappaport Hovav 1995: 34, 41):
A resultative phrase may be predicated of the immediately postverbal NP, but may not be predicated of a subject or of an oblique complement.

Following our syntactic analysis of the Stative in the previous section, we see that this generalisation applies precisely to the Akkadian Stative: Only internal arguments (i.e., the immediately postverbal NPs) are predicated by the form. Contrary to the English resultative construction, however, the Akkadian Stative does not rely on secondary predication. Rather, it draws both the result state as well as the CoS event either directly from the ROOT or from the underlying *v*P layer of its base.

In (19) we see three agentive transitive Statives for 'pay' (*eṭir*), 'give' (*nadin*), and 'face/confront' (*maḫir*). All three Stative forms are predicated on the internal argument of the underlying verb, in this case *šīm eqlišu* 'the price of his field'.

(19) Transitive Stative example

šīm	*eql-i=šu*	*kî*	*kasap*	*gamir-t-i*
price.CSTR	field-GEN=his	as/like	silver.CSTR	complete-FEM-GEN

eṭir	***nadin***	***maḫir***
√*ʔṭr* 'pay'	√*ndn* 'give'	√*mḫr* 'face'
pay.STAT.3SG.M	give.STAT.3SG.M	face.STAT.3SG.M

'The price of his field, as one complete (payment in) silver, has been paid, handed over, and received.' (Scheil 1927: 38, 19)

Similarly, unaccusatives, too, form Statives predicating the internal argument, as has been shown in previous examples, for instance (11a), but also in (20) below.

(20) Unaccusative Statives

šutt-u	... *lu*	***ḫalq-at***	*lu*	***naḫarmuṭ-at***
		√*ḫlq*.'disappear'		√*nḫrmṭ* 'dissolve/melt'
dream-NOM	... MOD	disappear.VA-STAT.3SG.F	MOD	dissolve.VA-STAT.3SG.F

'(So) let the dream be gone, let it be dissolved.' (KAR 2 339:22)

Now, following the DOR's requirement for an *internal* argument, it is predicted that unergative ROOTS, which derive their argument externally, cannot form resultative phrases, as illustrated also in Levin & Rappaport Hovav 1995: 35f. In none of the examples in (21) can the adjective be read as the result of the verb that precedes it. In other words, unergative predicates do not naturally entail a post-eventive result state.

(21) Ungrammatical unergative resultatives (Levin & Rappaport Hovav 1995: 36)

   a. **We yelled hoarse.*
   b. **My mistress grumbled calm.*
   c. **The officers laugh helpless.*

Akkadian Statives appear to again more or less pattern with this observation. For one, VAs formed from unergative ROOTS are rarer and also semantically different, denoting durative stage-level modifiers, rather than resultative ones (compare with (7) above). With the exception of √*wšb* 'sit', which is more commonly found in the non-agentive transitive meaning of ‘dwell in’, hardly any unergative Statives may be found in the literature.

One questionable case found is given in (22) below. Note also that a Stative-formation is only possible for the ROOT *√ṣʔḫ* ‘laugh’ under its alternative meaning ‘be alluring, act coquettishly’. It remains unclear to me whether the ROOT is inherently unergative or rather a PC ROOT.

(22) Unergative Stative example
kê ***ṣîḫ-āku*** *ana naḫš-i*
√ṣʔḫ ‘laugh’
as/like laugh.VA-STAT.1SG DAT lusty.M-GEN
‘How I entice (my) lusty boy.’ (KAR 1 158 r. *ii* 7)

We maintain thus that forming Statives from unergative ROOTS is problematic, though admittedly not entirely impossible. As will be shown in the discussion of secondary Statives in Section 4 further below, the Stative suffix appears to have, at some point in the diachronic history of Akkadian, reanalysed to independently attach to nominal and adjectival *stems*, too. On the basis of this pattern, I suggest that the very rare formation of unergative Statives patterns with the very rare formation of secondary Statives. Unergative Statives would thereby be derived from the lexicalised adjectival formations of these ROOTS and are thus not deradical.

In any case, for the formal motivation of the Stative as a resultative, we see that the primary Stative patterns with the syntactic generalisations made for the English resultative construction. I would argue that the majority of translations provided for Statives would also back up this claim, as has been shown for the few examples illustrated in this Section.

The core difference between the English resultative and the Akkadian one is the necessity for secondary predication in the English construction (Levin & Rappaport Hovav 1995: 54), while the Akkadian Stative inherits the CoS that leads up to the resultant state directly from its ROOT. This could also explain why primary unergative Statives are practically unattested: Their arguments are not naturally affected or changed by the event.

Statives inheriting their CoS directly from the ROOT might pose an issue for adjectival/PC ROOTS deriving Statives. As Levin & Rappaport Hovav (1995: 40) note, however, adjectival passives are known to form resultative phrases, too (see also Keyser & Roeper 1984, Roberts 1987, Stroik 1992), as for instance ‘*the spun-dry clothes*’. We can see similar behaviour with Akkadian adjectival ROOTS inserted into the Stative as in (23).

(23) Adjectival ROOT-derived Statives

a. ***ṣiḫ̮r-ēt*** *ūl* ***eṭl-ēt*** *ūl* *šārt-um*
√*ṣḫ̮r* 'small' √*ʔṭl* 'be.manly'
small.VA-STAT.2SG.M NEG be manly.VA-STAT.2SG.M NEG hair-NOM
ina *līt-i=ka*
in/on cheek-GEN=your.M.SG
'Are you a baby? Are you no man? Is there no beard on your cheek?' (ARM 1 108:6)

b. ***maruṣ*** ***kabit***
√*mrṣ* 'ill' √*kbt* 'heavy'
ill.VA.STAT.3SG.M heavy.VA.STAT.3SG.M
'He is seriously ill.' (Goetze 1958: 68, No.43:16)

In (23a), Šamši-Addû, king of Upper Mesopotamia, scolds his younger son Yasmaḫ̮-Addû for his lazy behaviour, comparing him to his exemplary 'manly' older brother Išme-Dagan. In the context of the comparison between the two brothers, both kings of cities, both 'grown men with wives', the younger brother's behaviour (though not out of character for the individual in question) is out of character for the expectations set for him. (23a) should thus rather be read as 'Are you a baby now? Are you no man all of a sudden?'.

Another example in given in (23b) wherein the state of an ill person is described with two adjectival Statives. Two readings are possible here: a., the person is sick (*maruṣ*) and the sickness is heavy (i.e., severe), or b., the person has become sick and as a result quite severe or difficult to treat. Both Statives appear to affect each other.

As far as the syntactic behaviour of the adjectival Statives can be addressed, Levin & Rappaport Hovav (1995: 40) suggest that a verb's direct internal argument undergoes lexical externalisation in the formation of adjectival passives, a process that appears to happen for Akkadian VAs, too, and therefore also manifests in their respective Stative formation, given the similarity between the syntactic processes of the two constructions.

As with the result ROOTS, PC ROOTS indicate the CoS event by themselves, e.g. 'become red', 'become sick', etc. As noted by Levin & Rappaport Hovav (1995: 54), but also by Dowty (1979), Pustejovsky (1991), Grimshaw & Vikner (1993), the activity denoted in an independently derived verb (i.e., *not* in the resultative) is the CoS event leading to the eventuation of the state denoted by the resultative.

Based on the discussions of the past two Subsections, I now revise the categorisation schema given in Table 6 once more into the schema in Table 7.

Table 7: Basic morphological Stative categorisation (revised again)

Primary Statives					Secondary Statives	
XaY(i)Z- + suffix					Stem + suffix	
Verbal ROOT				Adj. ROOT	Adj. stem	Nom. stem
Unerg.	Unacc.	Agent.-tr.	Exp.			
No Stat.	Root-inherited CoS					

To summarise, this section attempted to motivate the resultative nature of the Stative by drawing on syntactic parallels to the English resultative construction as well as presenting case examples for the resultative contexts the form appears in. A final important argument for the Stative's resultative nature will be made in the next Section 3.2.4, to separate it from what linguists may suspect a "perfect" conjugation to be. As Beavers et al. (2021: 3) note, an important component of CoS events is their interaction with a given verb's temporal features (see also Hay et al. 1999, Kennedy & Levin 2008, Beavers 2012). For a resultative state to come into being, it must be inherently associated with a *preceding* CoS event that has led to the eventuation of said resultative state. This association of a completed event and a subsequent state that follows as a direct consequence of the event is a property often ascribed to perfects. As I will show in the next section, Statives are not perfects, however.

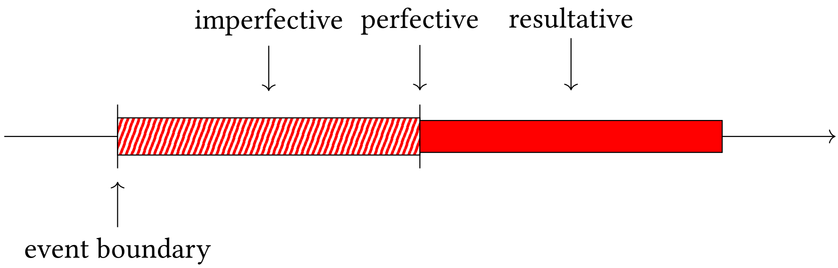
3.2.4 Notes on the aspectual position of the Stative

As was shown in various previous studies before (e.g., Cohen 1984), the Akkadian Stative is diachronically related to the Semitic perfect. It is thus a valid thought to consider the conjugation a perfect, rather than resultative/Stative.

I argue against this categorisation of the conjugation on the basis of the greater aspectual system of Akkadian, as outlined in Kamil (2025), Tanza-Kamil (In preparation). Imperfective, perfective, and Stative are seen here to be the three core aspect conjugations employed in Akkadian, with the perfective serving as a type of alternant to imperfective and Stative respectively.

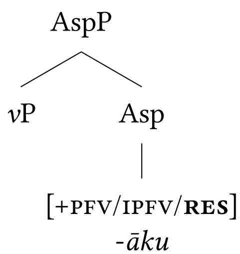
The imperfective denotes an ongoing dynamic event. It is contrasted with the perfective, which denotes a punctual or completed event. The perfective implies a resultant state to come about from the completion of its denoted event. The perfective then again is contrasted with the Stative/resultative, which denotes a static state, implied to have resulted from a preceding completed event. This is illustrated in (24).

(24) Aspectual frames in Akkadian



The core idea here is that the imperfective and the perfective both denote *events*, but while one focuses on an event's eventuation, the other focuses on its completion. The perfective and the Stative both entail an event leading to a CoS and a resultant state, but while the perfective denotes the event and implies the state, the Stative denotes the state and implies the event. On the basis of these alternations, I treat imperfective, perfective, and Stative as alternants on AspP, as illustrated in (25). For further elaboration, I refer the reader to Tanza-Kamil (In preparation).

(25) Higher structure of the Stative



With the primary Stative now covered, I move on to incorporate the secondary Stative into this theory.

4 The secondary Stative

4.1 Morphology

The secondary Stative is made up of a nominal/adjectival stem and the Stative-specific suffixes, (26). This means that the base of the secondary Stative is not derived deradically and does not conform to any one template. Crucially, this differentiates them from primary Statives in that secondary Statives are, in fact, deadjectival/denominal.

(26) Basic morphological make-up of the secondary Stative:
nominal/adjectival stem + Stative suffix

Examples of the secondary Stative are given in bold script in (27). Morphologically, there appear to be no restrictions as to what may be Stativised. Next to the primary Statives in (27a) (e.g., in *naˀdu* 'praised; famous', *gašru* 'strong', and *kabtu* 'heavy', and the INTENSIVE VA *šurruḫu* 'glorious') we see a non-VA adjective (*ašaridu* 'supreme') and two nouns, *šarru* 'king' and *bēlu* 'lord', form secondary Statives.

(27) Secondary Stative examples

a.

šarr-āku	***bēl-āku***	*naˀd-āku*	*gašr-āku*
		√*nʔd* 'praised'	√*gšr* 'strong'
king-STAT.1SG	lord-STAT.1SG	praised.VA-STAT.1SG	strong.VA-STAT.1SG

kabt-āku	*šurruḫ-āku*	***ašarid-āku***
√*kbt* 'heavy'	√*šrḫ* 'glorious'	
heavy.VA-STAT.1SG	D.glorious.VA-STAT.1SG	supreme-STAT.1SG

'I am king, I am lord, I am famous, I am strong, I am important, I am glorious, I am supreme.' (RIMA 2 A.0.101.1 *i* 32)

b.

šumma	*sinništ-u*	*karš-i*	*libb-i*	*raš-at*
				√*ršʔ* 'acquire'
if	woman-NOM	belly-OBL	inside-OBL	acquire.VA-STAT.3SG.F

ilān-ât
godly-STAT.3SG.F

'If a woman has a round belly, she is lucky.' (KAR 2 206 *ii* 6')

As will be discussed in Section 4.3, the present study found there to be no morphological restrictions as to which adjectives could be Stativised. This means that no particular template, consonant count, or amount of derivational affixes on an adjectival stem prevent it from taking the Stative suffix. As an example, (27b) features the adjective *ilānû* 'favoured by god', itself made up of three elements: *il-* 'god', *-ān-*, an adverbial/adjectival derivational suffix, and *-ī-*, an adjectival derivational suffix, which can be seen "merging" with the following vowel of the Stative suffix *-at*.

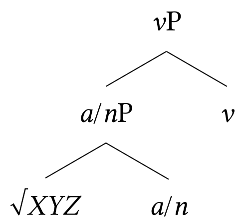
This pattern generally indicates a full reanalysis of the Stative suffix to productively verbalise, in principle, any lexical stem, making the secondary Stative a definitely deadjectival/denominal verb.[18] While no such morphological statements may be made for the Stativisation of nouns yet, it is reasonable to assume similar patterns to be found with nouns, too.

4.2 The problem of semantics

Secondary Statives being deadjectival/denominal implies that, unlike primary Statives, they do not contain an eventive *v*P at their base. This is insofar a problem for our analysis of all Statives as resultative states as secondary Statives have no eventive base to draw their CoS event from.

As I argued in Kamil (2023), secondary Statives can nevertheless be shown to be verbal, as evidenced by their syntactic clause-final position,[19] their combination with verbal suffixes such as the subjunctive and *ma*-conjunction, and their ability to be "middled". Moreover, they may not be used attributively, but rather only predicatively. Based on these morphological and syntactic generalisation, one may thus postulate the following primitive structure for secondary Statives, as given in (28).

(28) Primitive secondary Stative structure



The question now is how a secondary Stative may license resultative readings if it may not provide a CoS event itself. In most occasions found, secondary Statives appear in contexts that imply externally caused CoS resulting in the described state. For instance, in (27a) above, an excerpt from an Aššurnasirpal-inscription, the king highlights a decree that was proclaimed following his coronation (Aššurnasirpal II 1: *i* 31ff.), indicating that prior to this speech, he was not in fact king, lord, famous, important, glorious, or supreme. Among the many attributes, *šarrāku* 'I am king', *bēlāku* 'I am lord', and *ašaridāku* 'I am supreme' are

[18] I attempt no hypothesis as to when or how this reanalysis came about.

[19] Akkadian is generally very strictly SOV. In epic literature, VSO is often found, in which case Statives appear in clause-initial position.

secondary Statives. Through the context they are used in, potentially strengthened by the parallelism with the primary Statives in the same phrase, they, too, receive a resultative reading.

Two modal examples of usage are given in (29) below. Especially (29a) provides a good parallel: The adjective *imnû* '(on the) right' is derived from *imnu* 'right' + *-ī*, an adjectival suffix. While the observations about the gall bladder are being made, note that *imnû* is being used in its regular adjectival form. Only once the practitioner has established a knowledge of the gall bladder's position, is the adjective Stativised within a statement made about the gall bladder's position. In other words, the CoS event is the practitioner's understanding of the gall bladder's state.

(29) Secondary Statives headed by *šumma* 'if'

a. *šumma mart-u imn-û ina panī=ka*
if gall.bladder-NOM right-NOM in/at face.GEN=your.SG.M
mazzāz-u padān-u … imnû mart-i
station-NOM path-NOM right.CSTR gall.bladder-GEN
šakn-ū=ma mart-u ***imn-ât***
√*škn* 'put'
put.VA-STAT.3PL.M=CONJ gall.bladder-NOM right-STAT.3SG.F
ta-qabbi-Ø
√*qbʔ* 'speak'
2-speak.IPFV-SG.M
'If the gall bladder is right-side up (?) before you and the "station", the "path" (and other ominous parts) (all) lie at the right side of the gall bladder, you may say "the gall bladder is right" (i.e. favourable).' (CT 28 46:5f.)

b. *šumma ṣeḫr-u* NE (*lā*) ***ḫaḫḫaš***
if small-NOM fever NEG hot.STAT.3SG.M
'If the temperature of a small child is not very high, (his intestines are constricted, the *buʔšānu* disease has seized him).' (Labat TDP 228:101)

It is notable that many, if not most (adjectival) secondary Statives are found in modal phrases headed by either *šumma* 'if' or the precative marker *lu* 'may'. Two examples for Statives headed by *lu* are given in (30).

(30) Statives headed by *lu*

a. *lu* ***balṭ-āta*** *lu* ***šalm-āta*** *lu*
√*blṭ* 'live' √*šlm* 'whole/complete' MOD
MOD live.VA-STAT.2SG.M MOD whole/complete.VA-STAT.2SG.M
dari-āta
everlasting-STAT.2SG.M
'May you live, may you be healthy, may you (~ 'your life') be everlasting.' (YOS 2 119:7)

b. *tālitt-i* *alp-ē=ka* *lu* ***kayān-at***
calving.F-OBL cattle-PL.OBL=your.SG.M. may constant-STAT.3SG.F
'May the calving of your cattle be normal.' (Sidersky 1920: 568, 20)

The modal contexts here imply that the imagined/pre-conceived state has not yet been achieved. This means that between the time of speech and the possible described state, a CoS event can be presumed to happen. I thus suggest that secondary Statives license their resultativity by inheriting their CoS event from the context of the utterance they appear in. This happens most commonly syntactically, through modal particles/complementisers.

To highlight the resultative nature of the secondary Statives one may also draw a comparison with similar utterances utilising predicative adjectives, as in (31). Just like in (27a), (31) gives a number of predicative adjectives, but while in (27a), the adjectives describe a newly crowned Aššurnasirpal, the adjectives in (31) describe a deity whose state of having these attributes is not new but has always been true.[20]

(31) *anāku Asalluḫi dabr-u šagapūr-u bēl-u ša māt-i*
I Asalluḫi fierce-NOM majestic-NOM lord-NOM REL land-GEN
'I am Asalluḫi, fierce, majestic, lord of the land.' (Lambert 1954: 313, C 8)

In order to verify these assumptions, I conducted a small study on adjectival secondary Statives. The following section will show that among the adjectives surveyed, less than 40 derived secondary Statives. Among those, almost all Statives were found in modal contexts.

[20]The addition of the IPP *anāku* is not, however, a diagnostic for the predicative use of adjectives as it is very well also found with verbs, including Statives, as a means of emphasis. Note that Akkadian is a pro-drop language and thus does not require the utterance of pronominal subjects.

4.3 Adjectival derivations

4.3.1 Akkadian adjective formation

Akkadian adjective formation can be categorized into three general types: non-concatenative, concatenative, and other. Non-concatenative adjectival derivation refers to the root-and-template derivation mechanism wherein a usually triradical ROOT *XYZ* is inserted into a template. Concatenative derivations refer to derivations wherein derivational suffixes are added to a typically nominal, adjectival, or adverbial stem to derive adjectives. Finally, the category "other" refers to loans and contractions of multiple elements that have been lexicalised as adjectives.

The non-concatenative derivation method is by far the most common one in Akkadian. It comprises all VAs in all template patterns (SIMPLE, INTENSIVE, CAUSATIVE) and voices (active, middle, passive). The general paradigm of the VA bases for the 3SG.M is given in Table 8. But the non-concatenative adjectives include also (lexicalised) participles (e.g, *XāYiZ-*), SIMPLE adjectives derived with "intensive" doubling of the middle radical (e.g, *XaYYiZ-*), and other templates featuring attested roots.

Table 8: VA paradigm

	SIMPLE	INTENSIVE	CAUSATIVE
Active	*XaY(i)Z-*	*XuYYuZ-*	*šuXYuZ-*
Middle	*XitYuZ-*	*XutaYYuZ-*	*šutaXYuZ-*
Passive	*naXYuZ-*		

The concatenative derivations are mostly derived with the standard adjectival suffix *-ī*, or with the adverbial-adjectival suffix *-ān-*. Combinations of the two are also attested, thereby always in the order *-ān-ī-*. More rarely, the nominal suffix *-i-* (fem. *īt-*) is also attested to derive adjectives, mostly of the numerical type.

Finally, Akkadian features a number of adjectives from mostly Sumerian, Hurrian, Elamite, or Hittite, but also Amorite, Egyptian, and Kassite origin. These adjectives typically do not conform to any templatic forms, nor do they tend to feature the derivational suffixes described above. Very few adjectives are derived from contractions of two or more lexical elements, as for instance *laššu* 'absent' < *lā* 'no(t)' + *išû* 'is'.

4.3.2 Survey of adjectival statives

The present study intended to shed light on two questions, supplementing the semantic discussion of the secondary Stative: 1) Can any nominal element be Stativised?, and 2) how is the resultativity of secondary Statives licensed?

As a first step, all adjectives listed in the *Concise Dictionary of Akkadian* (CDA) were extracted according to the following set of diagnostics: a) Adjectives that conform to the VA templates or any of the other known templates outlined in von Soden (1995), b) adjectives that agree with the noun they describe in terms of case and φ-features, and c) adjectives that follow nouns and precede verbs in non-predicative usages. Following the clearance of faulty entries, the total count of adjectives in the CDA amounted to 1433 adjectives.

The adjectives were then sorted according to the following set of parameters: their template, whether they are primary or secondary adjectives, whether or not they are VA, whether or not they are derived from adjectival/PC ROOTS, and whether or not they form a Stative.

Out of the 1433 adjectives, 838 adjectives are VA (182 derived from PC roots), 9 adjectives are very likely VA but feature uncharacteristic peculiarities such as vowel differences, 70 adjectives could not be clearly identified nor rejected as VA, and finally 516 adjectives were not VAs (results summarised in Figure 1 below).

The non-VA adjectives moreover fall into several classes: adjectival participles (i.e., participles that have become lexicalised as adjectives),[21] *-ī/-ān*-derived adjectives, and loans. In total, 113 adjectives were derived from participles of all template patterns (i.e., SIMPLE, INTENSIVE, CAUSATIVE / ACTIVE, MIDDLE, PASSIVE), 97 adjectives were intensive adjectives, and a total of 149 adjectives were derived concatenatively through the suffixation of *-ī* (102 adjectives), *-ān-* (22 adjectives), or the combination of both (25 adjectives). The remaining adjectives were loans (Figure 1).

With regard to which adjectives could derive Statives, the only semantic restriction found related to adjectives denoting ethnicities/geographic origins (e.g., *amurrû* 'Amorite', *elamû* 'Elamite', etc.). Perhaps owing to the contexts during which they are used being overwhelmingly of archival nature (i.e., inventory lists with items from said places), these adjectives were not found to derive Statives.

All other adjective types derived at least one Stative, as shown in (32). Example (32a) shows a Stative from a participle, (32b) shows the Stative derived from a denominal adjective derived with both *-ān-* and *-ī-*.

[21] Note that "regular" productive (active) participles like 'seer', 'eater', 'giver', have been excluded from this study. The participles that have lexicalised as adjectives have usually taken on meanings more akin to 'seeing', 'eating', 'giving'. Not all verbs derive participles of this type.

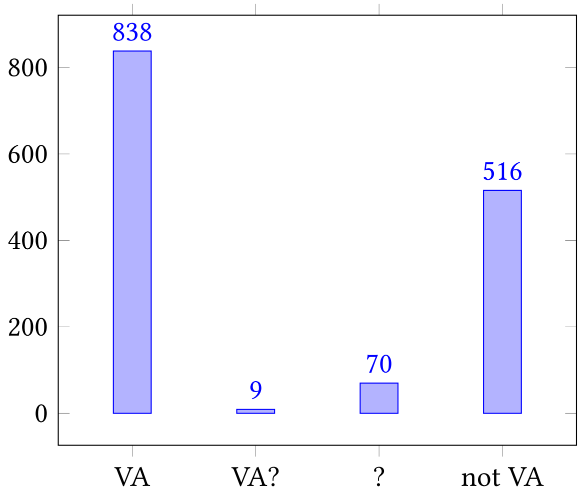


Figure 1: Distribution of VA among the total of 1433 adjectives

(32) More secondary Stative examples

a. *šumma liqpî=šu* ***šābul***

√*ʔbl* ‘dry’

if palate=his.SG.M CAUS.dry.PTCP.STAT.3SG.M

‘If his palate is dry (and covered with a deposit) …’ (Labat TDP 64:54)

b. *šumma imitt-a tirk-u* ***lumnānî***

if right-OBL dark.spot-NOM evilly.inclined.STAT.3SG.M

‘If there is a black spot on the right, he is prone to misfortune.’ (CT 28 29:14)

As initially expected in Section 4.2 above, the vast majority of Statives appears in conditional *šumma*-contexts. Further, the majority of Statives appears in the 3SG.M form, i.e., they do not feature any Stative-suffixes. As they also lack case-marking, I exclude the possibility of predicatively used “regular” adjectives.

The attestation of (adjectival) secondary Statives is incredibly rare vis-à-vis the attestation of primary Statives. As Kraus (1984) also noted in his study of the old Babylonian letters corpus (*altbabylonische Briefe*), out of 1200 Statives attested in the selected volumes investigated, only 20 Statives were formed of nominal or adjectival stems, i.e., of secondary Statives. In my present study, comprising the attestations provided by the *Chicago Assyrian Dictionary*, less than forty adjectives were found to certainly derive Stative forms.

From this statistical observation, I infer that the inheritance of a CoS event from the ROOT is a necessity for Stative formation. Drawing a CoS from a pragmatic/modal context appears to have been more costly, and thus so rarely observed. This necessity to inherit eventive information from a ROOT could also be argued to serve as further evidence for both the verbal, but particularly the resultative nature of the state described by the Stative. We may now revise the schema in Table 9 for the last time.

Table 9: Basic morphological Stative categorisation (final revision)

Primary Statives					Secondary Statives	
XaY(i)Z- + suffix					Stem + suffix	
Verbal ROOT				Adj. ROOT	Adj. stem	Nom. stem
Unerg.	Unacc.	Agent.-tr.	Exp.			
No Stat.	Root-inherited CoS				External CoS event	

I leave the question of diachrony open at this point, hypothesising that the resultative/Stative attaching to adjectival or nominal stems potentially derived from parallel contexts of primary Statives and adjectival/nominal stems in modal constructions, particularly in omens and medical texts. Given the scope of the present paper, this shall remain an endeavour for future studies to embark on.

5 Conclusion

This paper's goals were twofold: First, it set out to argue against the synchronic interpretation of all Statives as deadjectival verbs by presenting the allosemic Experiencer-subject alternation in primary Statives. This alternation could be accounted for by assuming that different functional heads, namely *a* and *v*, set different s-selectional requirements for the syntax to filter out the arguments necessary for their derivation. Ultimately, it shows that primary Statives are not syntactically deadjectival.

Secondly, this paper sought to motivate the Stative's function as a resultative state through cross-linguistic parallels of resultative constructions. I argued that the referenced CoS event is implicitly encoded in the ROOT of primary Statives. The deadjectival secondary Statives, which lack eventive encoding, must draw their CoS context from external syntactic or semantic/pragmatic sources, most commonly from modal particles/complementisers. The Stative requiring access to eventive encoding could serve as further evidence for its resultative function as well as an explanation for the scarcity of secondary Statives.

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Abbreviations

ACC	accusative
CAUS	causative
COM	common gender
CONJ	conjunction
CoS	change of state
CSTR	construct state
D	*Dopplungstamm* / INTENSIVE pattern
DAT	dative
DOR	direct-object restriction
F(EM)	feminine
G	*Grundstamm* / SIMPLE pattern
GEN	genitive
IPFV	imperfective
IPP	independent personal pronouns
M(ASC)	masculine
MOD	modal
N	*N-Stamm* / PASSIVE pattern
NEG	negation
NOM	nominative
OBL	oblique
PC	property concept
PFV	perfective
PN	personal name
PTCP	participle
PL	plural
RN	regnal name
Š	*Š-Stamm* / CAUSATIVE pattern
SG	singular
STAT	Stative
TAM	tense/aspect/mood
VA	verbal adjective

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Name index

Book index

Abbreviations used in book index

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CV Cleasby, Richard & Guðbrandur Vigfússon. 1874. *An Icelandic-English dictionary.* Oxford: Clarendon.

EDG Beekes, Robert. 2010. *Etymological dictionary of Greek.* Leiden: Brill.

eDiAna Hackstein, Olav, Jared Miller, Elisabeth Rieken & Ilya Yakubovich (eds.). N.d. *Digital philological-etymological dictionary of the minor Anatolian corpus languages.* https://www.ediana.gwi.uni-muenchen.de (05.09.2025).

eDIL N. A. 2019. *An electronic dictionary of the Irish Language, based on the Contributions to a Dictionary of the Irish Language.* https://dil.ie/ (23 November, 2023).

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EDPG Kroonen, Guus. 2013. *Etymological dictionary of Proto-Germanic.* Leiden: Brill.

ESJS Havlová, Eva, Ilona Janyšková and Adolf Erhart (ed.). 1989–2023. *Etymologický slovník jazyka staroslověnského.* Praha: Academia - Nakladatelství Československé Akademie věd.

EVP Renou, Louis. 1955–1969. *Études védiques et pāṇinéennes*, vols. I–XVII. Paris: Institut de Civilisation Indienne.

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EWGP Heidermanns, Frank. 1993. *Etymologisches Wörterbuch der germanischen Primäradjektive.* Berlin: De Gruyter.

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KEWA III Mayrhofer, Manfred. 1976. *Kurzgefasstes etymologisches Wörterbuch des Altindischen*, vol. III. Heidelberg: Universitätsverlag Winter.

Labat TDP Labat, René. 1951. *Traité akkadien de diagnostics et prognostics médicaux*, vol. 1 (Collections de travaux de l'académie internationale d'histoire des sciences 7). Leiden: Brill.

Lambert BWL Lambert, Wilfred G. 1996. *Babylonian wisdom literature.* Winona Lake: Eisenbrauns.

LEIA Vendryes, Joseph, E. Bachellery & Pierre-Yves Lambert (eds.). 1959-1996. *Lexique etymologique de l'irlandais ancien.* 7 vols. Dublin: Institute for Advanced Studies.

LIV2 Rix, Helmut. 2001. *Lexikon der indogermanischen Verben.* Bearbeitet von Martin Kümmel, Thomas Zehnder, Reiner Lipp und Brigitte Schirmer. 2nd ed. Wiesbaden: Reichert.

LIV2 Addenda Kümmel, Martin J. 2023. *Addenda und Corrigenda zu LIV2.* Version 11.08.2023 16:56. http://www.martinkuemmel.de/liv2add.pdf.

LKG Ulvydas, Kazys (ed.). 1971. *Lietuvių kalbos gramatika.* 2: Morfologija. Vilnius: Mintis.

LKŽ Lietuvių Kalbos Institutas. 1941–2002. *Lietuvių kalbos žodynas.* Vilnius: Lietuvių kalbos ir literatūros institutas. https://www.lkz.lt.

LP Wills, Tarrin (ed.). 2024. *Lexicon poeticum.* https://lexiconpoeticum.org (10 October, 2024).

MhdGr Klein, Thomas, Hans-Joachim Soms & Klaus-Peter Wegera. 2009. *Mittelhochdeutsche Grammatik.* Teil 3: Wortbildungslehre. Boston: De Gruyter.

MLLVG Sokols, Ēvalds, Anna Bergmane, Rūdolfs Grabis & Milda Lepika (eds.). 1959. *Mūsdienu latviešu literārās valodas gramatika* 1: Fonētika un morfoloģija. Rīga: Latvijas PSR Zinātņu akadēmijas Latviešu valodas institūts.

NIL Wodtko, Dagmar S., Britta Irslinger & Carolin Schneider (eds.). 2008. *Nomina im indogermanischen Lexikon*. Heidelberg: Winter.

OgnS Fritzner, Johan. 1867. *Ordbog over det gamle norske Sprog*. Kristiania: Feilberg & Landmarks.

ONP Sigurðardóttir, Aldís (ed.) 2005–. *Old Norse prose dictionary*. https://onp.ku.dk/onp/onp.php.

RIMA 2 Grayson, A. Kirk. 1991. *Assyrian Rulers of the Early First Millennium BC*, vol. 1, (1114–859 BC) (The Royal Inscriptions of Mesopotamia 2). Toronto: University of Toronto Press.

TIM 2 Cagni, Luigi. 1980. *Briefe aus dem Iraq Museum* (Texts in the Iraq Museum (TIM) 2). Leiden: Brill.

WAP Baetke, Walter. 1965–1968. *Wörterbuch zur altnordischen Prosaliteratur*. Berlin: Akademie Verlag.

YOS 2 Lutz, Henry F. 1917. *Early Babylonian letters from Larsa* (Yale Oriental Series 2). New Haven: Yale University Press.

Subject index

Form index

Order of languages:
Proto-Indo-European,
Anatolian (Proto-Anatolian, Palaic, Hittite, Cuneiform Luwian, Hieroglyphic Luwian, Lycian, Lydian),
Tocharian (Common Tocharian, Tocharian A, Tocharian B),
Indo-Iranian (Proto-Indo-Iranian, Sanskrit, Proto-Iranian, Avestan, Khotanese, Ossetic, Old Persian, Middle Persian, Pashto),
Armenian (Proto-Armenian, Classical Armenian),
Greek (Proto-Greek, Mycenaean Greek, Ancient Greek, Modern Greek),
Italic (Oscan, Latin, Italian, Spanish),
Celtic (Proto-Celtic, Proto-Goidelic, Gaulish, Ogam Irish, Old Irish, Middle Irish, Modern Irish, Proto-British, Old British, Old Breton, Middle Breton, Modern Breton, Old Cornish, Middle Cornish, Modern Cornish, Old Welsh, Middle Welsh, Modern Welsh),
Germanic (Proto-Germanic, Gothic, Langobardic, Proto-Norse, Old (West) Norse, Old Danish, Old Swedish, Modern Norwegian, Old English, Middle English, Old Frisian, Old Saxon, Old High German, Middle High German, Early New High German, New High German, Middle Low German),
Balto-Slavic (Proto-Balto-Slavic, Proto-Baltic, Old Prussian, Old Lithuanian, Modern Lithuanian, Latvian, Proto- and Common Slavic, Old Church Slavonic, Russian Church Slavonic, Old Russian, Modern Russian, Czech, Polish, Serbo-Croatian),
Albanian (Proto-Albanian, Albanian),
Akkadian,
Hebrew.

English alphabetical order is followed unless otherwise indicated.

Proto-Indo-European The alphabetical order follows that of LIV[2]. Syllabic liquids and nasals are ordered after their non-syllabic counterparts.

Deadjectival verb formation in Indo-European and beyond

The lexicalization of property concepts (PCs) — basic adjectival states such as *big*, *small*, *hard*, *soft*, *black*, *white*, *fast*, *quick* — has long been a central concern of both typology and linguistic theory. In historical Indo-European linguistics, PCs are most often examined through the lens of the “Caland System”, a complex derivational network of suffixes that structure the expression of such concepts. Yet while the nominal and adjectival exponents of this system have been intensively studied, their verbal counterparts remain comparatively underexplored. This volume takes up that challenge by asking: which classes of adjectives serve as inputs to verbalization, what syntactic and semantic alternations emerge in these derived verbs, and what do these patterns reveal about the status of “adjectival roots” in Indo-European and beyond? The contributions collected here investigate the morphology, syntax, and semantics of deadjectival verbs across the older Indo-European languages. By situating these formations within the broader typology of adjective derivation, property concept encoding, and primary vs. secondary verb formation, the volume forges new connections between typological generalizations, theoretical models, and historical-comparative evidence. In doing so, it not only illuminates the dynamics of verbal derivation in Indo-European but also advances our understanding of how the pathways from property concepts to verbal predicates are structured cross-linguistically.